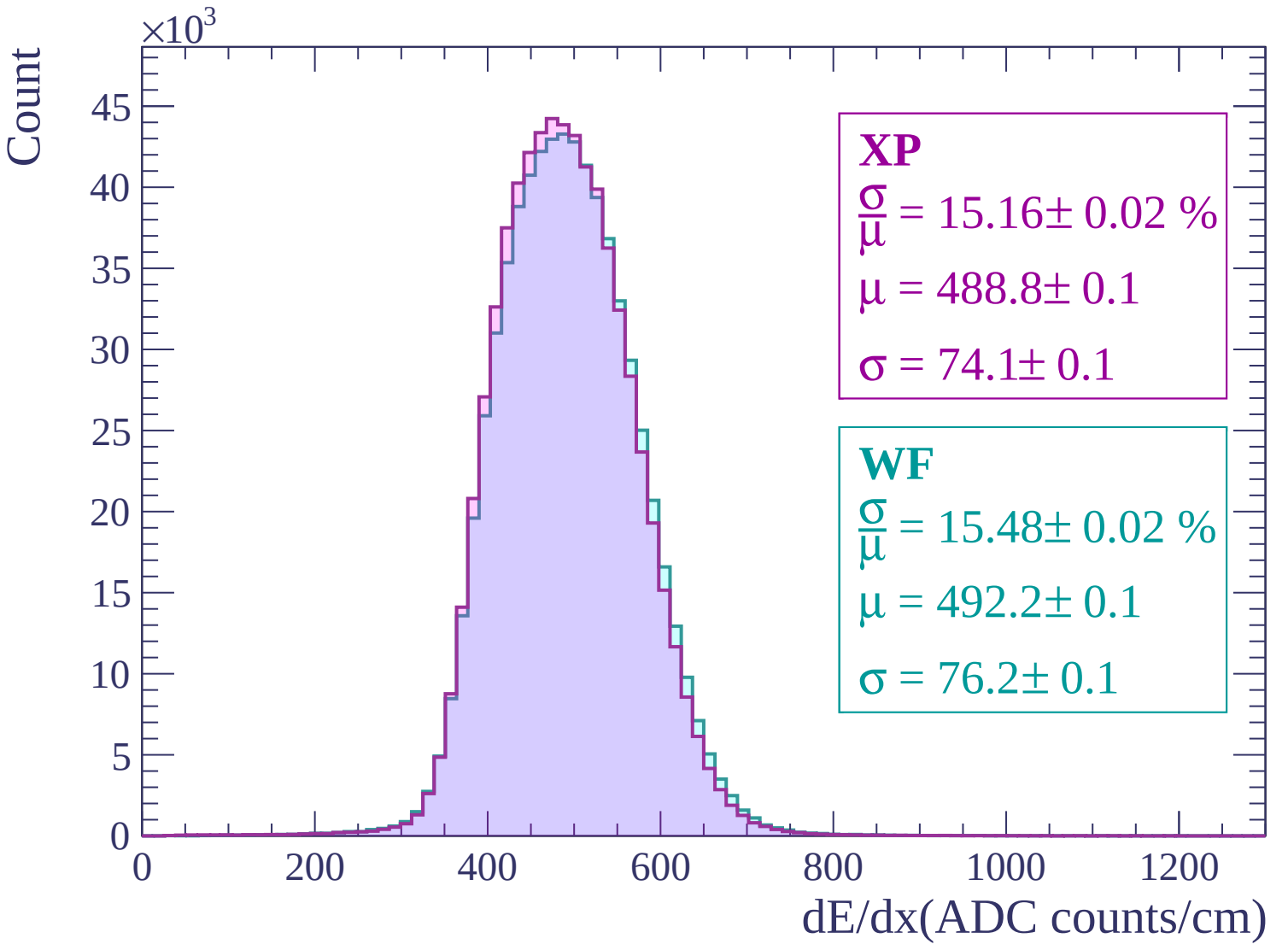
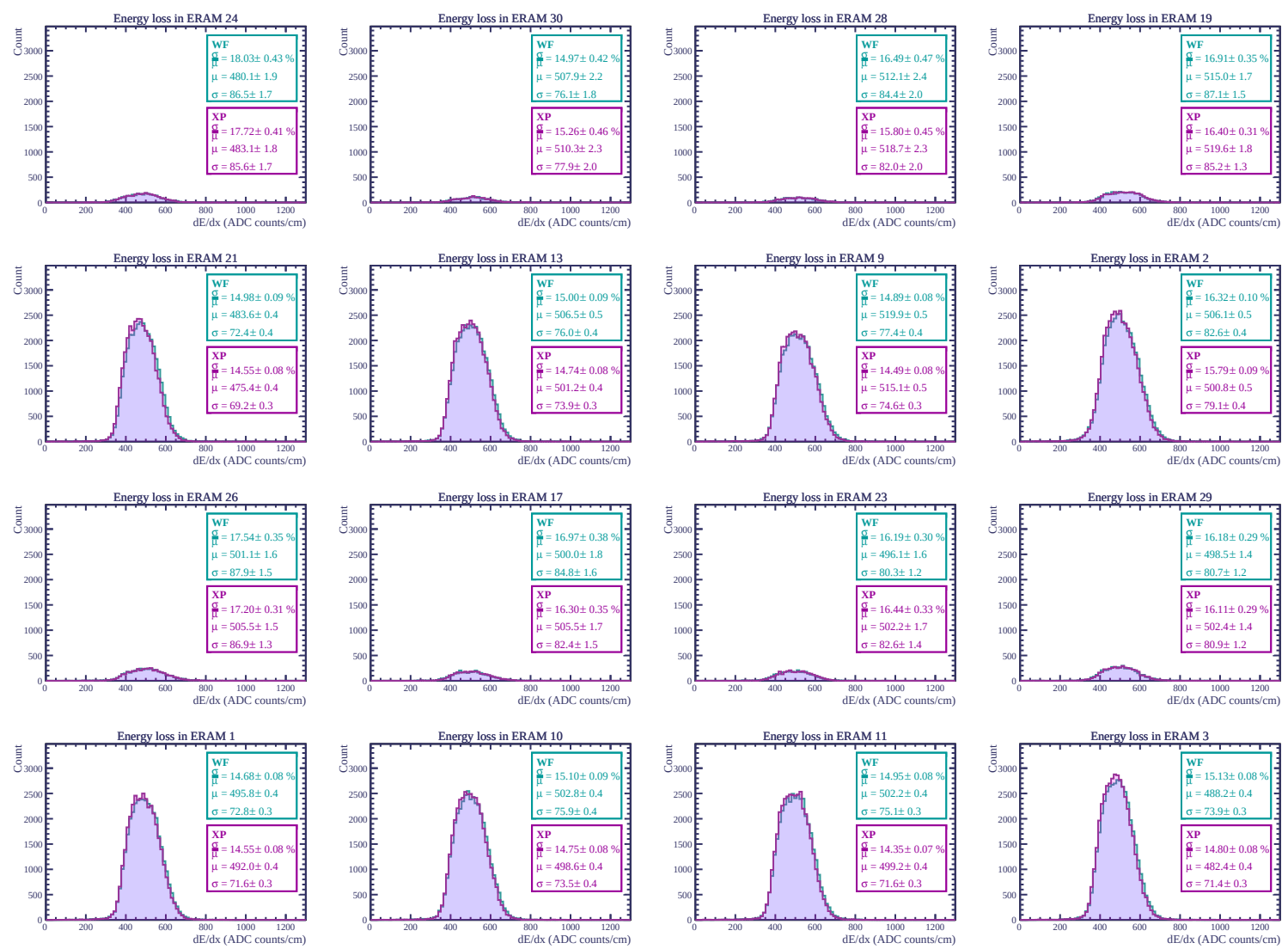
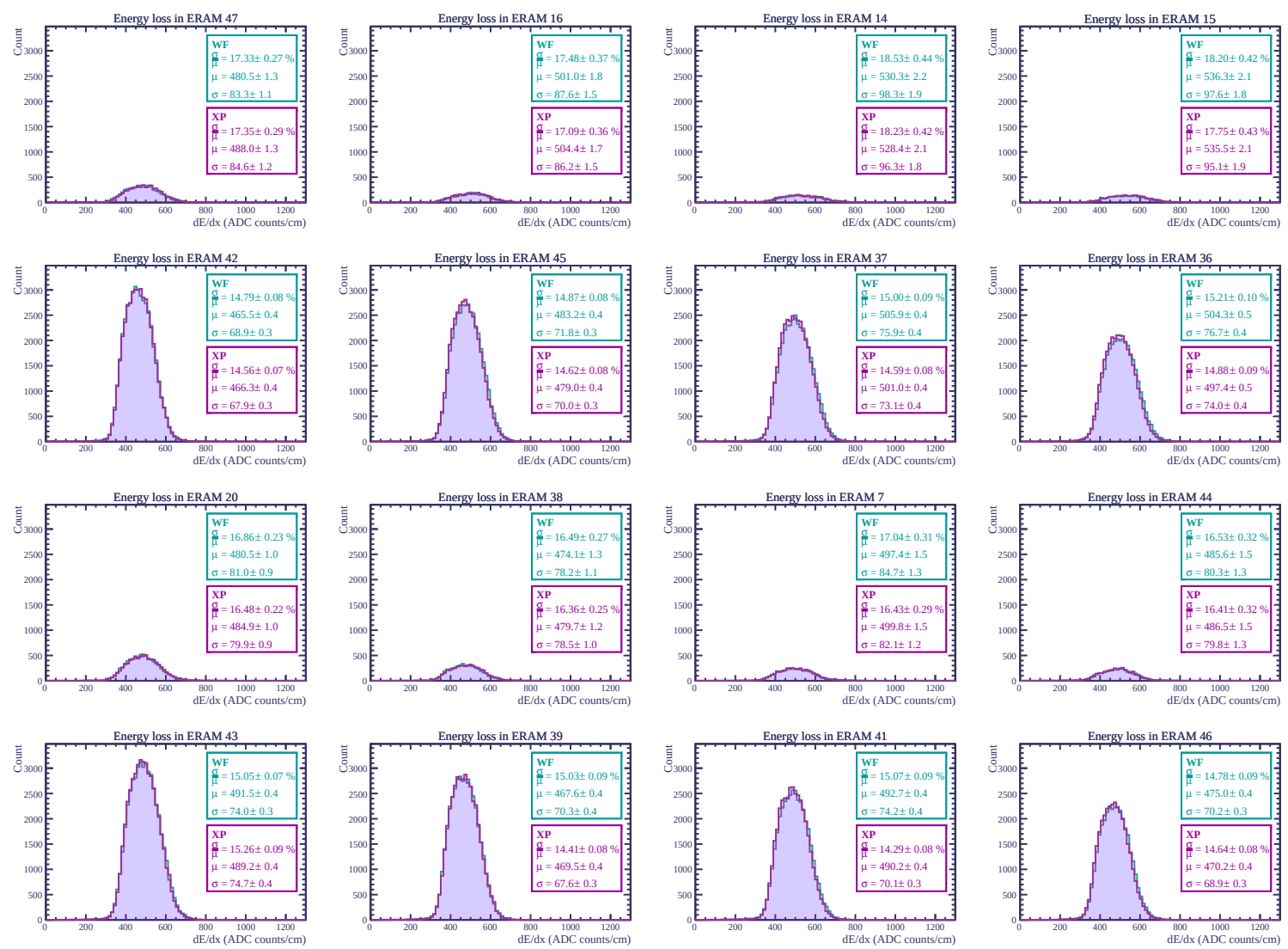
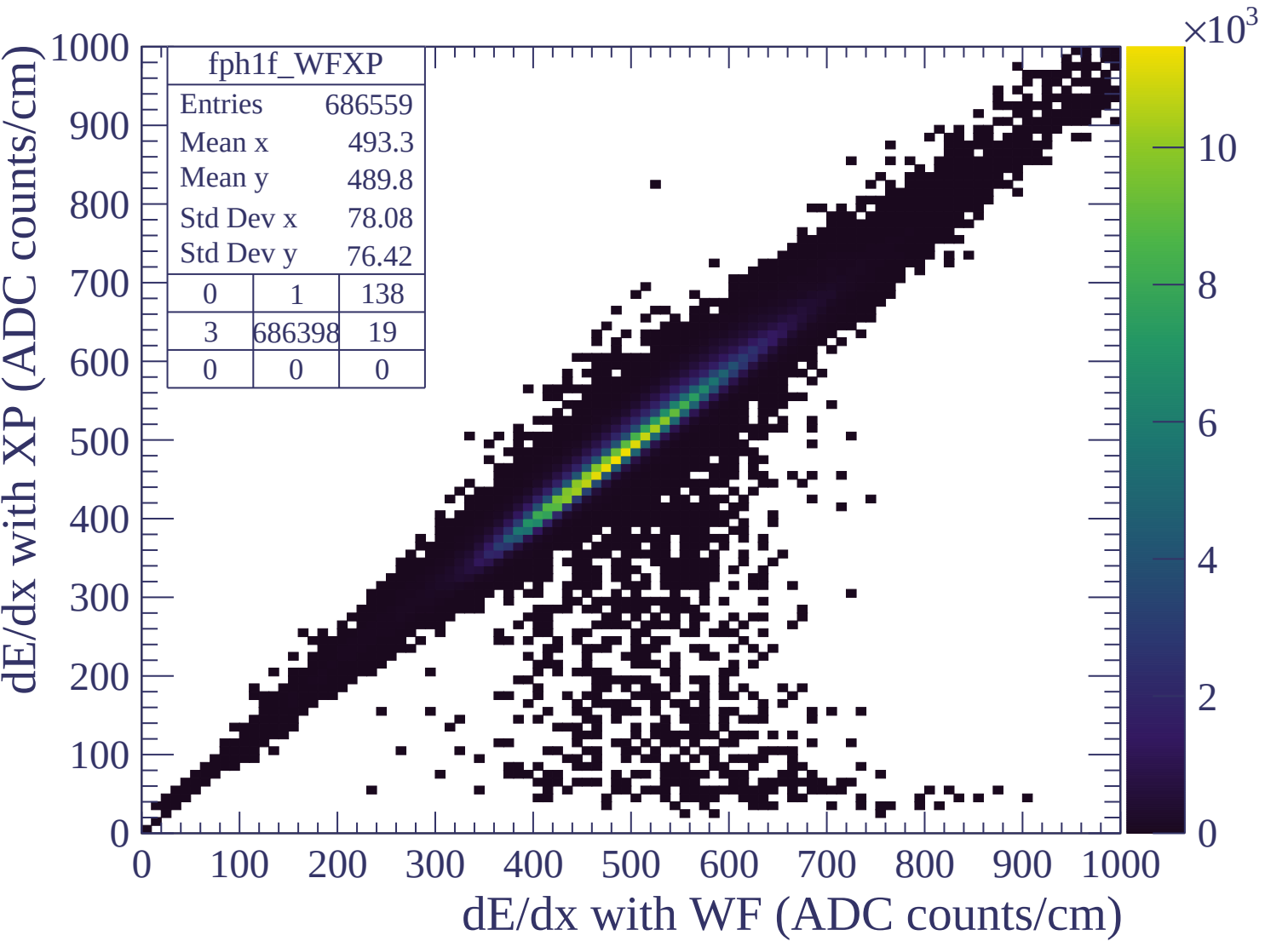


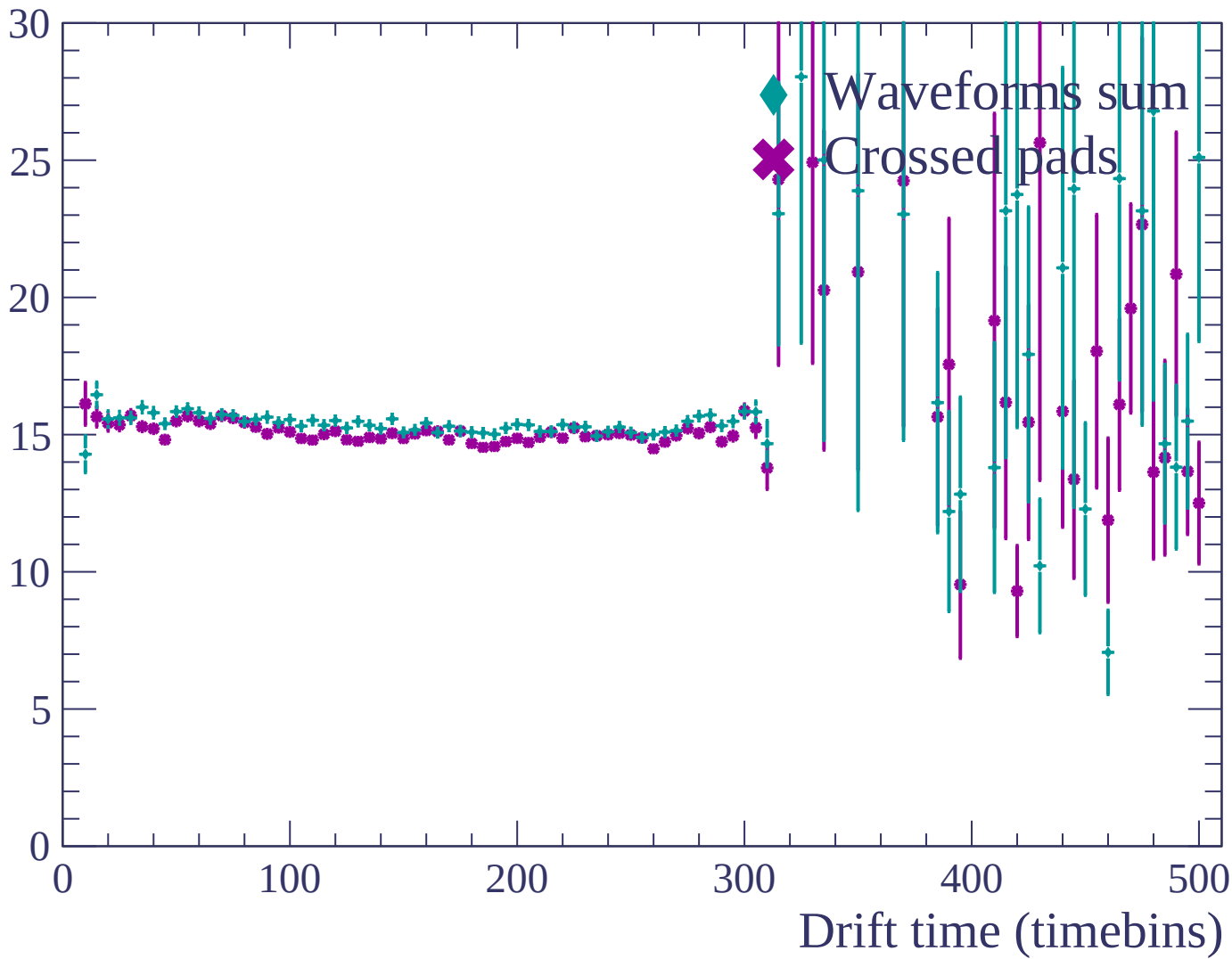


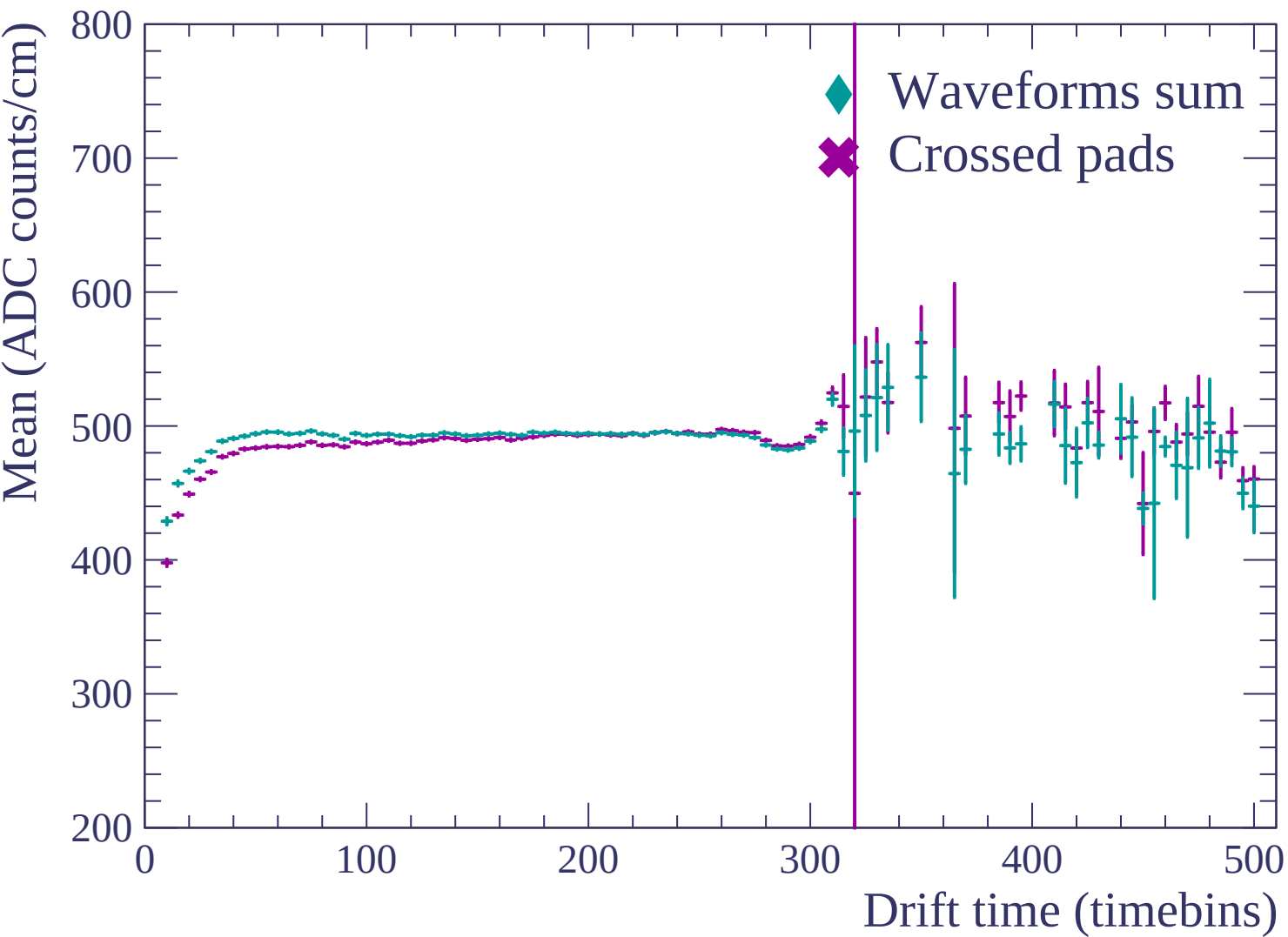


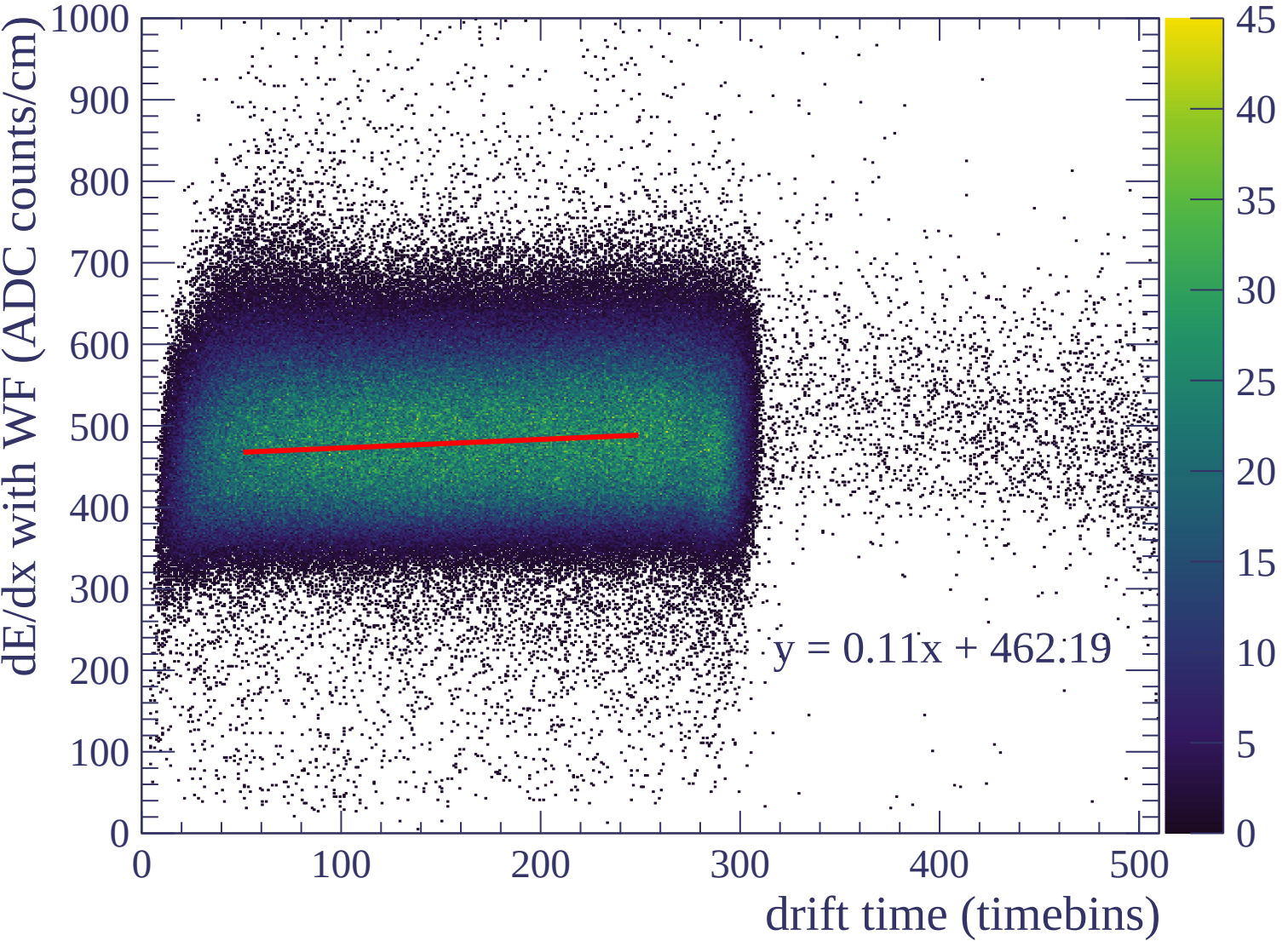


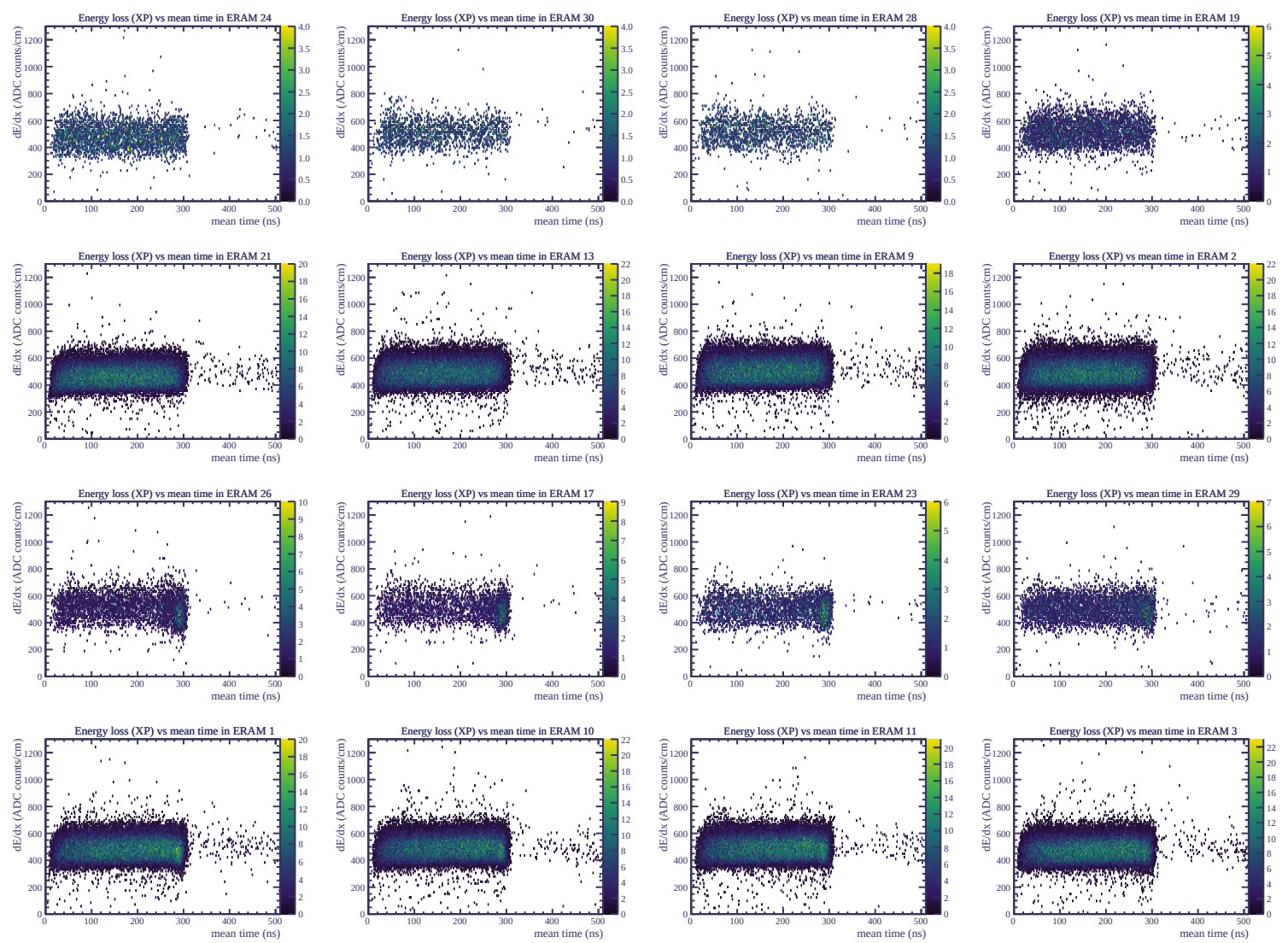


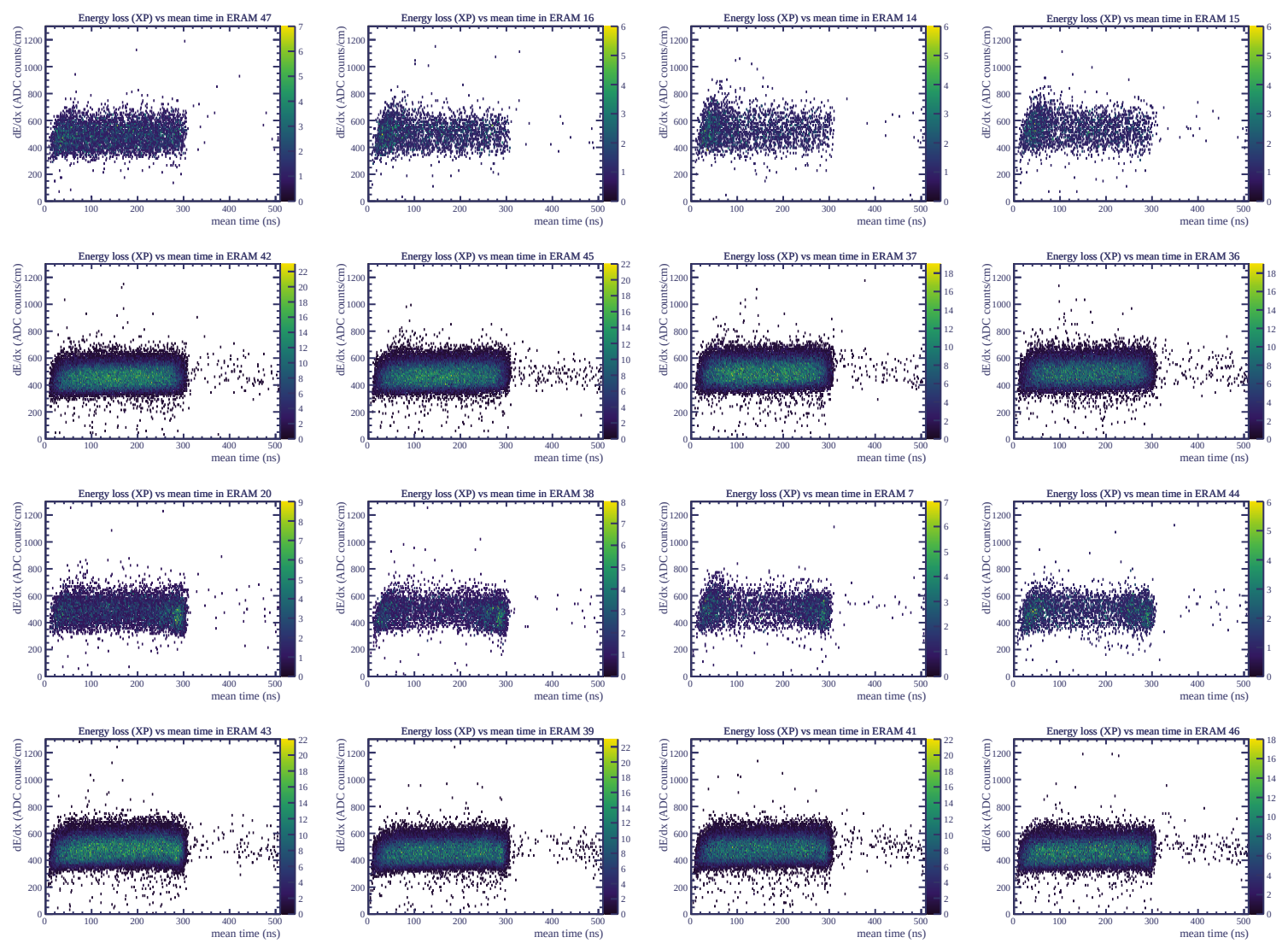
Resolution (%)

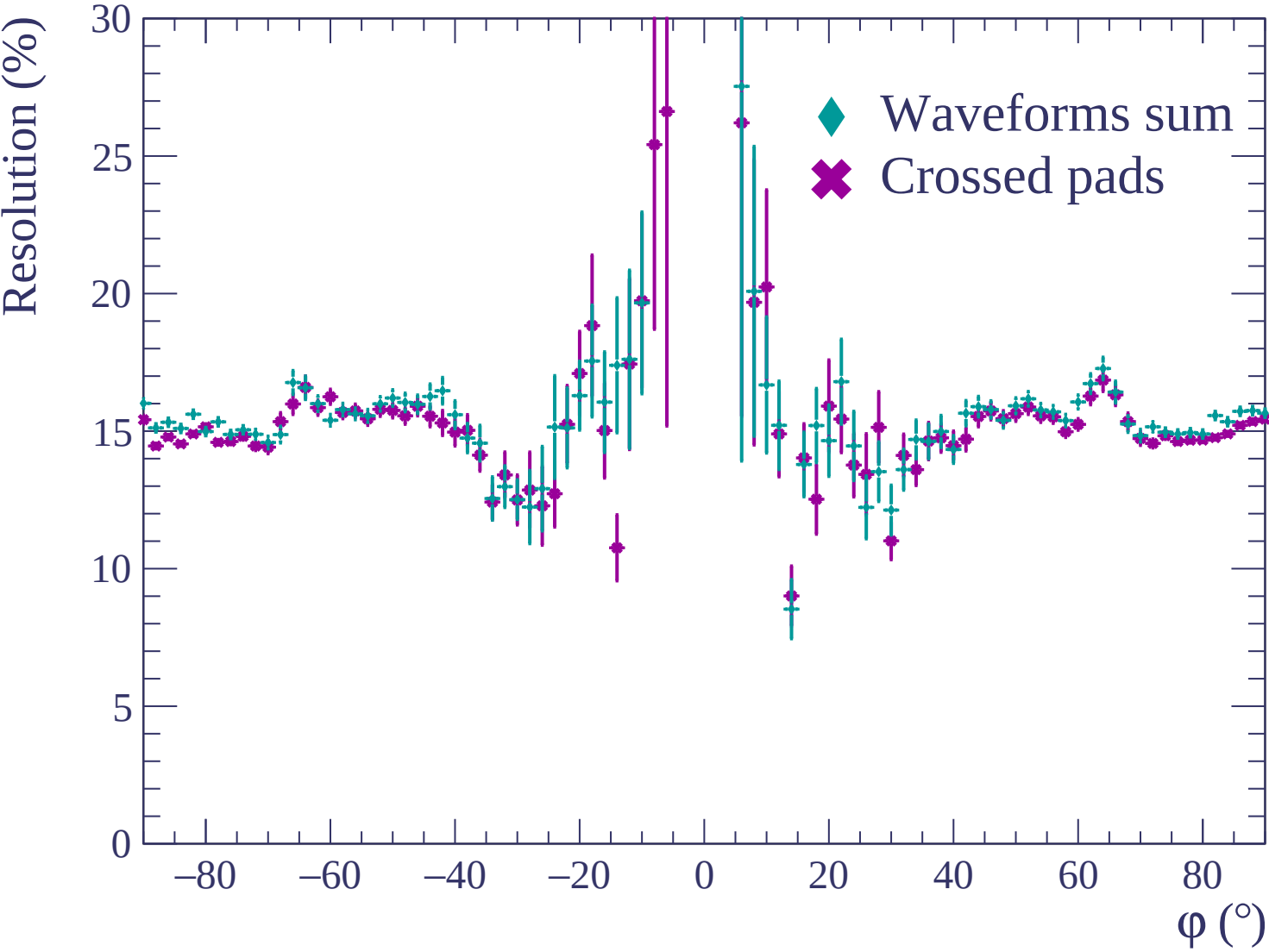


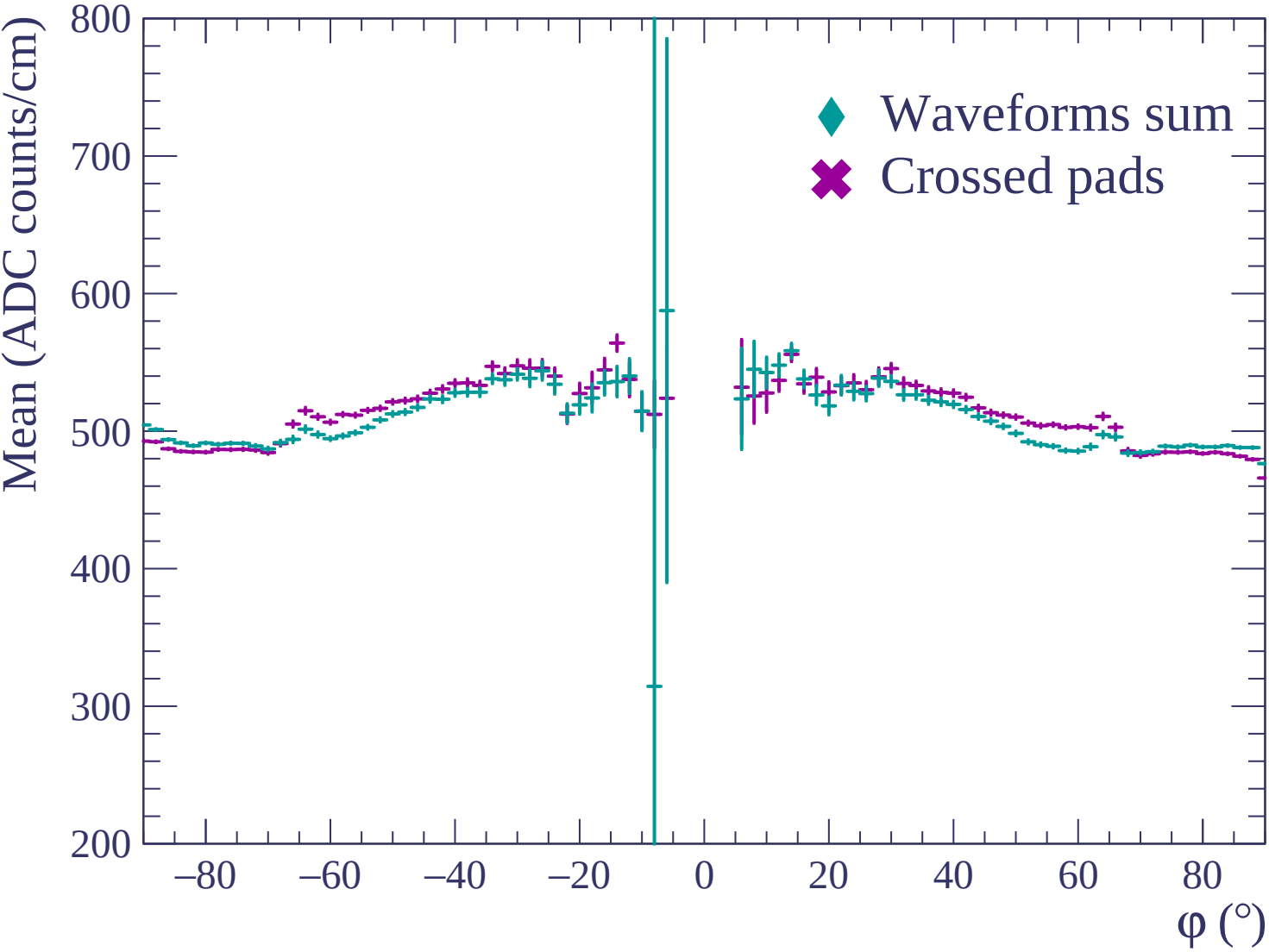


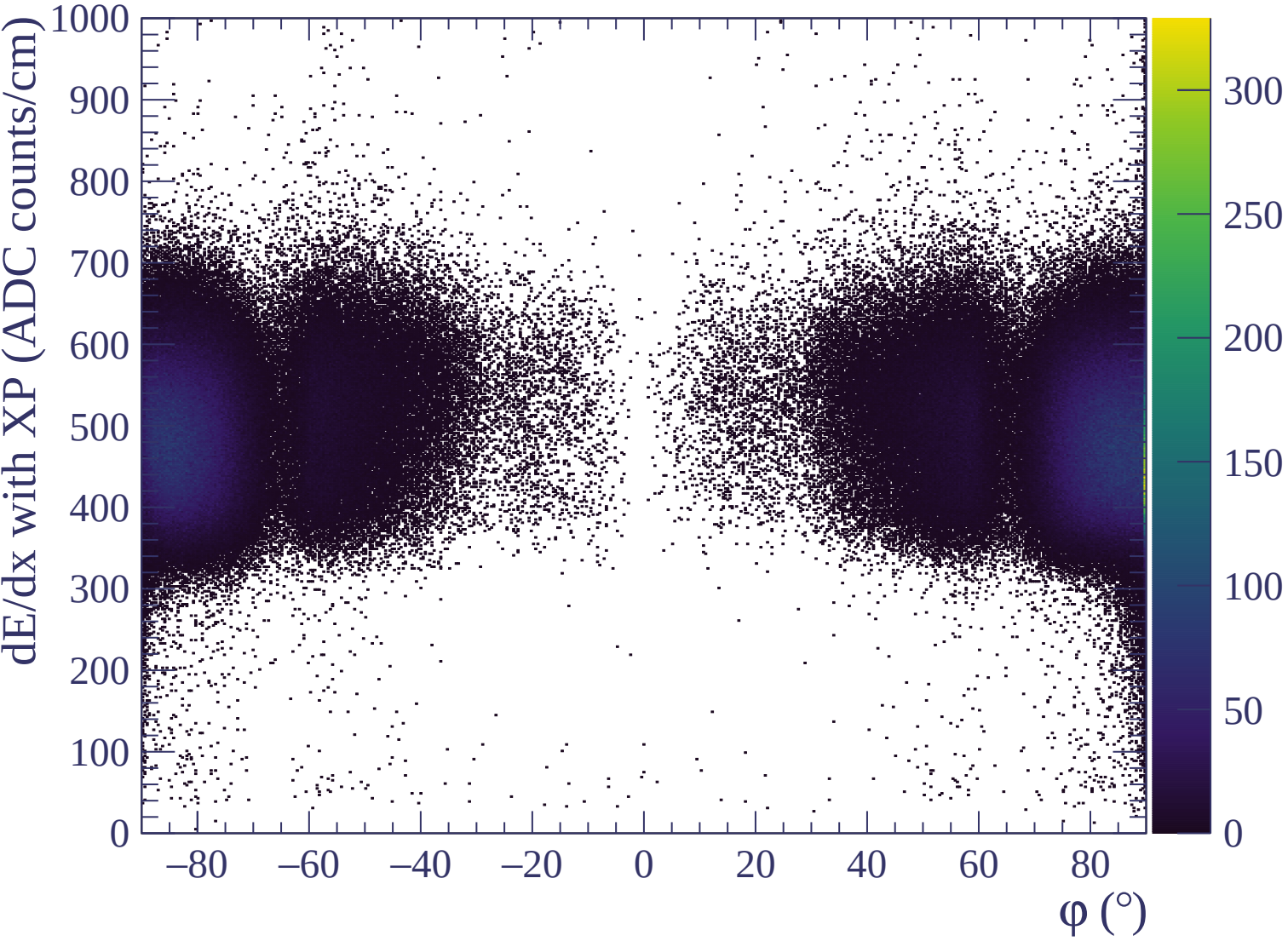


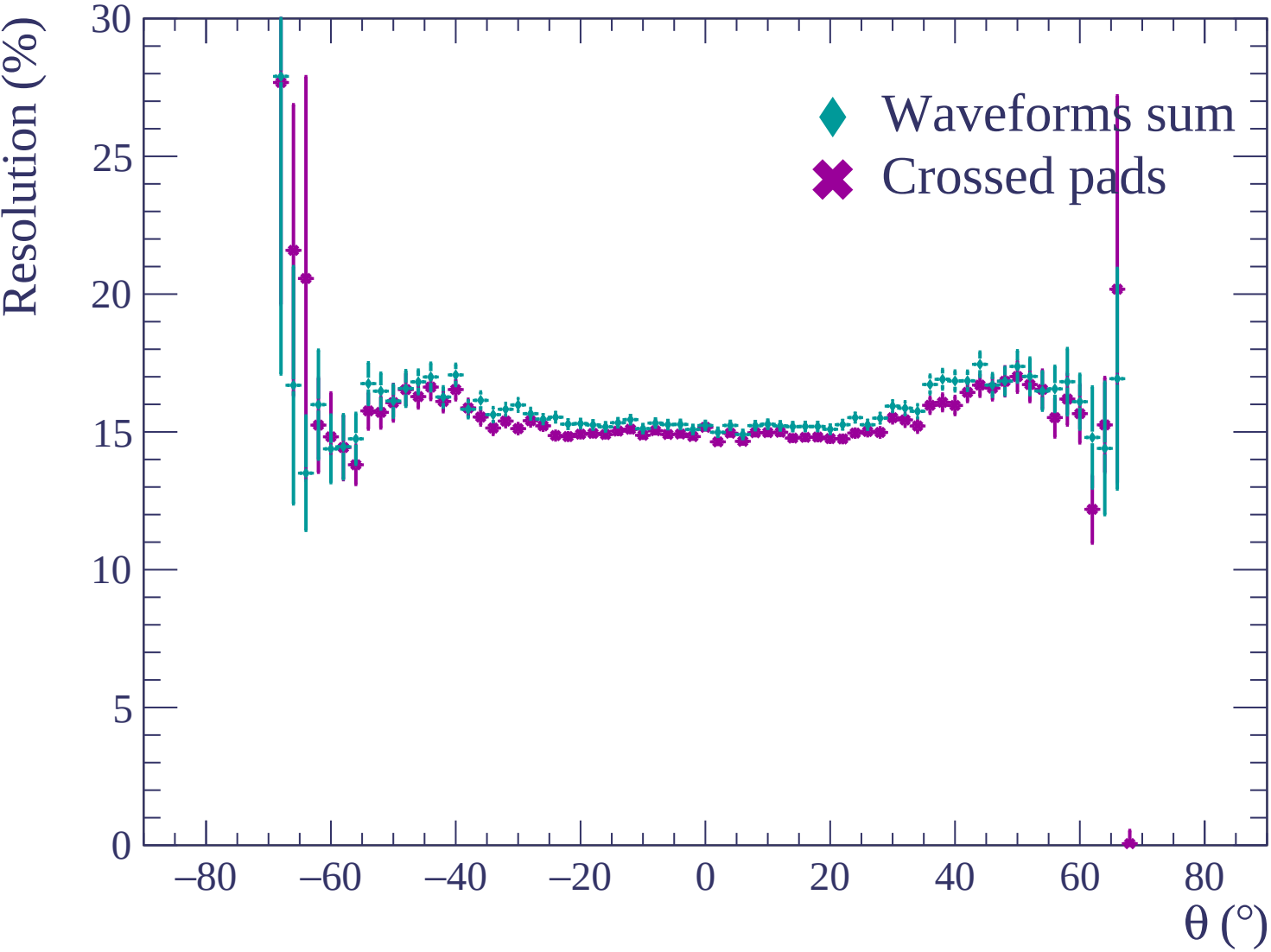


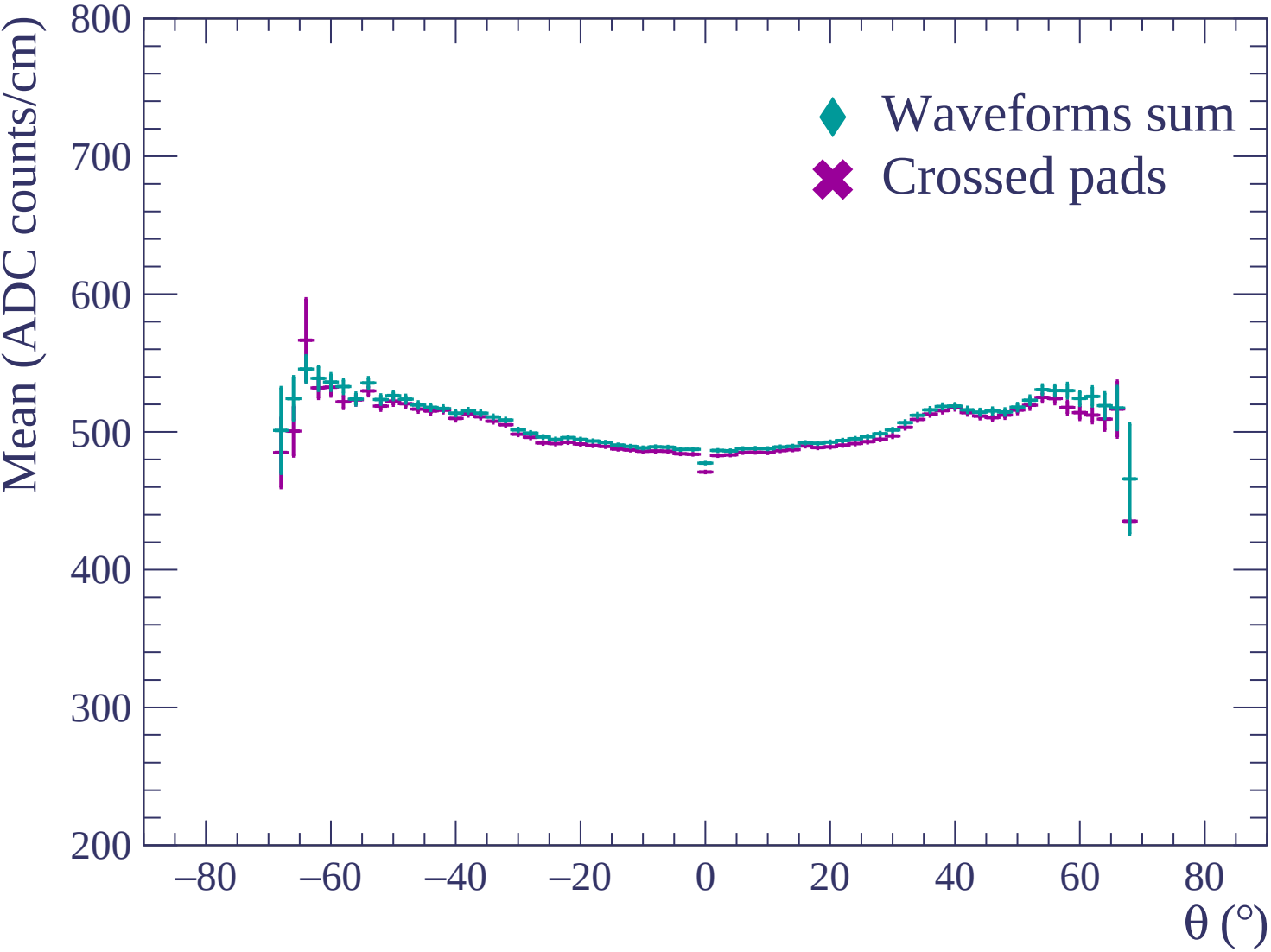


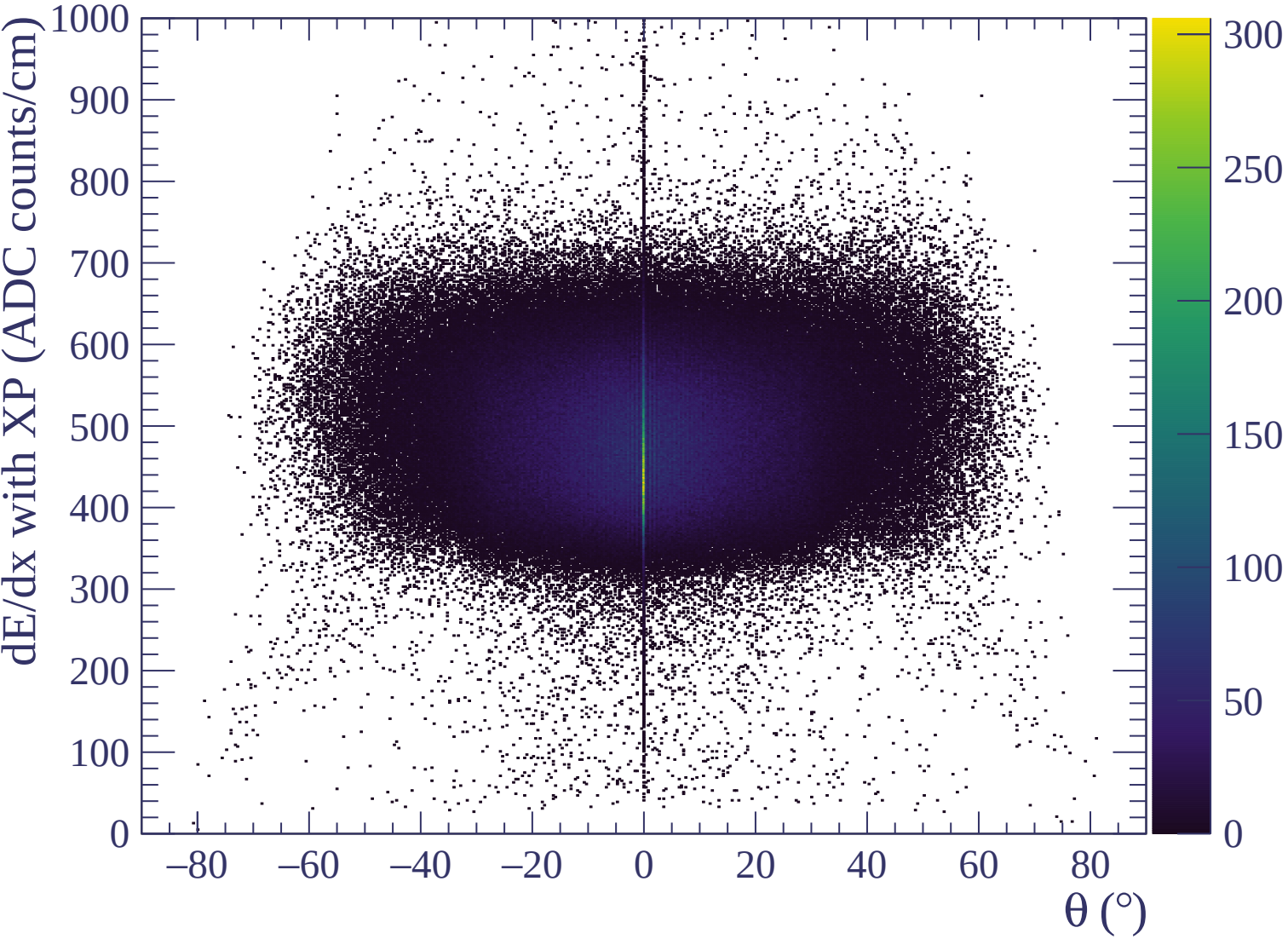


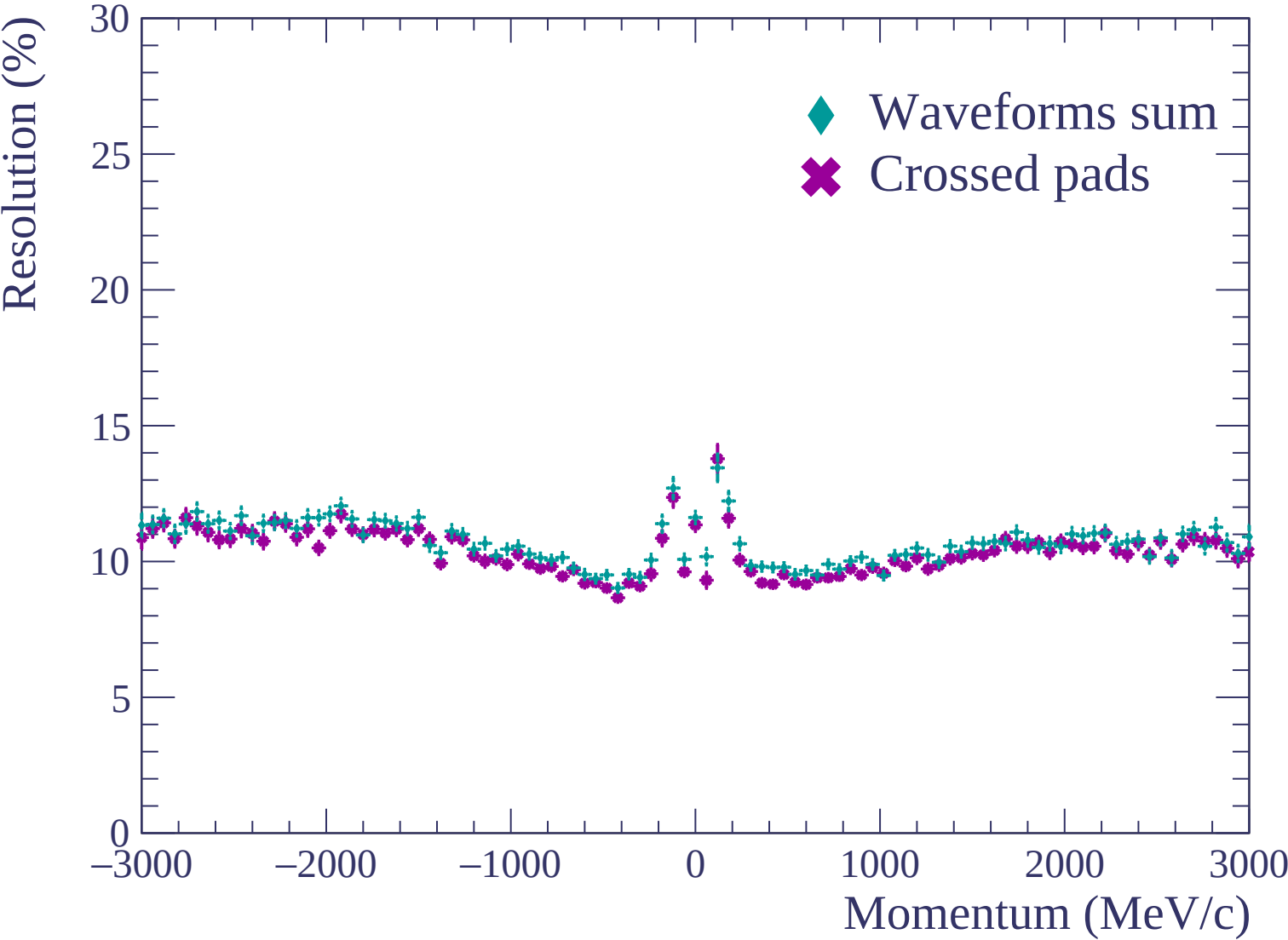


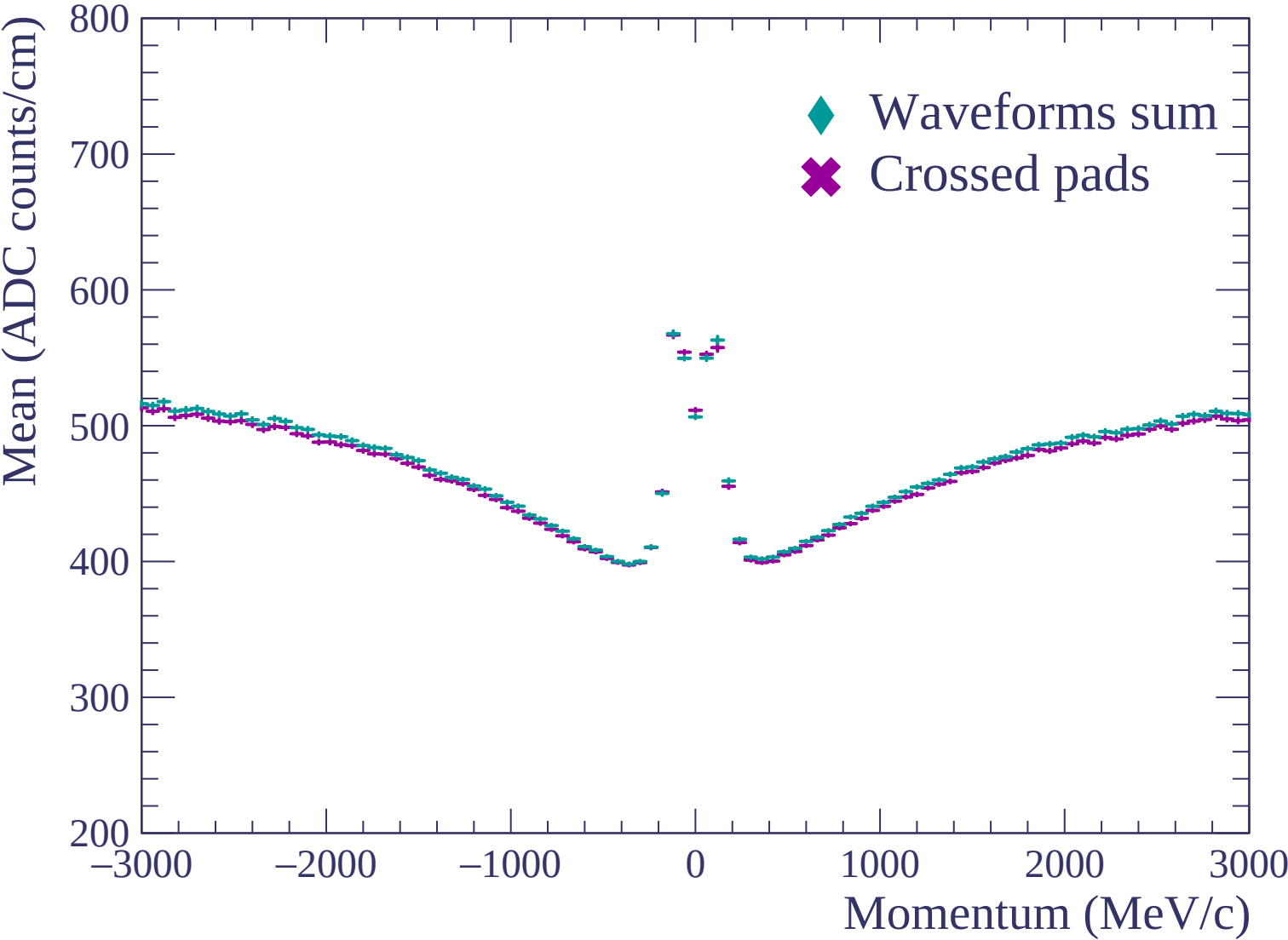


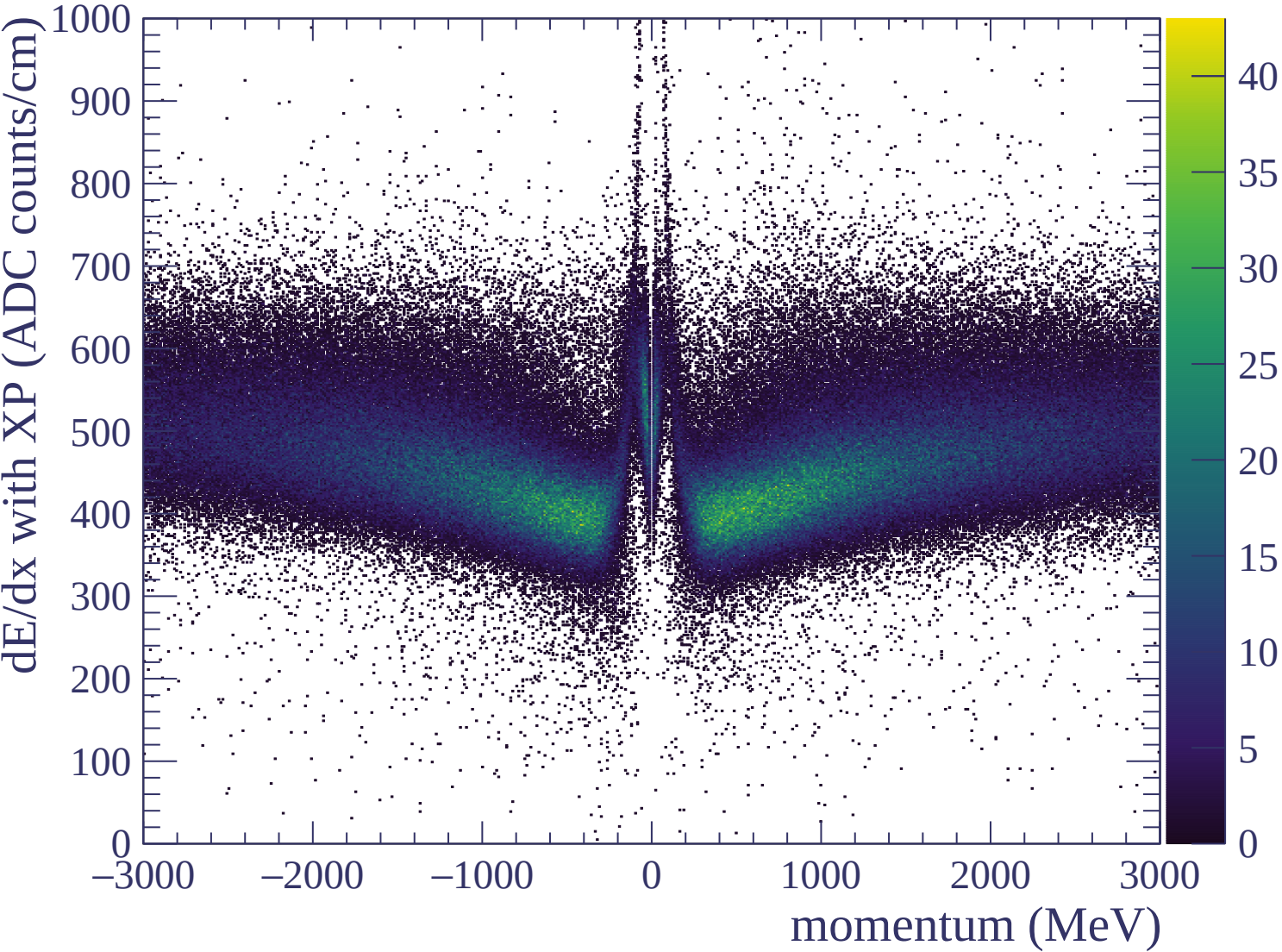


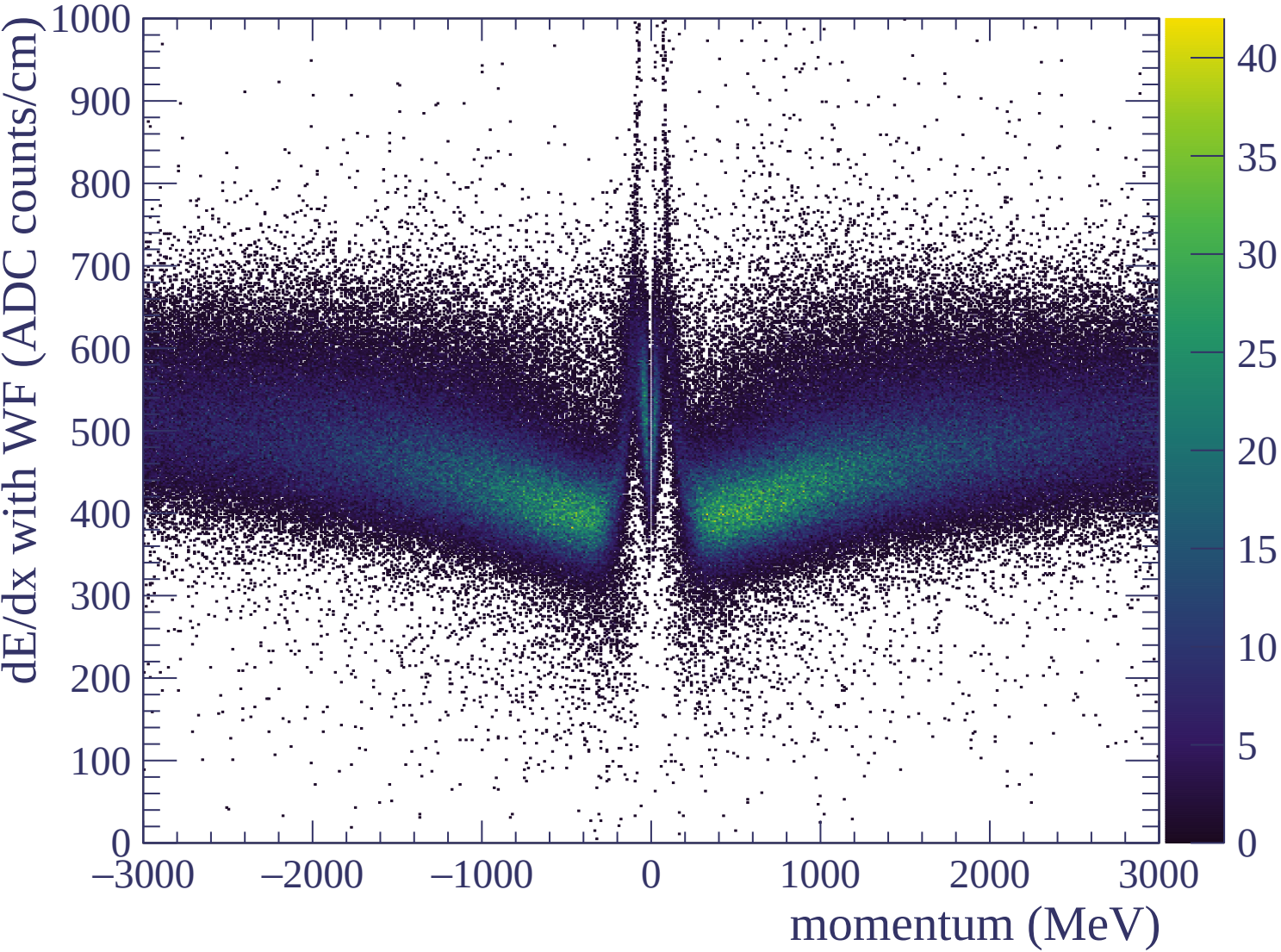


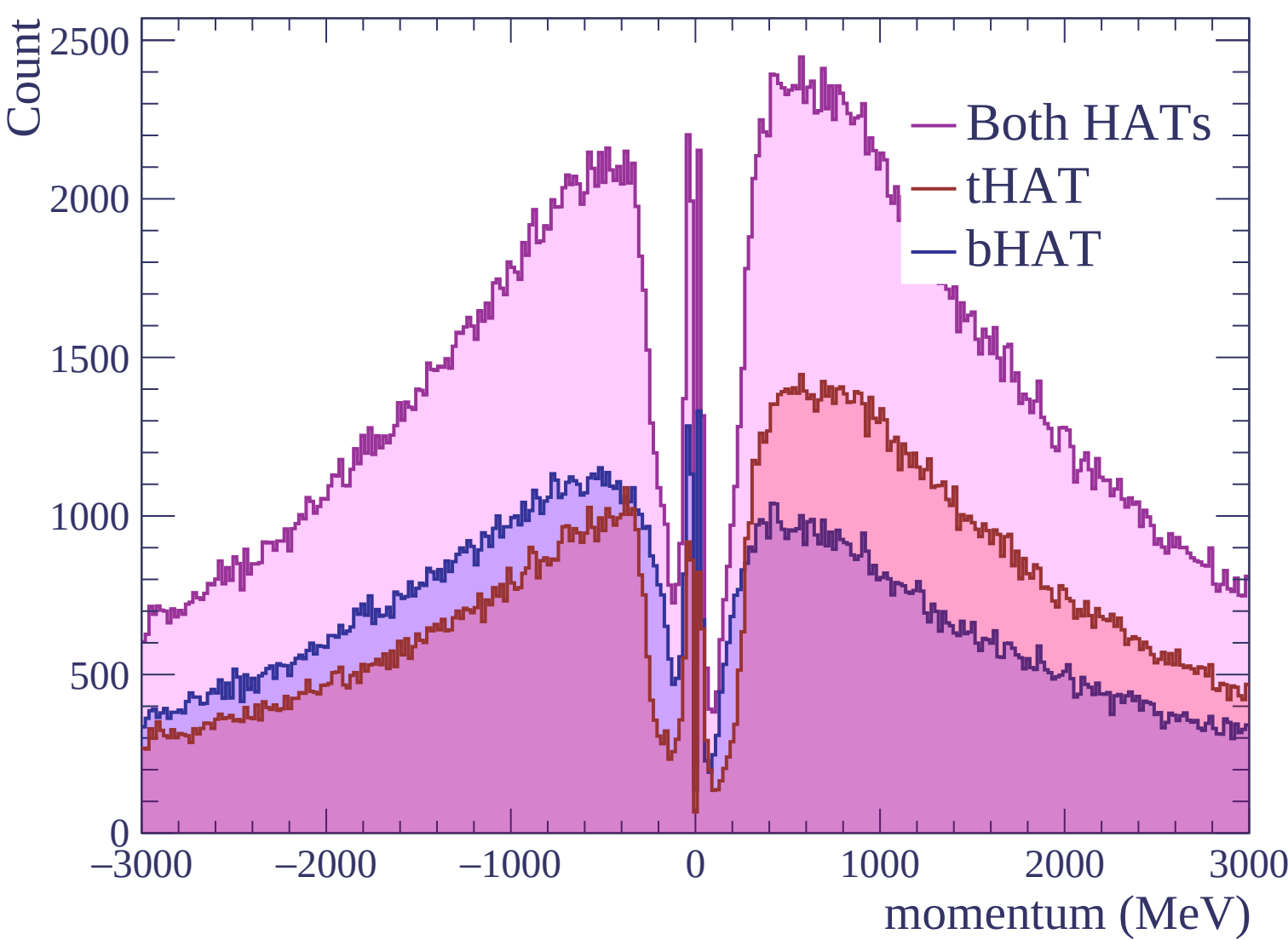


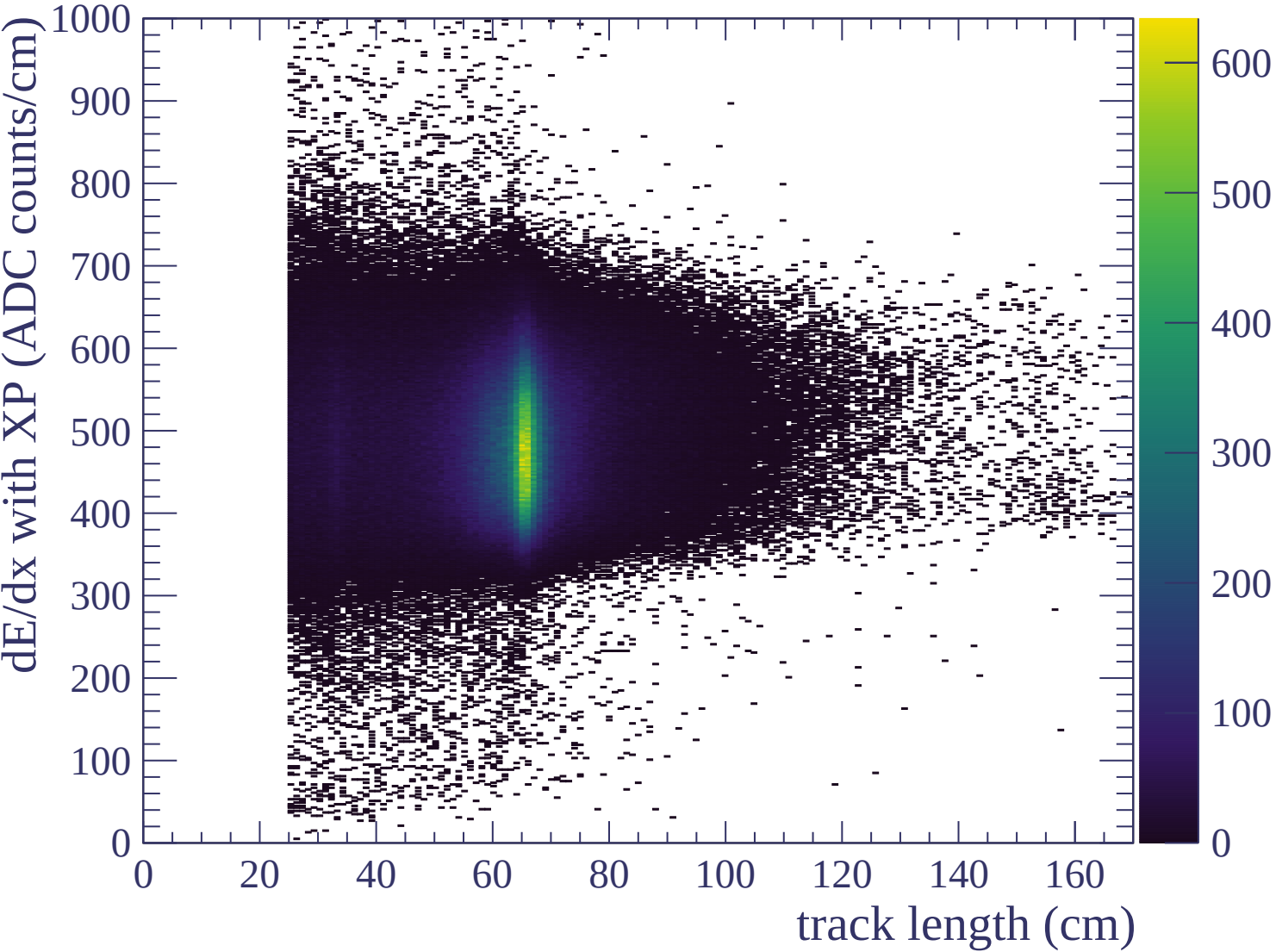


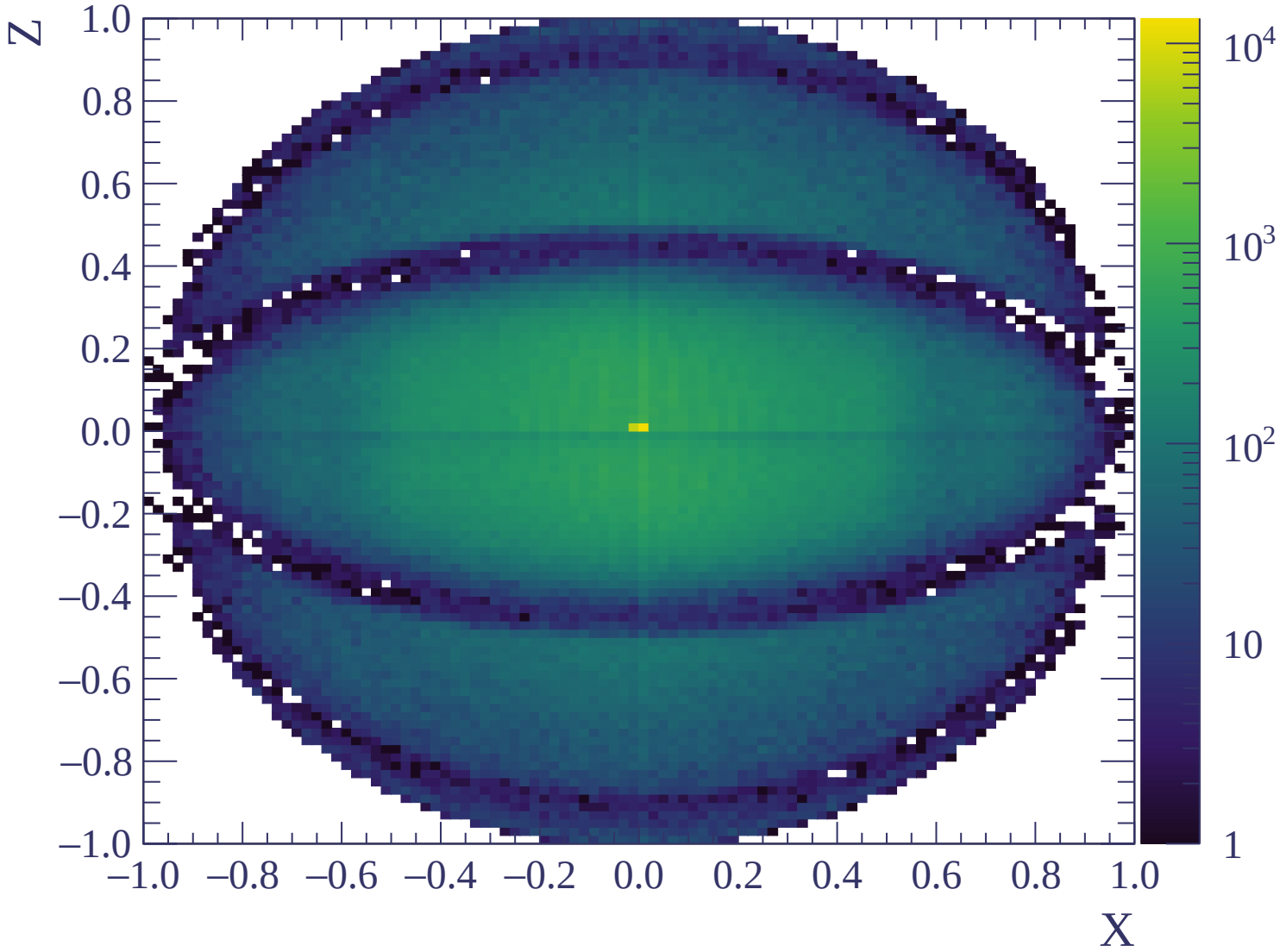


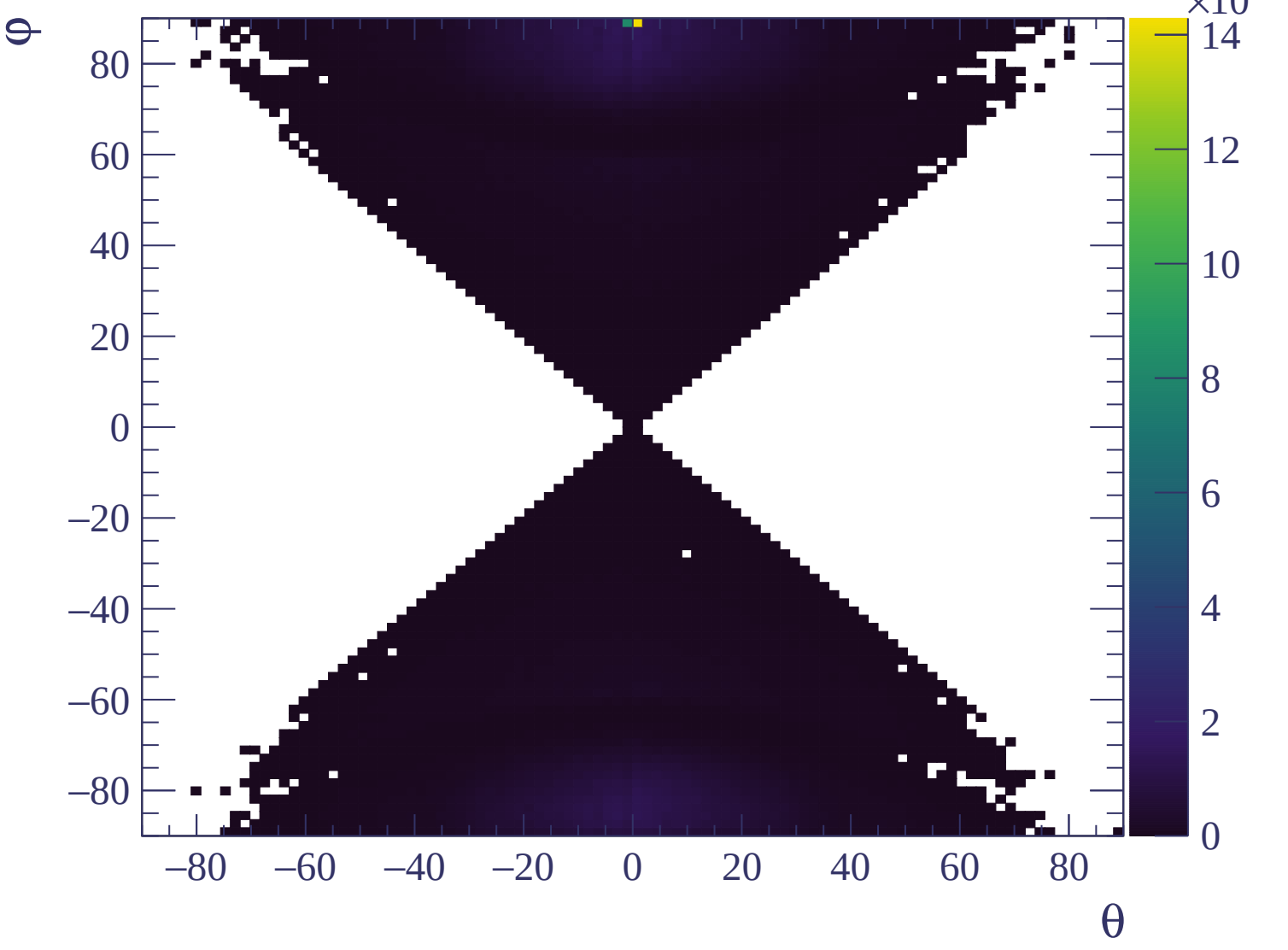


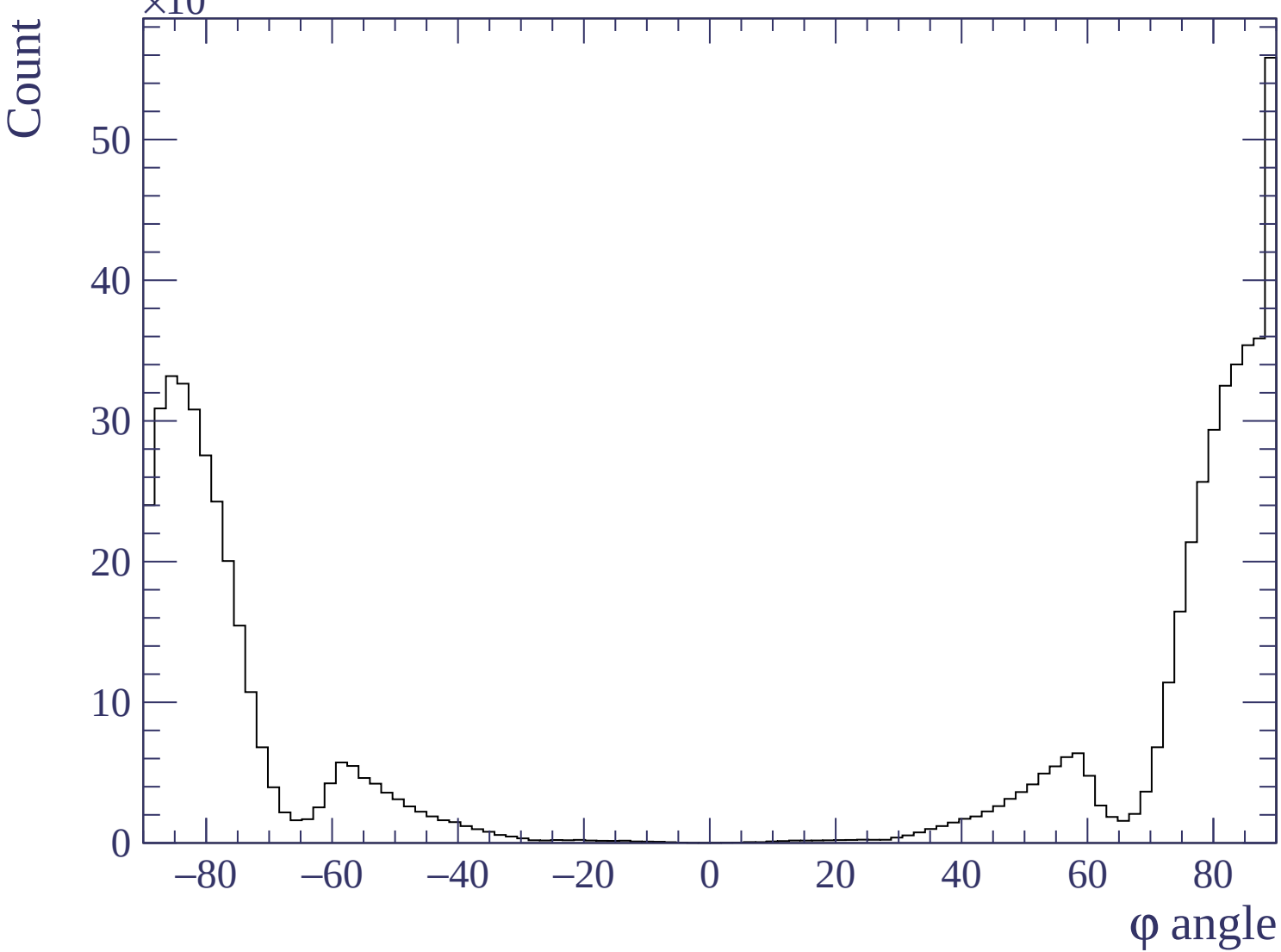


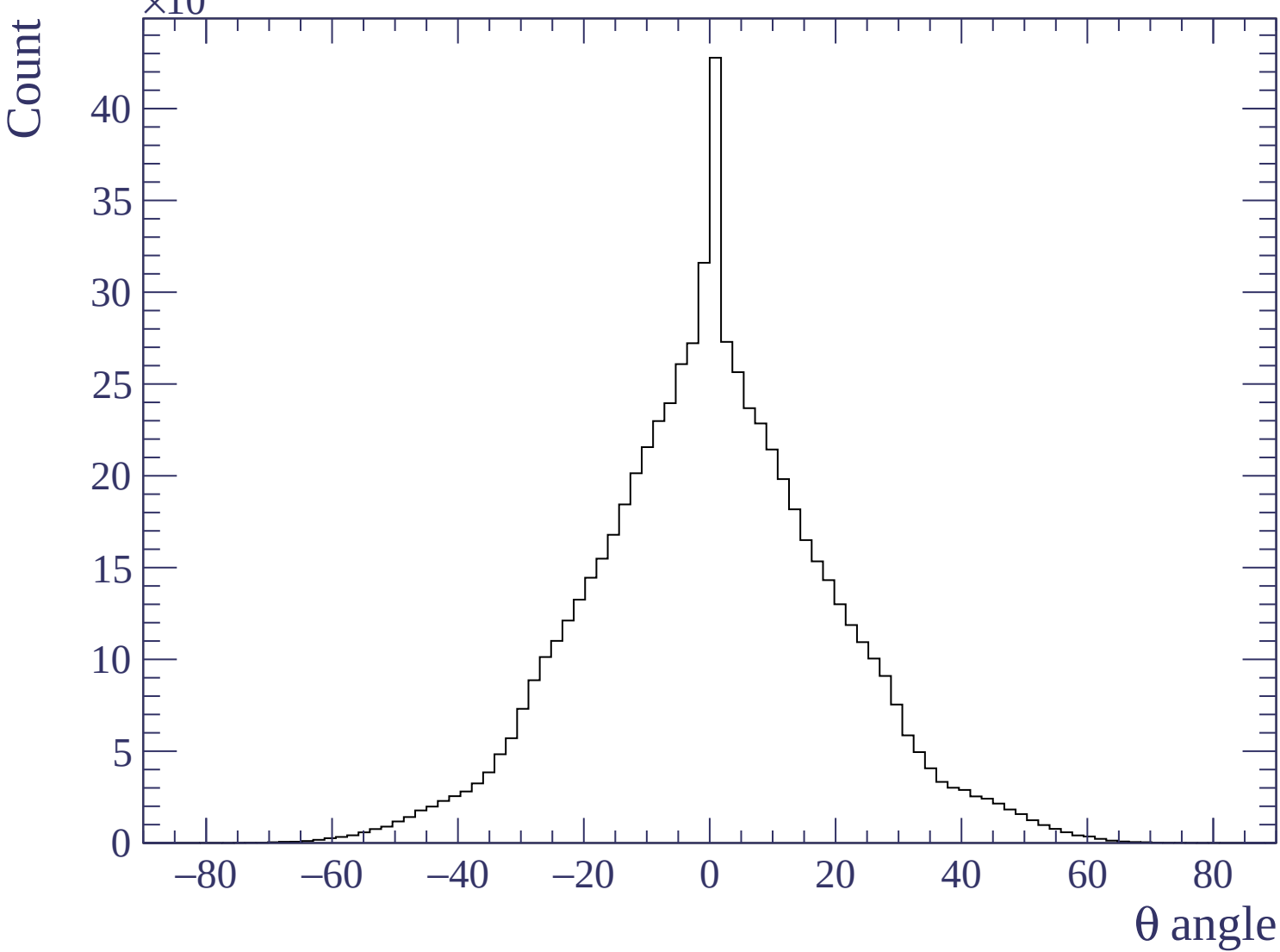


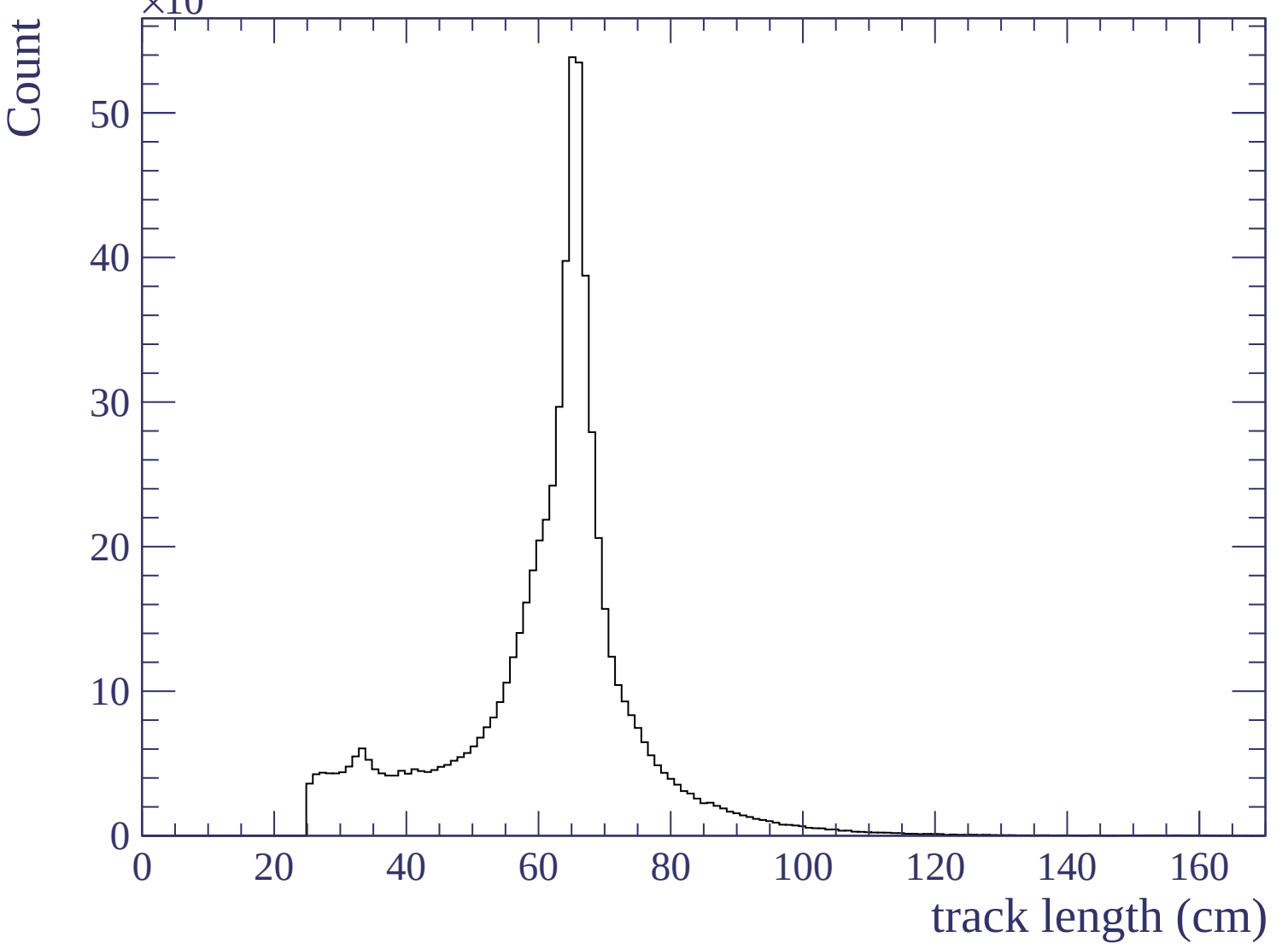


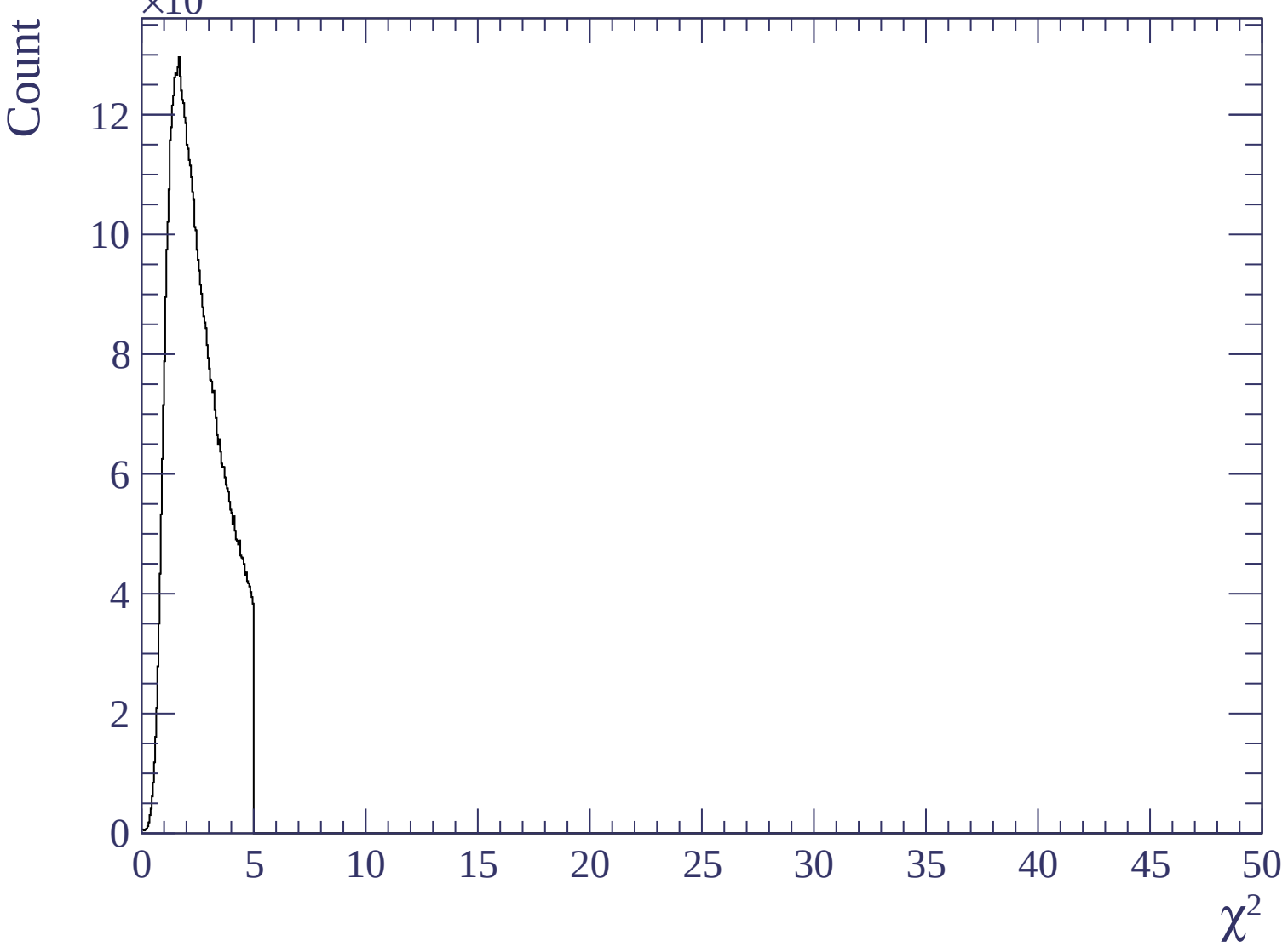


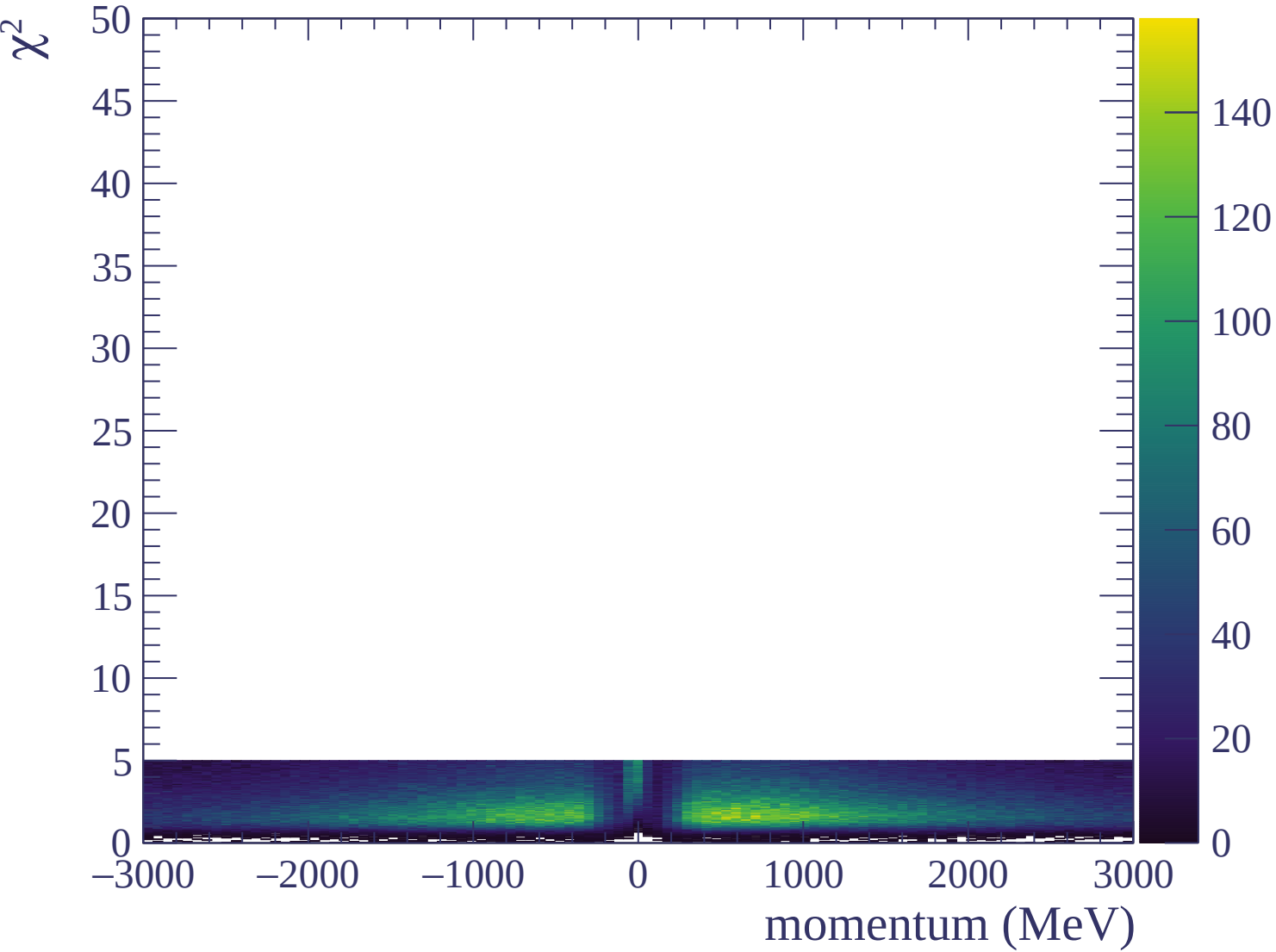


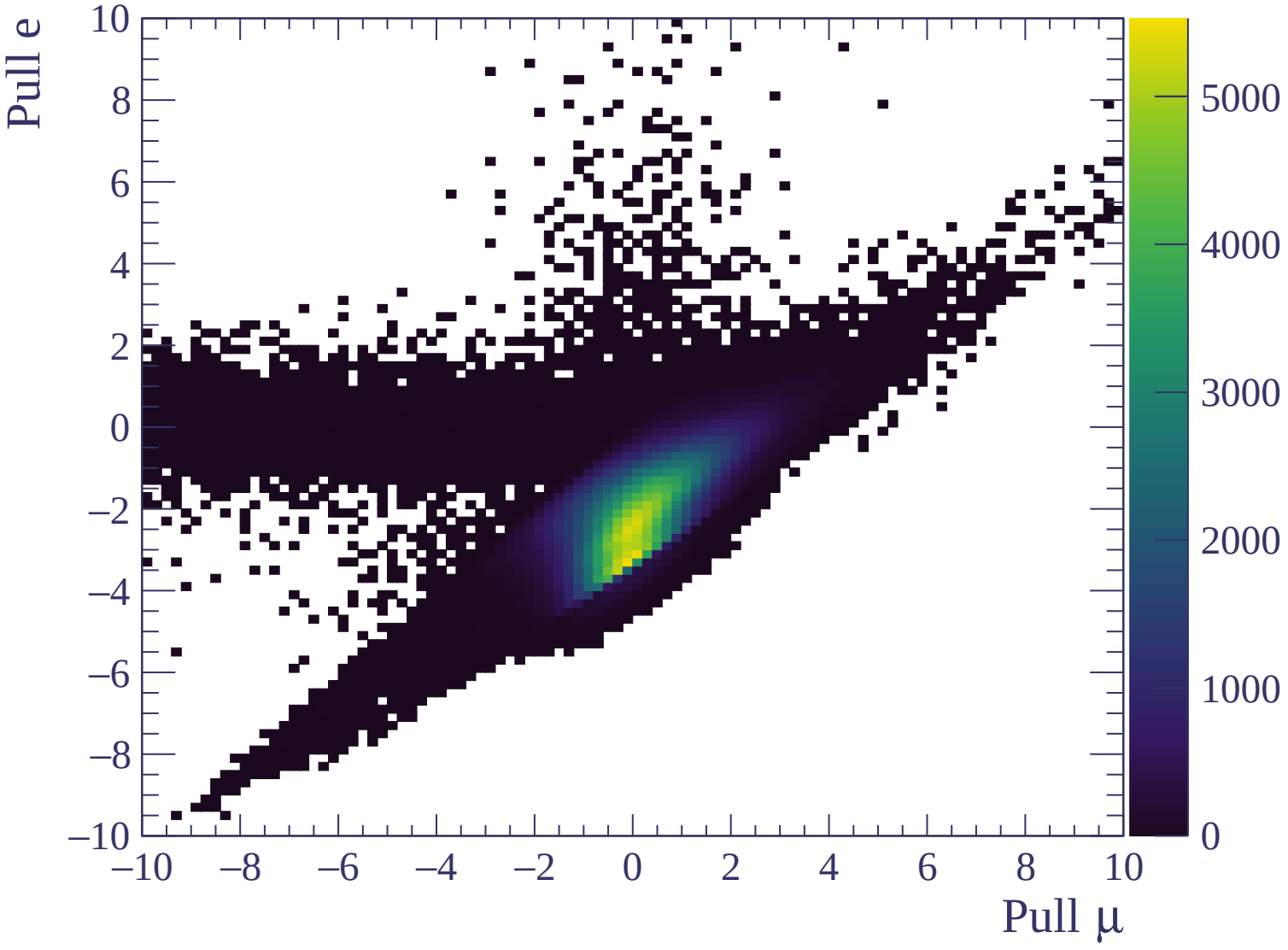


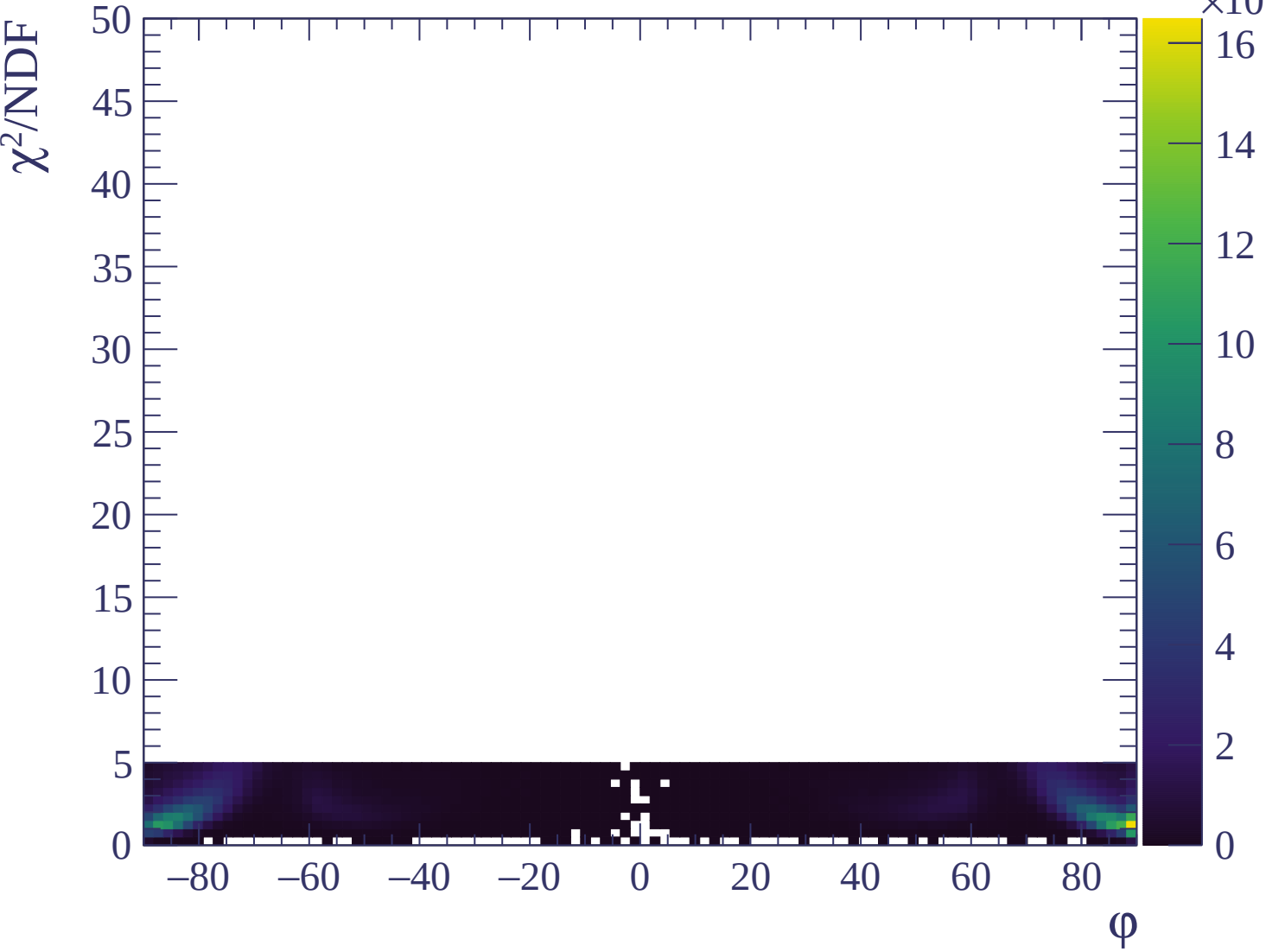


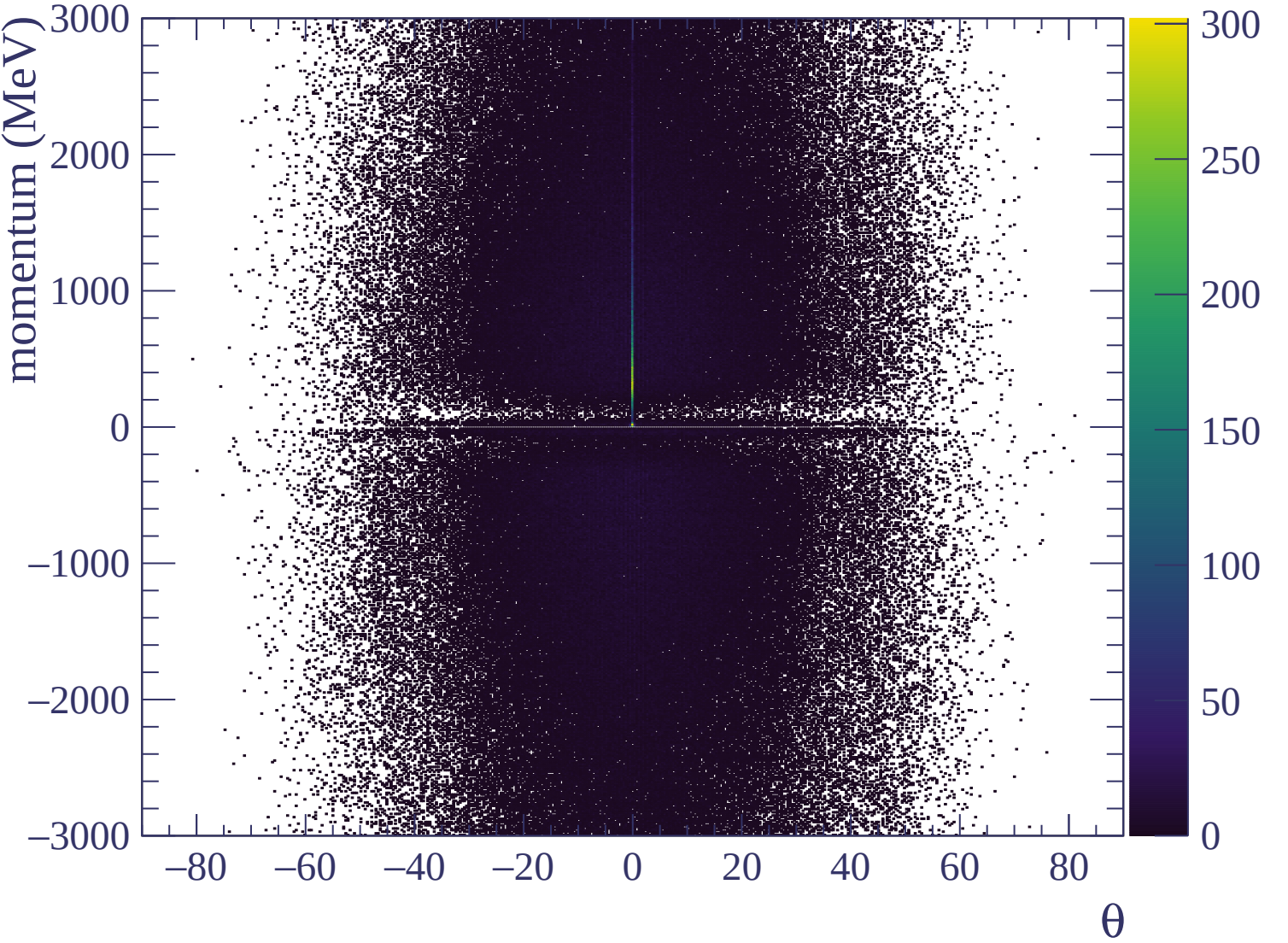


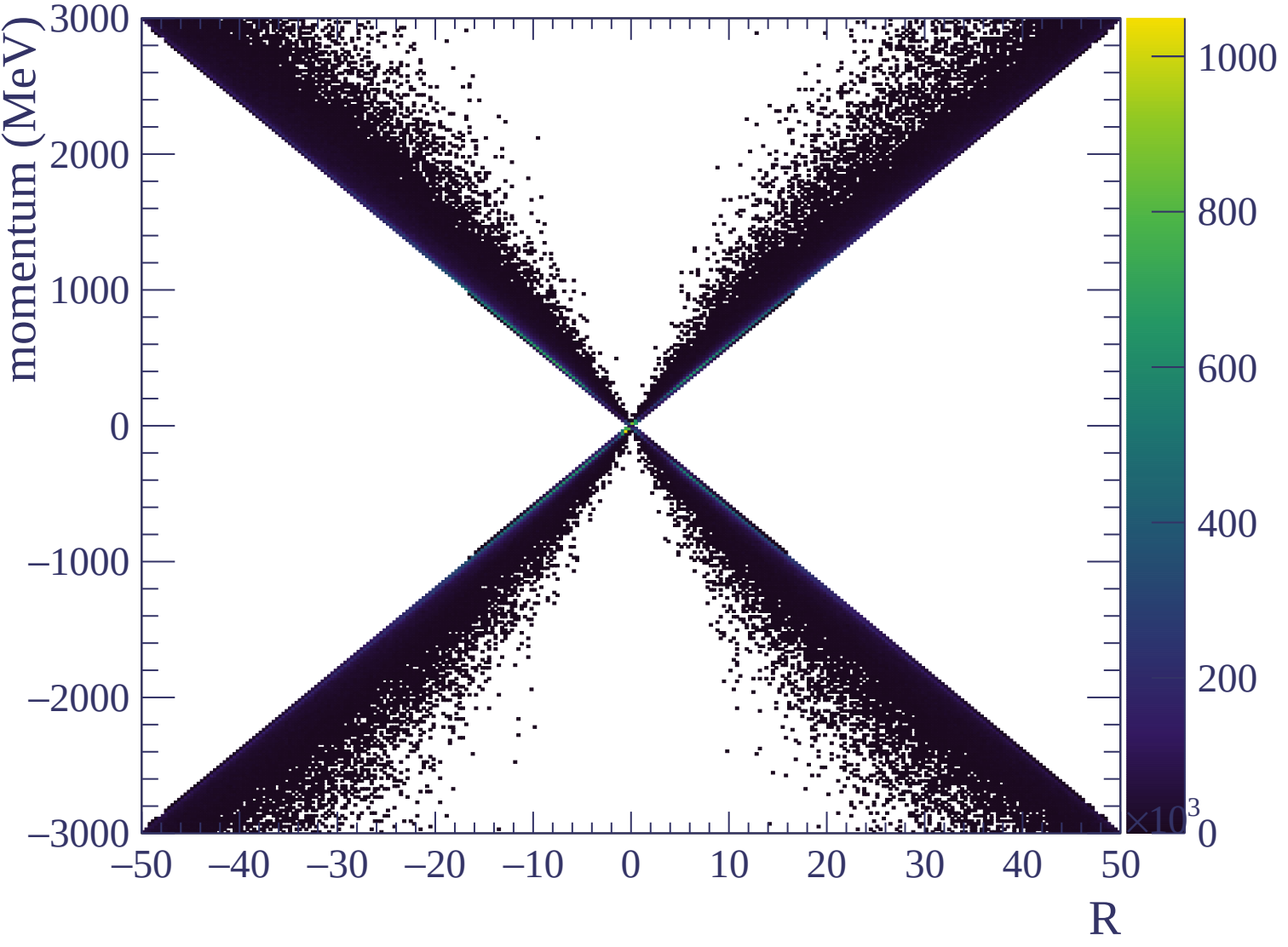


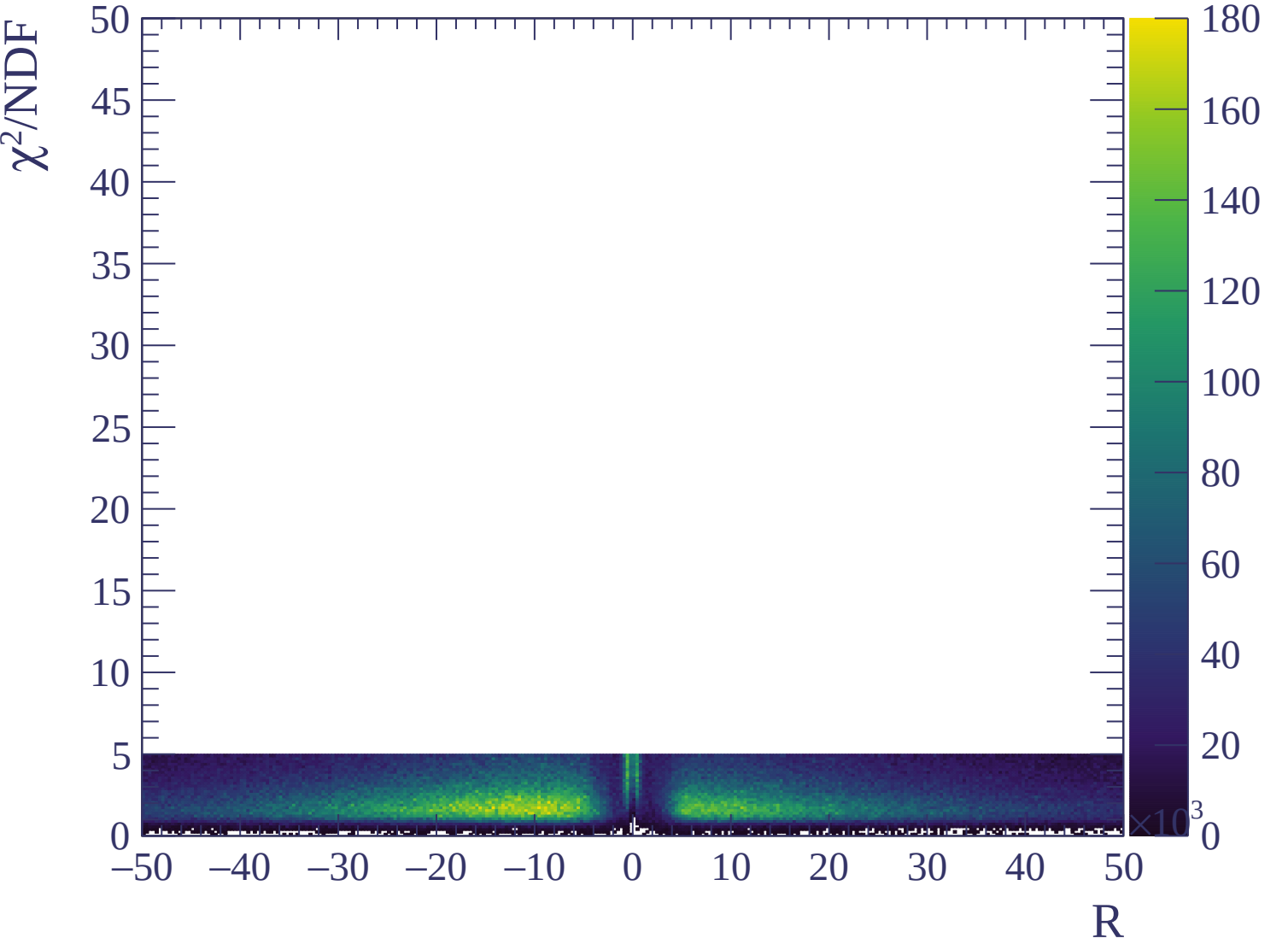


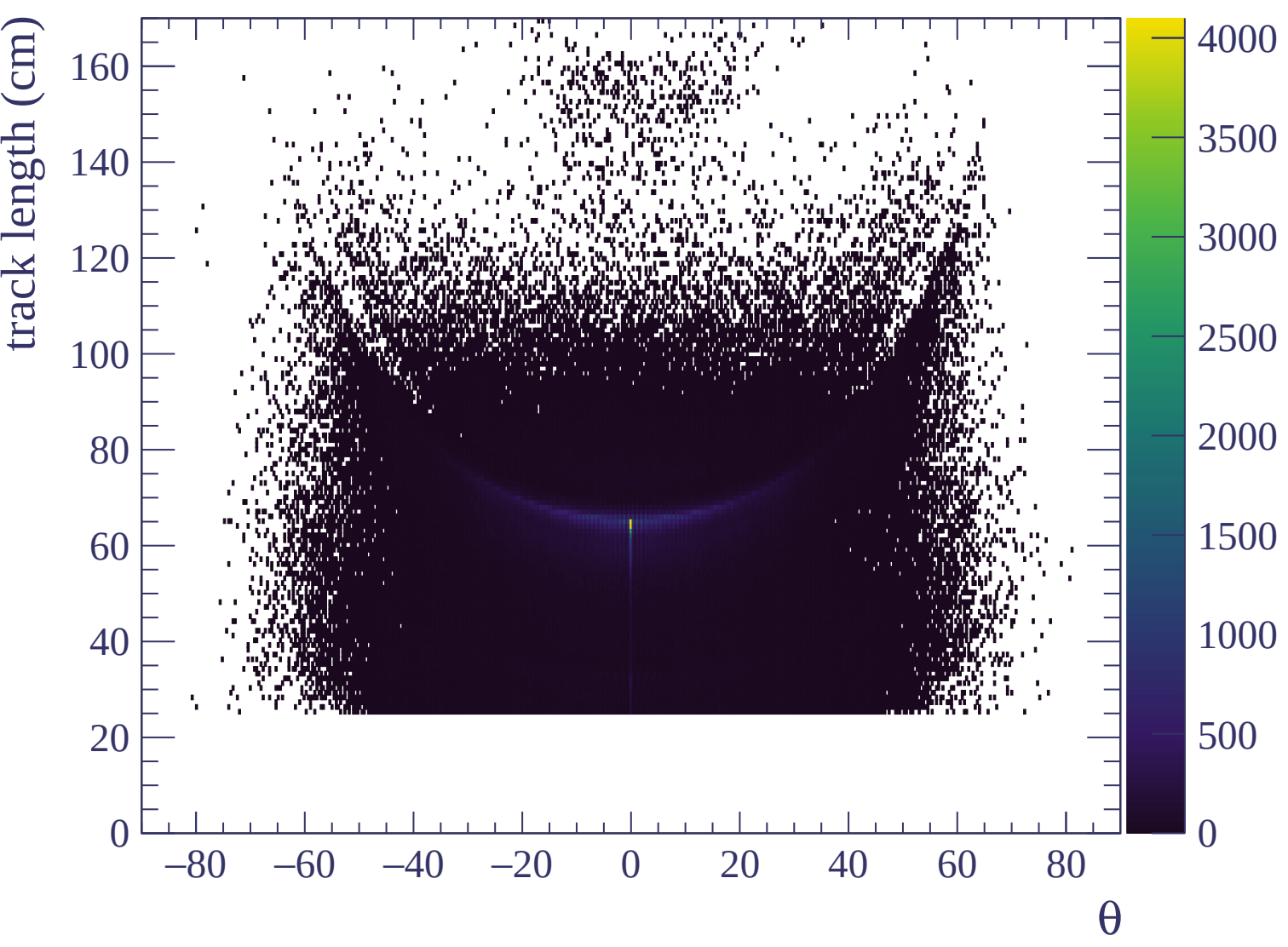


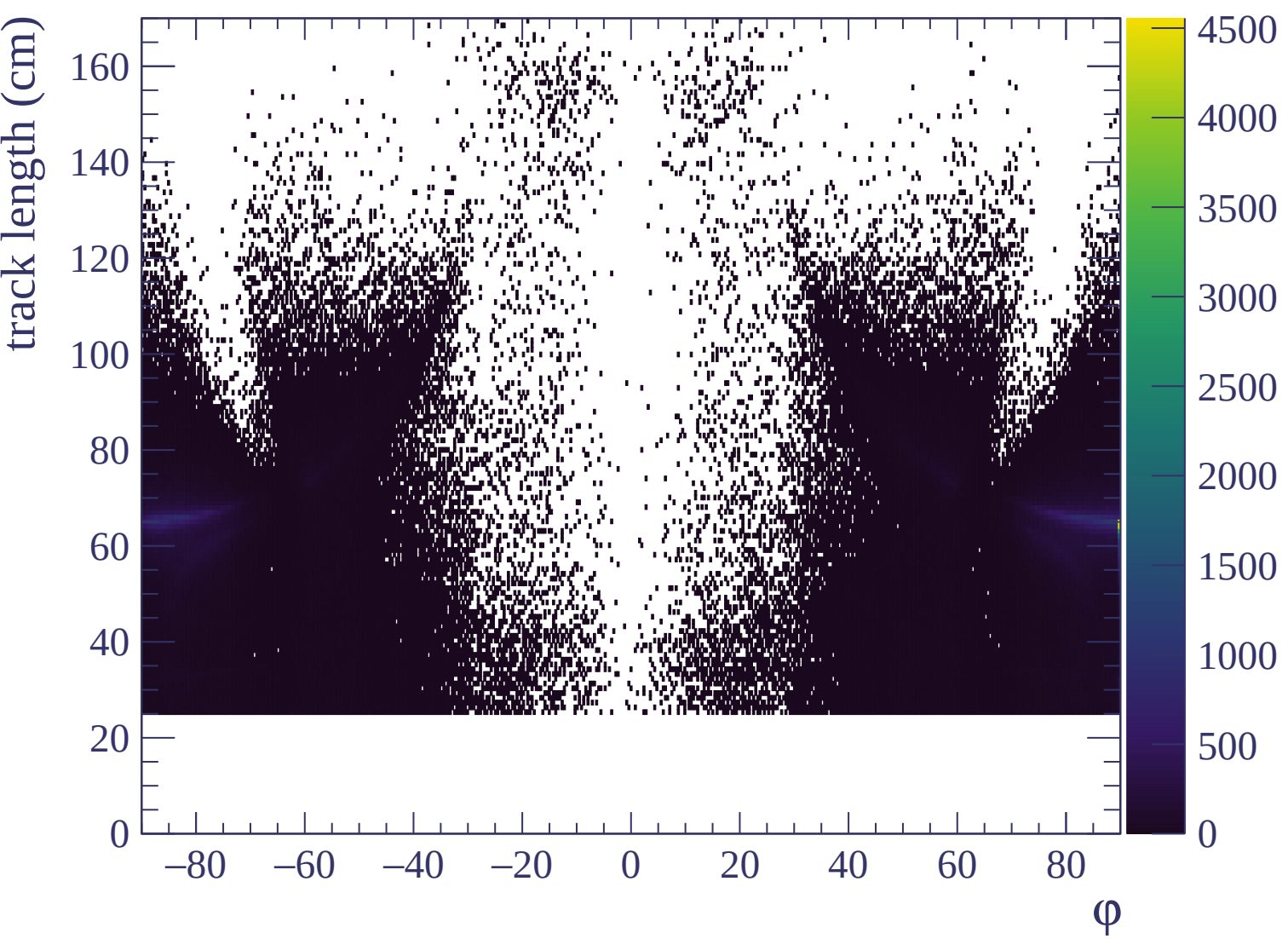






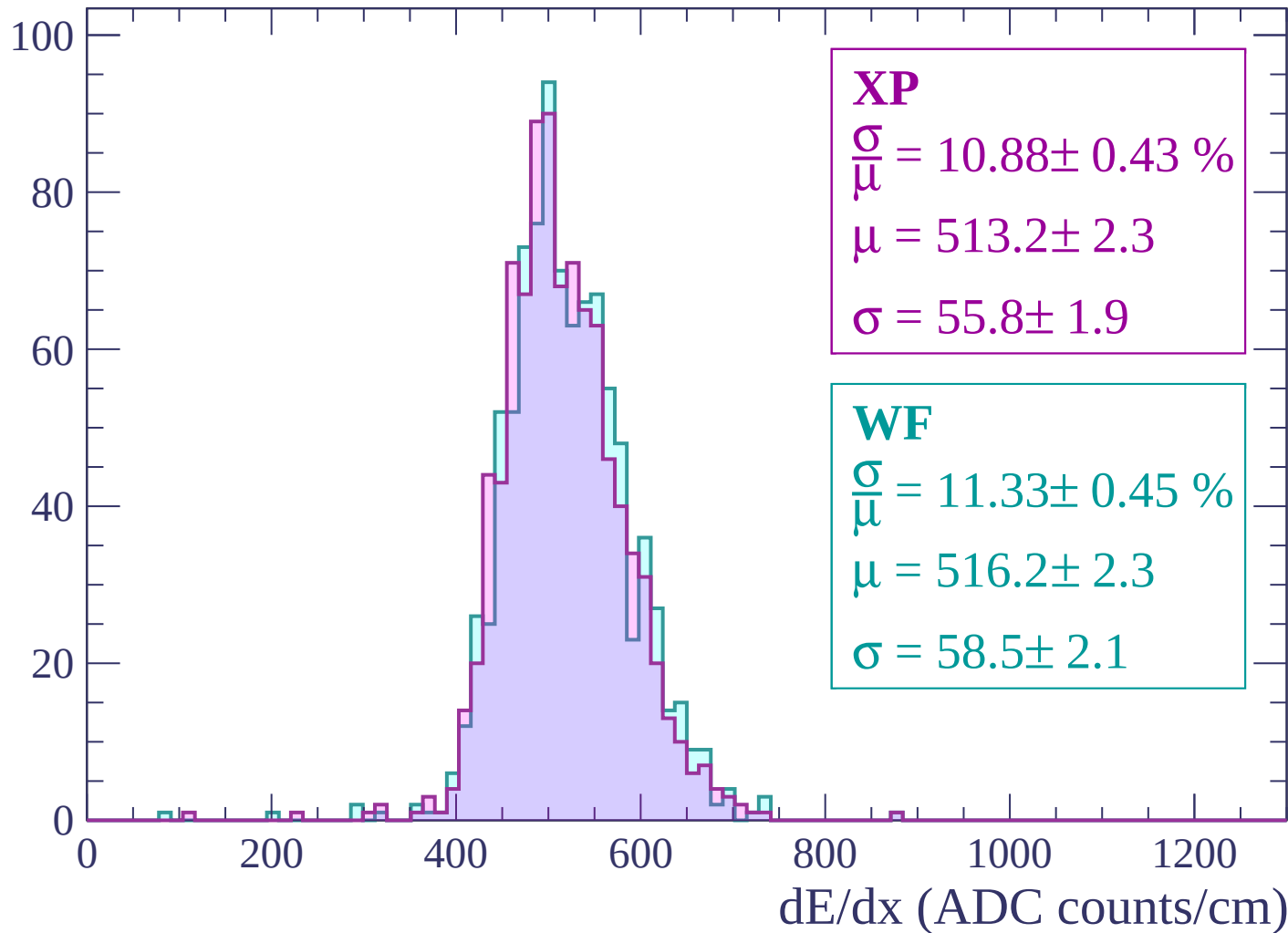






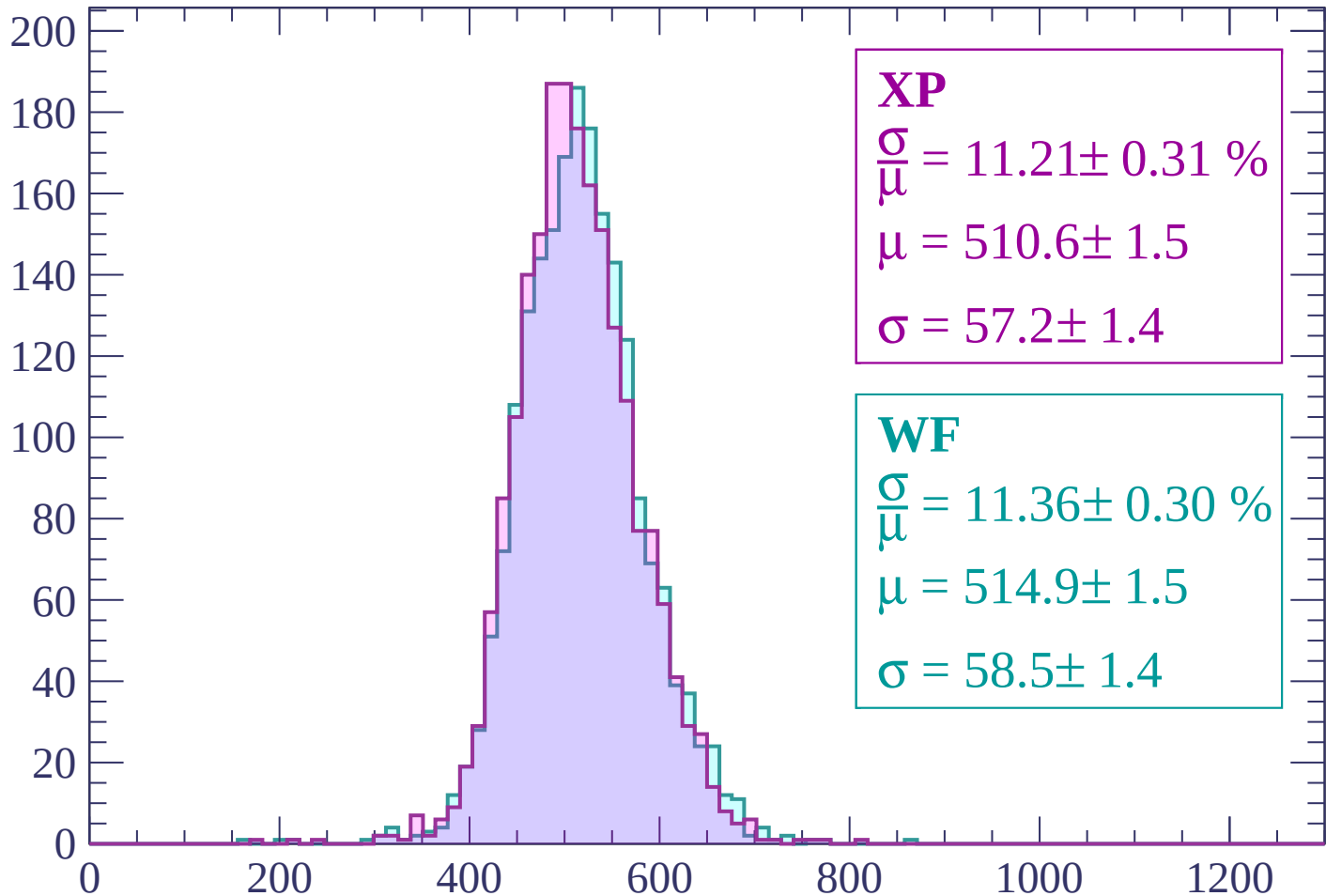
Energy loss | $-3000 < p < -2940$

Count



Energy loss | $-2940 < p < -2880$

Count



XP

$$\frac{\sigma}{\mu} = 11.21 \pm 0.31 \%$$

$$\mu = 510.6 \pm 1.5$$

$$\sigma = 57.2 \pm 1.4$$

WF

$$\frac{\sigma}{\mu} = 11.36 \pm 0.30 \%$$

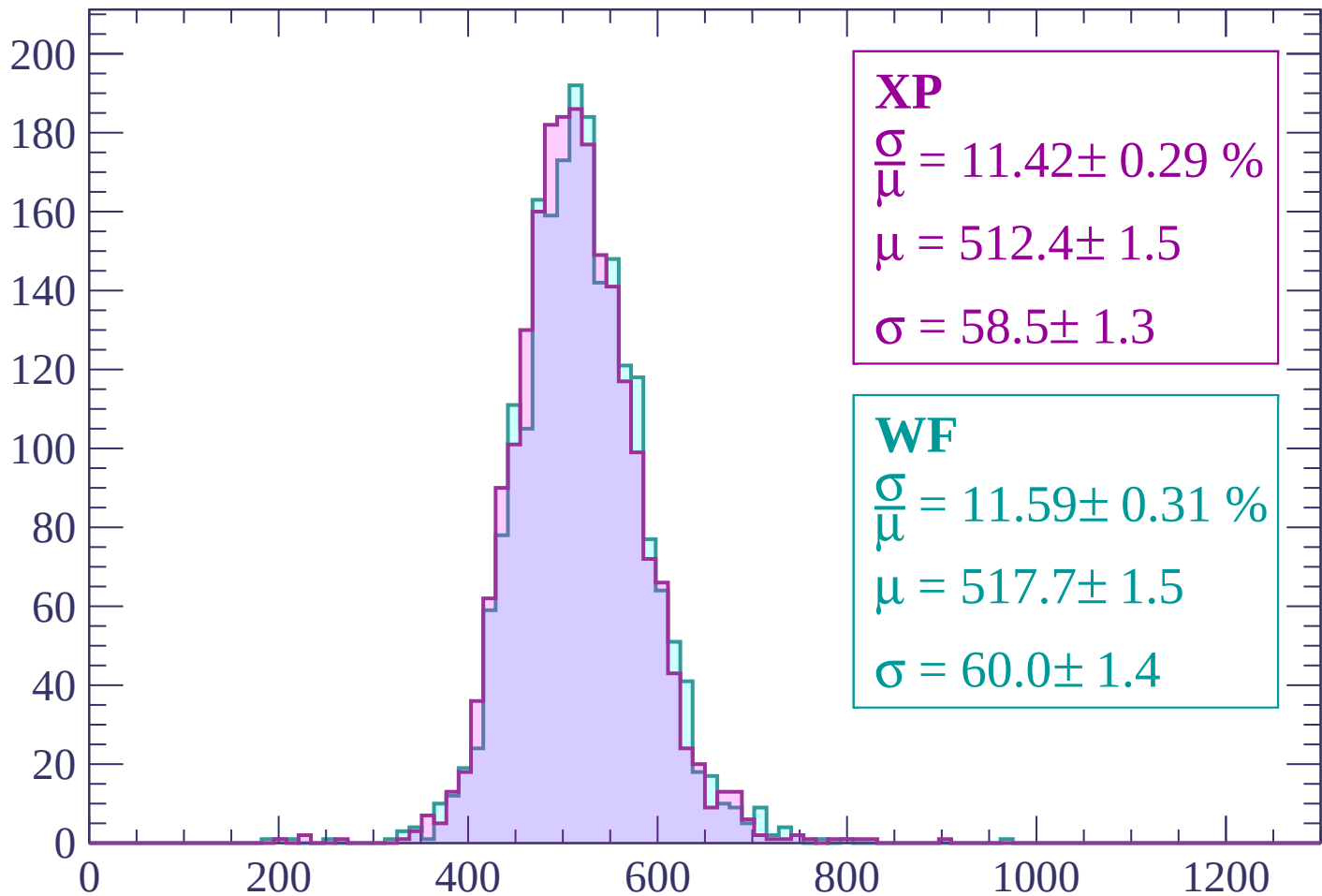
$$\mu = 514.9 \pm 1.5$$

$$\sigma = 58.5 \pm 1.4$$

dE/dx (ADC counts/cm)

Energy loss | $-2880 < p < -2820$

Count



XP

$$\frac{\sigma}{\mu} = 11.42 \pm 0.29 \%$$

$$\mu = 512.4 \pm 1.5$$

$$\sigma = 58.5 \pm 1.3$$

WF

$$\frac{\sigma}{\mu} = 11.59 \pm 0.31 \%$$

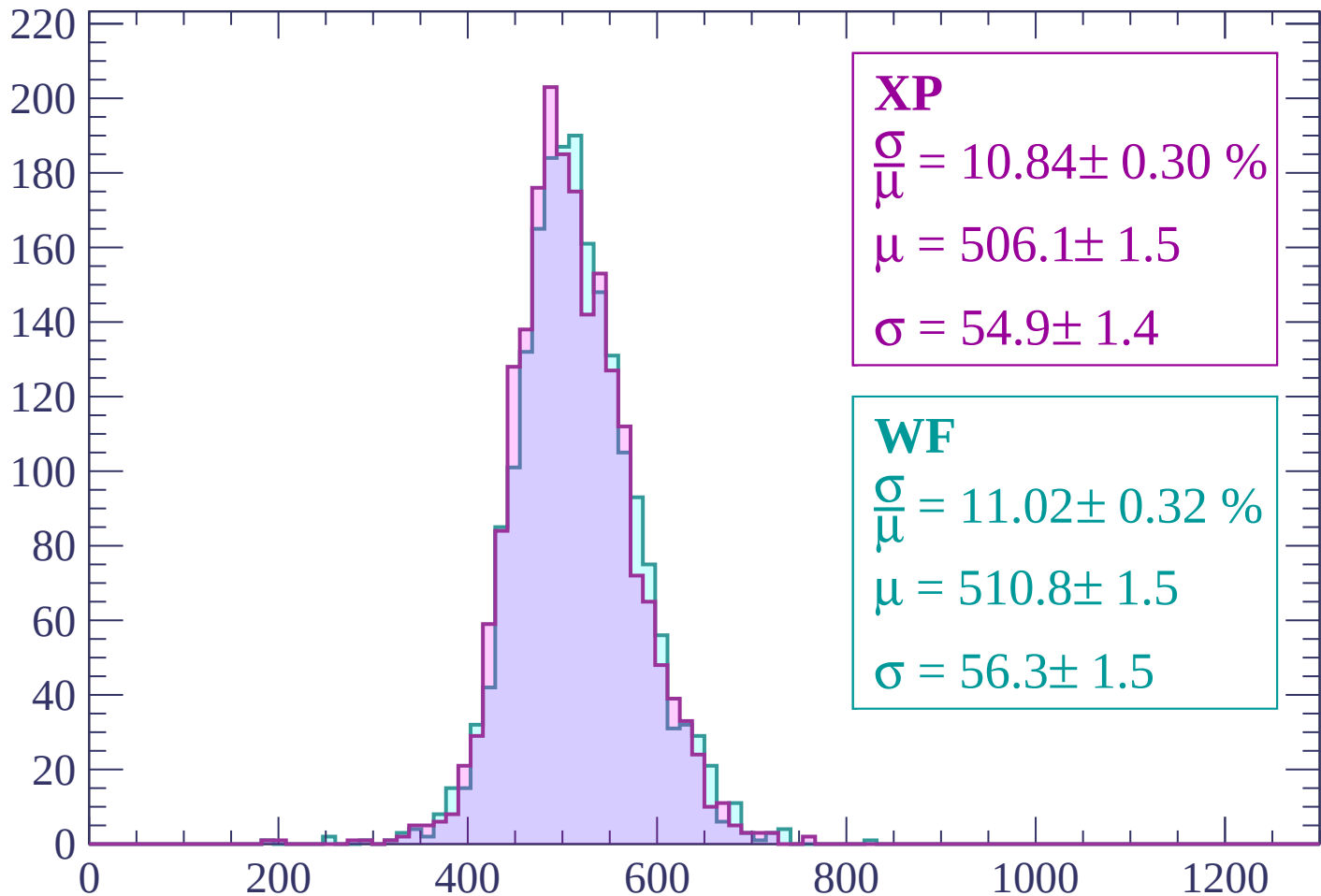
$$\mu = 517.7 \pm 1.5$$

$$\sigma = 60.0 \pm 1.4$$

dE/dx (ADC counts/cm)

Energy loss | -2820 < p < -2760

Count



XP

$$\frac{\sigma}{\mu} = 10.84 \pm 0.30 \%$$

$$\mu = 506.1 \pm 1.5$$

$$\sigma = 54.9 \pm 1.4$$

WF

$$\frac{\sigma}{\mu} = 11.02 \pm 0.32 \%$$

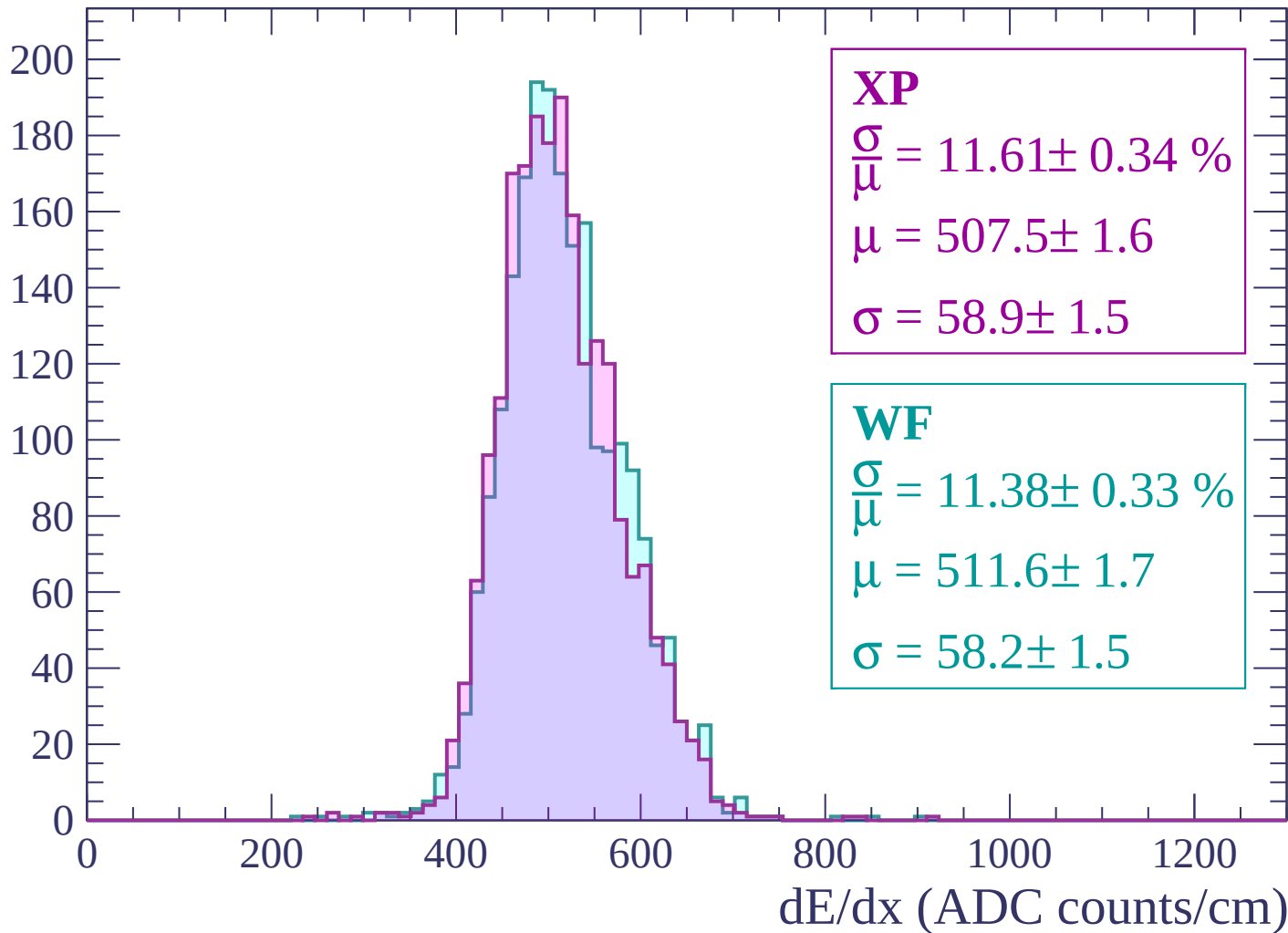
$$\mu = 510.8 \pm 1.5$$

$$\sigma = 56.3 \pm 1.5$$

dE/dx (ADC counts/cm)

Energy loss | -2760 < p < -2700

Count



Energy loss | -2700 < p < -2640

Count

250

200

150

100

50

0

0

200

400

600

800

1000

1200

dE/dx (ADC counts/cm)

XP

$$\frac{\sigma}{\mu} = 11.31 \pm 0.30 \%$$

$$\mu = 508.4 \pm 1.5$$

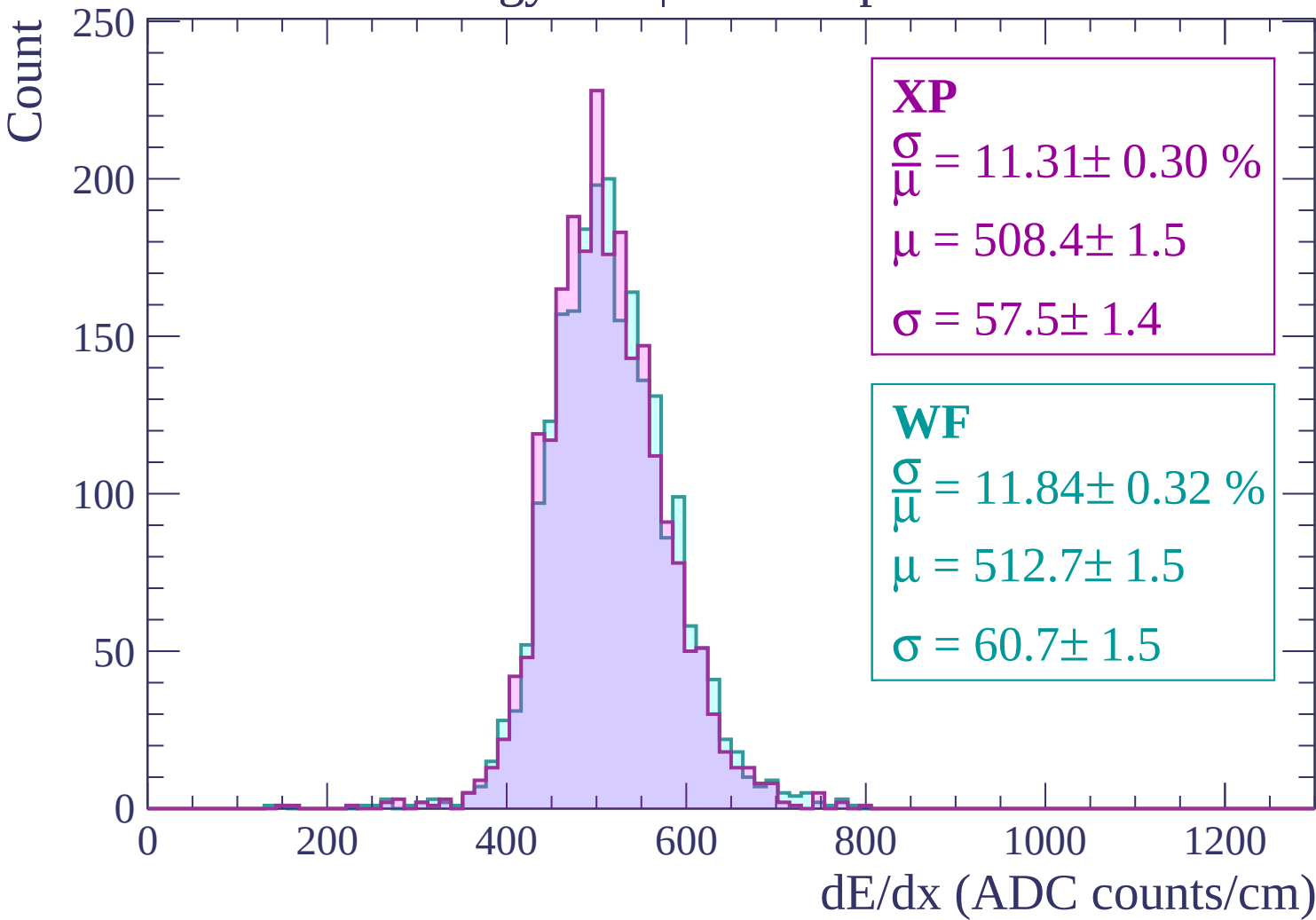
$$\sigma = 57.5 \pm 1.4$$

WF

$$\frac{\sigma}{\mu} = 11.84 \pm 0.32 \%$$

$$\mu = 512.7 \pm 1.5$$

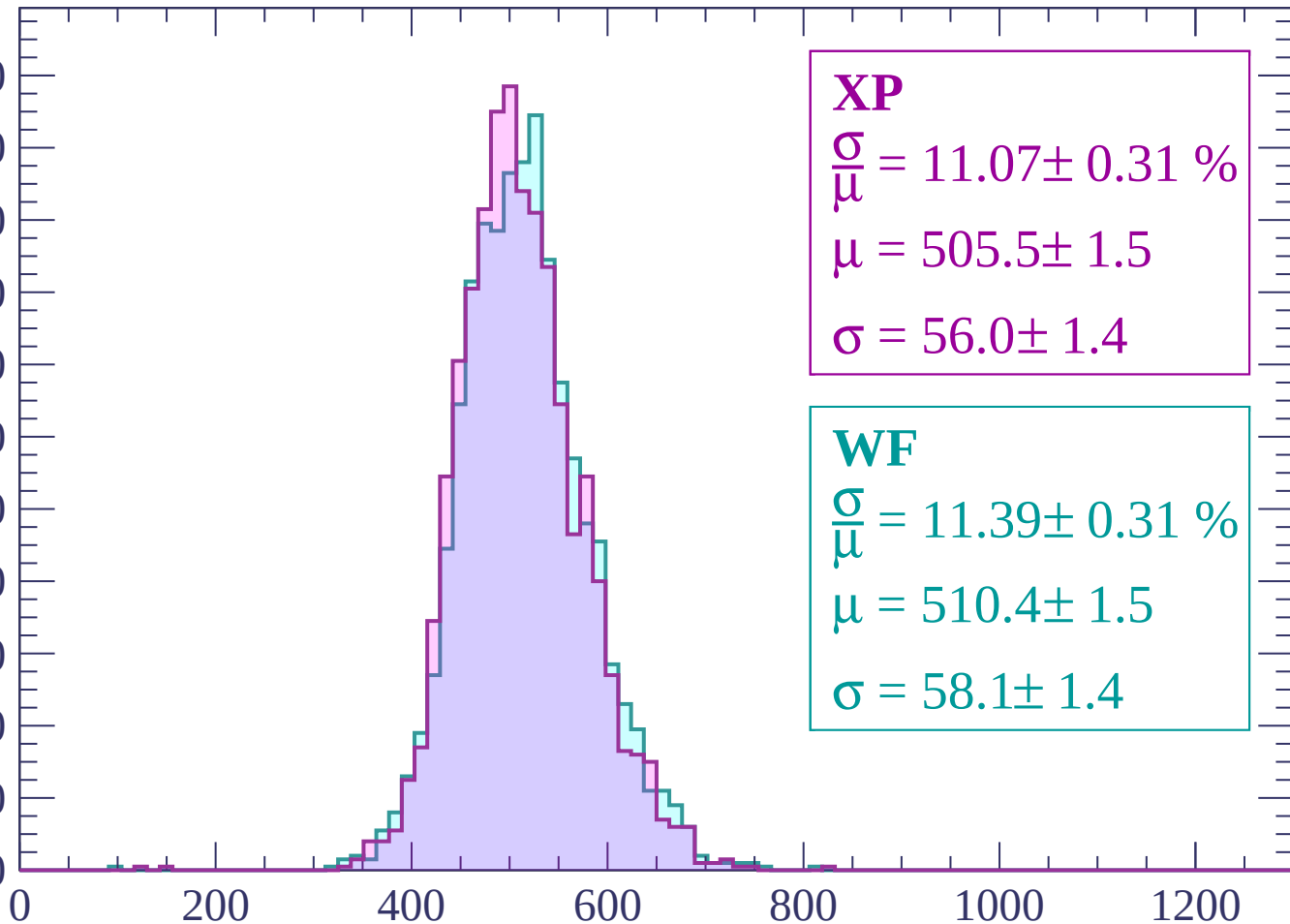
$$\sigma = 60.7 \pm 1.5$$



Energy loss | -2640 < p < -2580

Count

220
200
180
160
140
120
100
80
60
40
20
0



XP

$$\frac{\sigma}{\mu} = 11.07 \pm 0.31 \%$$

$$\mu = 505.5 \pm 1.5$$

$$\sigma = 56.0 \pm 1.4$$

WF

$$\frac{\sigma}{\mu} = 11.39 \pm 0.31 \%$$

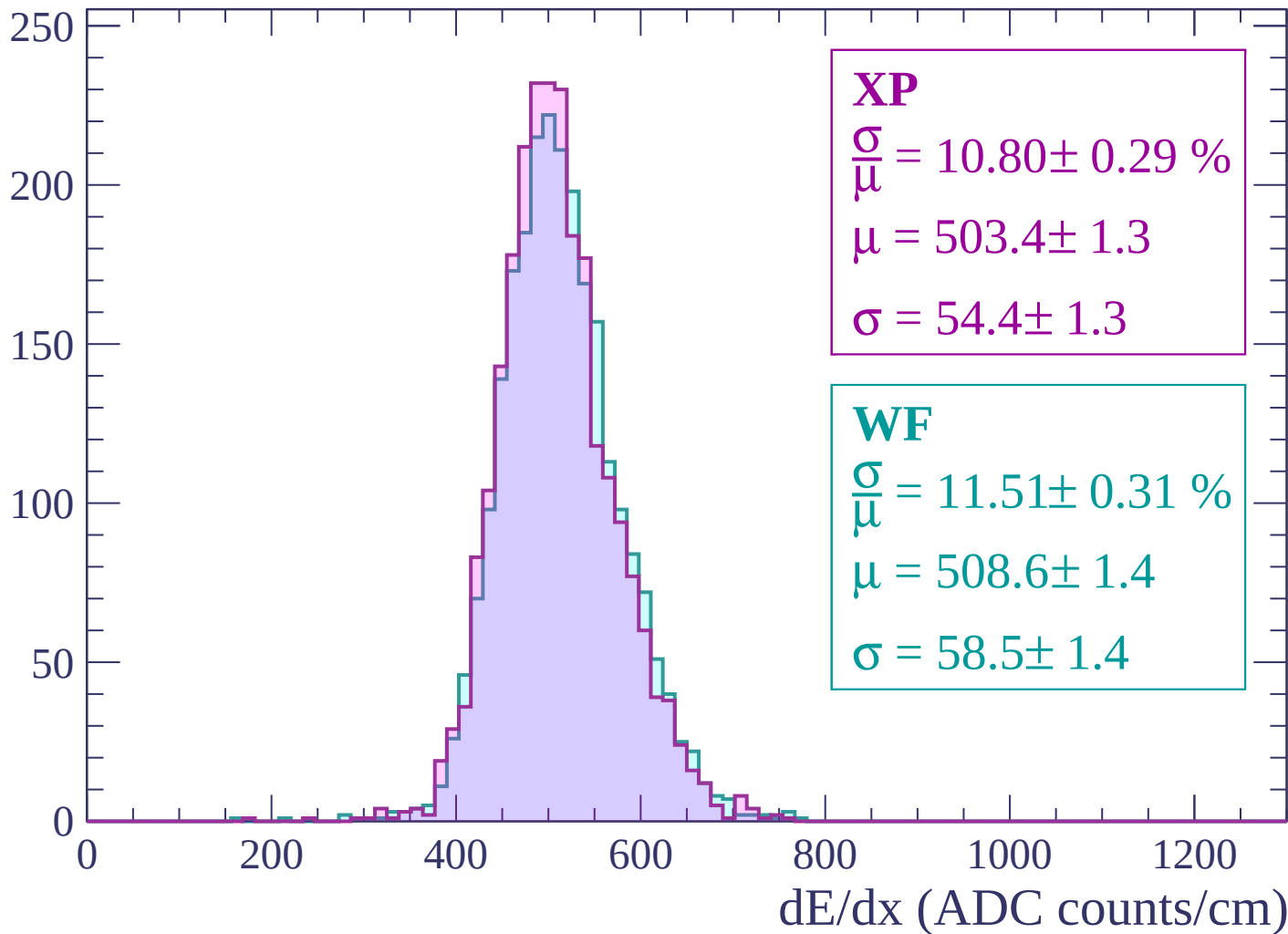
$$\mu = 510.4 \pm 1.5$$

$$\sigma = 58.1 \pm 1.4$$

dE/dx (ADC counts/cm)

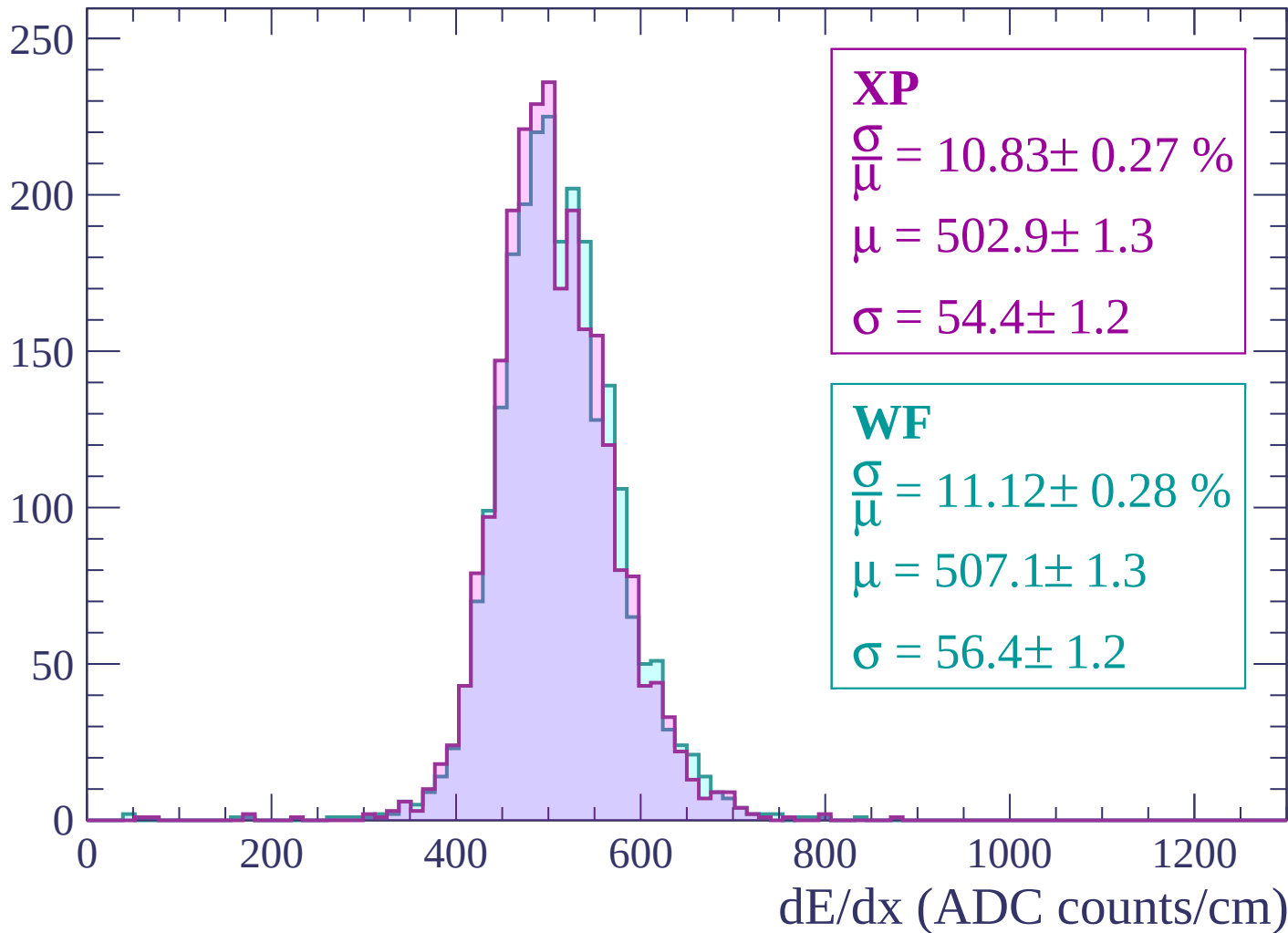
Energy loss | -2580 < p < -2520

Count



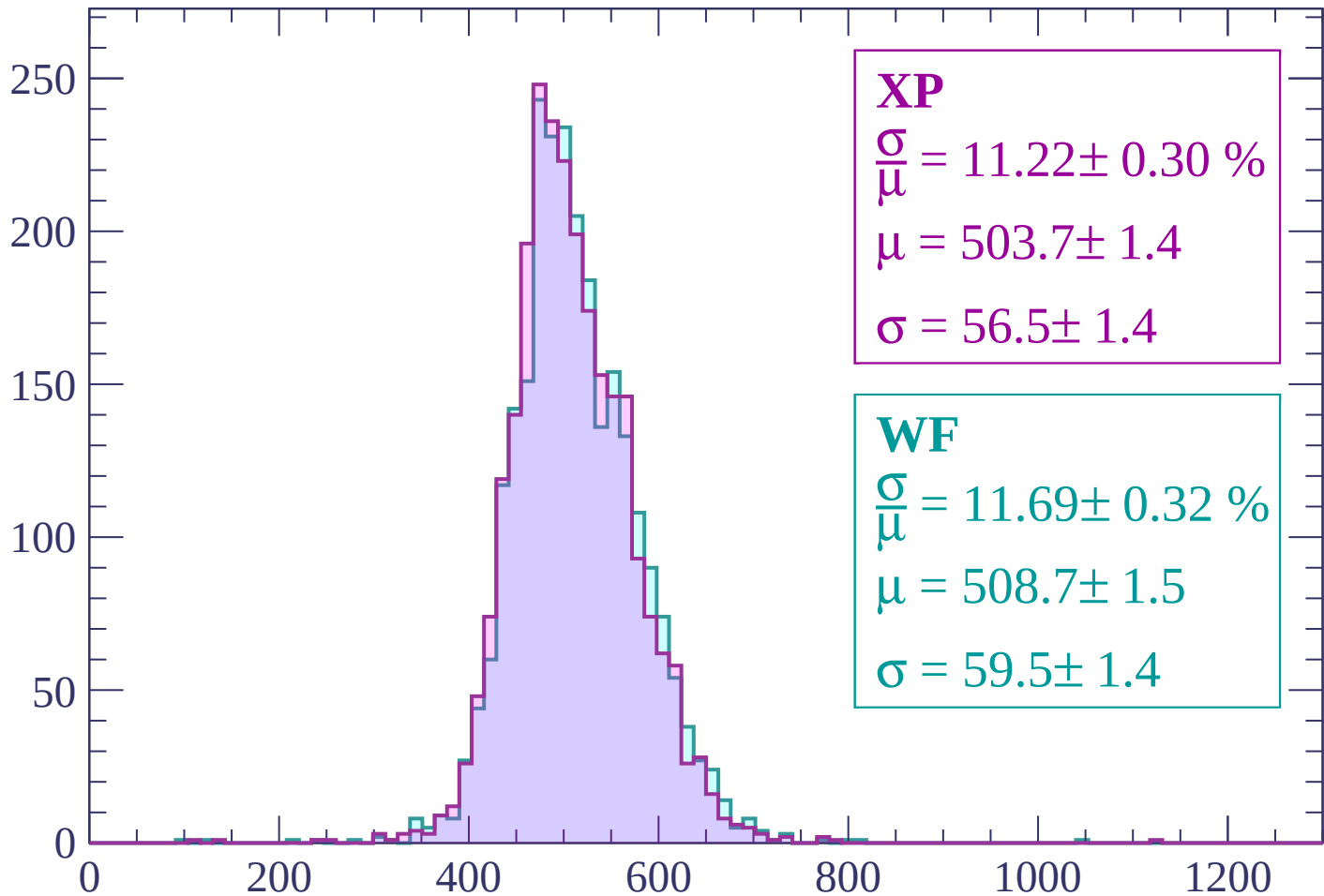
Energy loss | -2520 < p < -2460

Count



Energy loss | -2460 < p < -2400

Count



XP

$$\frac{\sigma}{\mu} = 11.22 \pm 0.30 \%$$

$$\mu = 503.7 \pm 1.4$$

$$\sigma = 56.5 \pm 1.4$$

WF

$$\frac{\sigma}{\mu} = 11.69 \pm 0.32 \%$$

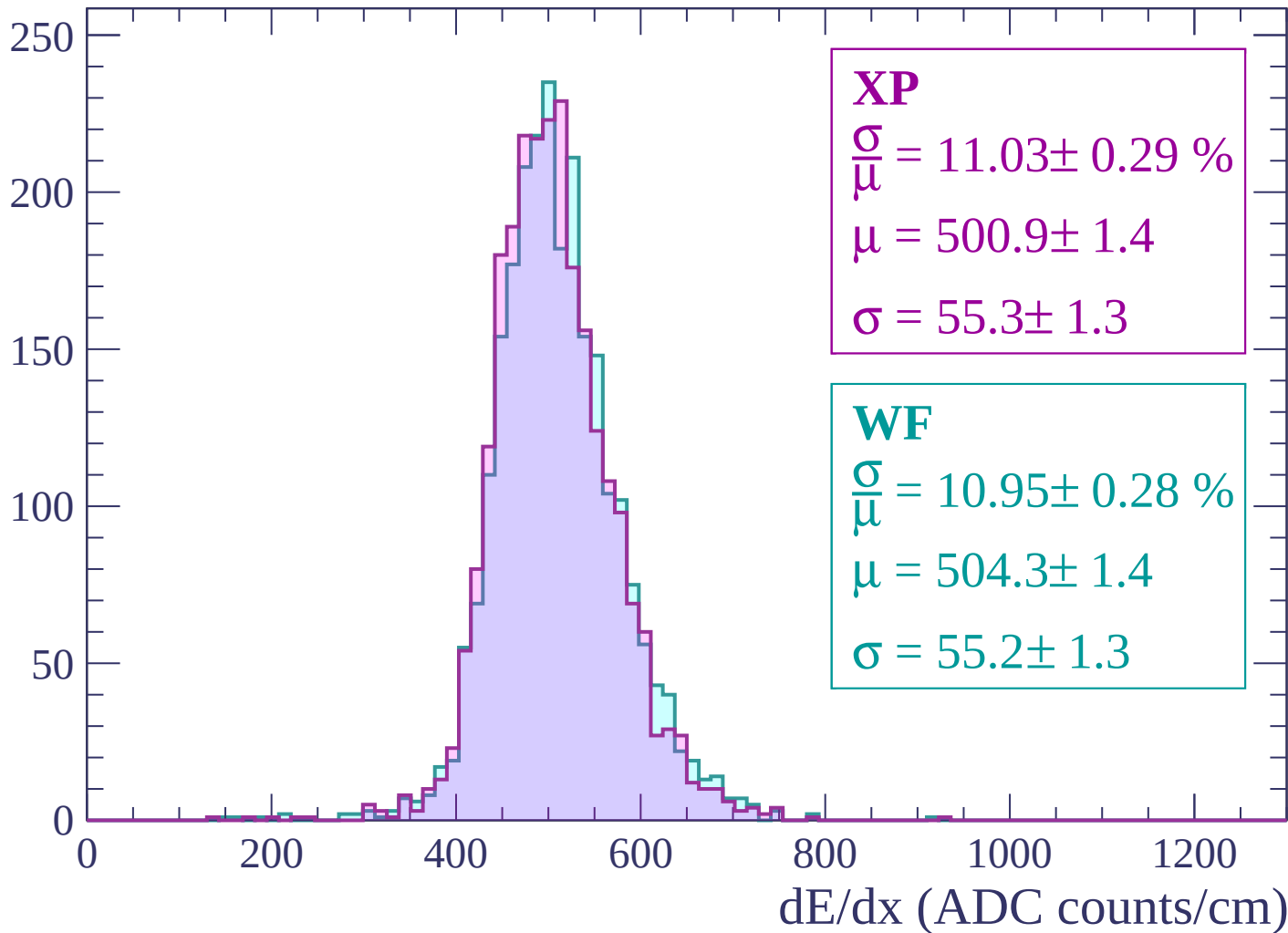
$$\mu = 508.7 \pm 1.5$$

$$\sigma = 59.5 \pm 1.4$$

dE/dx (ADC counts/cm)

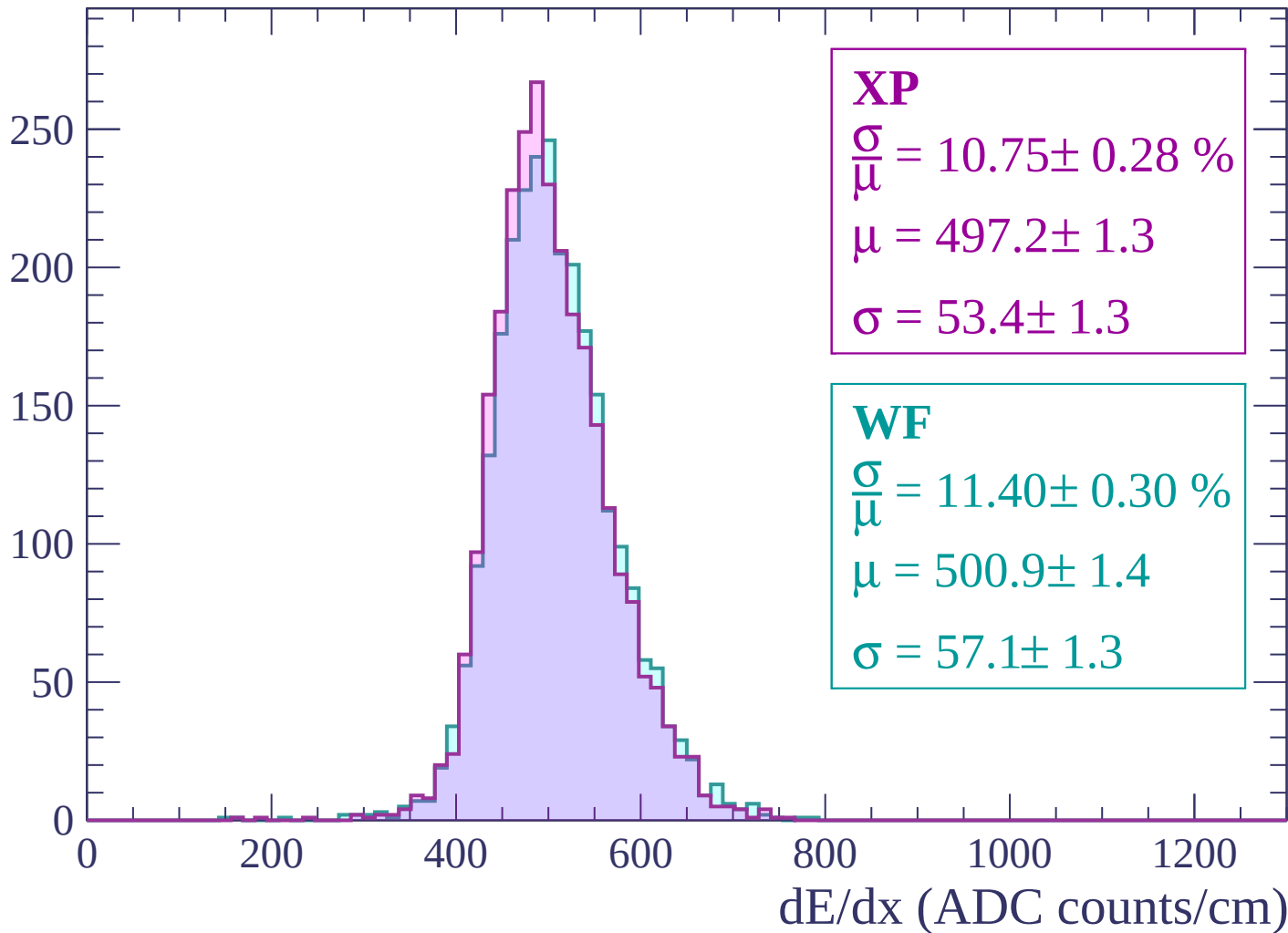
Energy loss | -2400 < p < -2340

Count



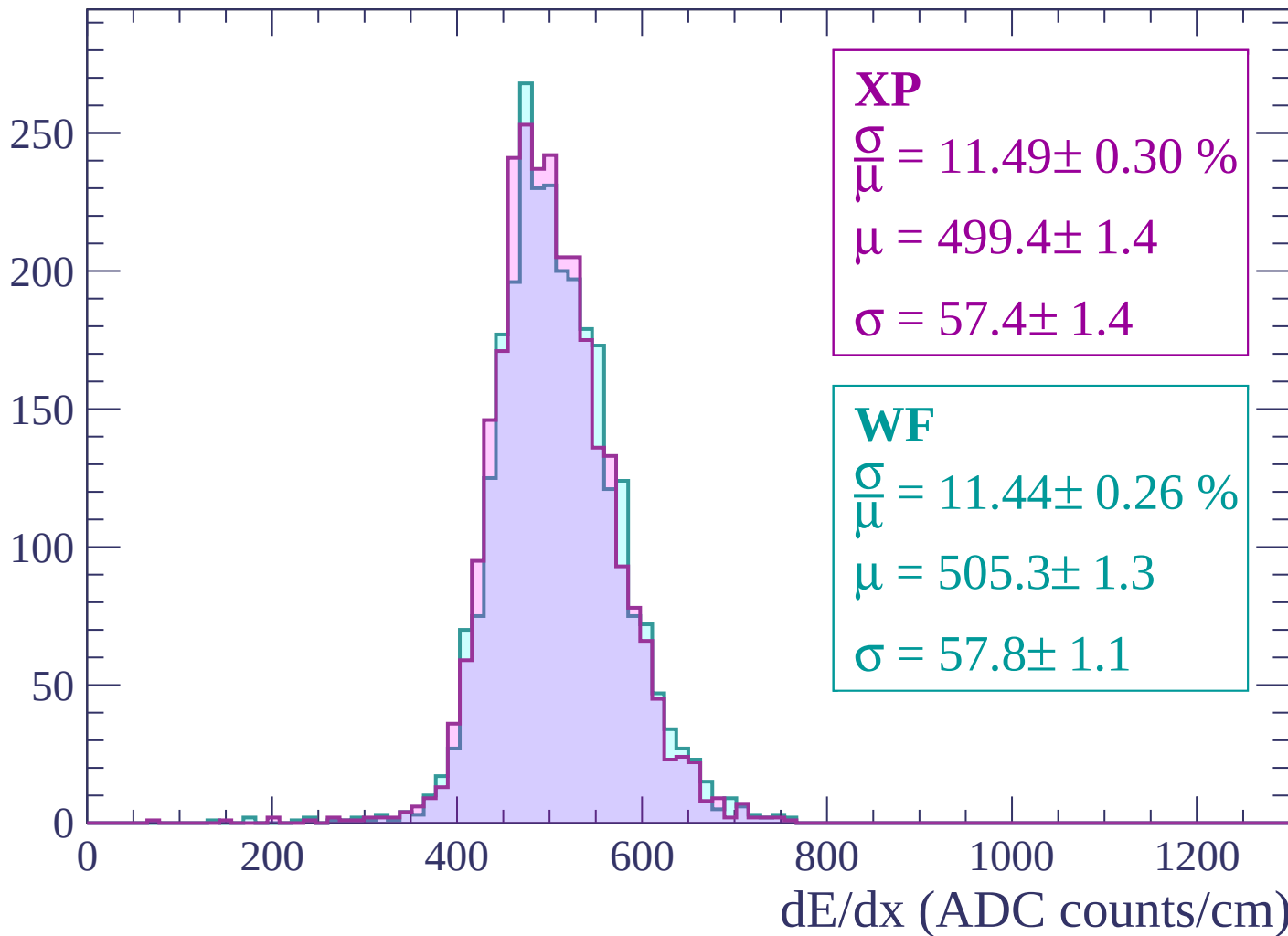
Energy loss | $-2340 < p < -2280$

Count



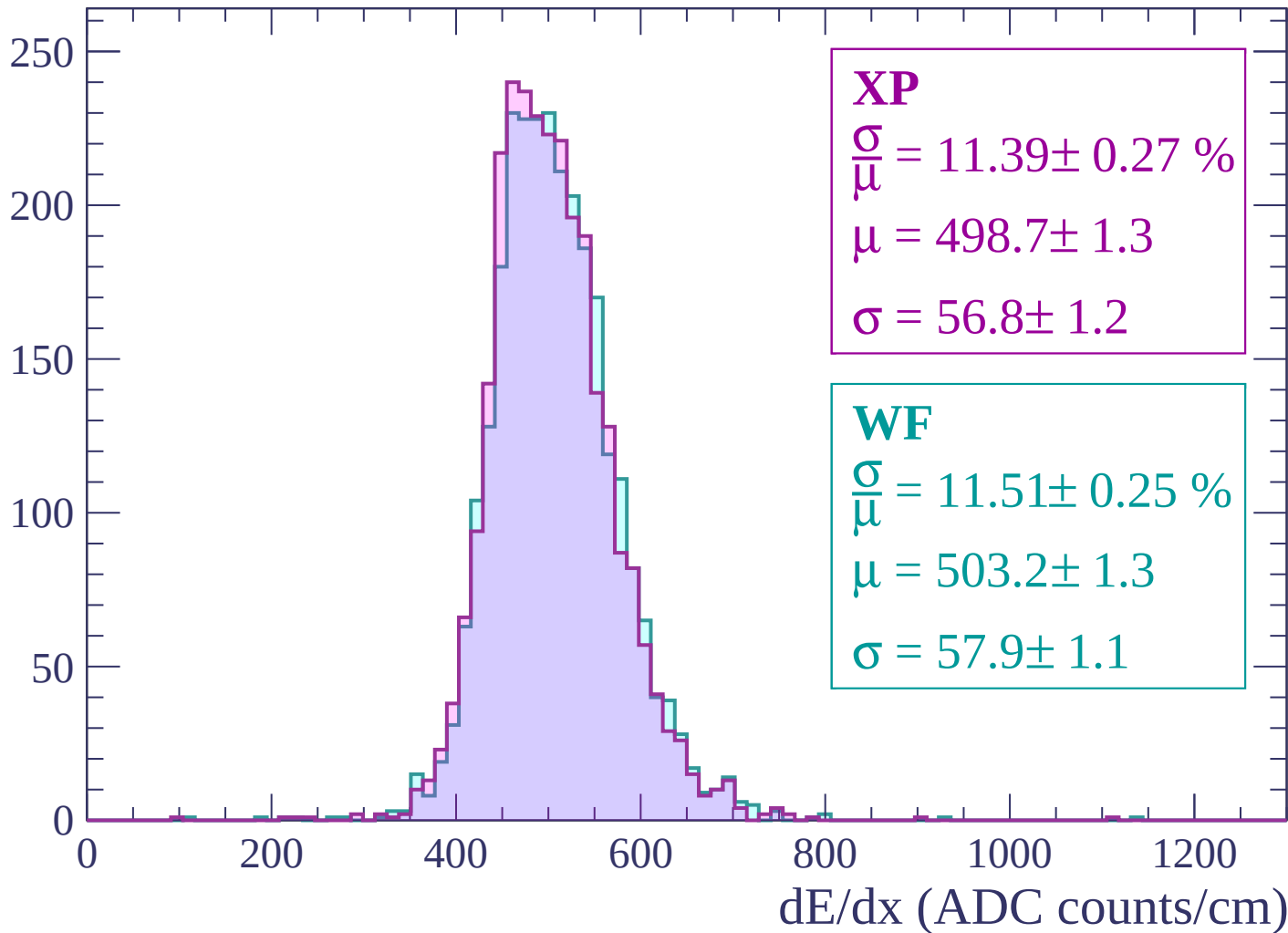
Energy loss | $-2280 < p < -2220$

Count



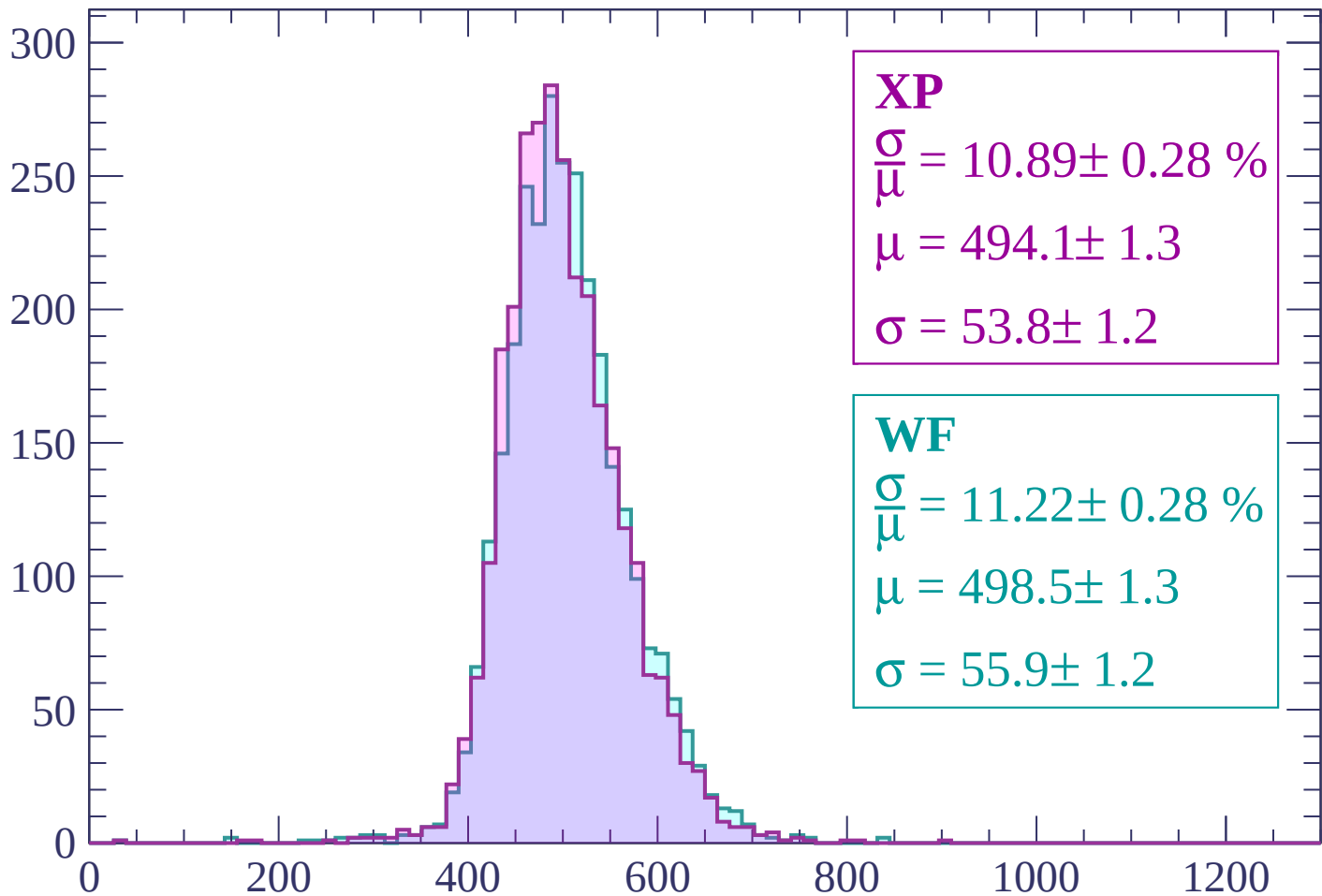
Energy loss | -2220 < p < -2160

Count



Energy loss | -2160 < p < -2100

Count



XP

$$\frac{\sigma}{\mu} = 10.89 \pm 0.28 \%$$

$$\mu = 494.1 \pm 1.3$$

$$\sigma = 53.8 \pm 1.2$$

WF

$$\frac{\sigma}{\mu} = 11.22 \pm 0.28 \%$$

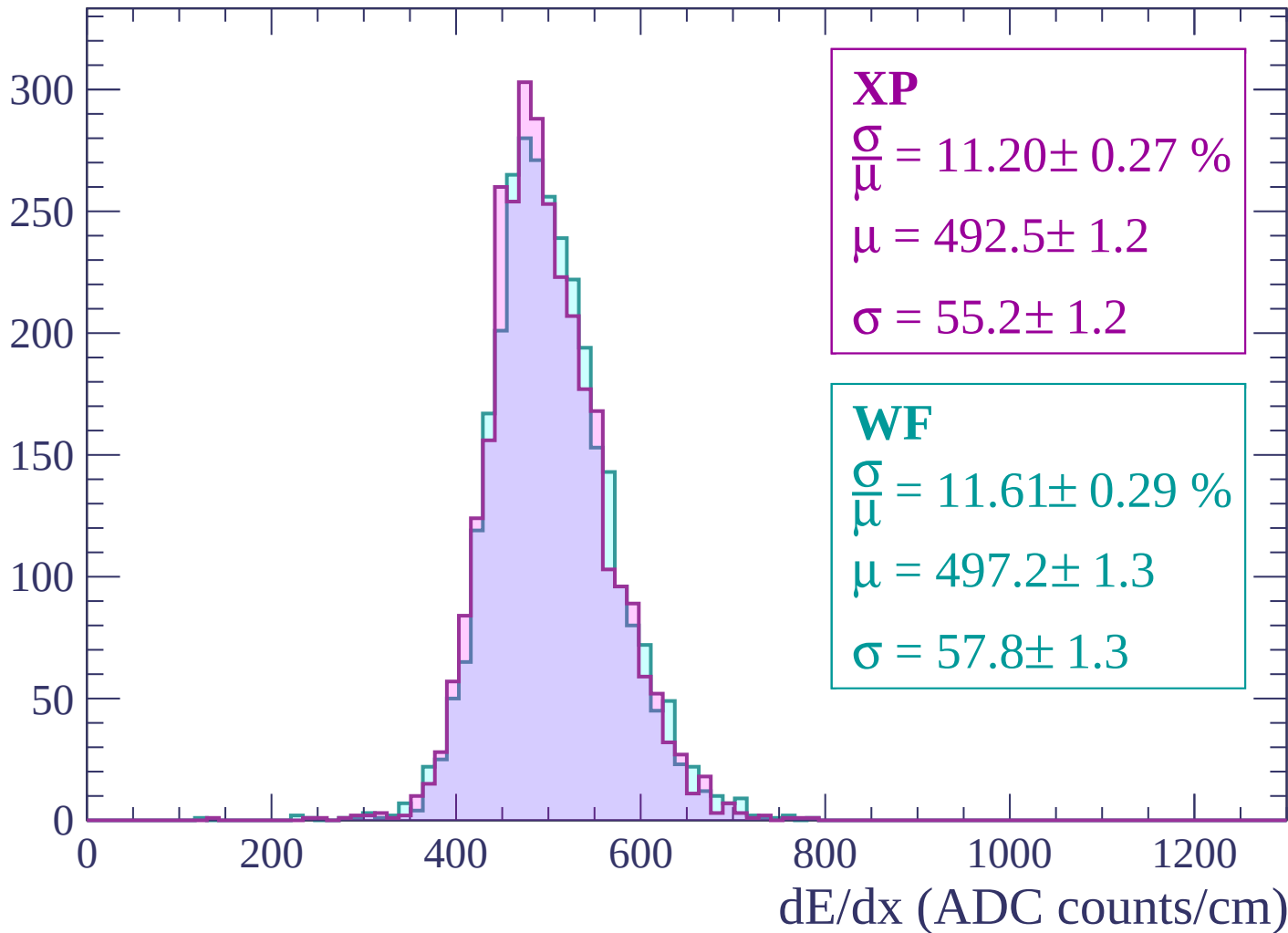
$$\mu = 498.5 \pm 1.3$$

$$\sigma = 55.9 \pm 1.2$$

dE/dx (ADC counts/cm)

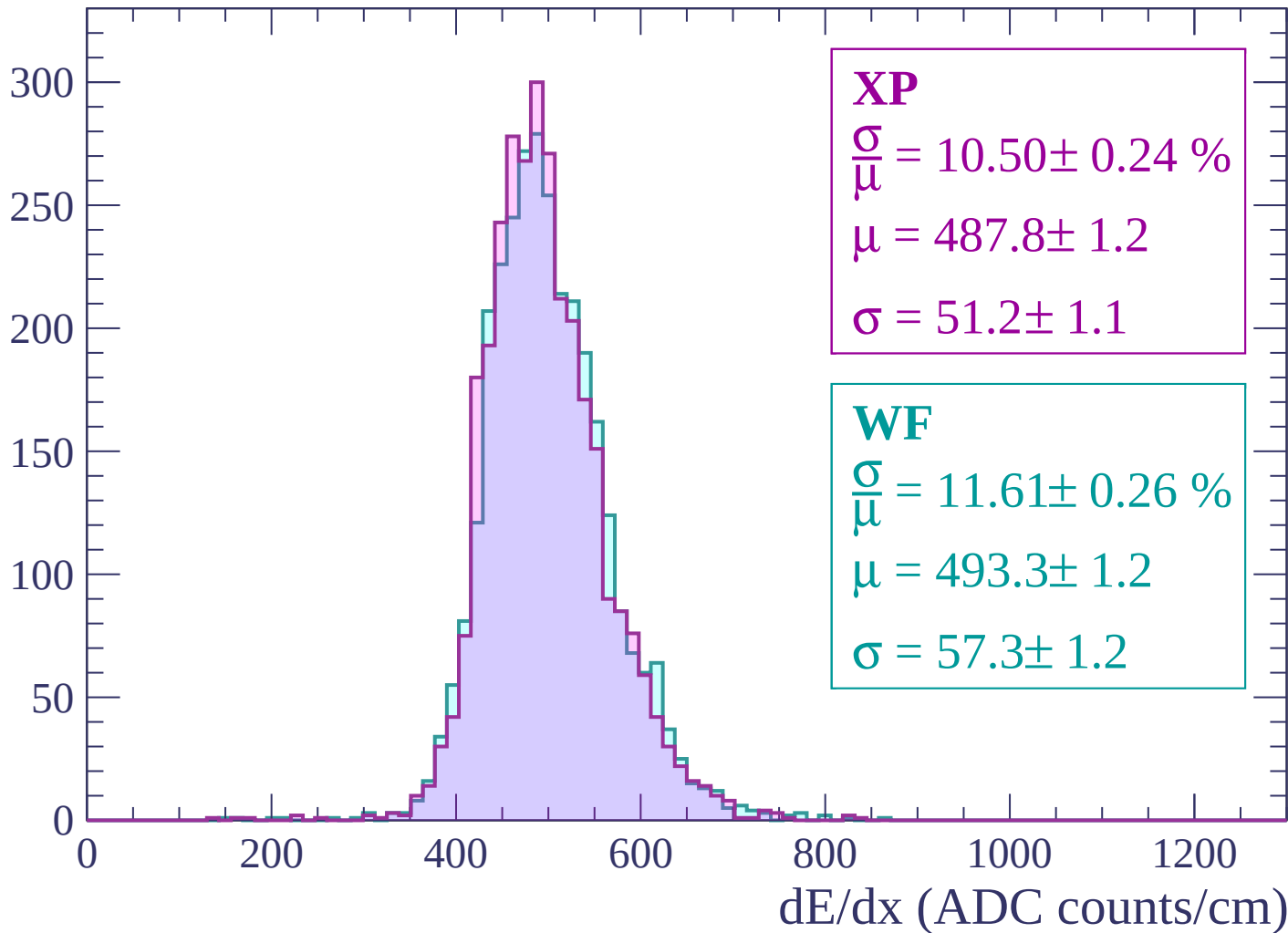
Energy loss | -2100 < p < -2040

Count



Energy loss | -2040 < p < -1980

Count



Energy loss | -1980 < p < -1920

Count

300

250

200

150

100

50

0

dE/dx (ADC counts/cm)

XP

$$\frac{\sigma}{\mu} = 11.13 \pm 0.24 \%$$

$$\mu = 488.0 \pm 1.2$$

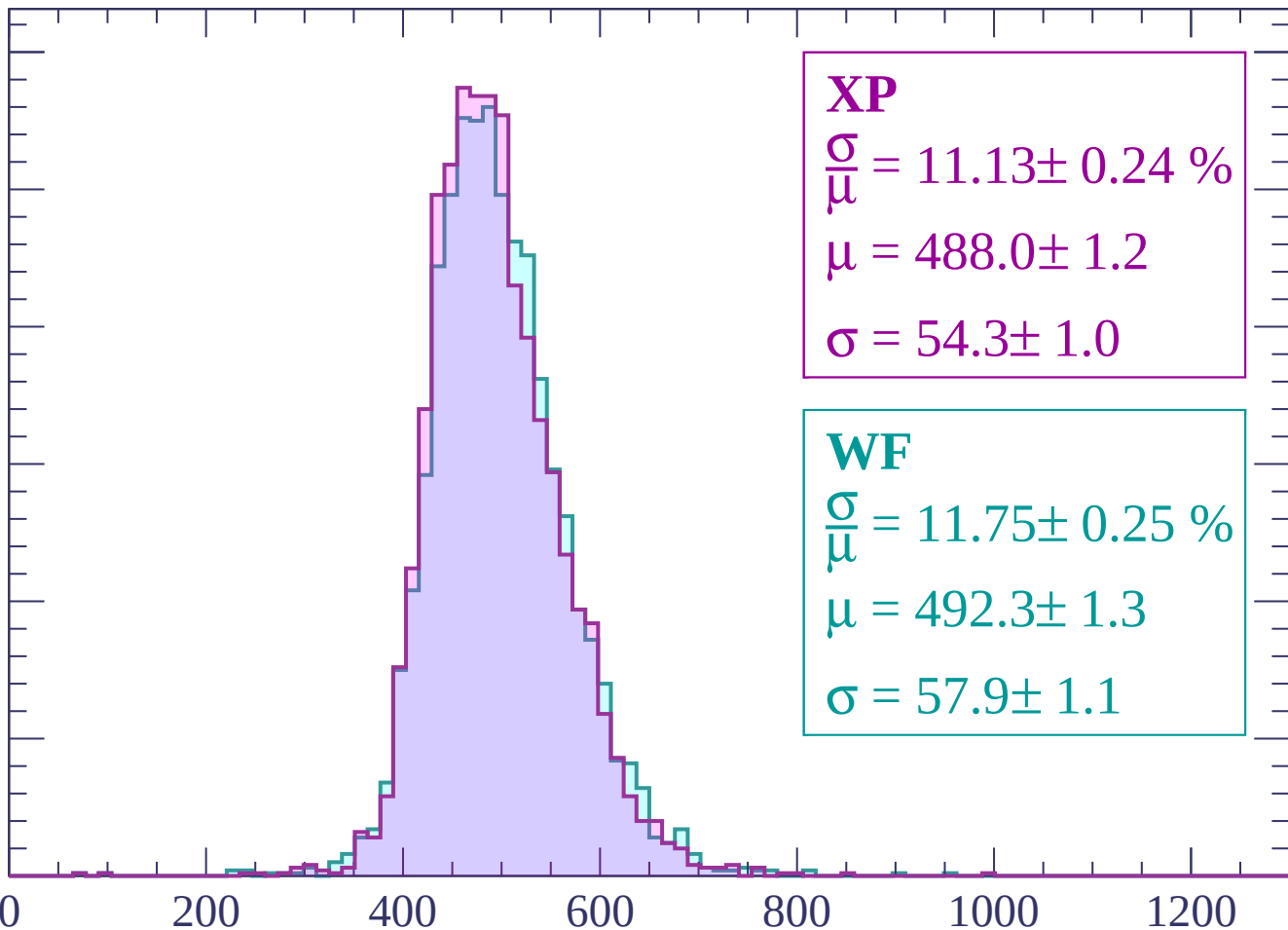
$$\sigma = 54.3 \pm 1.0$$

WF

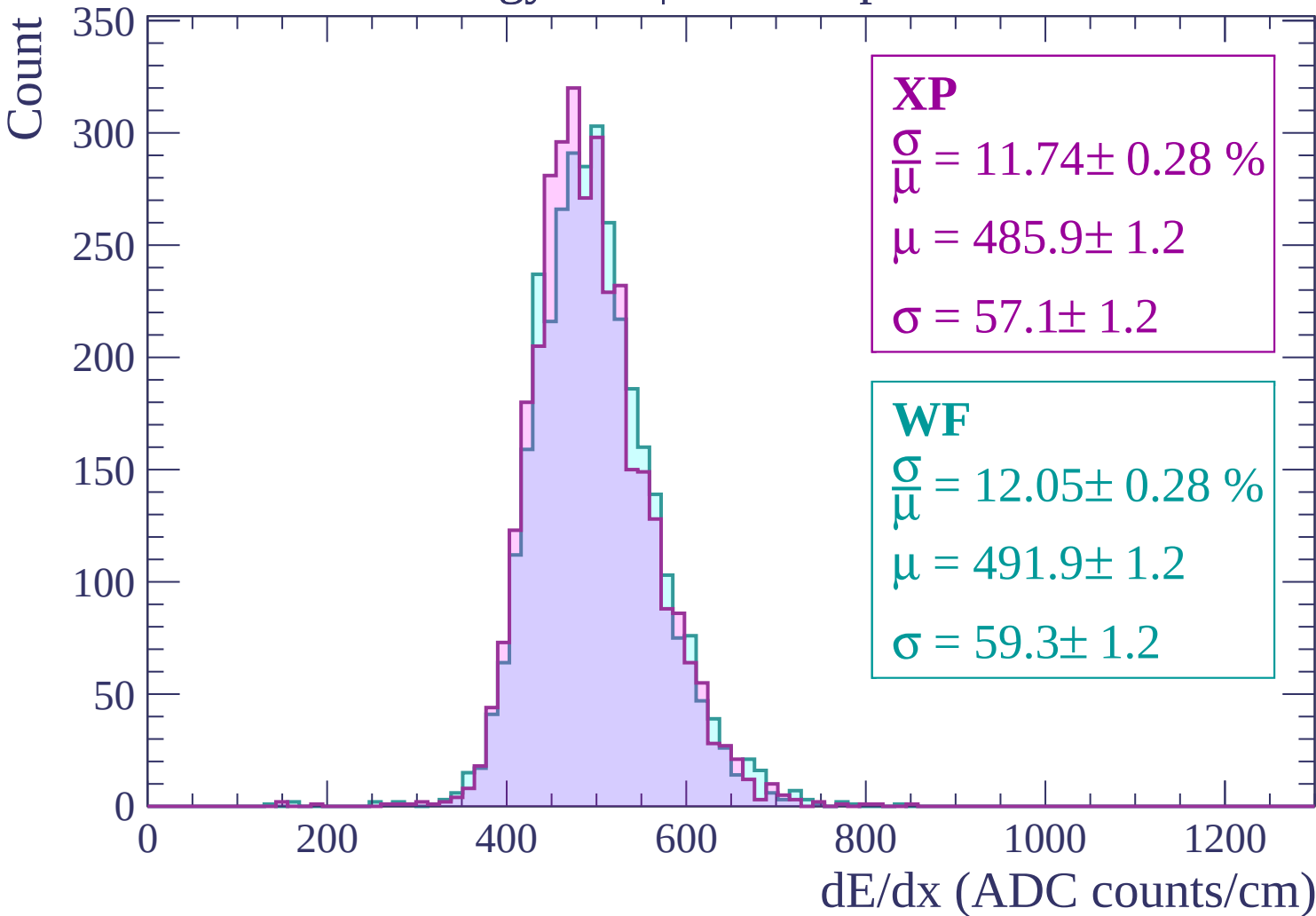
$$\frac{\sigma}{\mu} = 11.75 \pm 0.25 \%$$

$$\mu = 492.3 \pm 1.3$$

$$\sigma = 57.9 \pm 1.1$$

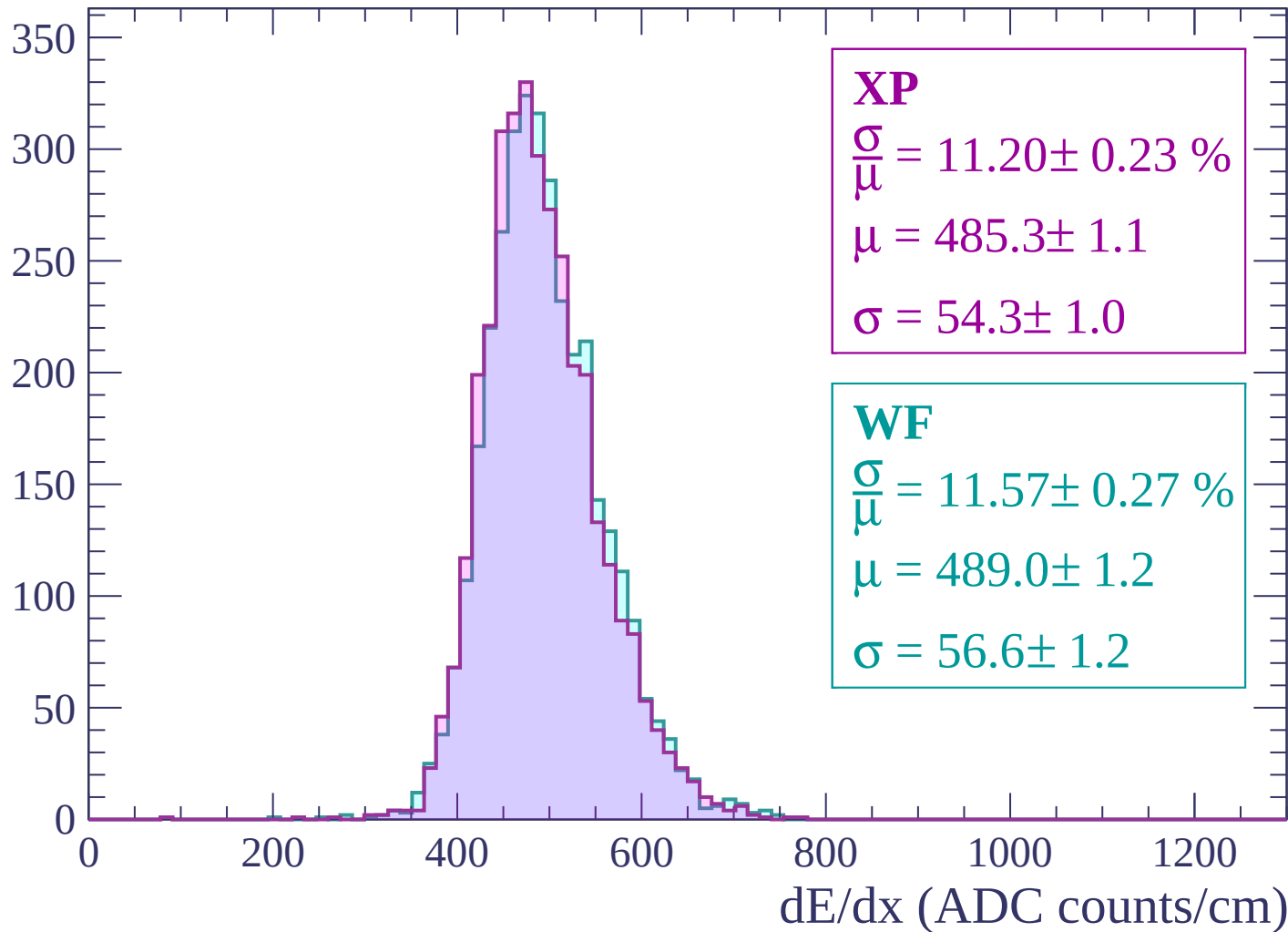


Energy loss | -1920 < p < -1860



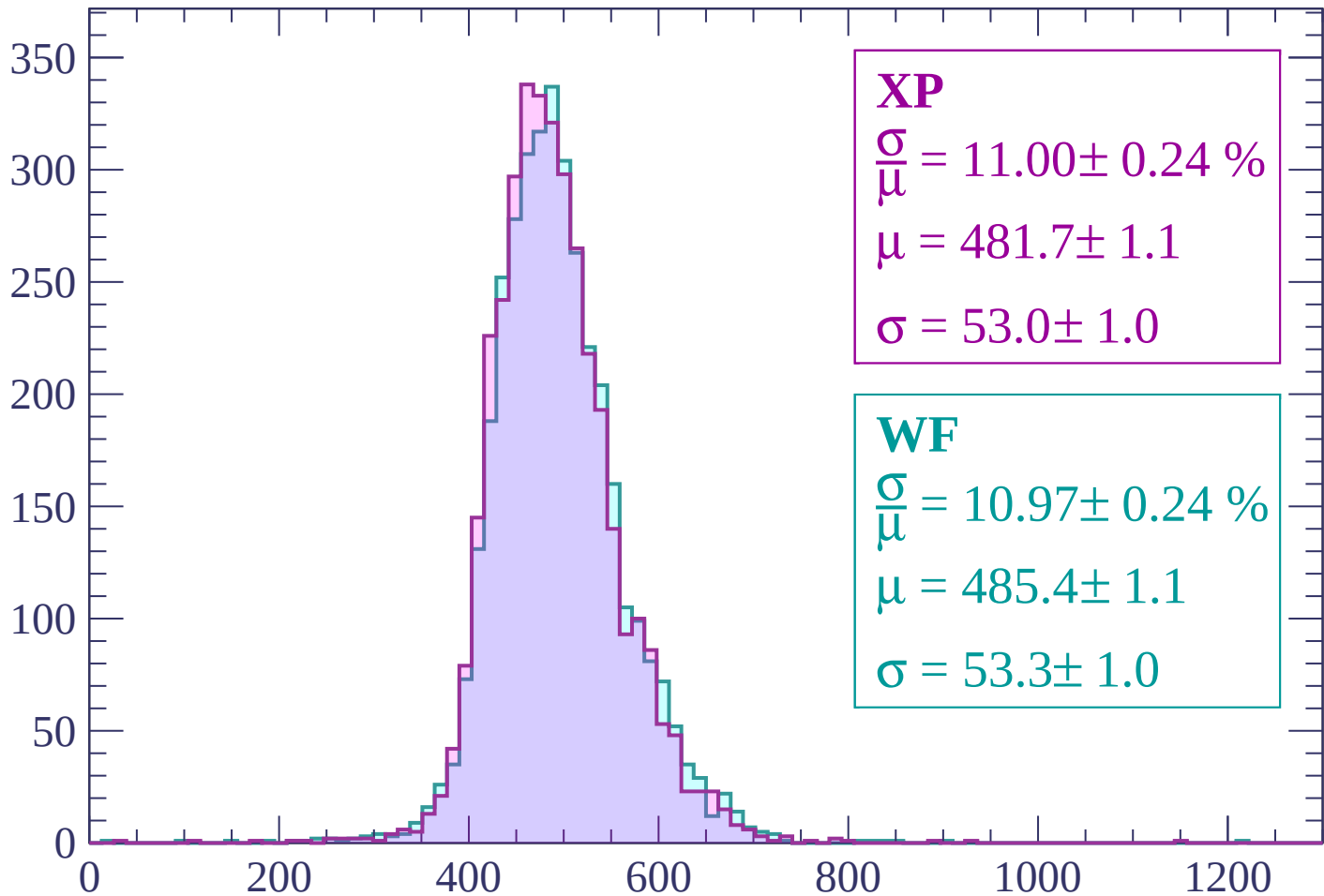
Energy loss | -1860 < p < -1800

Count



Energy loss | -1800 < p < -1740

Count



XP

$$\frac{\sigma}{\mu} = 11.00 \pm 0.24 \%$$

$$\mu = 481.7 \pm 1.1$$

$$\sigma = 53.0 \pm 1.0$$

WF

$$\frac{\sigma}{\mu} = 10.97 \pm 0.24 \%$$

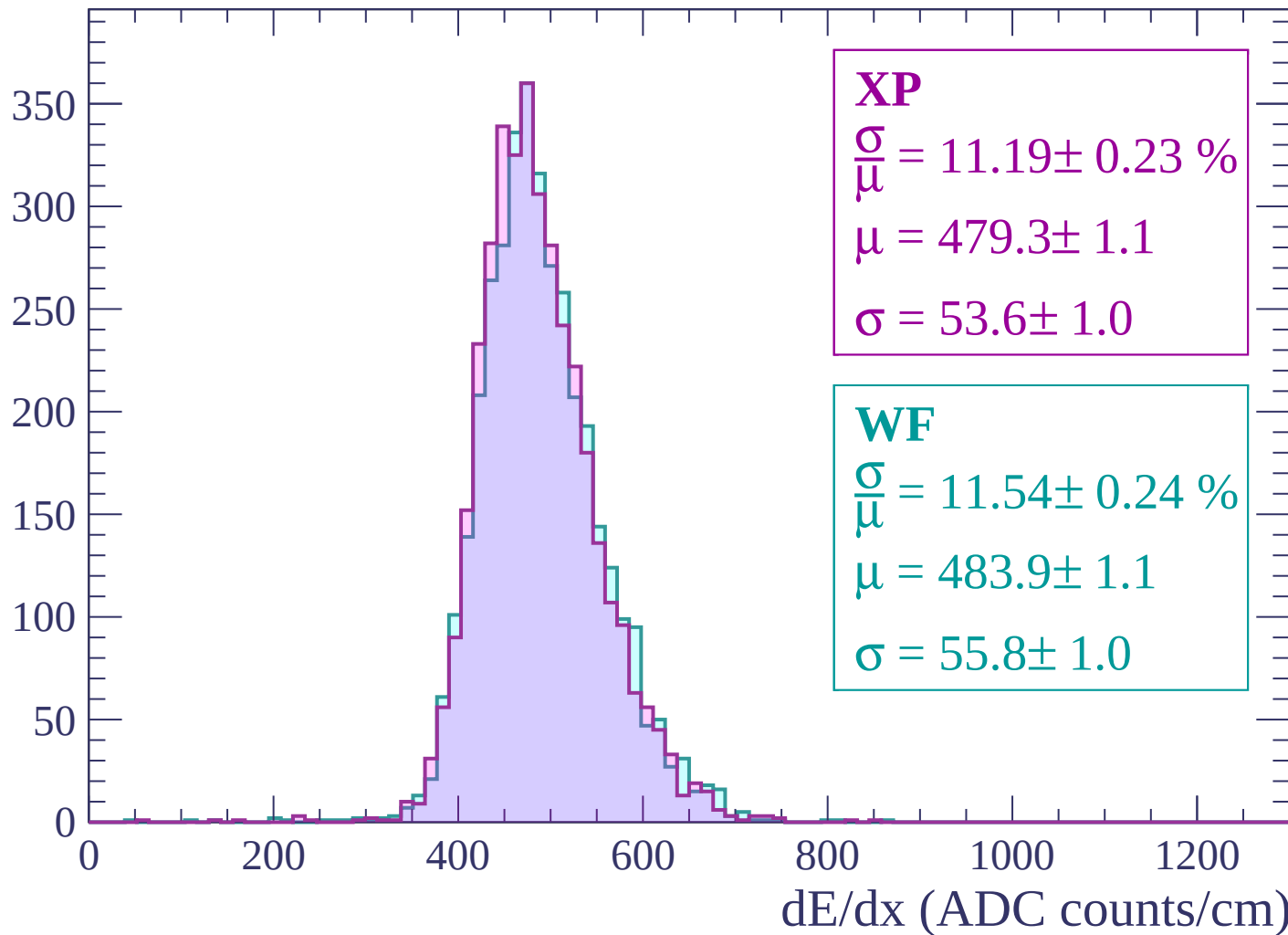
$$\mu = 485.4 \pm 1.1$$

$$\sigma = 53.3 \pm 1.0$$

dE/dx (ADC counts/cm)

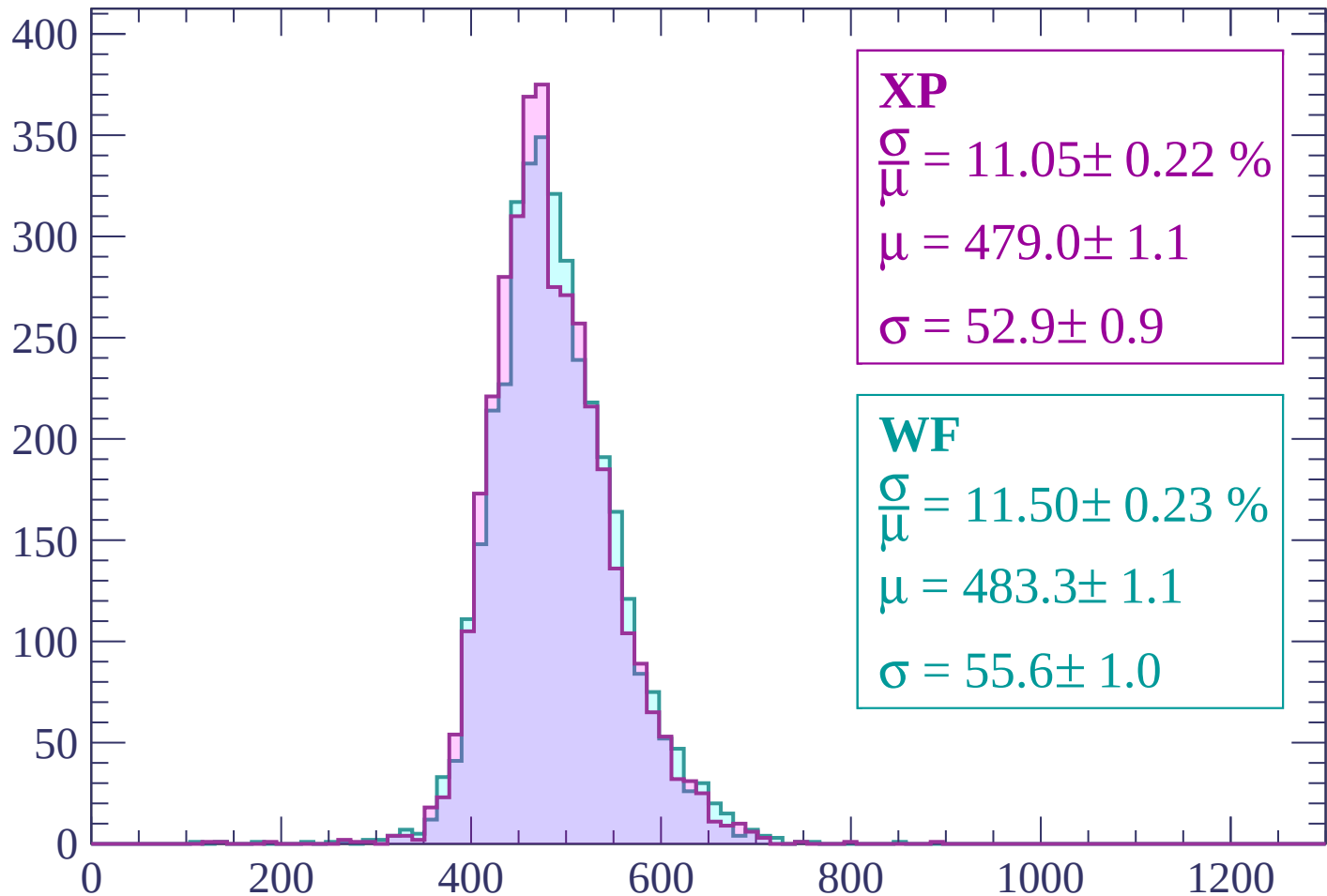
Energy loss | $-1740 < p < -1680$

Count



Energy loss | -1680 < p < -1620

Count



XP

$$\frac{\sigma}{\mu} = 11.05 \pm 0.22 \%$$

$$\mu = 479.0 \pm 1.1$$

$$\sigma = 52.9 \pm 0.9$$

WF

$$\frac{\sigma}{\mu} = 11.50 \pm 0.23 \%$$

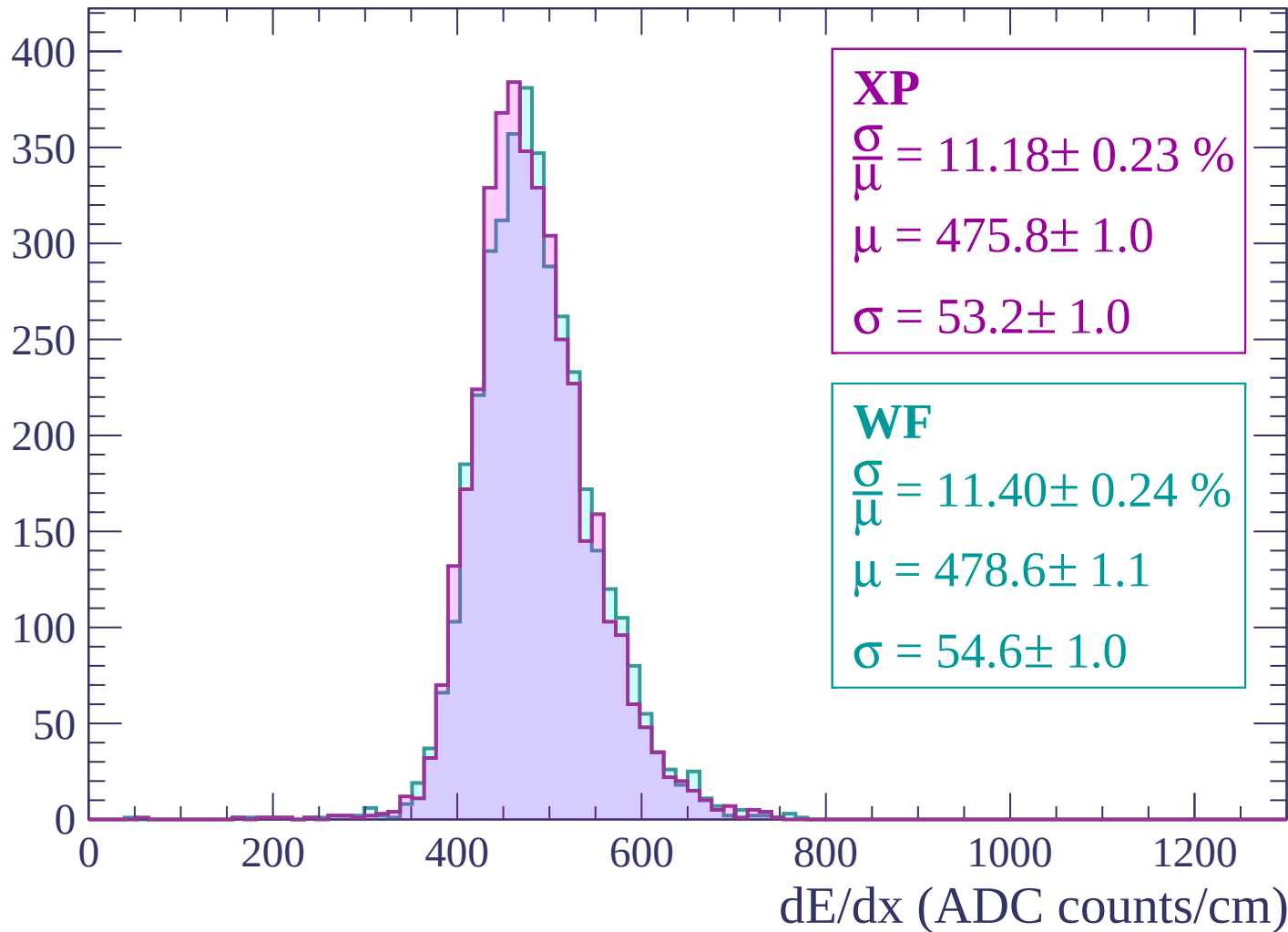
$$\mu = 483.3 \pm 1.1$$

$$\sigma = 55.6 \pm 1.0$$

dE/dx (ADC counts/cm)

Energy loss | -1620 < p < -1560

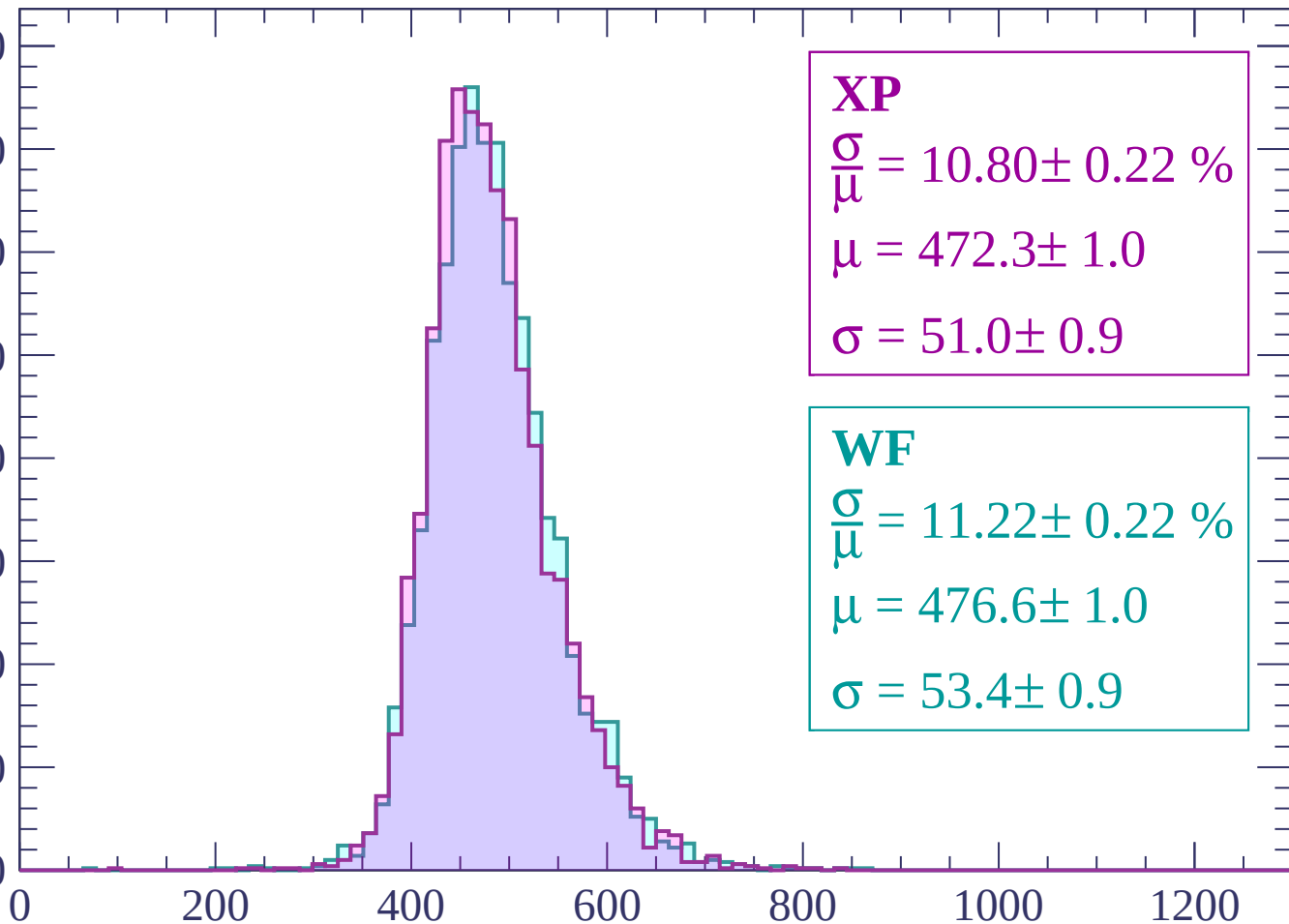
Count



Energy loss | -1560 < p < -1500

Count

400
350
300
250
200
150
100
50
0



XP

$$\frac{\sigma}{\mu} = 10.80 \pm 0.22 \%$$

$$\mu = 472.3 \pm 1.0$$

$$\sigma = 51.0 \pm 0.9$$

WF

$$\frac{\sigma}{\mu} = 11.22 \pm 0.22 \%$$

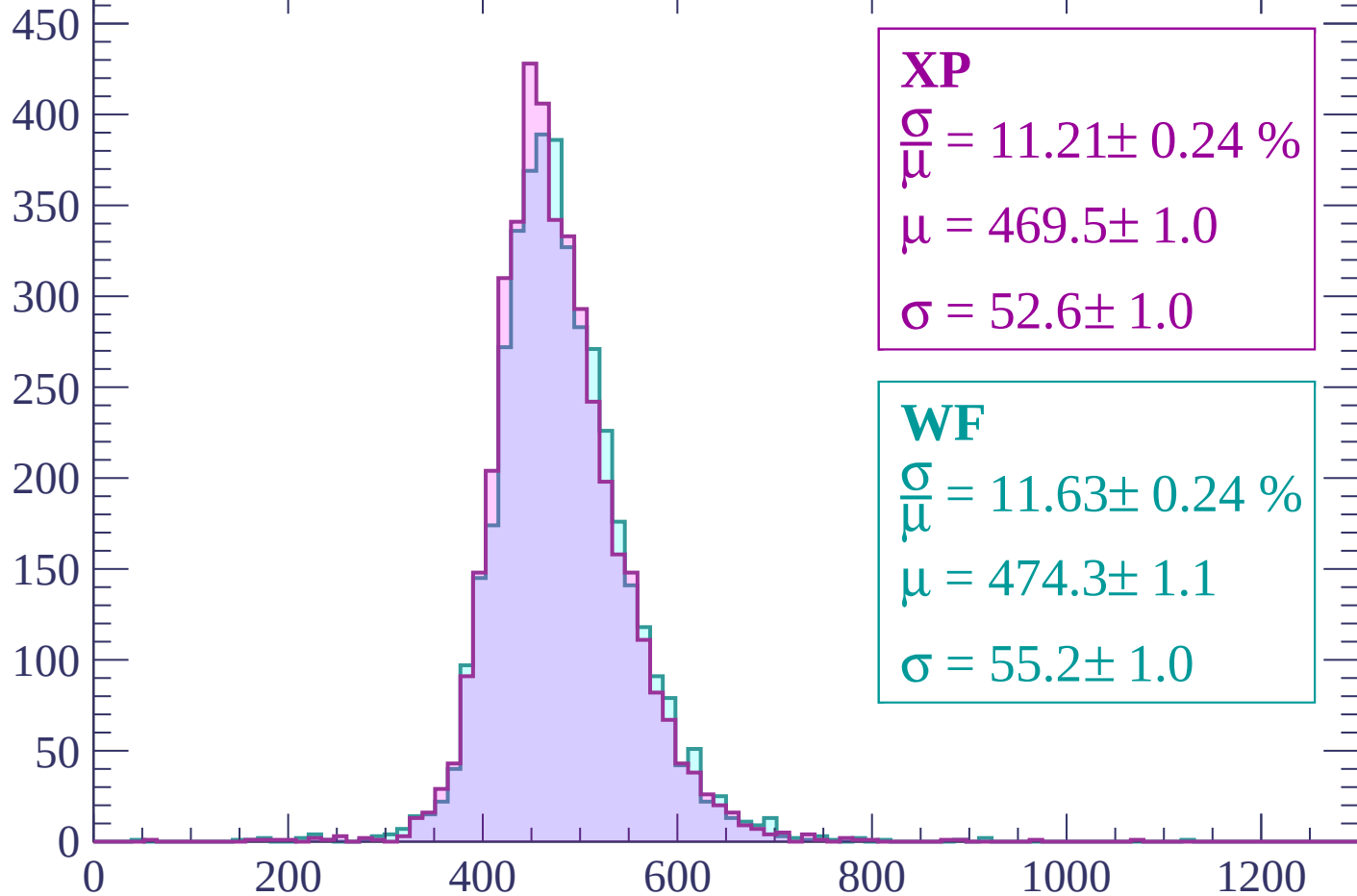
$$\mu = 476.6 \pm 1.0$$

$$\sigma = 53.4 \pm 0.9$$

dE/dx (ADC counts/cm)

Energy loss | -1500 < p < -1440

Count



XP

$$\frac{\sigma}{\mu} = 11.21 \pm 0.24 \%$$

$$\mu = 469.5 \pm 1.0$$

$$\sigma = 52.6 \pm 1.0$$

WF

$$\frac{\sigma}{\mu} = 11.63 \pm 0.24 \%$$

$$\mu = 474.3 \pm 1.1$$

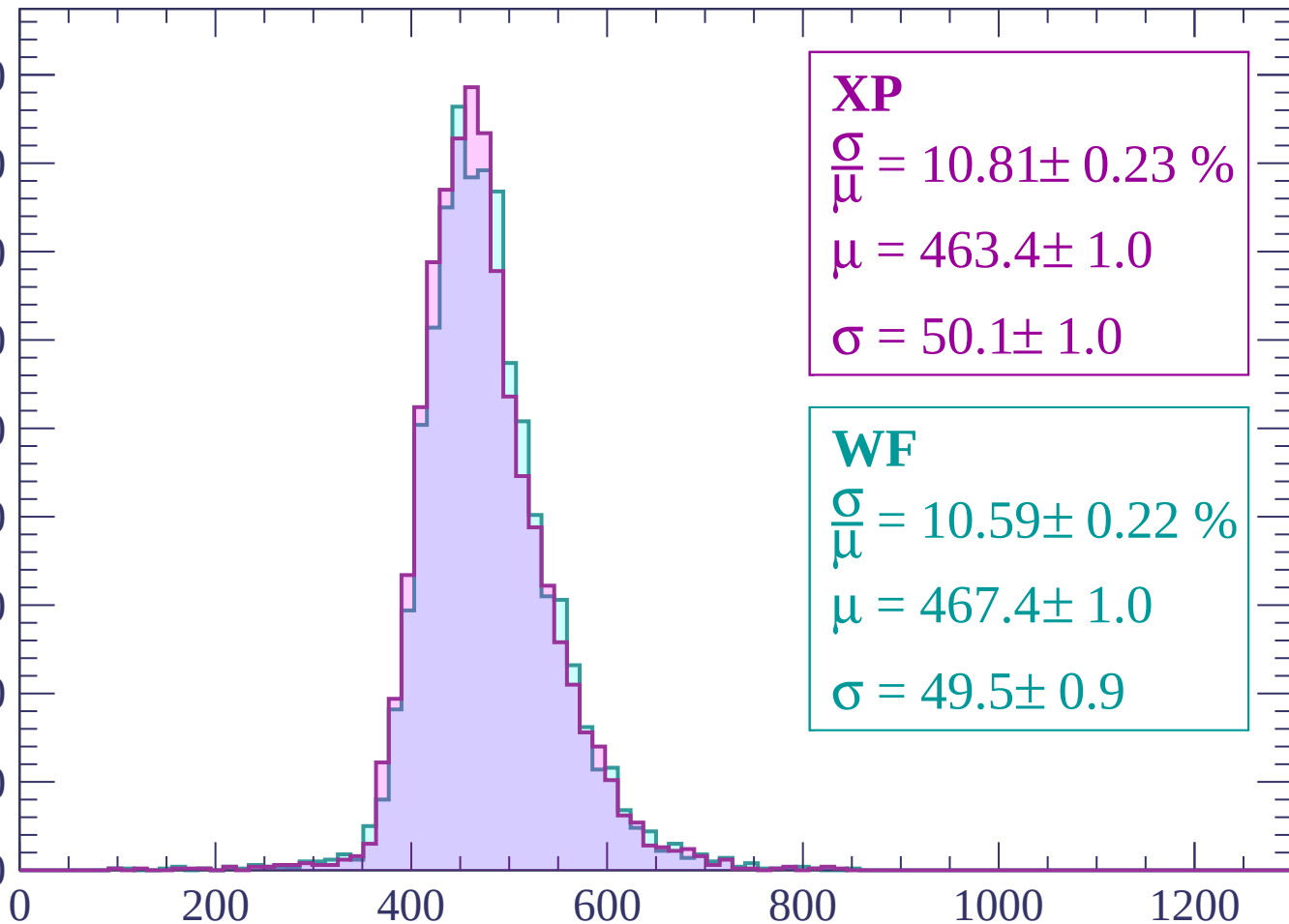
$$\sigma = 55.2 \pm 1.0$$

dE/dx (ADC counts/cm)

Energy loss | -1440 < p < -1380

Count

450
400
350
300
250
200
150
100
50
0



XP

$$\frac{\sigma}{\mu} = 10.81 \pm 0.23 \%$$

$$\mu = 463.4 \pm 1.0$$

$$\sigma = 50.1 \pm 1.0$$

WF

$$\frac{\sigma}{\mu} = 10.59 \pm 0.22 \%$$

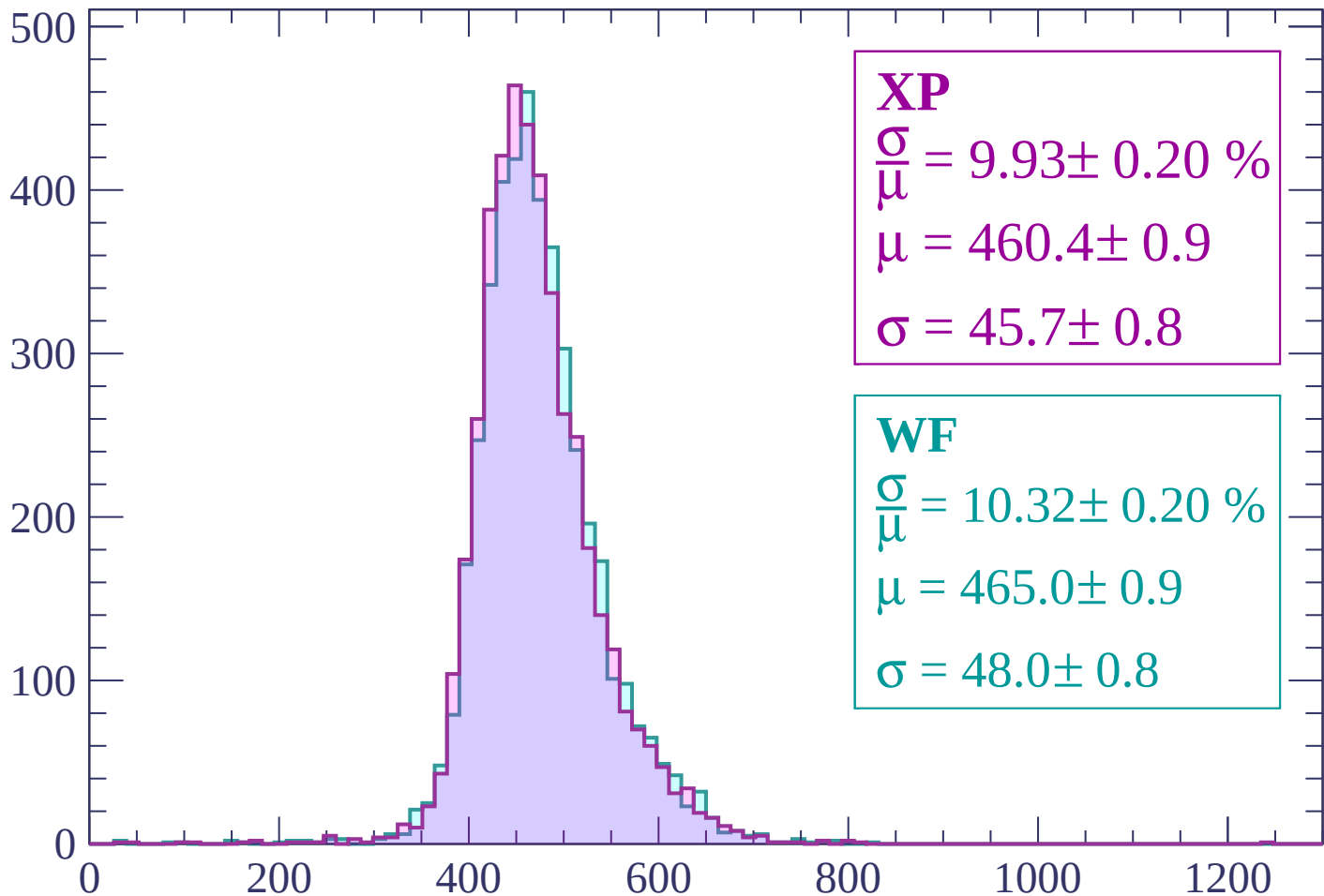
$$\mu = 467.4 \pm 1.0$$

$$\sigma = 49.5 \pm 0.9$$

dE/dx (ADC counts/cm)

Energy loss | -1380 < p < -1320

Count



XP

$$\frac{\sigma}{\mu} = 9.93 \pm 0.20 \%$$

$$\mu = 460.4 \pm 0.9$$

$$\sigma = 45.7 \pm 0.8$$

WF

$$\frac{\sigma}{\mu} = 10.32 \pm 0.20 \%$$

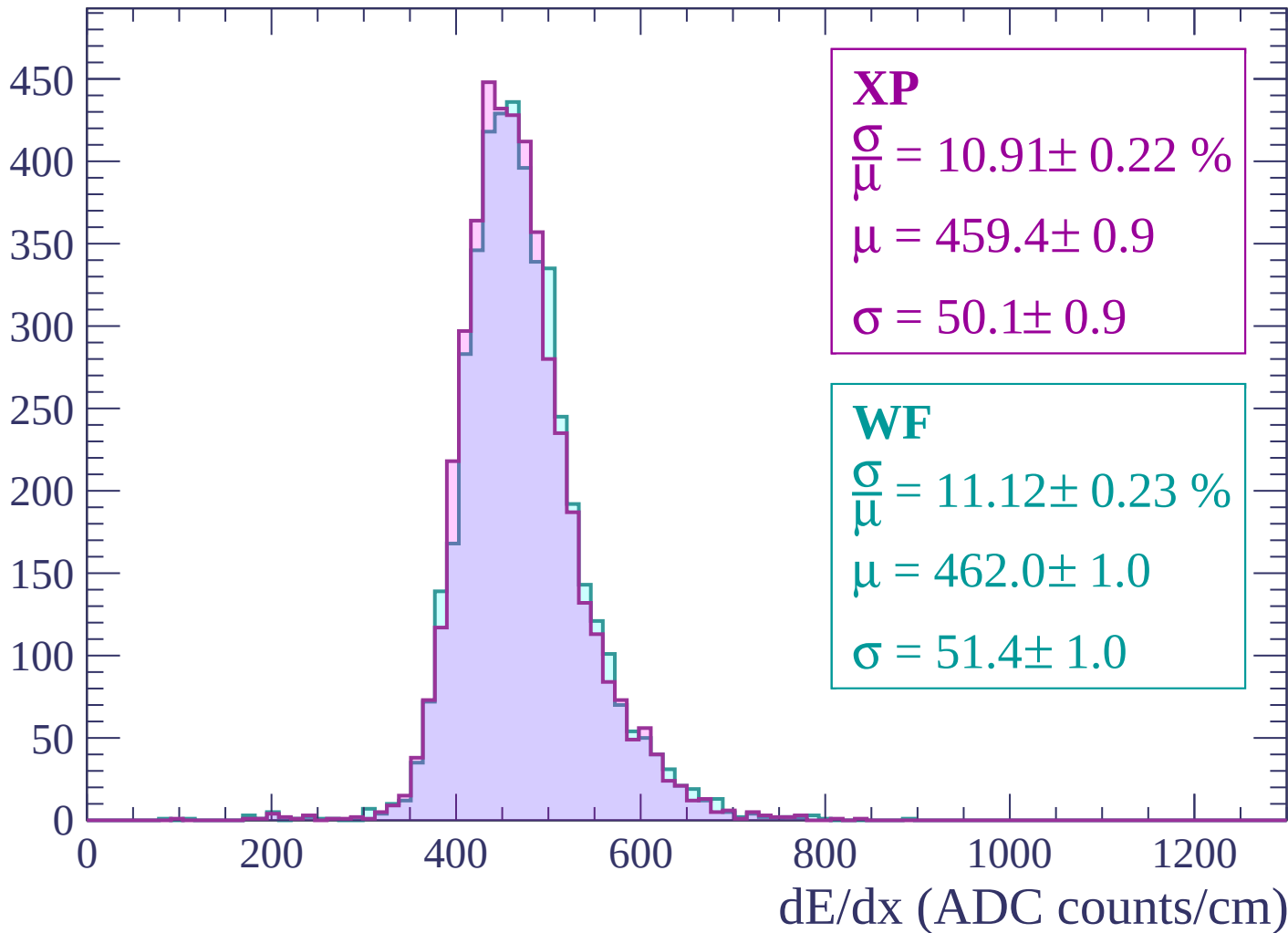
$$\mu = 465.0 \pm 0.9$$

$$\sigma = 48.0 \pm 0.8$$

dE/dx (ADC counts/cm)

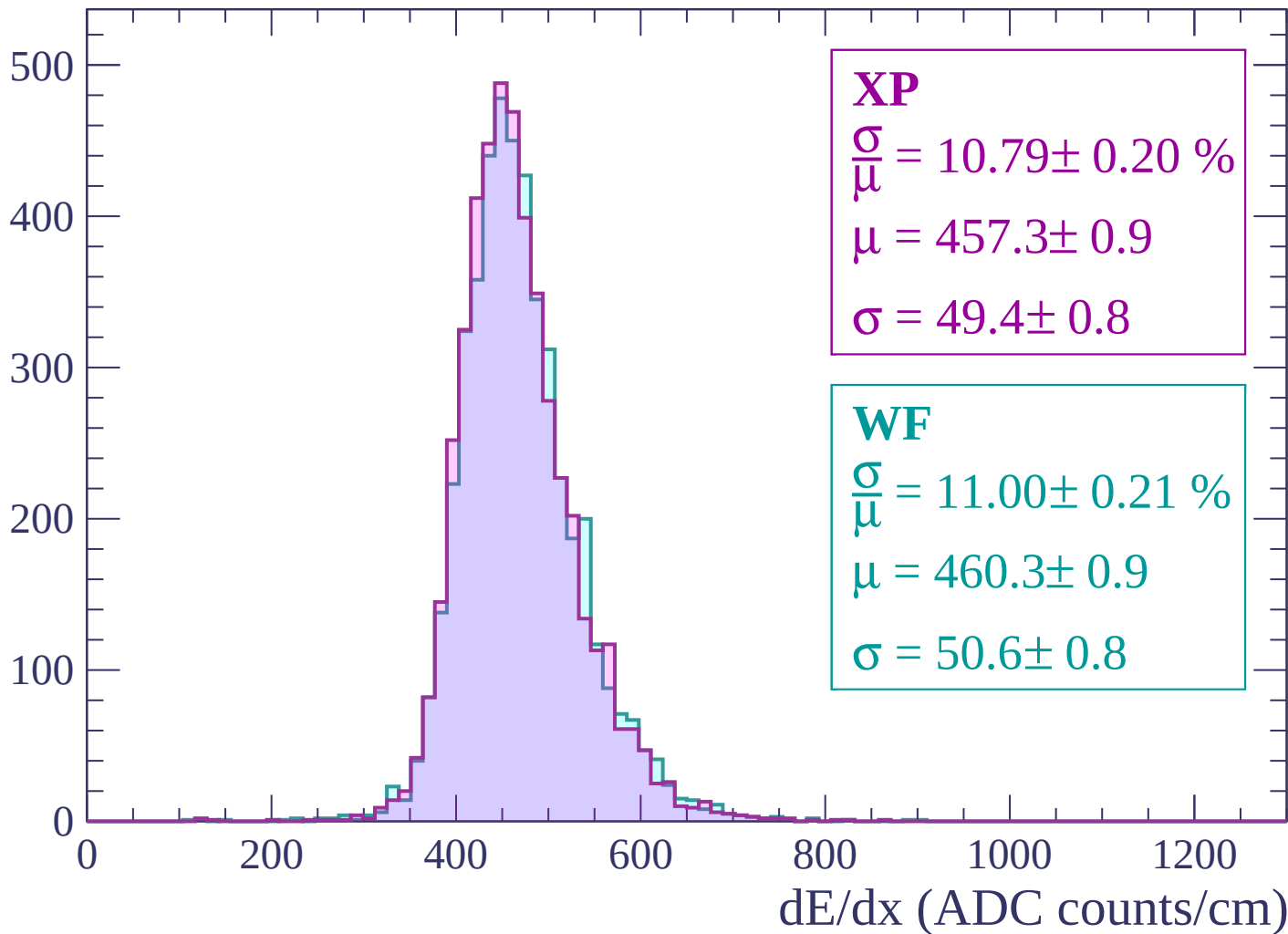
Energy loss | -1320 < p < -1260

Count



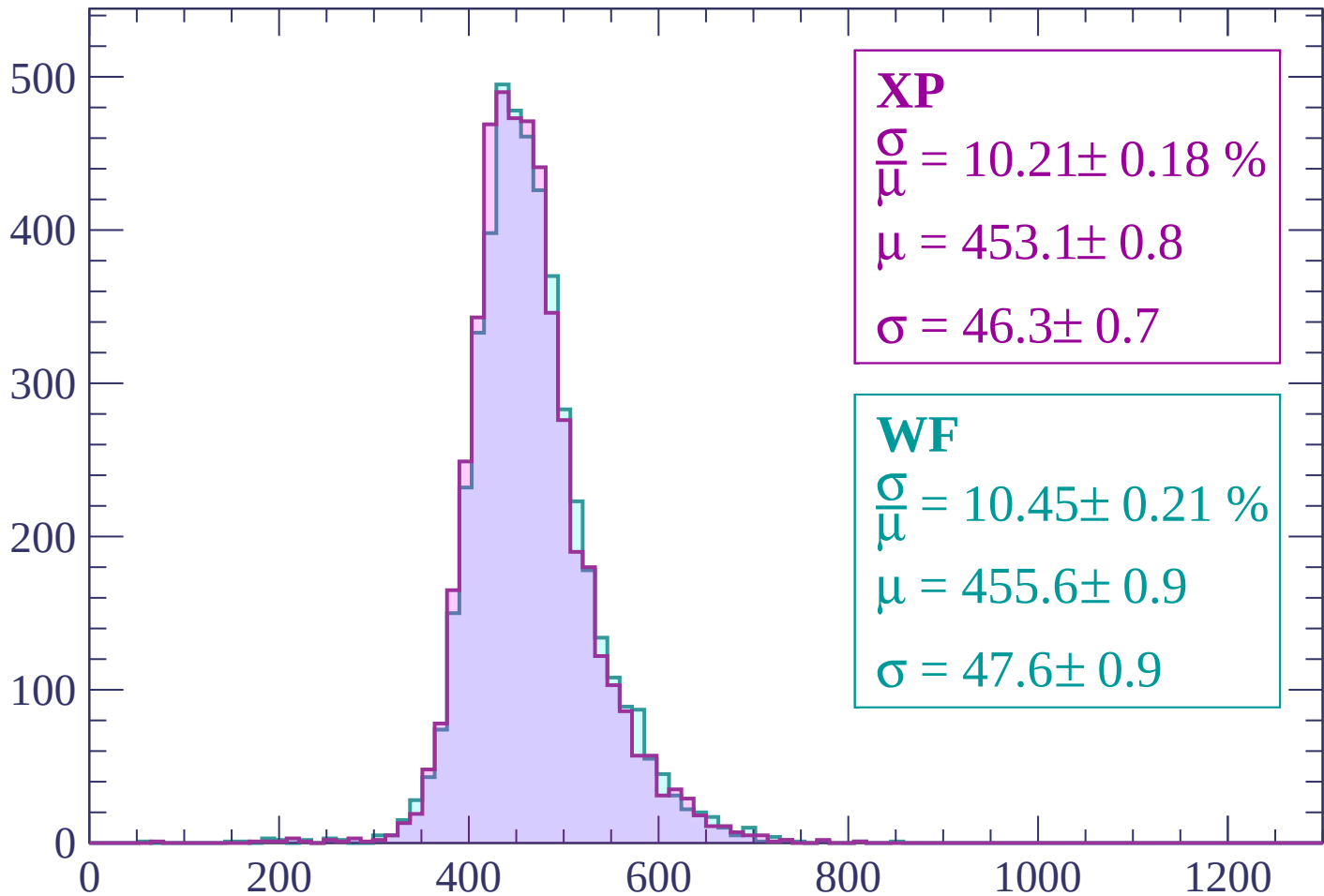
Energy loss | $-1260 < p < -1200$

Count



Energy loss | -1200 < p < -1140

Count



XP

$$\frac{\sigma}{\mu} = 10.21 \pm 0.18 \%$$

$$\mu = 453.1 \pm 0.8$$

$$\sigma = 46.3 \pm 0.7$$

WF

$$\frac{\sigma}{\mu} = 10.45 \pm 0.21 \%$$

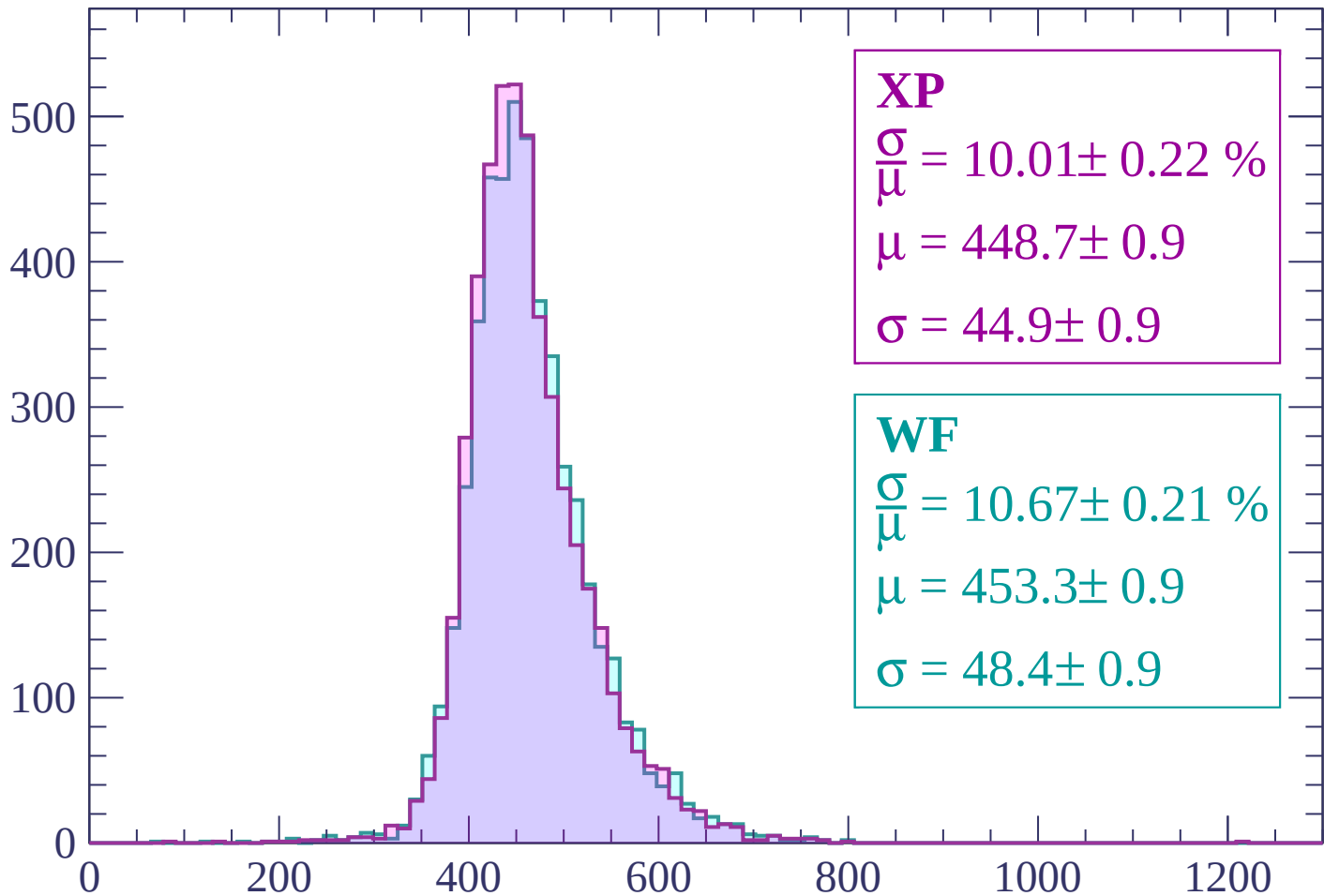
$$\mu = 455.6 \pm 0.9$$

$$\sigma = 47.6 \pm 0.9$$

dE/dx (ADC counts/cm)

Energy loss | -1140 < p < -1080

Count



XP

$$\frac{\sigma}{\mu} = 10.01 \pm 0.22 \%$$

$$\mu = 448.7 \pm 0.9$$

$$\sigma = 44.9 \pm 0.9$$

WF

$$\frac{\sigma}{\mu} = 10.67 \pm 0.21 \%$$

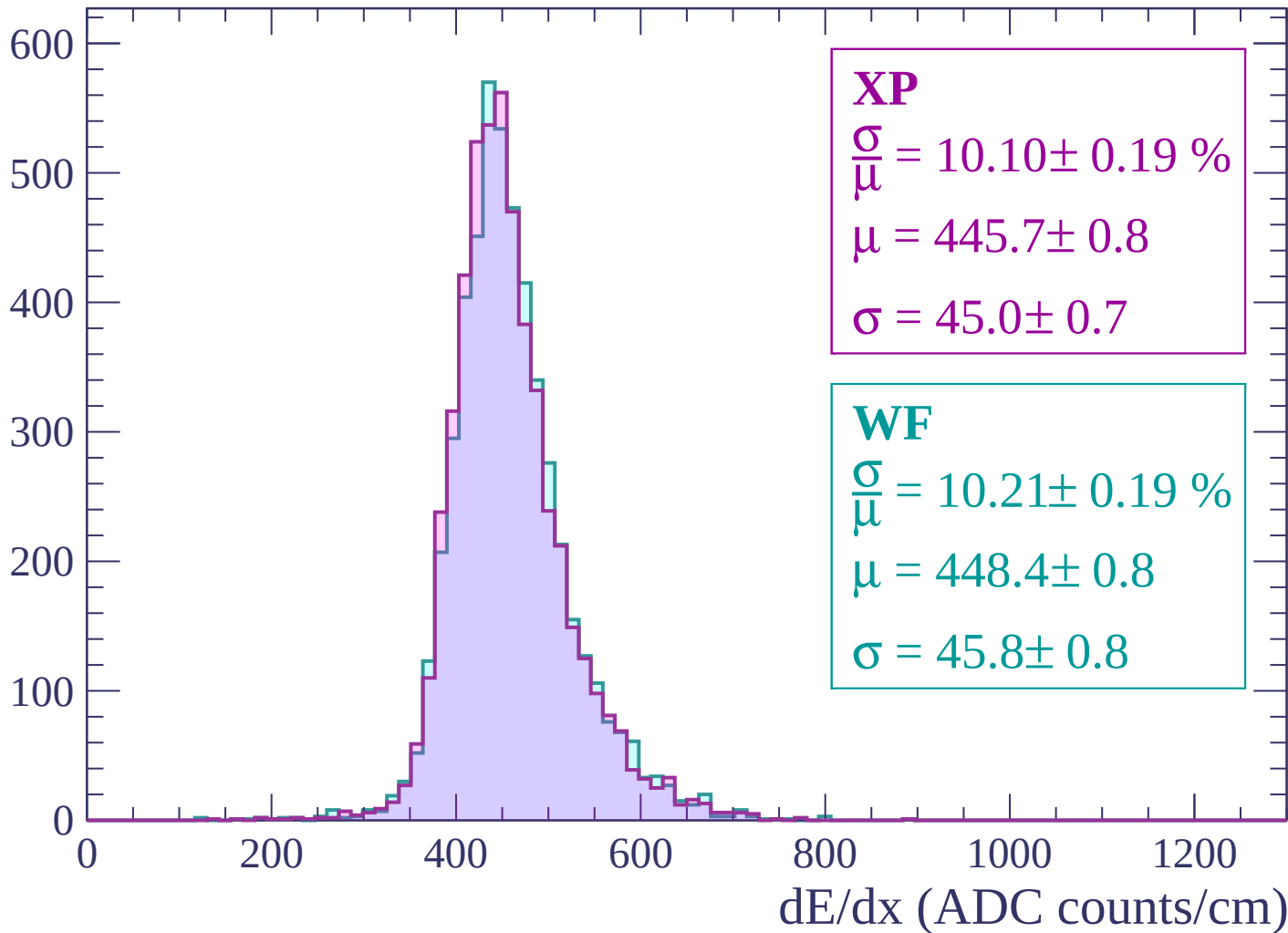
$$\mu = 453.3 \pm 0.9$$

$$\sigma = 48.4 \pm 0.9$$

dE/dx (ADC counts/cm)

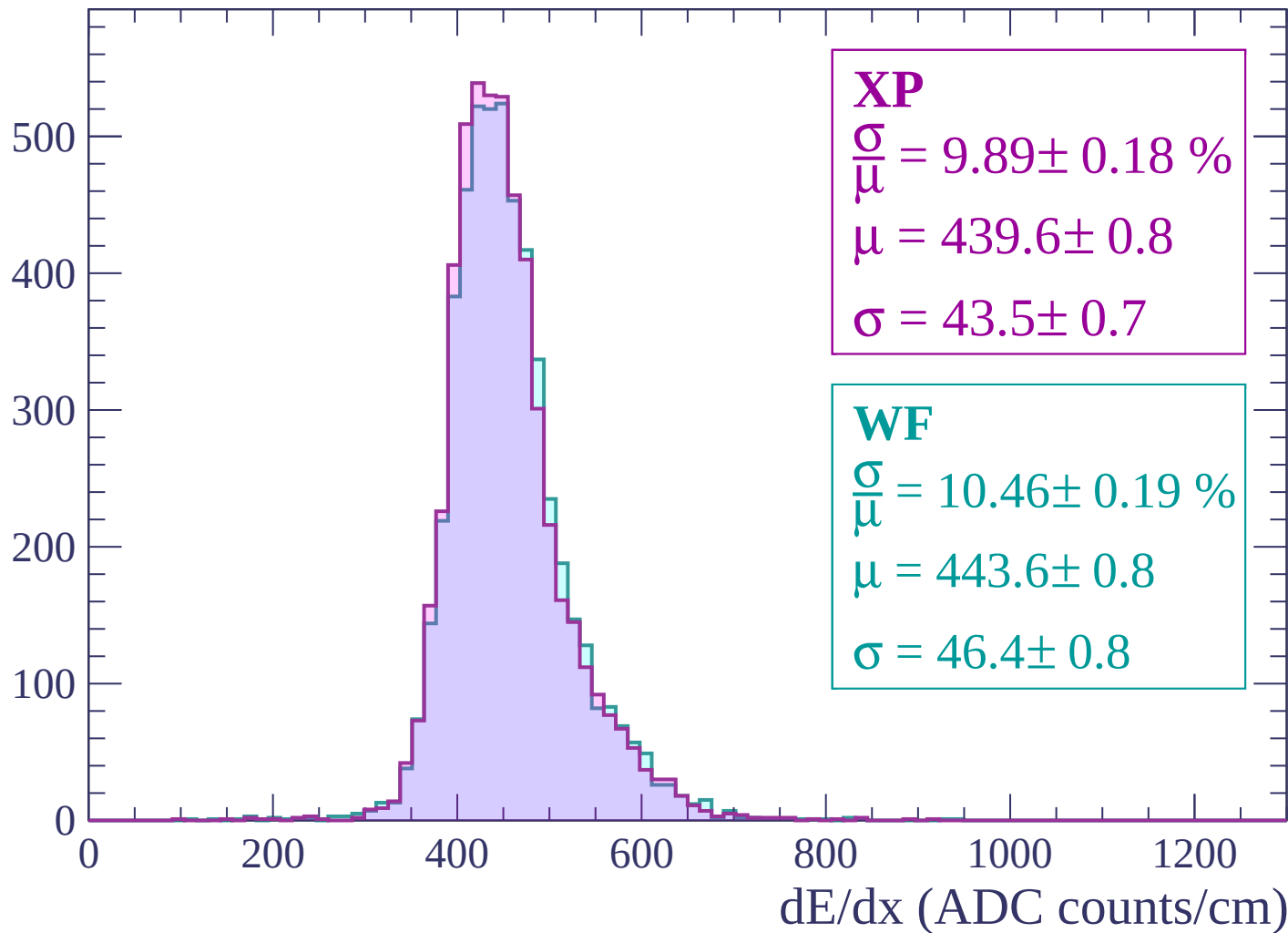
Energy loss | -1080 < p < -1020

Count



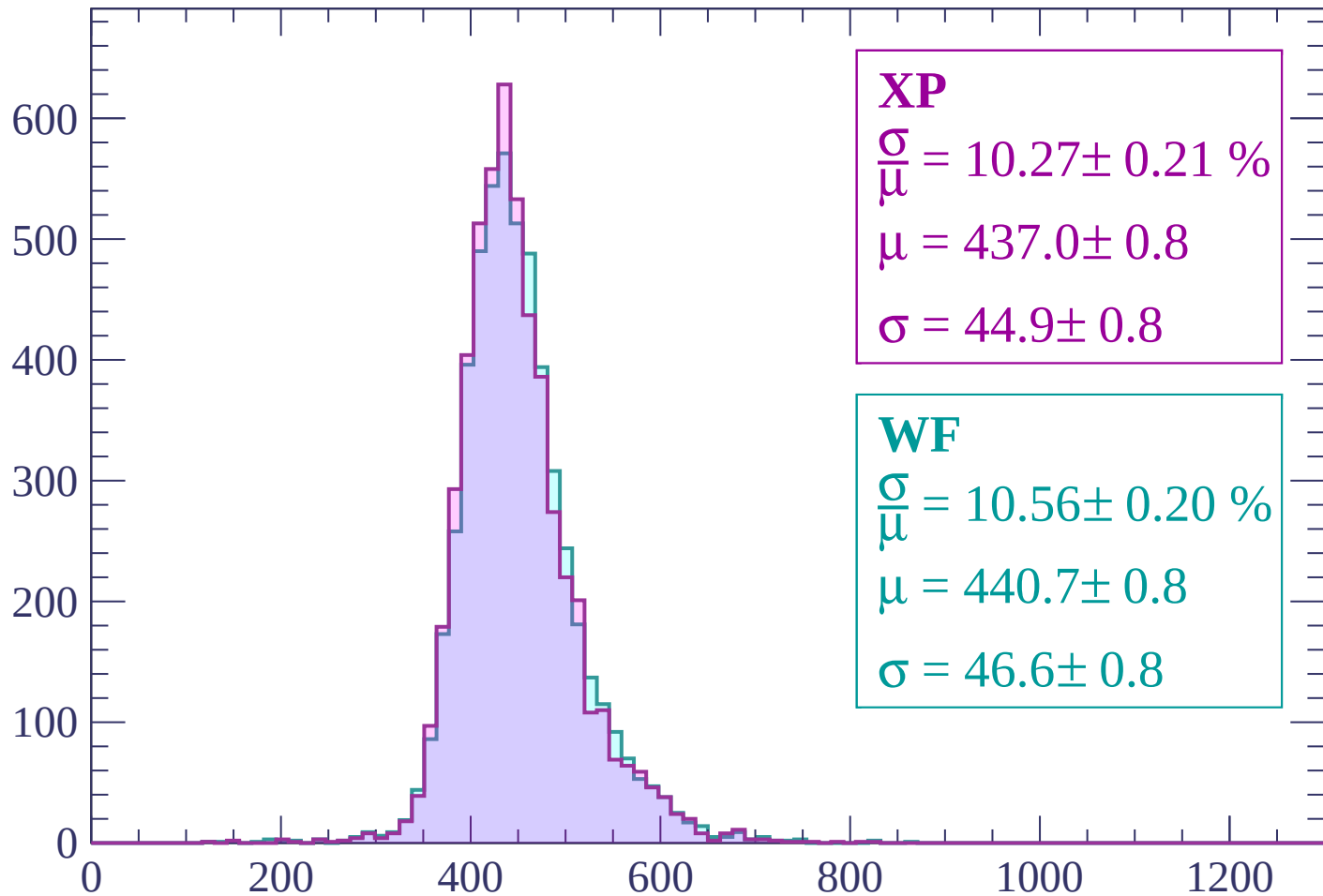
Energy loss | $-1020 < p < -960$

Count



Energy loss | $-960 < p < -900$

Count



XP

$$\frac{\sigma}{\mu} = 10.27 \pm 0.21 \%$$

$$\mu = 437.0 \pm 0.8$$

$$\sigma = 44.9 \pm 0.8$$

WF

$$\frac{\sigma}{\mu} = 10.56 \pm 0.20 \%$$

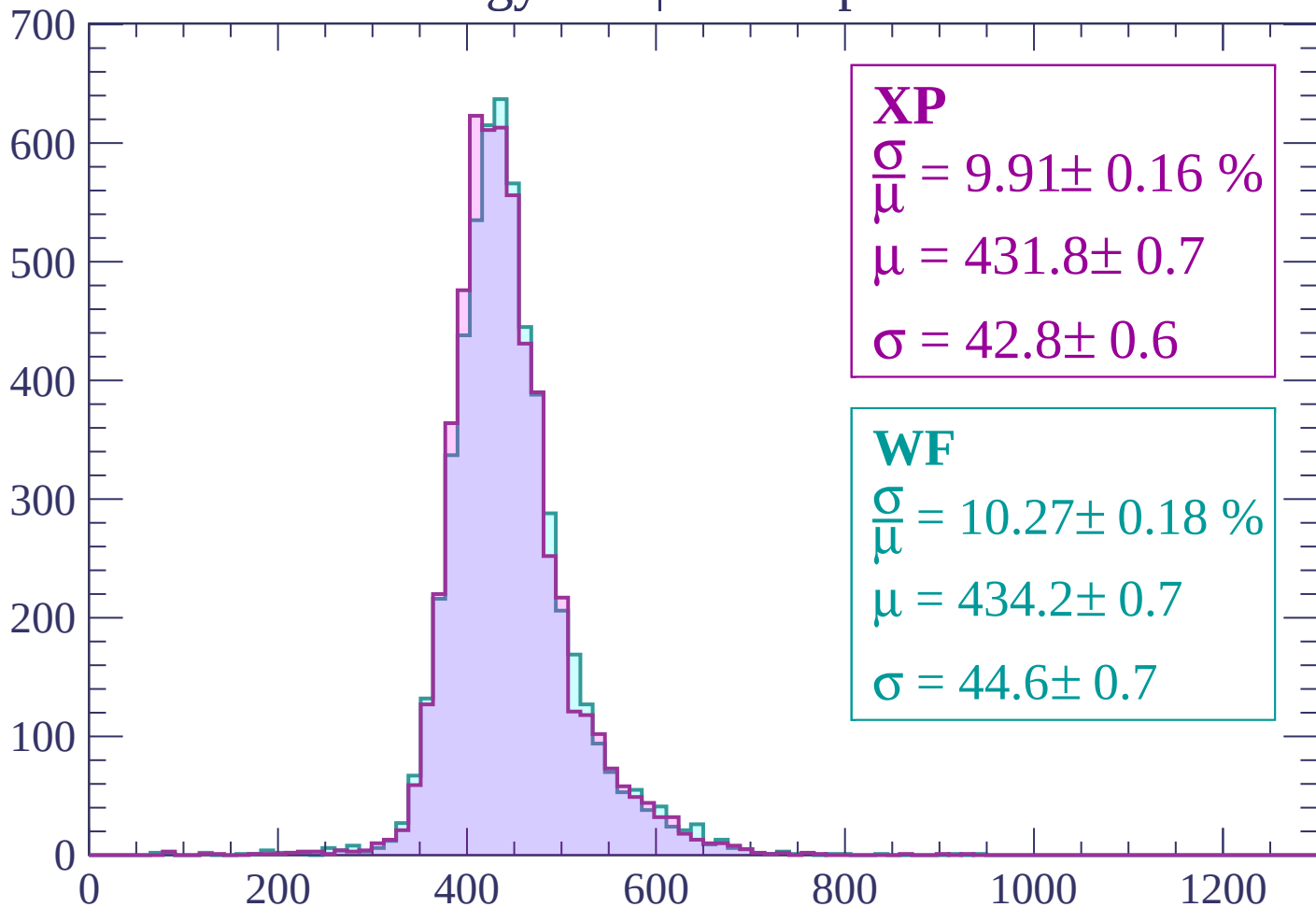
$$\mu = 440.7 \pm 0.8$$

$$\sigma = 46.6 \pm 0.8$$

dE/dx (ADC counts/cm)

Energy loss | $-900 < p < -840$

Count



XP

$$\frac{\sigma}{\mu} = 9.91 \pm 0.16 \%$$

$$\mu = 431.8 \pm 0.7$$

$$\sigma = 42.8 \pm 0.6$$

WF

$$\frac{\sigma}{\mu} = 10.27 \pm 0.18 \%$$

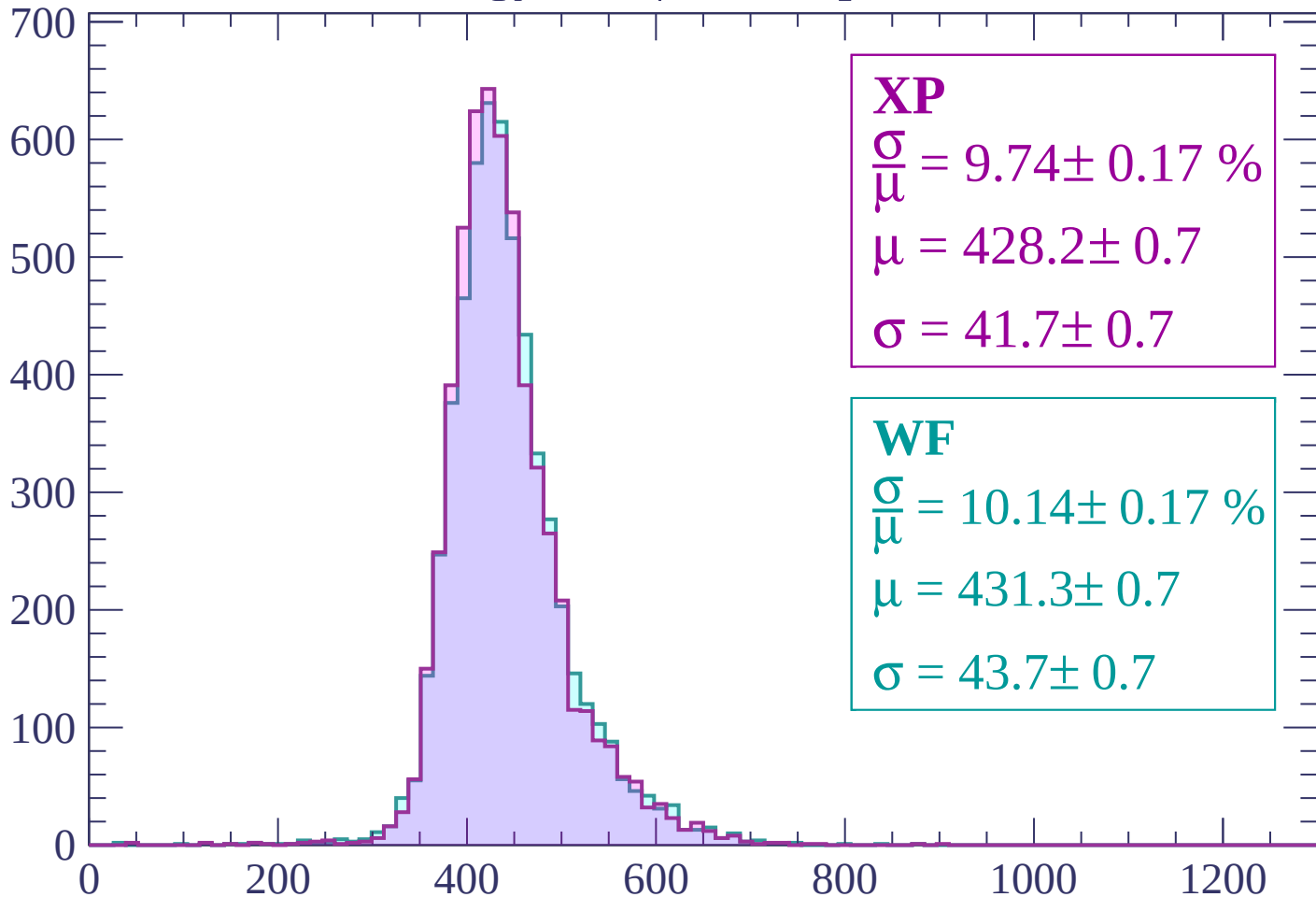
$$\mu = 434.2 \pm 0.7$$

$$\sigma = 44.6 \pm 0.7$$

dE/dx (ADC counts/cm)

Energy loss | $-840 < p < -780$

Count



XP

$$\frac{\sigma}{\mu} = 9.74 \pm 0.17 \%$$

$$\mu = 428.2 \pm 0.7$$

$$\sigma = 41.7 \pm 0.7$$

WF

$$\frac{\sigma}{\mu} = 10.14 \pm 0.17 \%$$

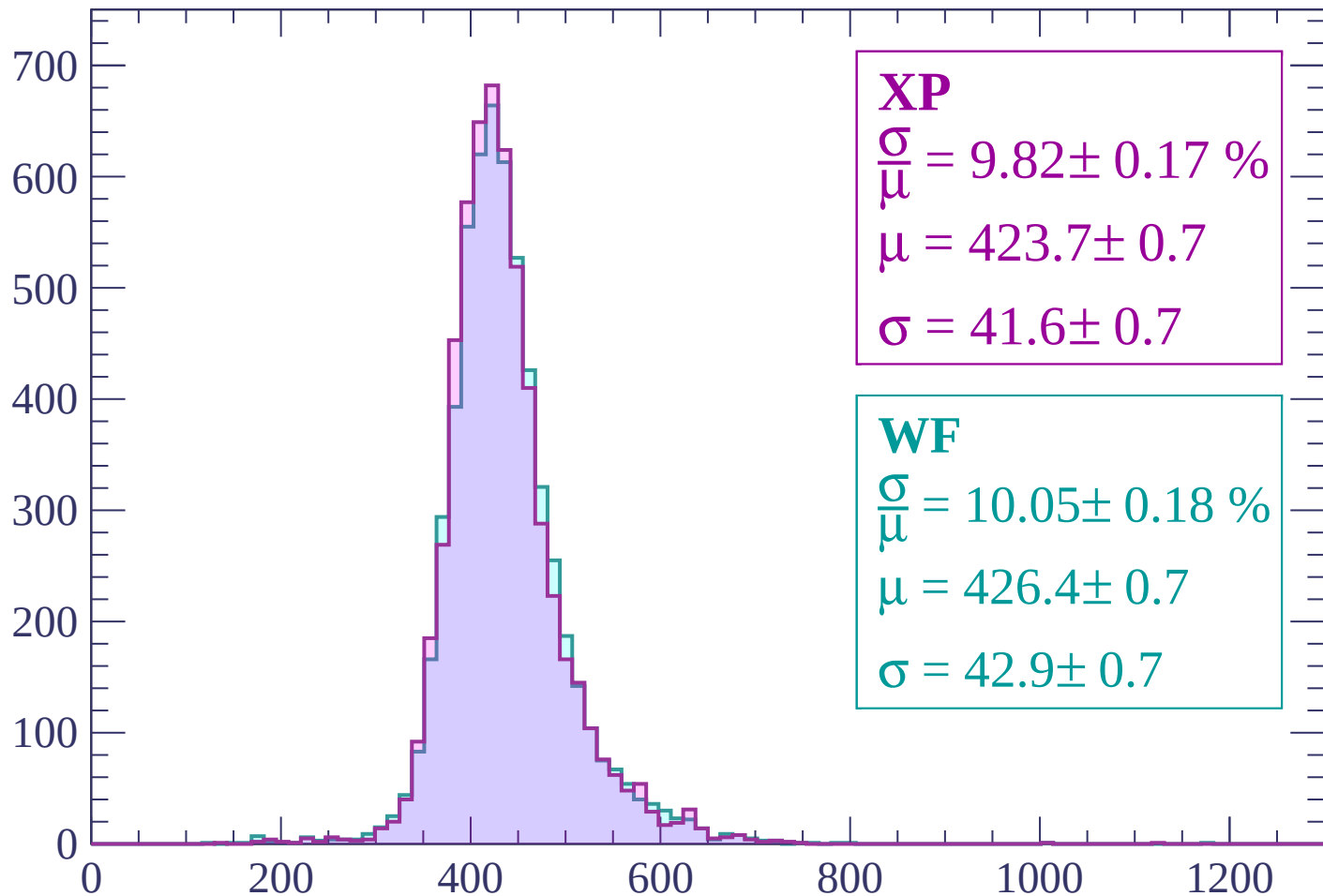
$$\mu = 431.3 \pm 0.7$$

$$\sigma = 43.7 \pm 0.7$$

dE/dx (ADC counts/cm)

Energy loss | $-780 < p < -720$

Count



XP

$$\frac{\sigma}{\mu} = 9.82 \pm 0.17 \%$$

$$\mu = 423.7 \pm 0.7$$

$$\sigma = 41.6 \pm 0.7$$

WF

$$\frac{\sigma}{\mu} = 10.05 \pm 0.18 \%$$

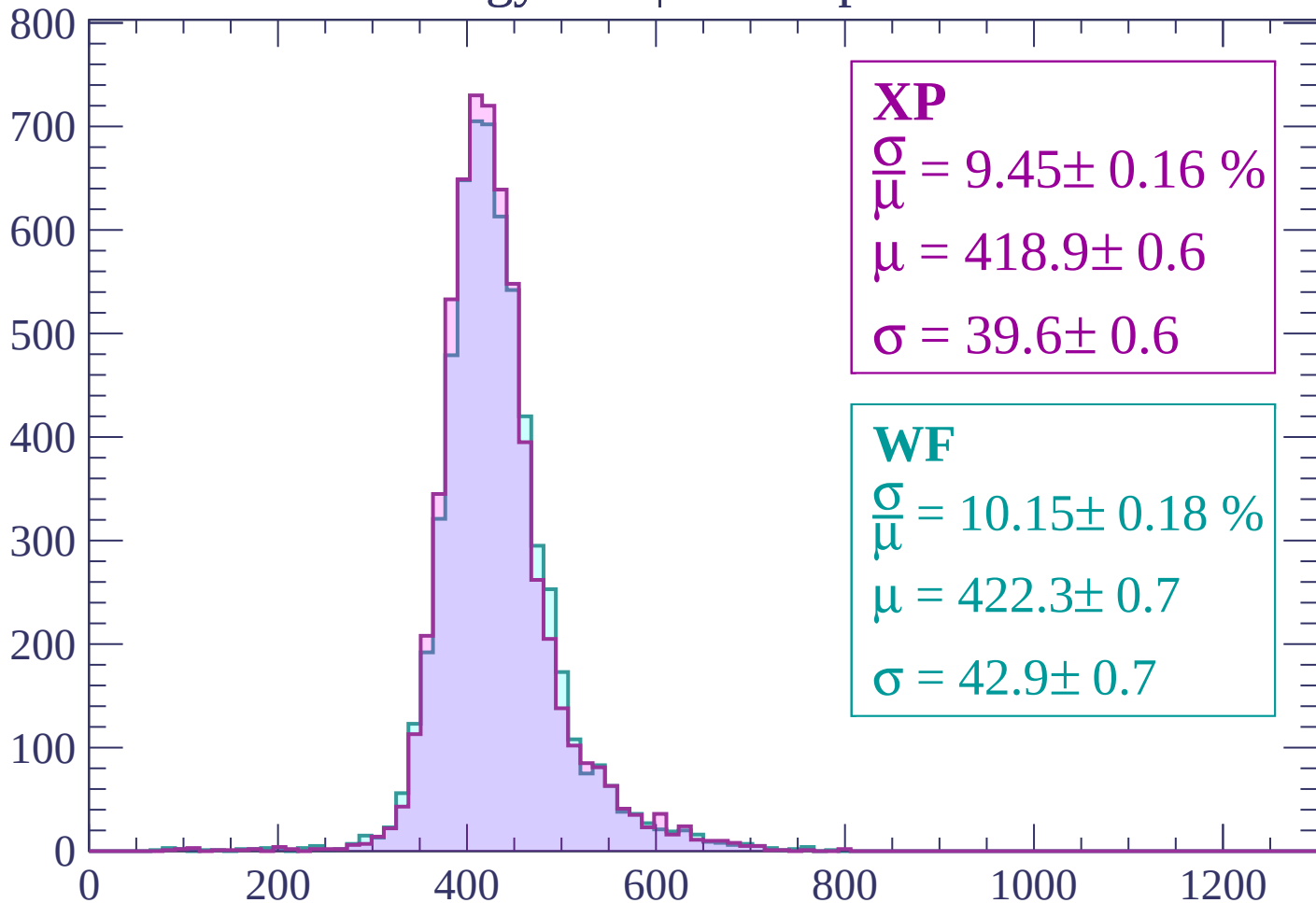
$$\mu = 426.4 \pm 0.7$$

$$\sigma = 42.9 \pm 0.7$$

dE/dx (ADC counts/cm)

Energy loss | $-720 < p < -660$

Count



XP

$$\frac{\sigma}{\mu} = 9.45 \pm 0.16 \%$$

$$\mu = 418.9 \pm 0.6$$

$$\sigma = 39.6 \pm 0.6$$

WF

$$\frac{\sigma}{\mu} = 10.15 \pm 0.18 \%$$

$$\mu = 422.3 \pm 0.7$$

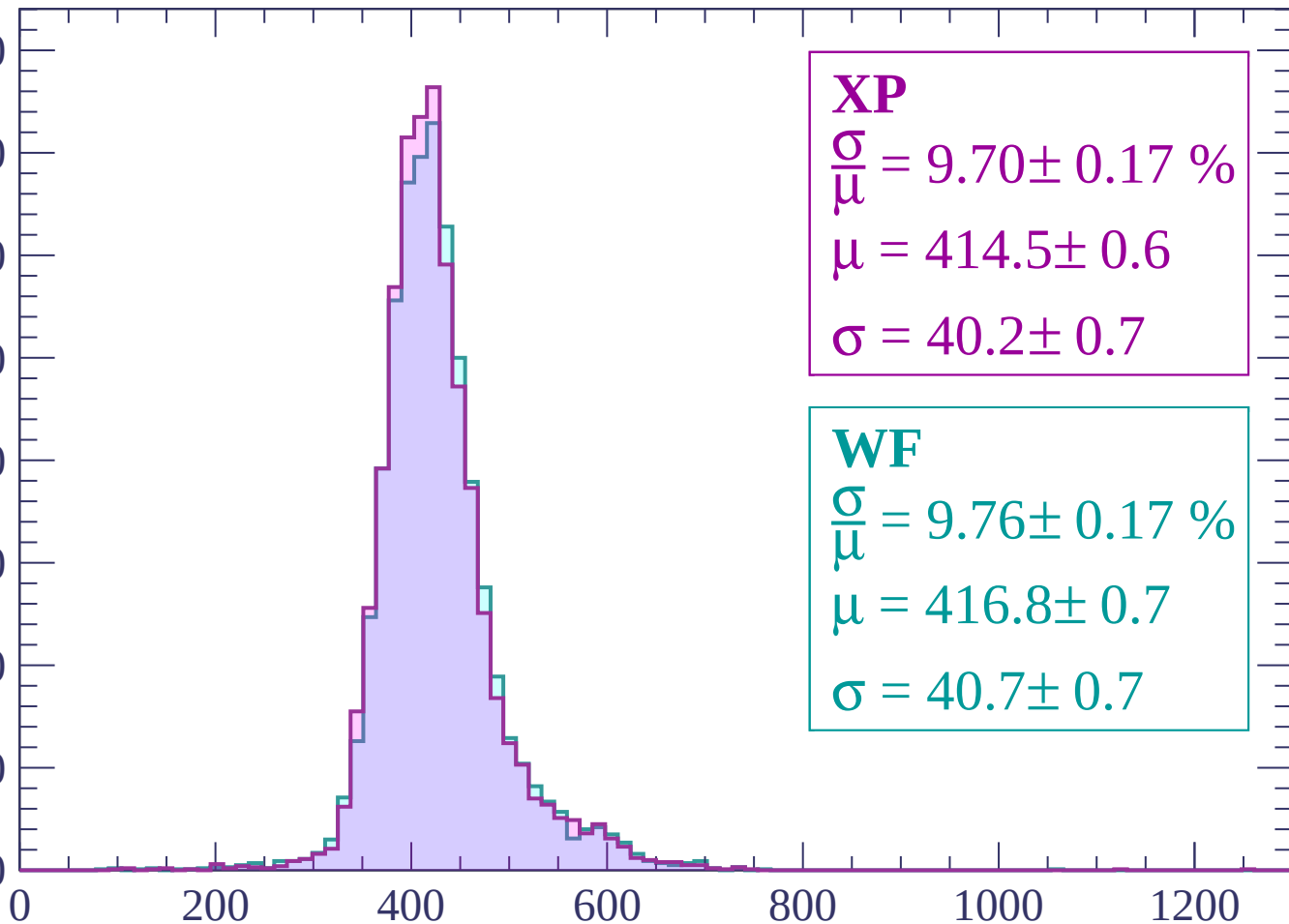
$$\sigma = 42.9 \pm 0.7$$

dE/dx (ADC counts/cm)

Energy loss | -660 < p < -600

Count

800
700
600
500
400
300
200
100
0



XP

$$\frac{\sigma}{\mu} = 9.70 \pm 0.17 \%$$

$$\mu = 414.5 \pm 0.6$$

$$\sigma = 40.2 \pm 0.7$$

WF

$$\frac{\sigma}{\mu} = 9.76 \pm 0.17 \%$$

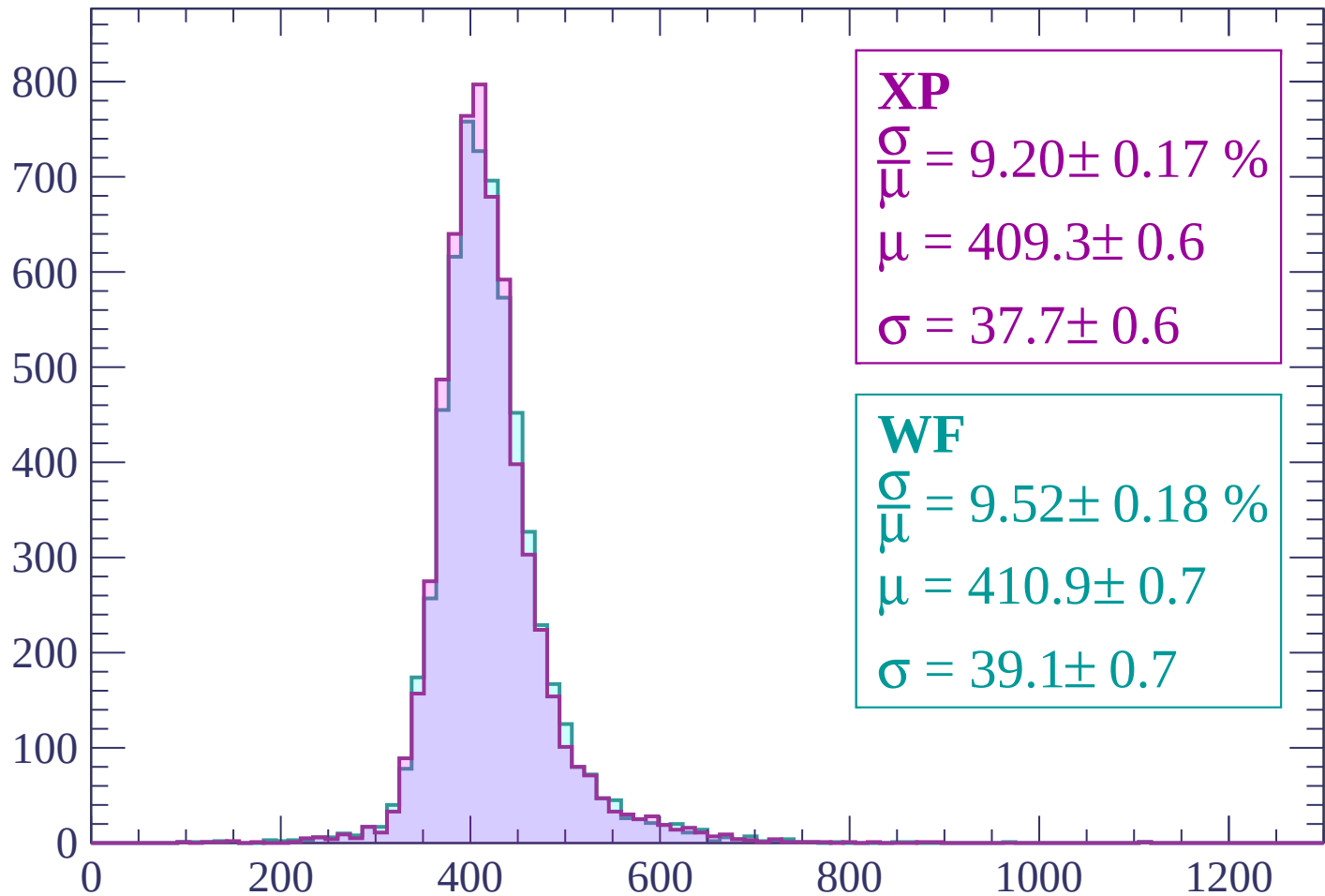
$$\mu = 416.8 \pm 0.7$$

$$\sigma = 40.7 \pm 0.7$$

dE/dx (ADC counts/cm)

Energy loss | $-600 < p < -540$

Count



XP

$$\frac{\sigma}{\mu} = 9.20 \pm 0.17 \%$$

$$\mu = 409.3 \pm 0.6$$

$$\sigma = 37.7 \pm 0.6$$

WF

$$\frac{\sigma}{\mu} = 9.52 \pm 0.18 \%$$

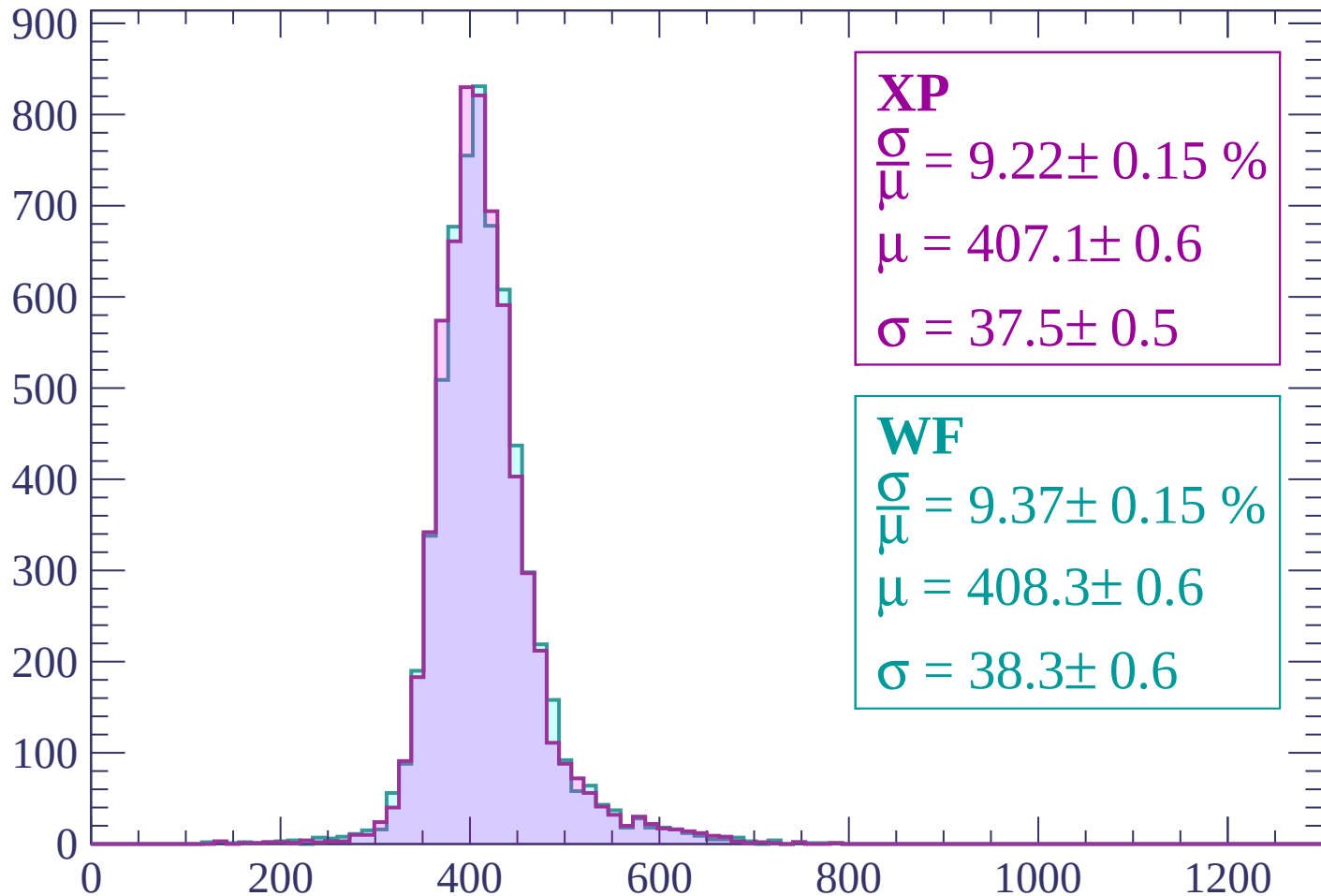
$$\mu = 410.9 \pm 0.7$$

$$\sigma = 39.1 \pm 0.7$$

dE/dx (ADC counts/cm)

Energy loss | $-540 < p < -480$

Count



XP

$$\frac{\sigma}{\mu} = 9.22 \pm 0.15 \%$$

$$\mu = 407.1 \pm 0.6$$

$$\sigma = 37.5 \pm 0.5$$

WF

$$\frac{\sigma}{\mu} = 9.37 \pm 0.15 \%$$

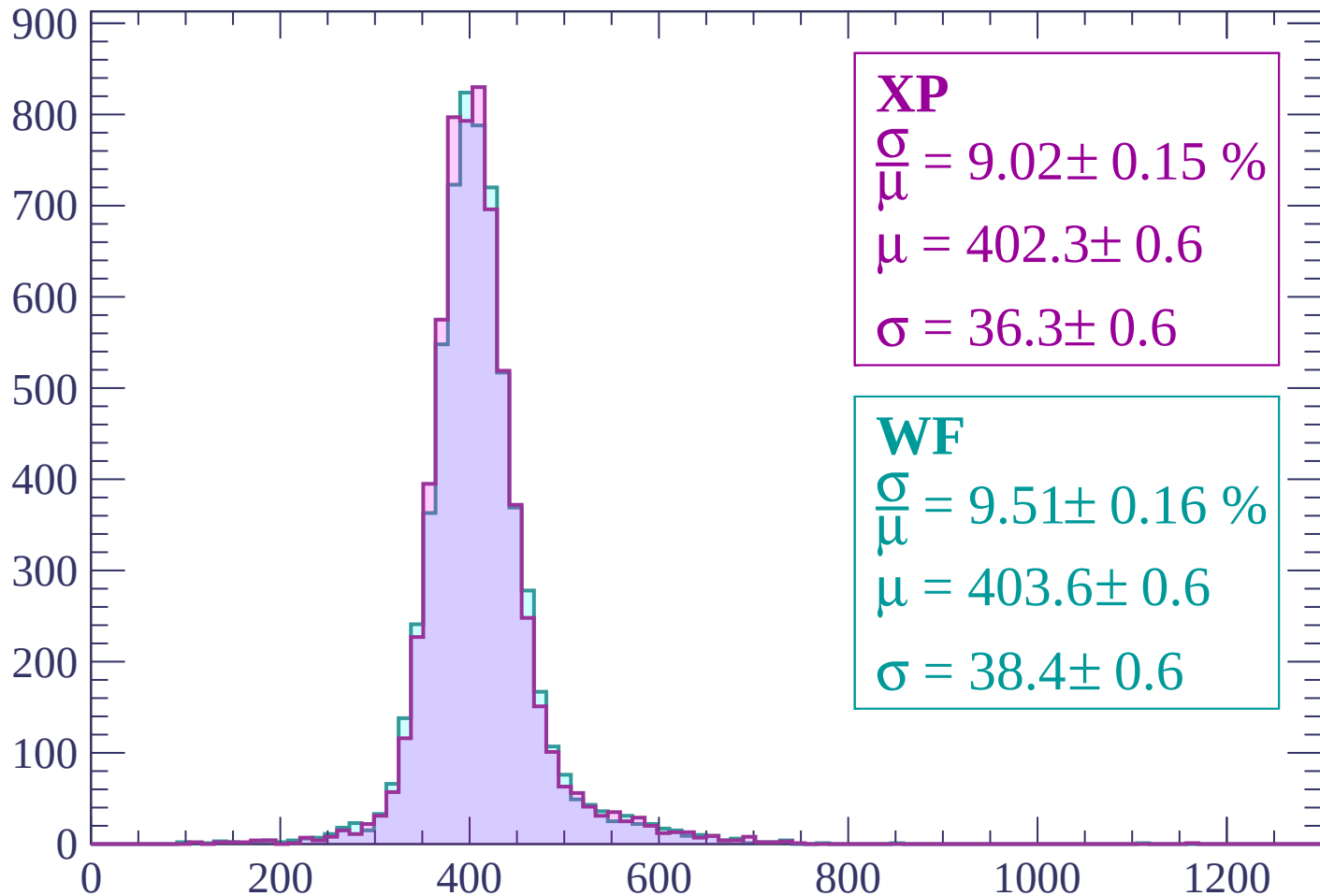
$$\mu = 408.3 \pm 0.6$$

$$\sigma = 38.3 \pm 0.6$$

dE/dx (ADC counts/cm)

Energy loss | $-480 < p < -420$

Count



XP

$$\frac{\sigma}{\mu} = 9.02 \pm 0.15 \%$$

$$\mu = 402.3 \pm 0.6$$

$$\sigma = 36.3 \pm 0.6$$

WF

$$\frac{\sigma}{\mu} = 9.51 \pm 0.16 \%$$

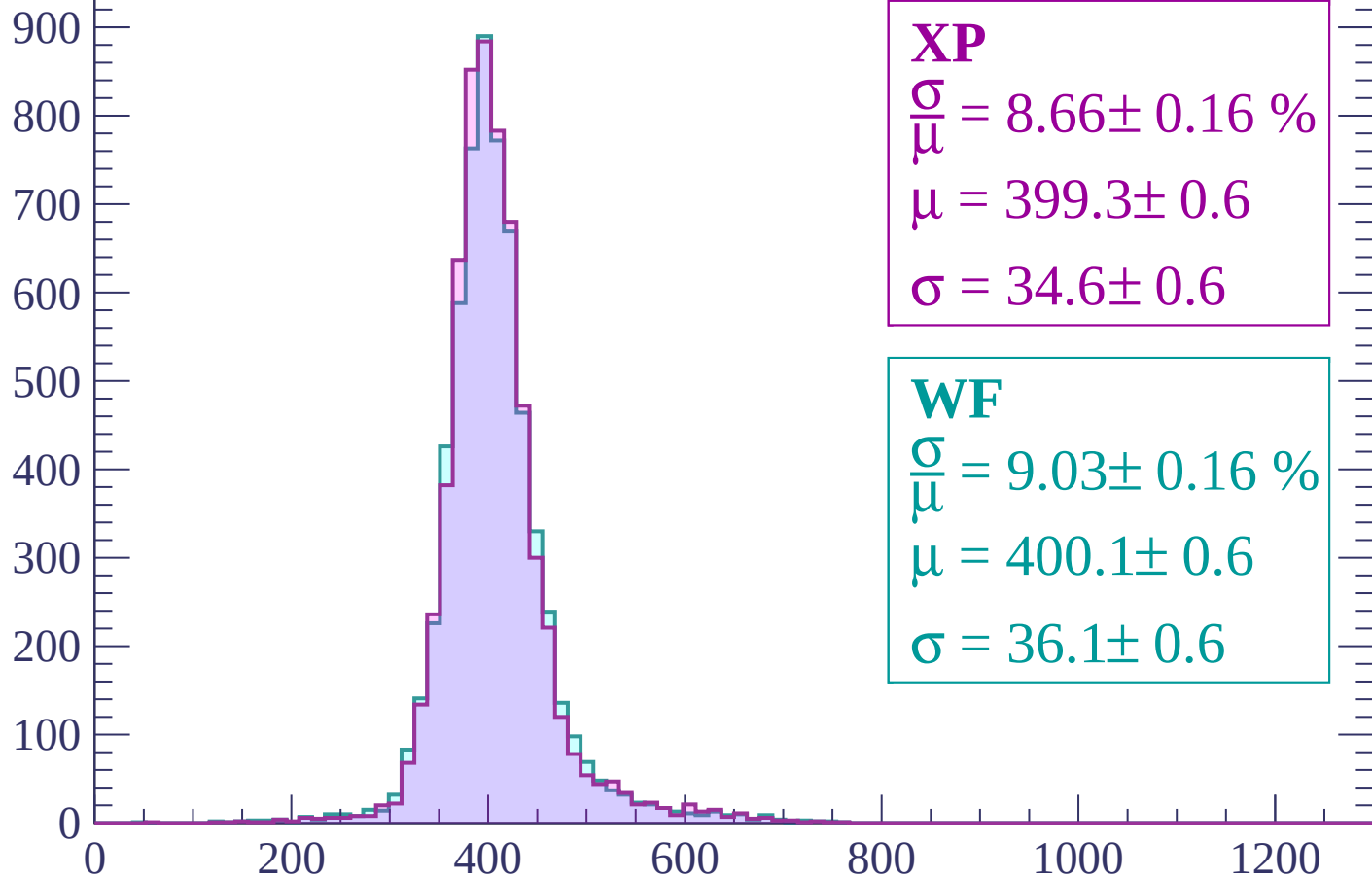
$$\mu = 403.6 \pm 0.6$$

$$\sigma = 38.4 \pm 0.6$$

dE/dx (ADC counts/cm)

Energy loss | $-420 < p < -360$

Count



XP

$$\frac{\sigma}{\mu} = 8.66 \pm 0.16 \%$$

$$\mu = 399.3 \pm 0.6$$

$$\sigma = 34.6 \pm 0.6$$

WF

$$\frac{\sigma}{\mu} = 9.03 \pm 0.16 \%$$

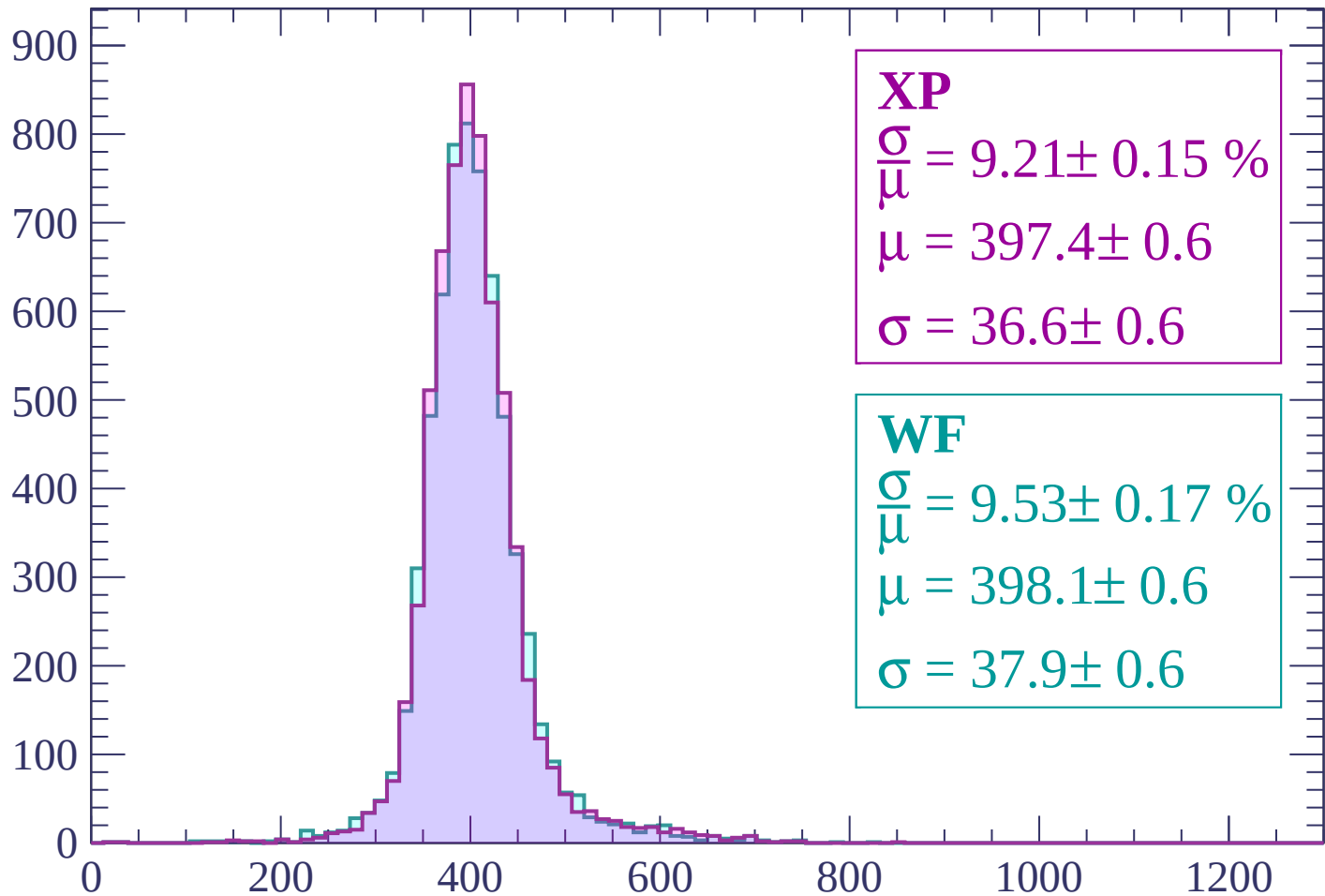
$$\mu = 400.1 \pm 0.6$$

$$\sigma = 36.1 \pm 0.6$$

dE/dx (ADC counts/cm)

Energy loss | $-360 < p < -300$

Count



XP

$$\frac{\sigma}{\mu} = 9.21 \pm 0.15 \%$$

$$\mu = 397.4 \pm 0.6$$

$$\sigma = 36.6 \pm 0.6$$

WF

$$\frac{\sigma}{\mu} = 9.53 \pm 0.17 \%$$

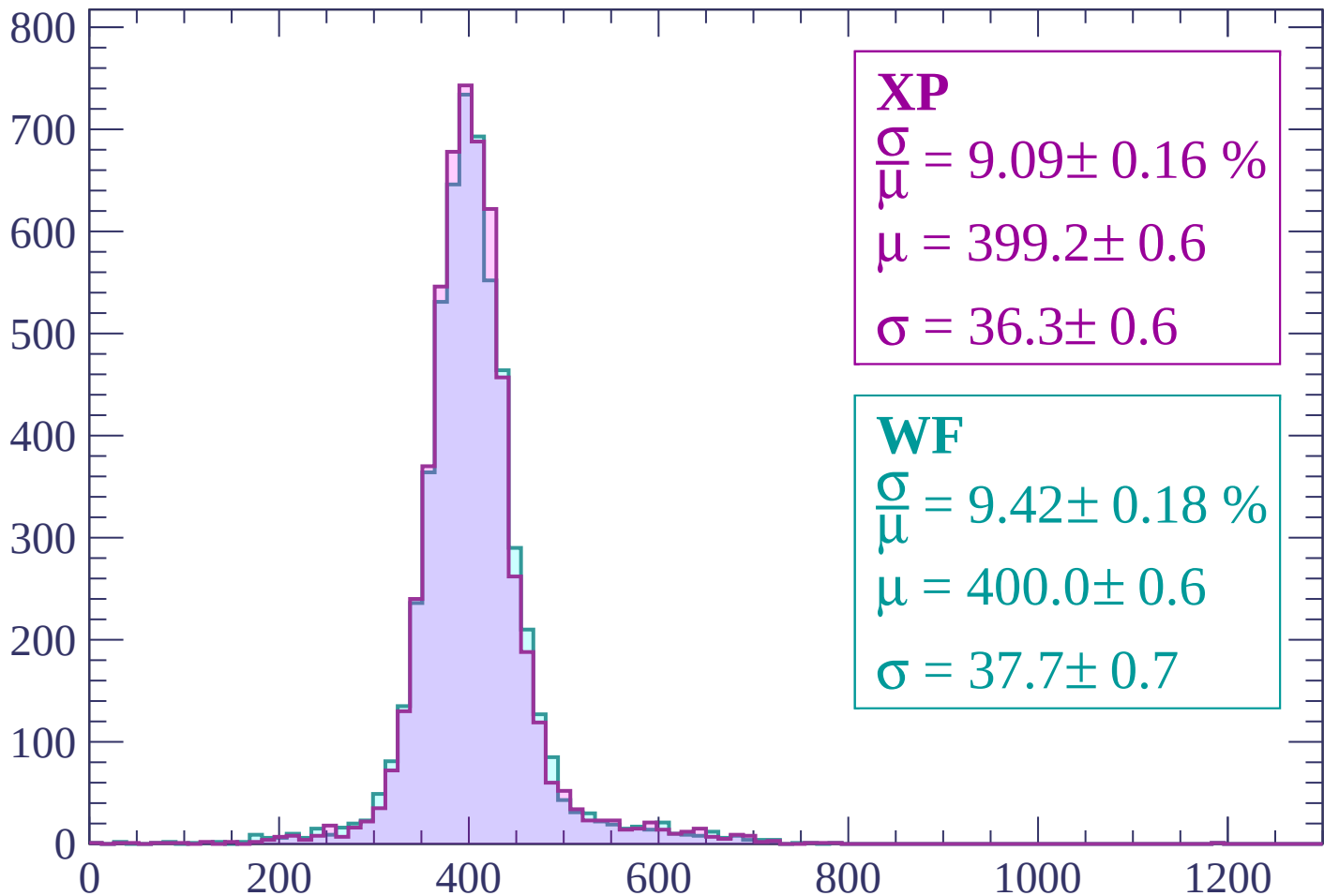
$$\mu = 398.1 \pm 0.6$$

$$\sigma = 37.9 \pm 0.6$$

dE/dx (ADC counts/cm)

Energy loss | $-300 < p < -240$

Count



XP

$$\frac{\sigma}{\mu} = 9.09 \pm 0.16 \%$$

$$\mu = 399.2 \pm 0.6$$

$$\sigma = 36.3 \pm 0.6$$

WF

$$\frac{\sigma}{\mu} = 9.42 \pm 0.18 \%$$

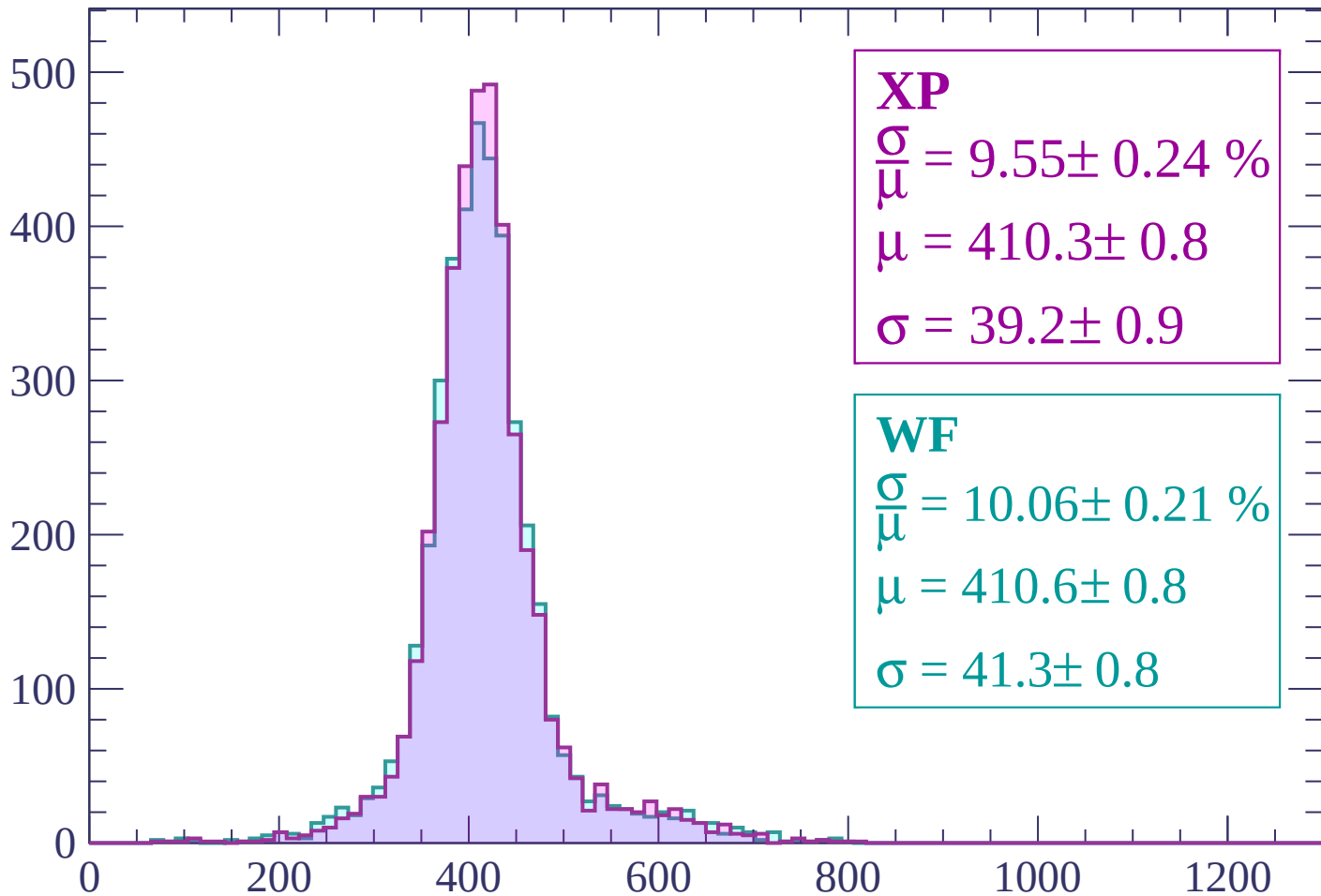
$$\mu = 400.0 \pm 0.6$$

$$\sigma = 37.7 \pm 0.7$$

dE/dx (ADC counts/cm)

Energy loss | $-240 < p < -180$

Count



XP

$$\frac{\sigma}{\mu} = 9.55 \pm 0.24 \%$$

$$\mu = 410.3 \pm 0.8$$

$$\sigma = 39.2 \pm 0.9$$

WF

$$\frac{\sigma}{\mu} = 10.06 \pm 0.21 \%$$

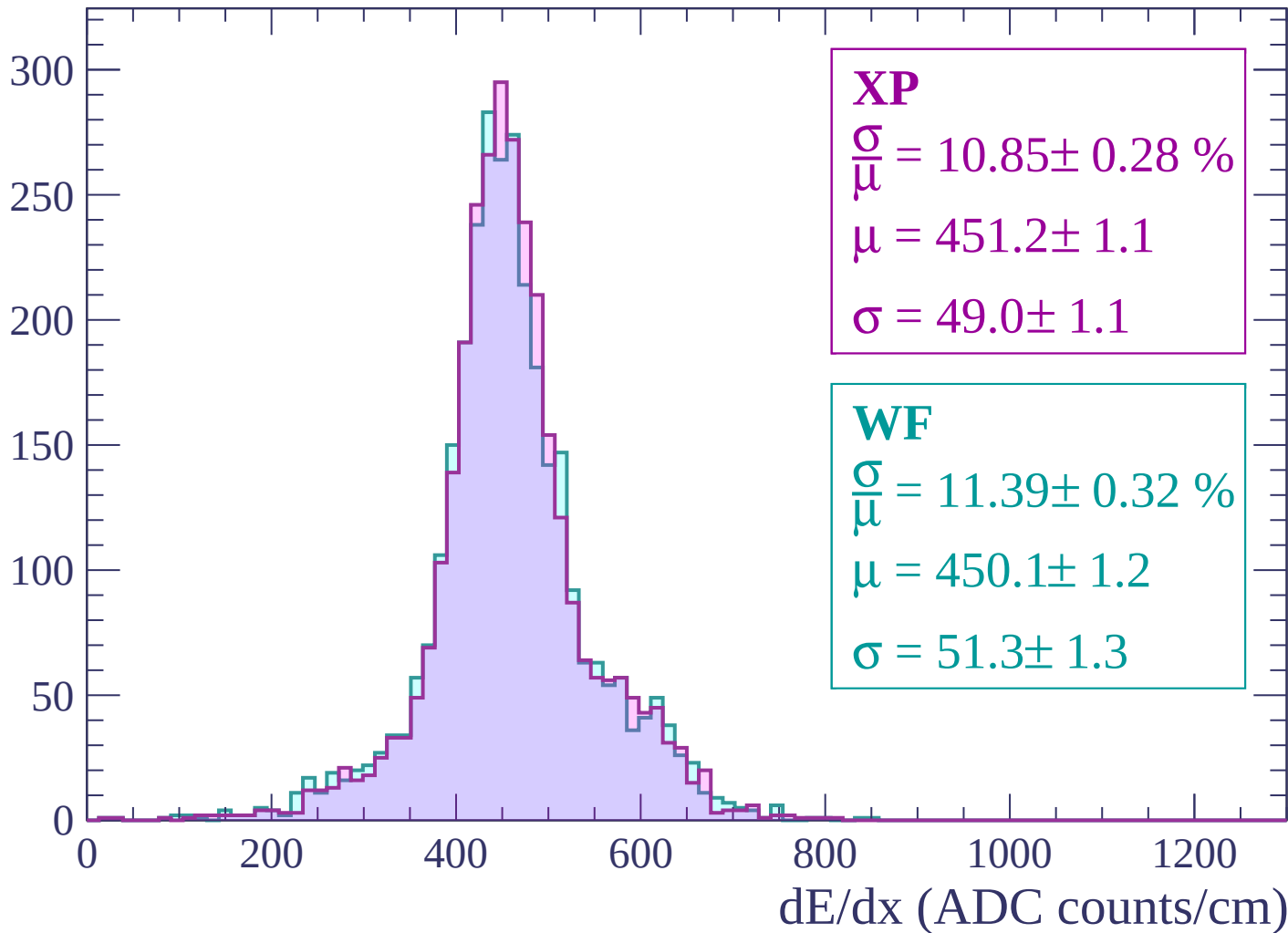
$$\mu = 410.6 \pm 0.8$$

$$\sigma = 41.3 \pm 0.8$$

dE/dx (ADC counts/cm)

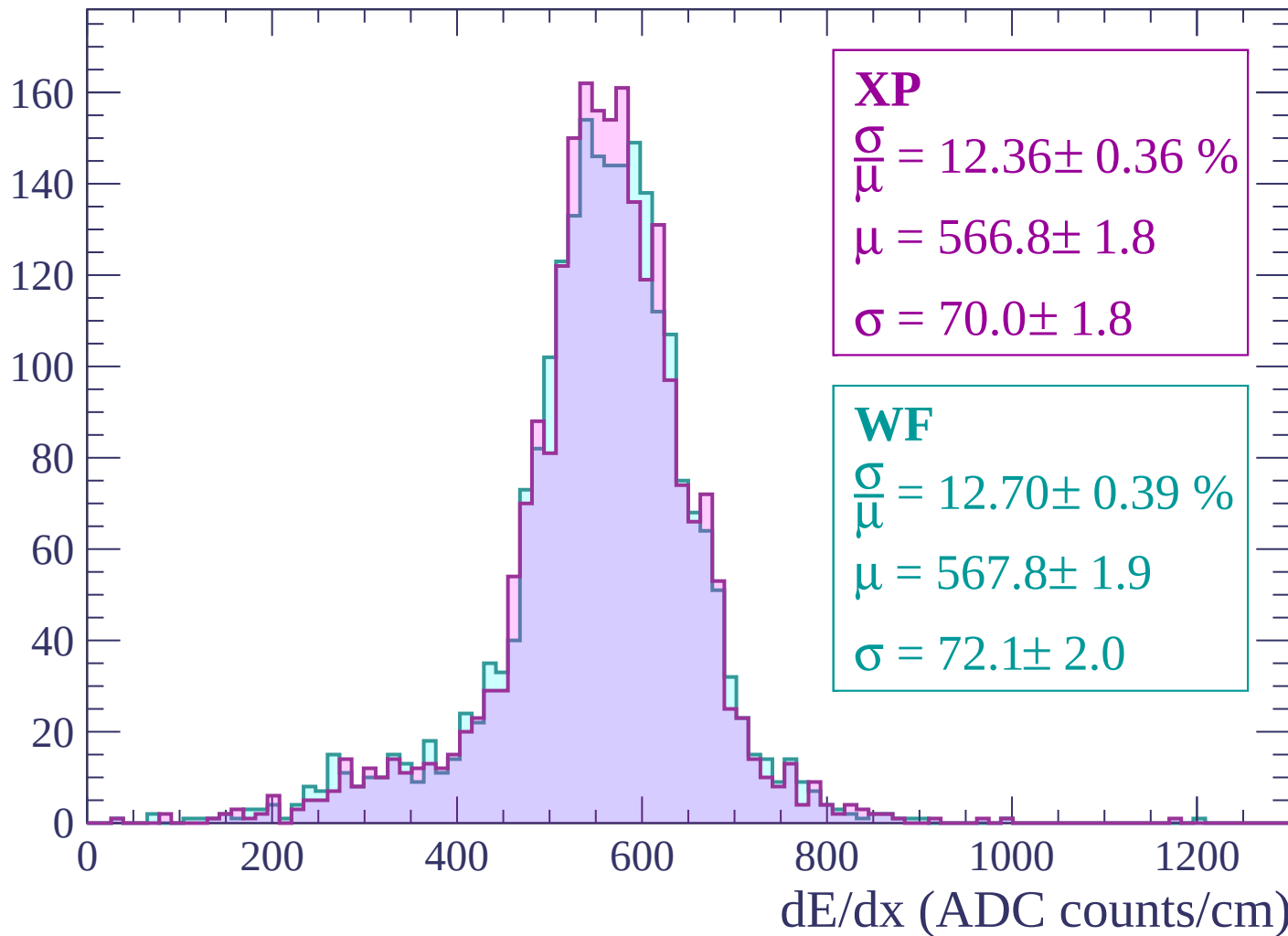
Energy loss | -180 < p < -120

Count



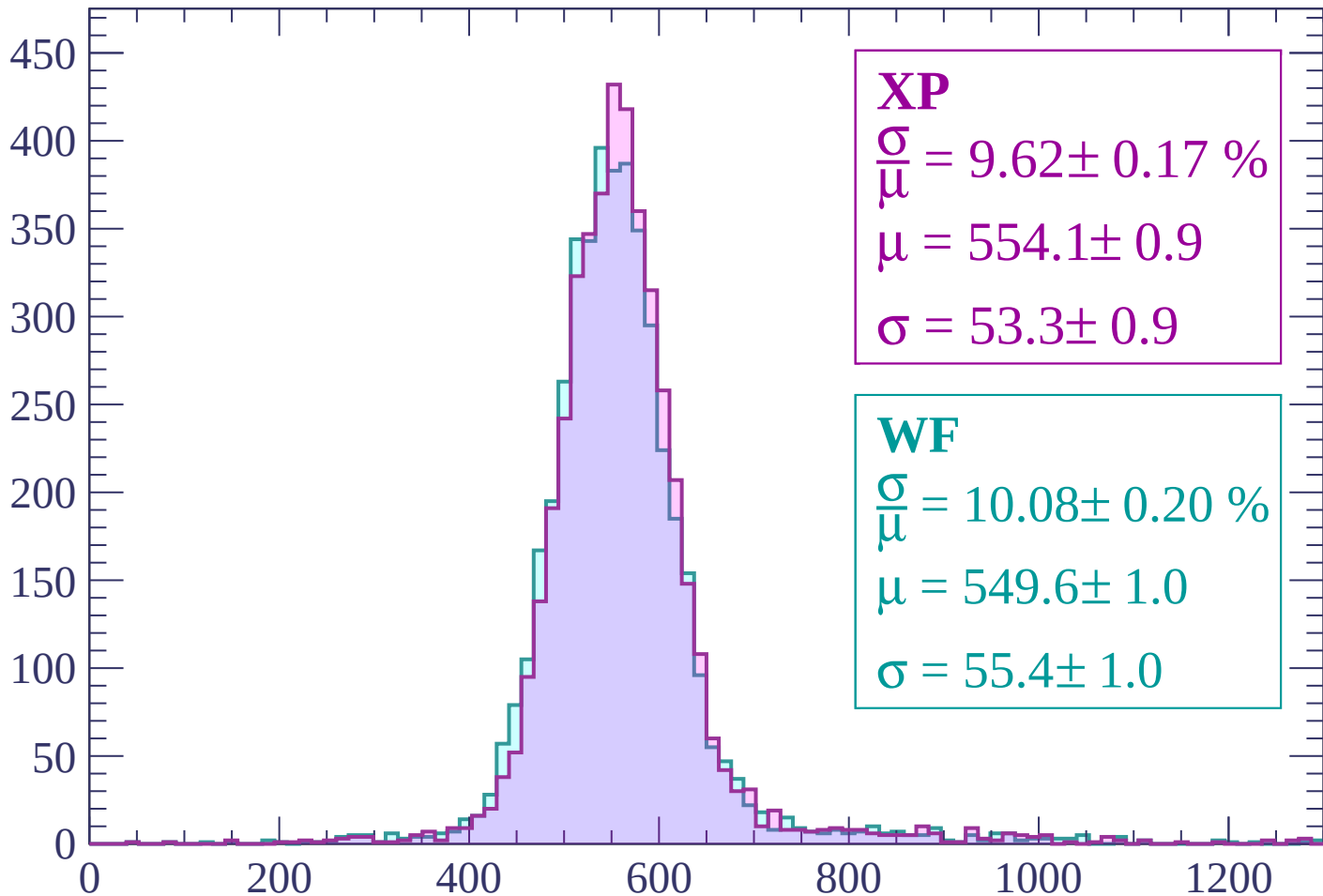
Energy loss | $-120 < p < -60$

Count



Energy loss | $-60 < p < 0$

Count



XP

$$\frac{\sigma}{\mu} = 9.62 \pm 0.17 \%$$

$$\mu = 554.1 \pm 0.9$$

$$\sigma = 53.3 \pm 0.9$$

WF

$$\frac{\sigma}{\mu} = 10.08 \pm 0.20 \%$$

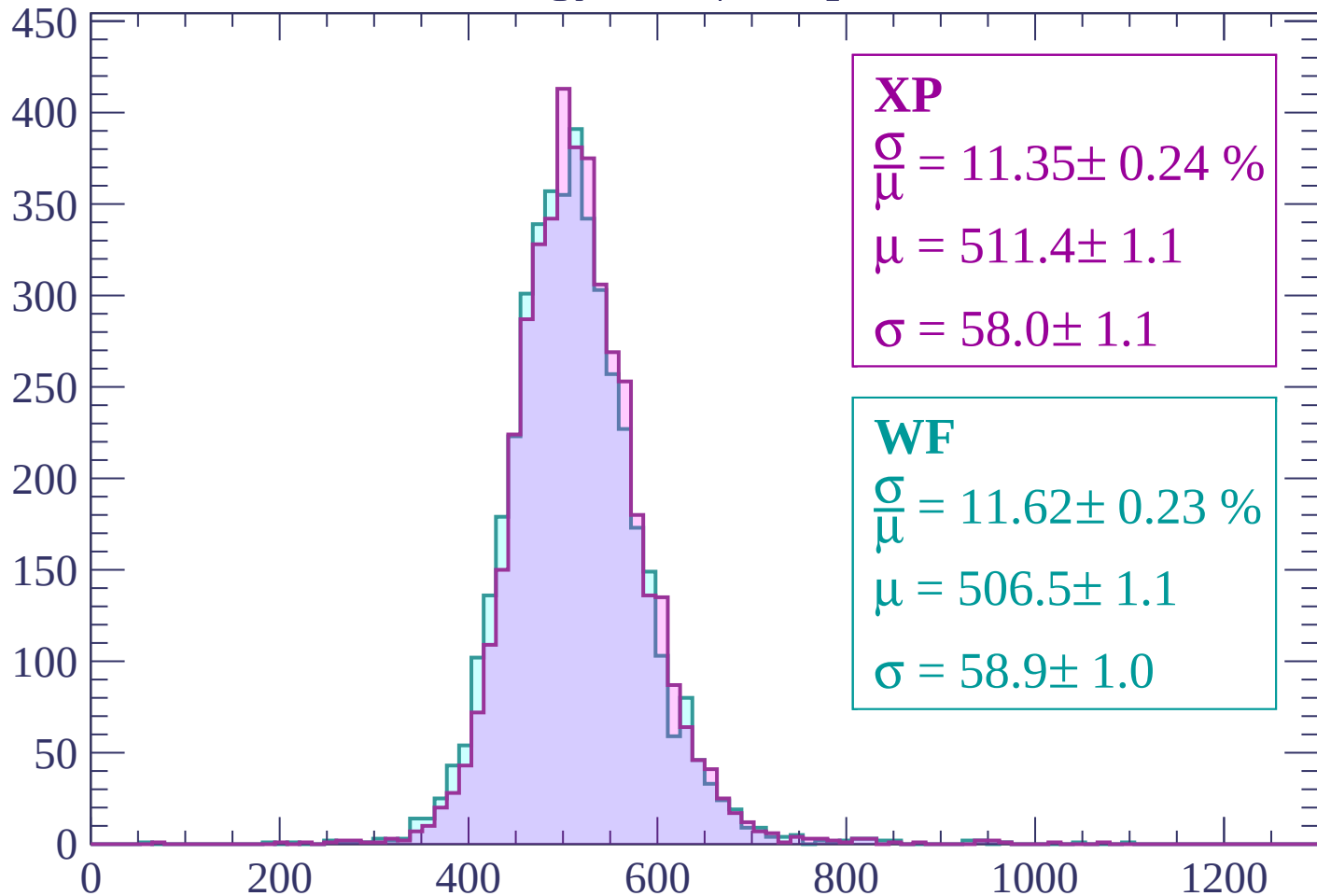
$$\mu = 549.6 \pm 1.0$$

$$\sigma = 55.4 \pm 1.0$$

dE/dx (ADC counts/cm)

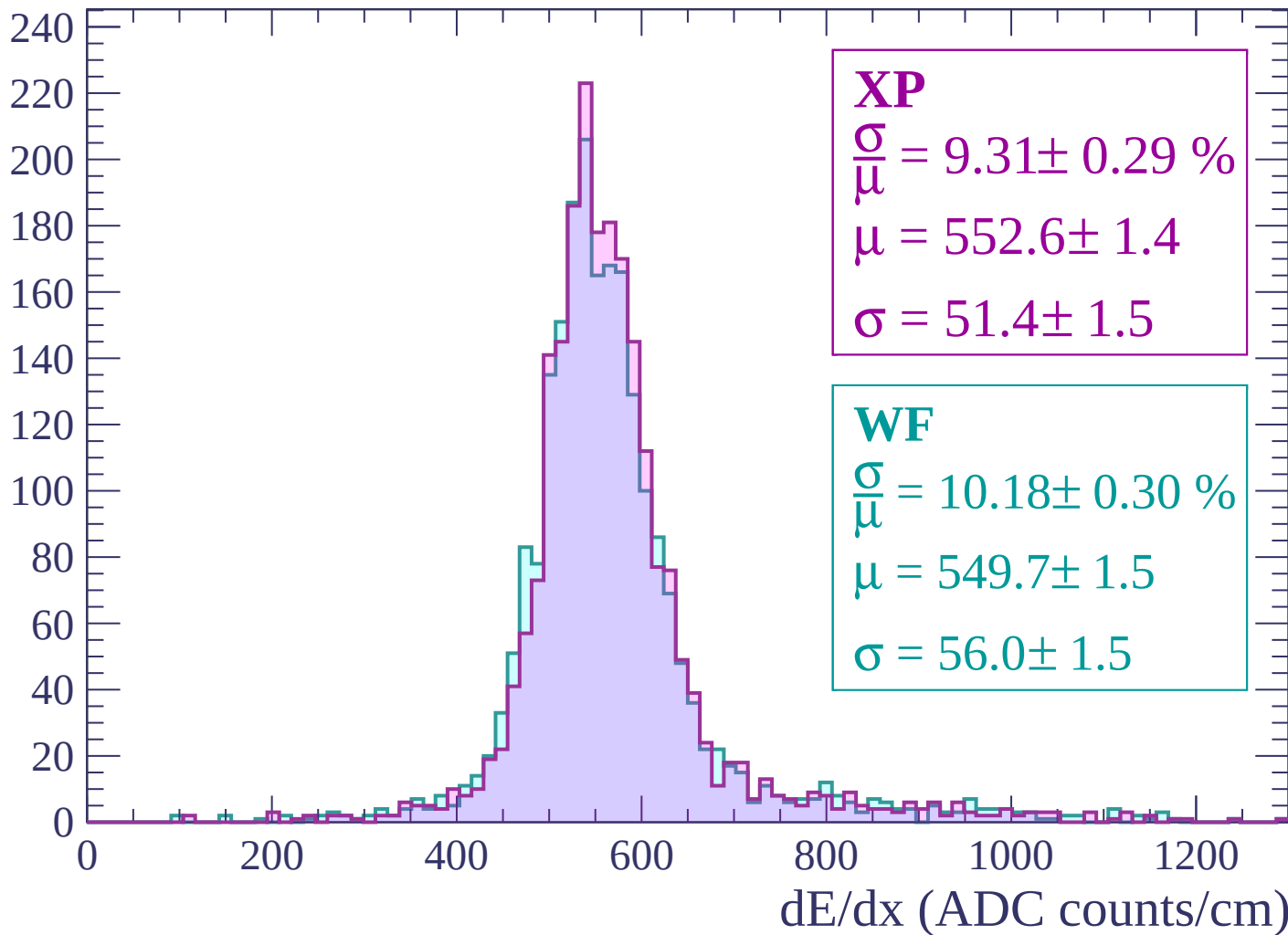
Energy loss | $0 < p < 60$

Count



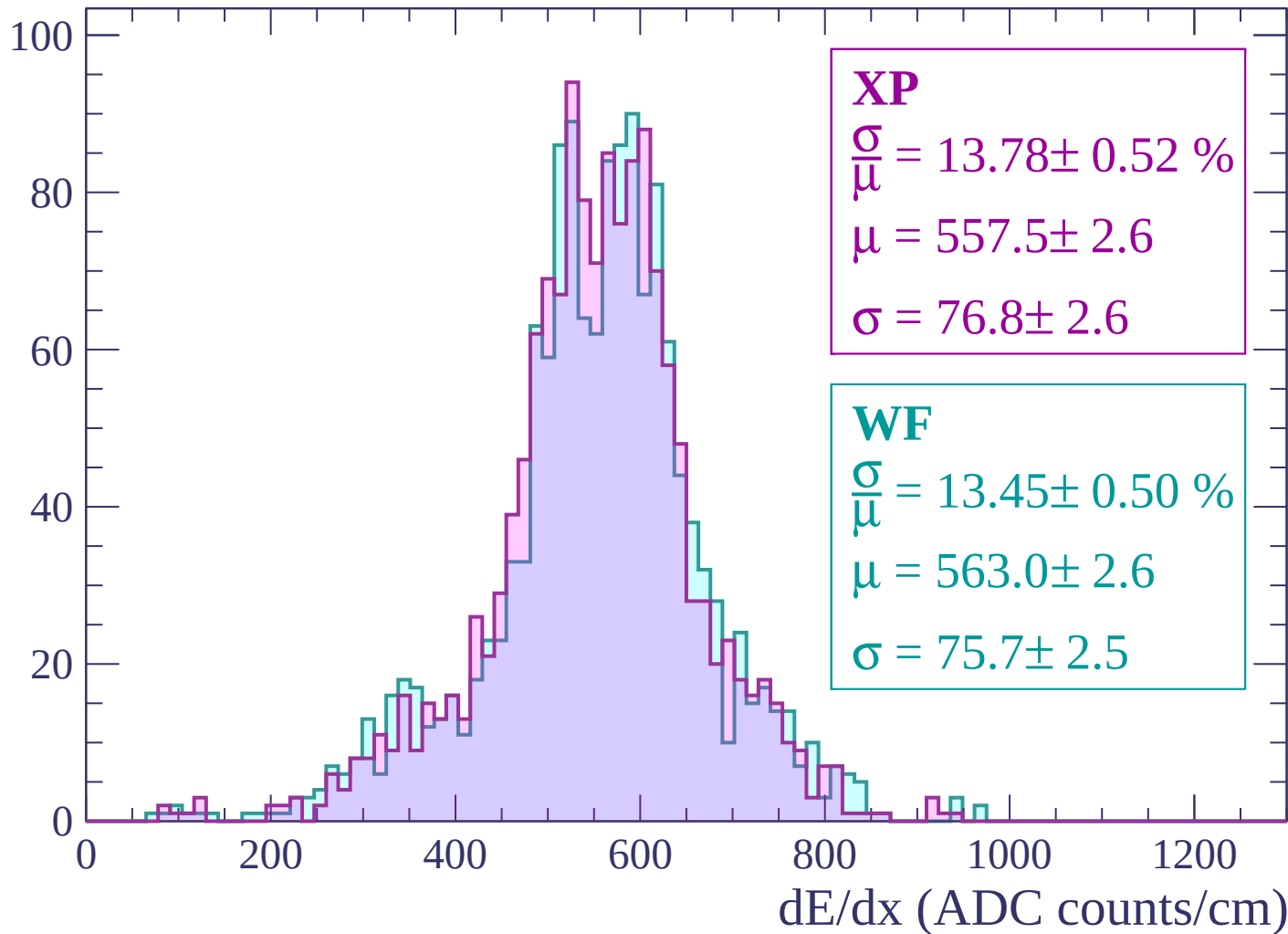
Energy loss | $60 < p < 120$

Count



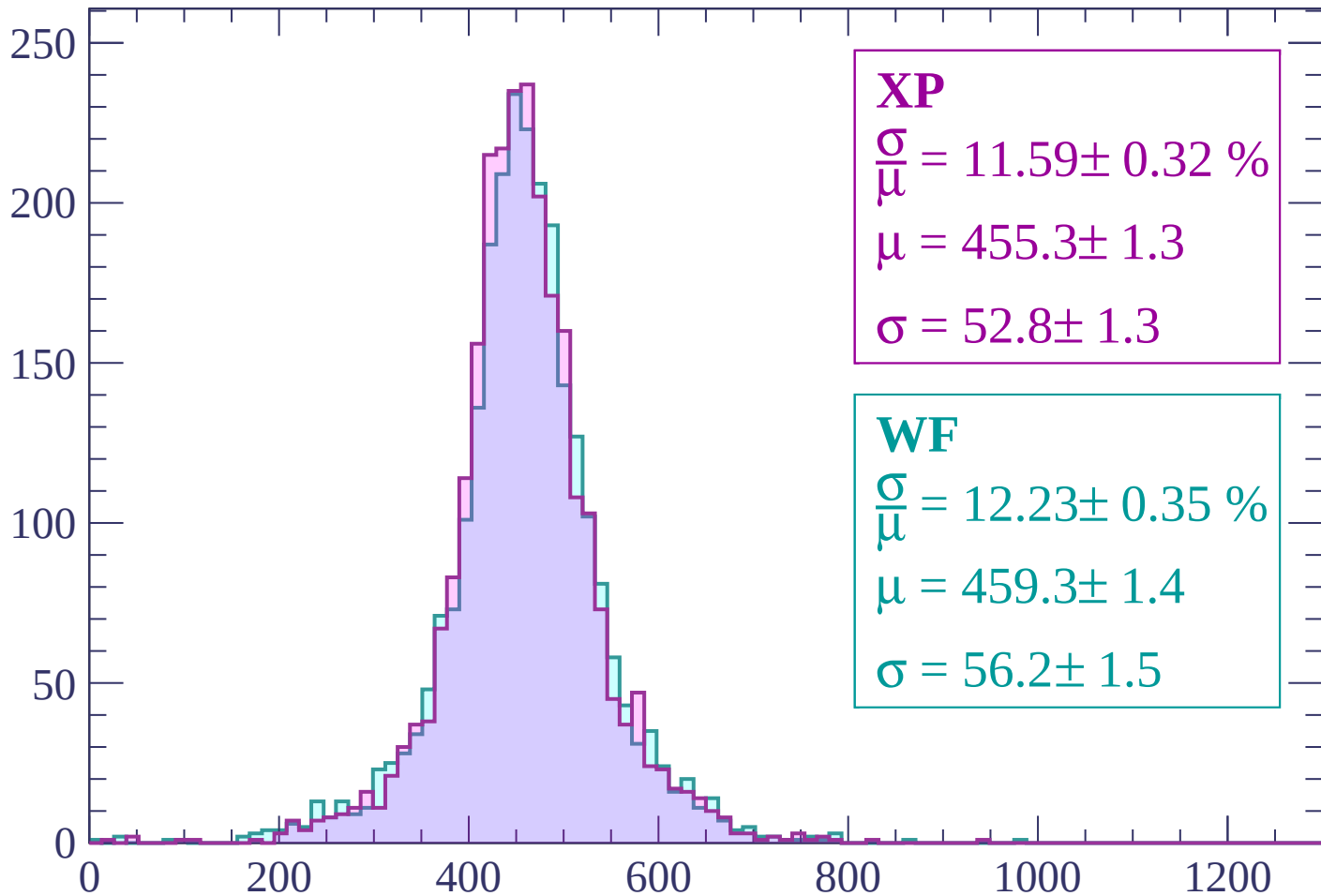
Energy loss | $120 < p < 180$

Count



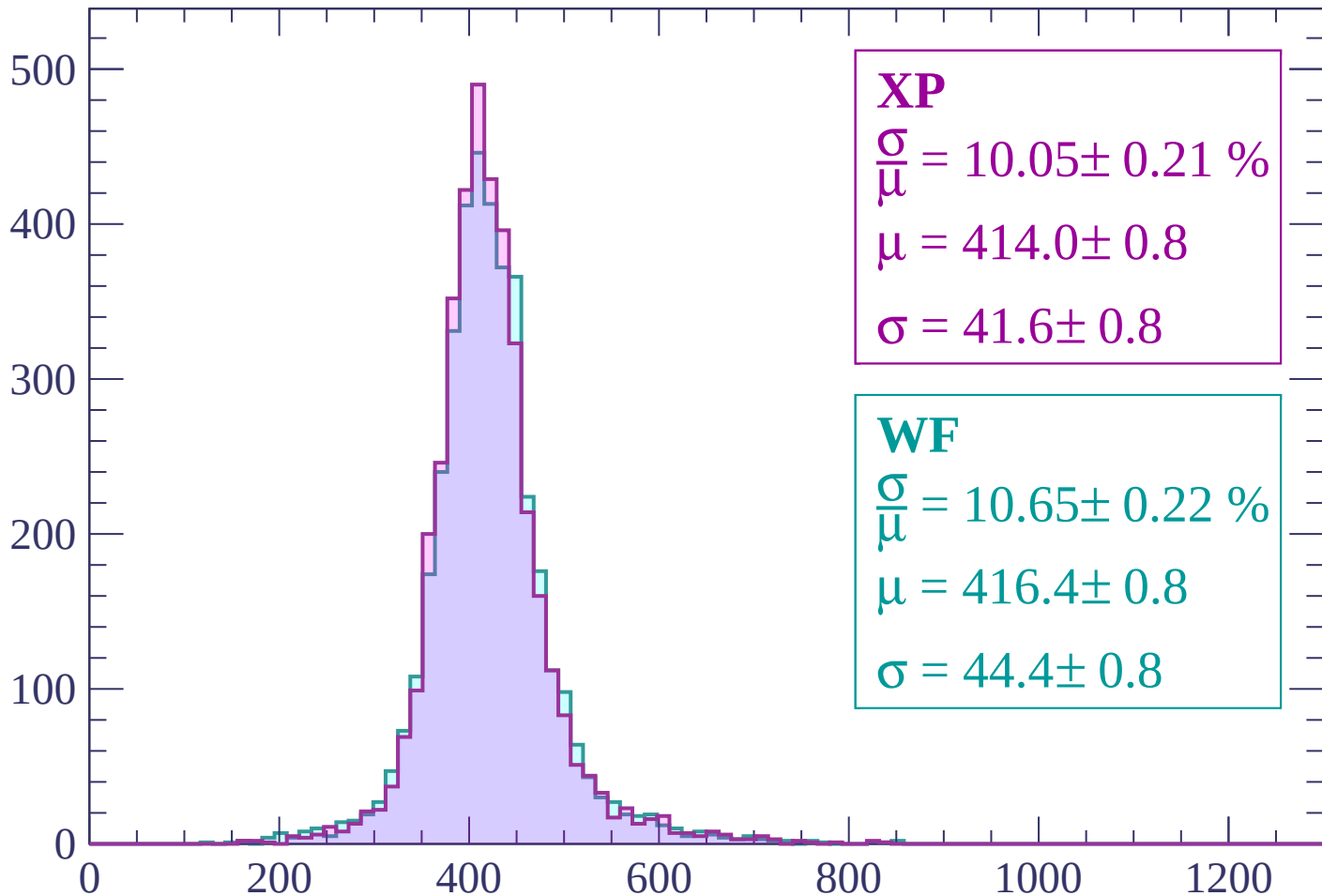
Energy loss | $180 < p < 240$

Count



Energy loss | $240 < p < 300$

Count



XP

$$\frac{\sigma}{\mu} = 10.05 \pm 0.21 \%$$

$$\mu = 414.0 \pm 0.8$$

$$\sigma = 41.6 \pm 0.8$$

WF

$$\frac{\sigma}{\mu} = 10.65 \pm 0.22 \%$$

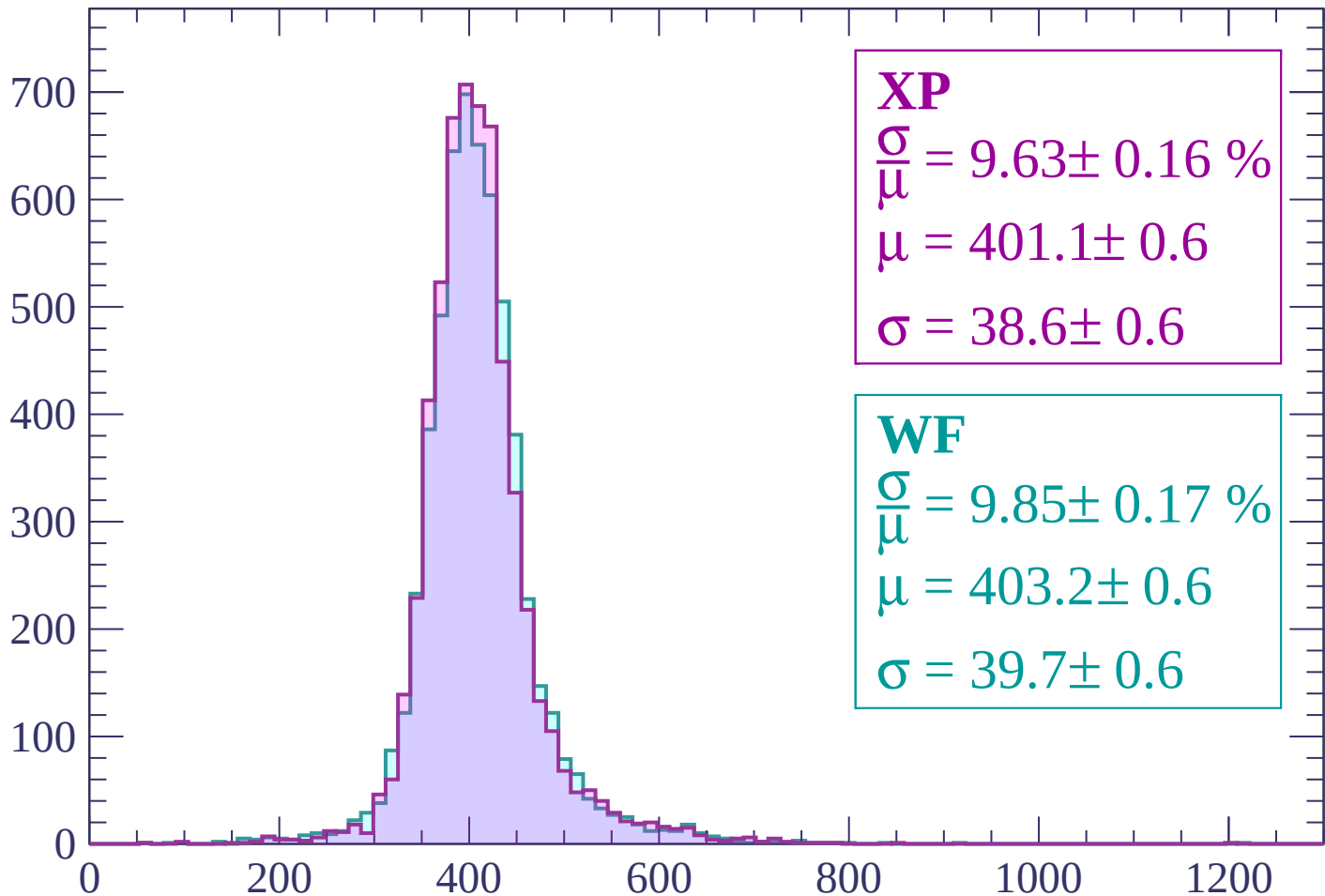
$$\mu = 416.4 \pm 0.8$$

$$\sigma = 44.4 \pm 0.8$$

dE/dx (ADC counts/cm)

Energy loss | $300 < p < 360$

Count



XP

$$\frac{\sigma}{\mu} = 9.63 \pm 0.16 \%$$

$$\mu = 401.1 \pm 0.6$$

$$\sigma = 38.6 \pm 0.6$$

WF

$$\frac{\sigma}{\mu} = 9.85 \pm 0.17 \%$$

$$\mu = 403.2 \pm 0.6$$

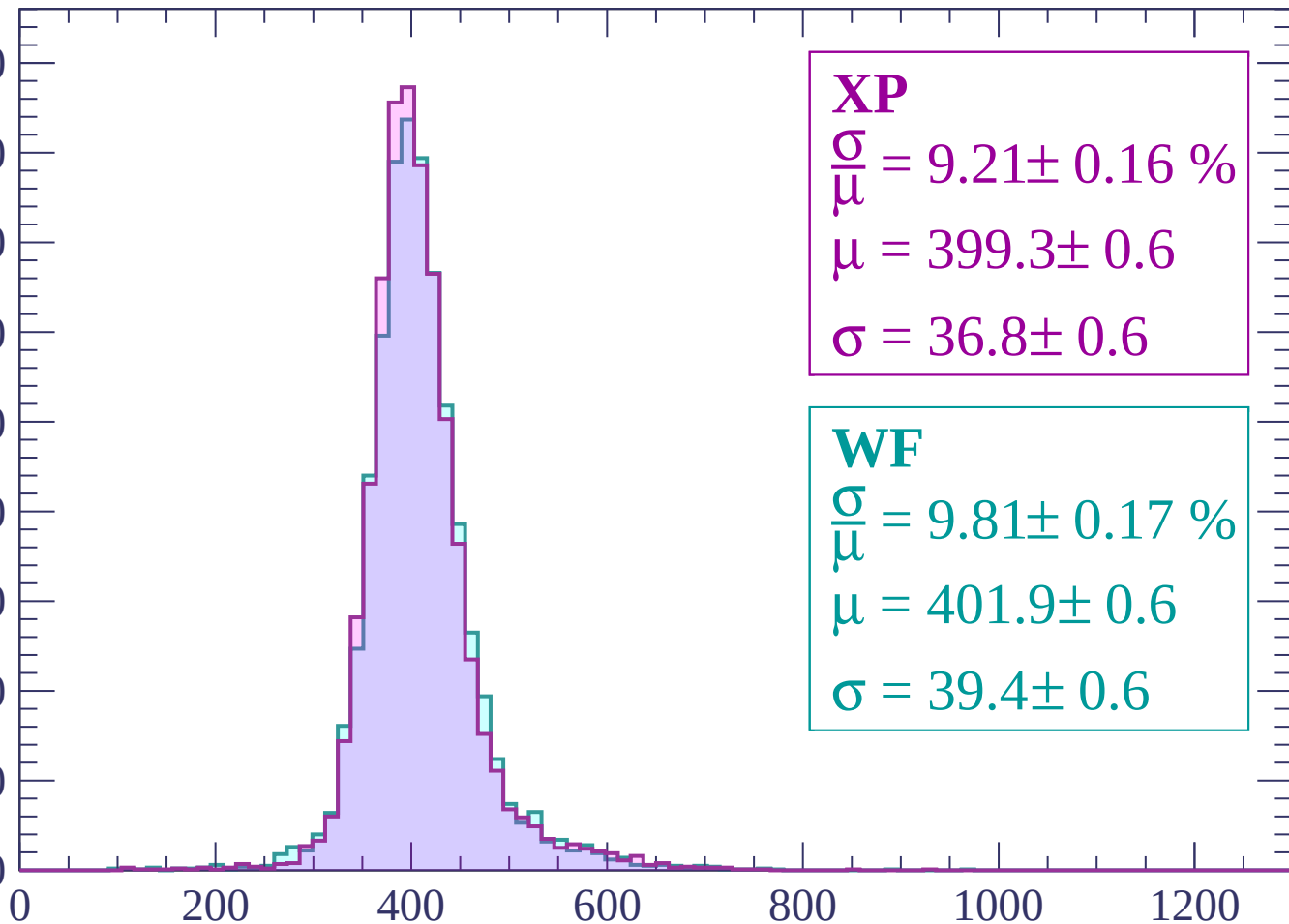
$$\sigma = 39.7 \pm 0.6$$

dE/dx (ADC counts/cm)

Energy loss | $360 < p < 420$

Count

900
800
700
600
500
400
300
200
100
0



XP

$$\frac{\sigma}{\mu} = 9.21 \pm 0.16 \%$$

$$\mu = 399.3 \pm 0.6$$

$$\sigma = 36.8 \pm 0.6$$

WF

$$\frac{\sigma}{\mu} = 9.81 \pm 0.17 \%$$

$$\mu = 401.9 \pm 0.6$$

$$\sigma = 39.4 \pm 0.6$$

dE/dx (ADC counts/cm)

Energy loss | $420 < p < 480$

Count

1000

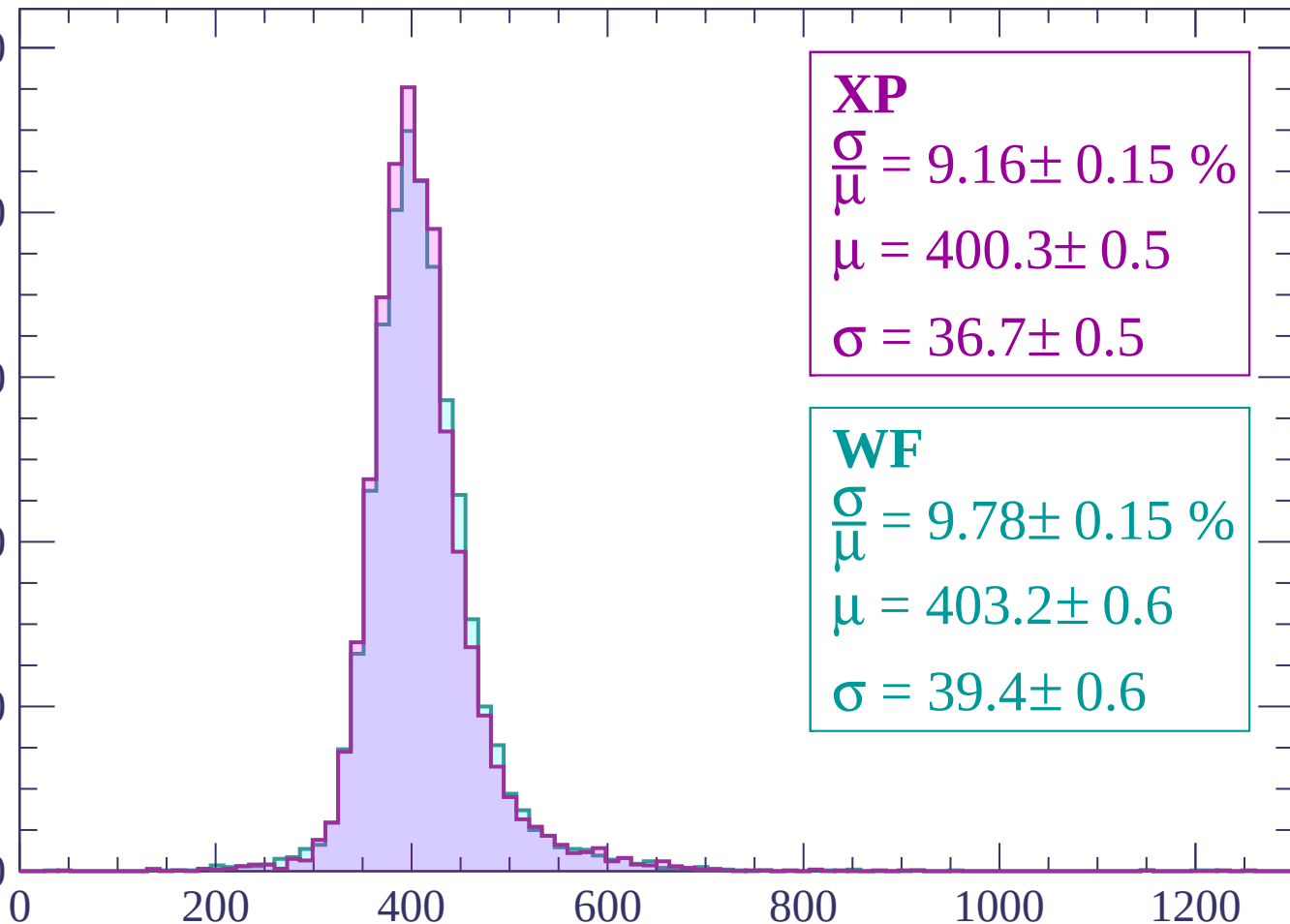
800

600

400

200

0



XP

$$\frac{\sigma}{\mu} = 9.16 \pm 0.15 \%$$

$$\mu = 400.3 \pm 0.5$$

$$\sigma = 36.7 \pm 0.5$$

WF

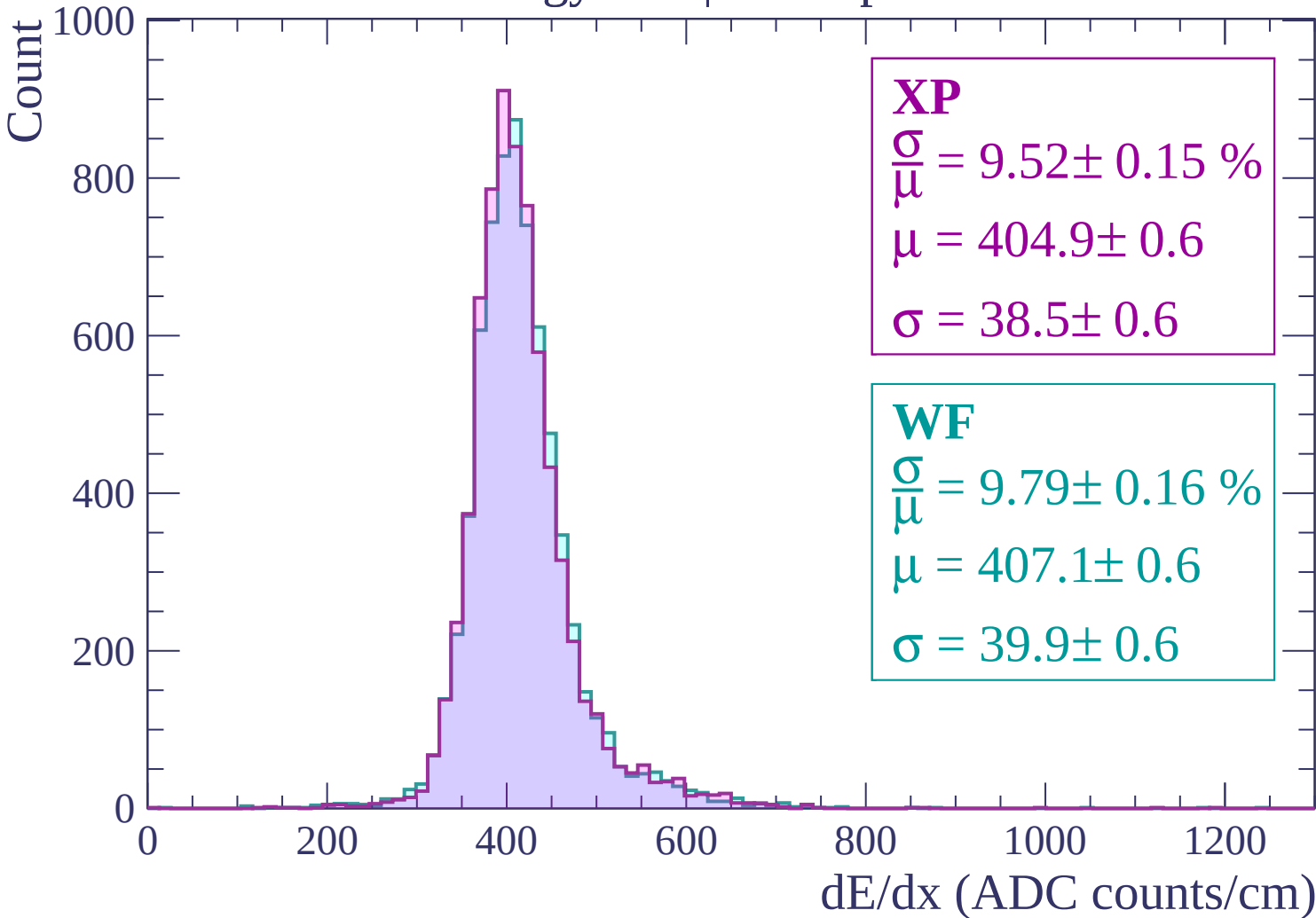
$$\frac{\sigma}{\mu} = 9.78 \pm 0.15 \%$$

$$\mu = 403.2 \pm 0.6$$

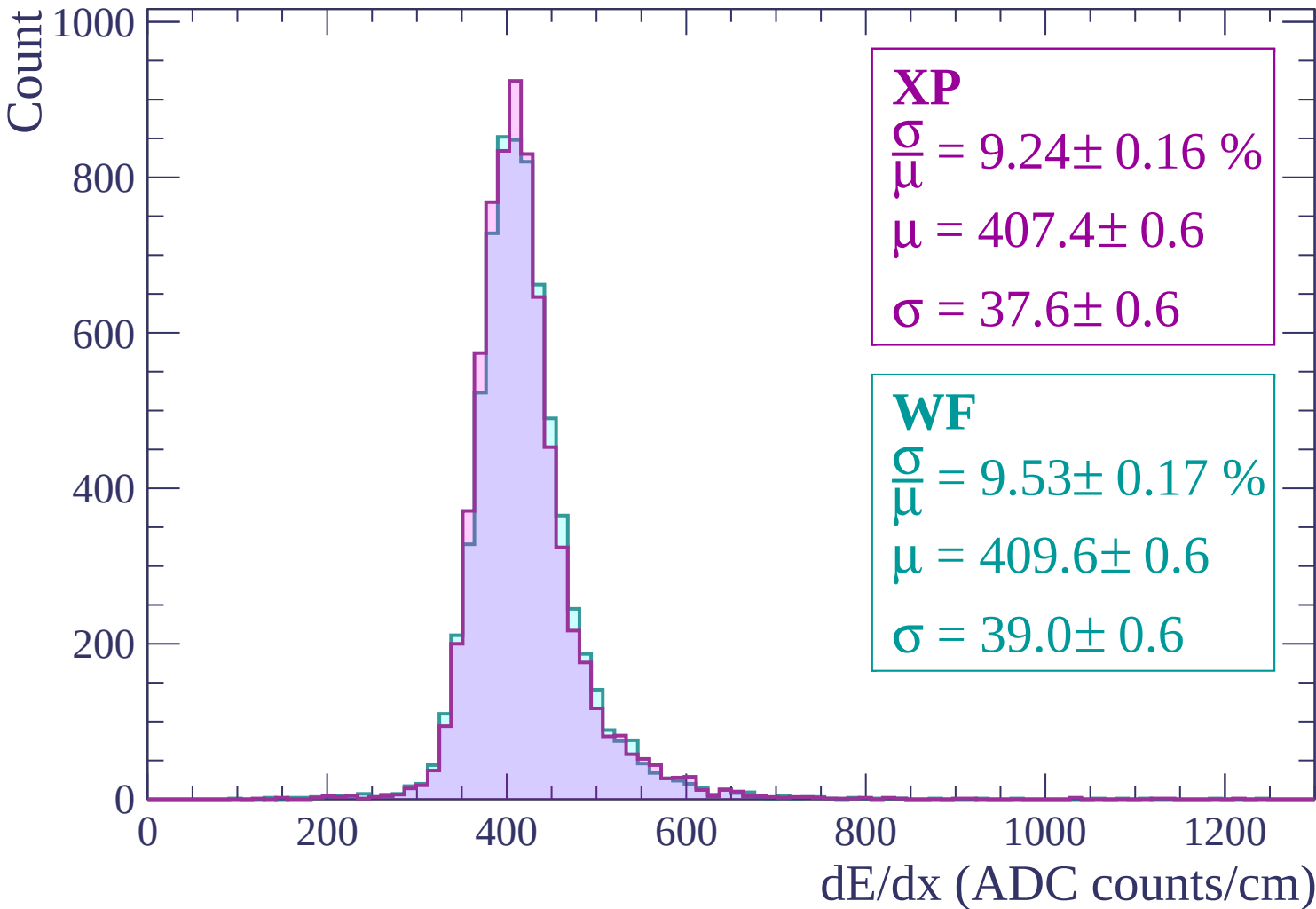
$$\sigma = 39.4 \pm 0.6$$

dE/dx (ADC counts/cm)

Energy loss | $480 < p < 540$



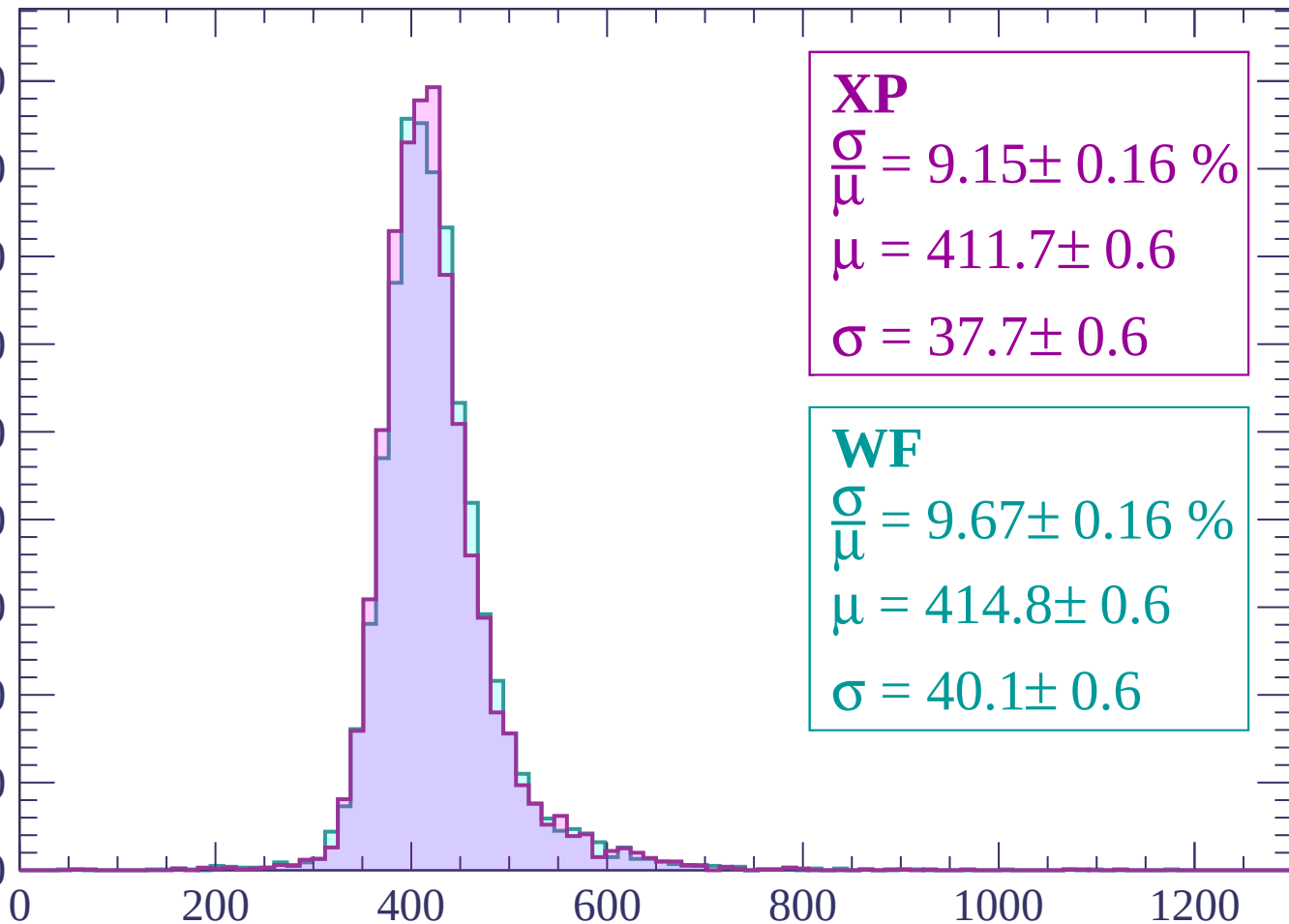
Energy loss | $540 < p < 600$



Energy loss | $600 < p < 660$

Count

900
800
700
600
500
400
300
200
100
0



XP

$$\frac{\sigma}{\mu} = 9.15 \pm 0.16 \%$$

$$\mu = 411.7 \pm 0.6$$

$$\sigma = 37.7 \pm 0.6$$

WF

$$\frac{\sigma}{\mu} = 9.67 \pm 0.16 \%$$

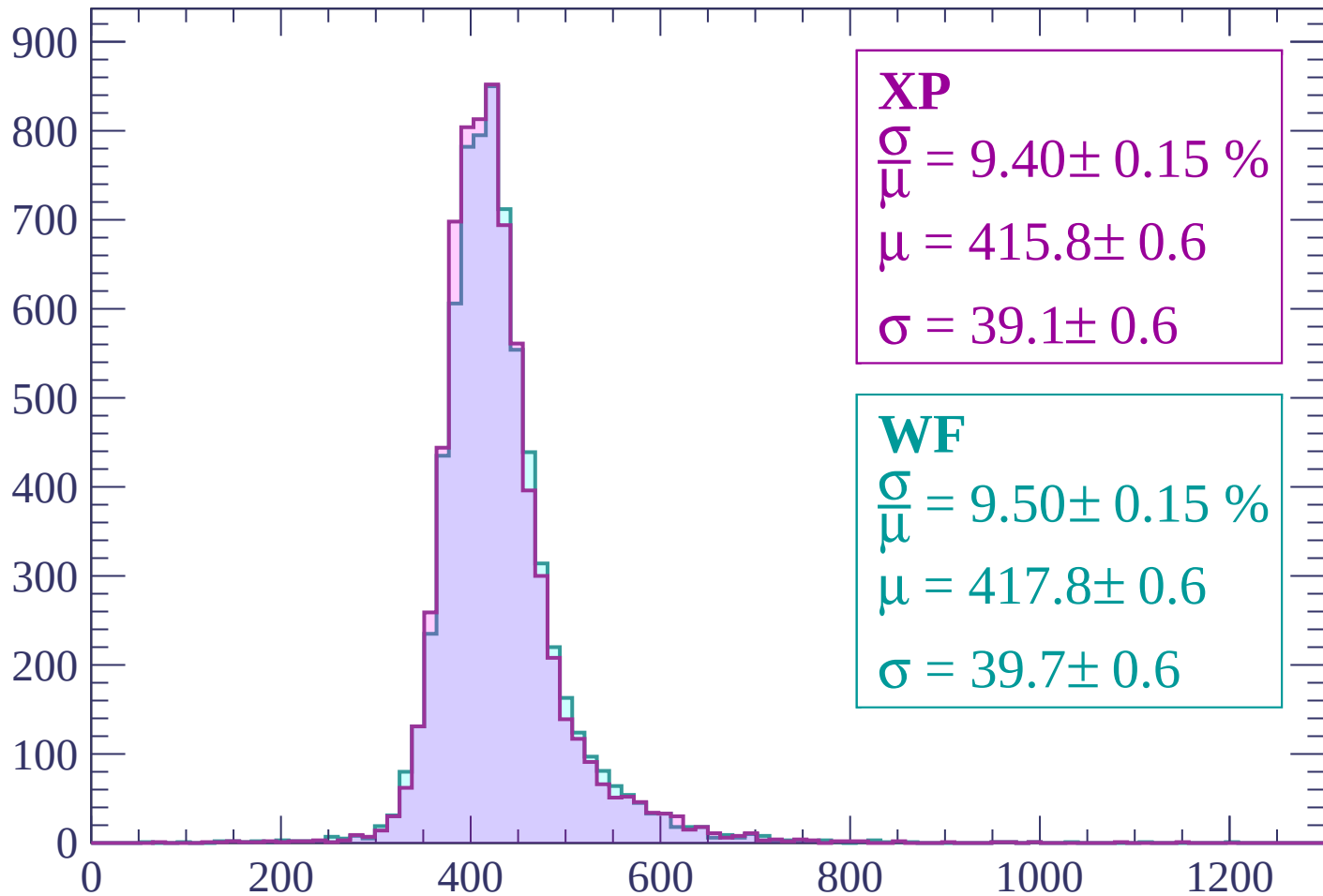
$$\mu = 414.8 \pm 0.6$$

$$\sigma = 40.1 \pm 0.6$$

dE/dx (ADC counts/cm)

Energy loss | $660 < p < 720$

Count



XP

$$\frac{\sigma}{\mu} = 9.40 \pm 0.15 \%$$

$$\mu = 415.8 \pm 0.6$$

$$\sigma = 39.1 \pm 0.6$$

WF

$$\frac{\sigma}{\mu} = 9.50 \pm 0.15 \%$$

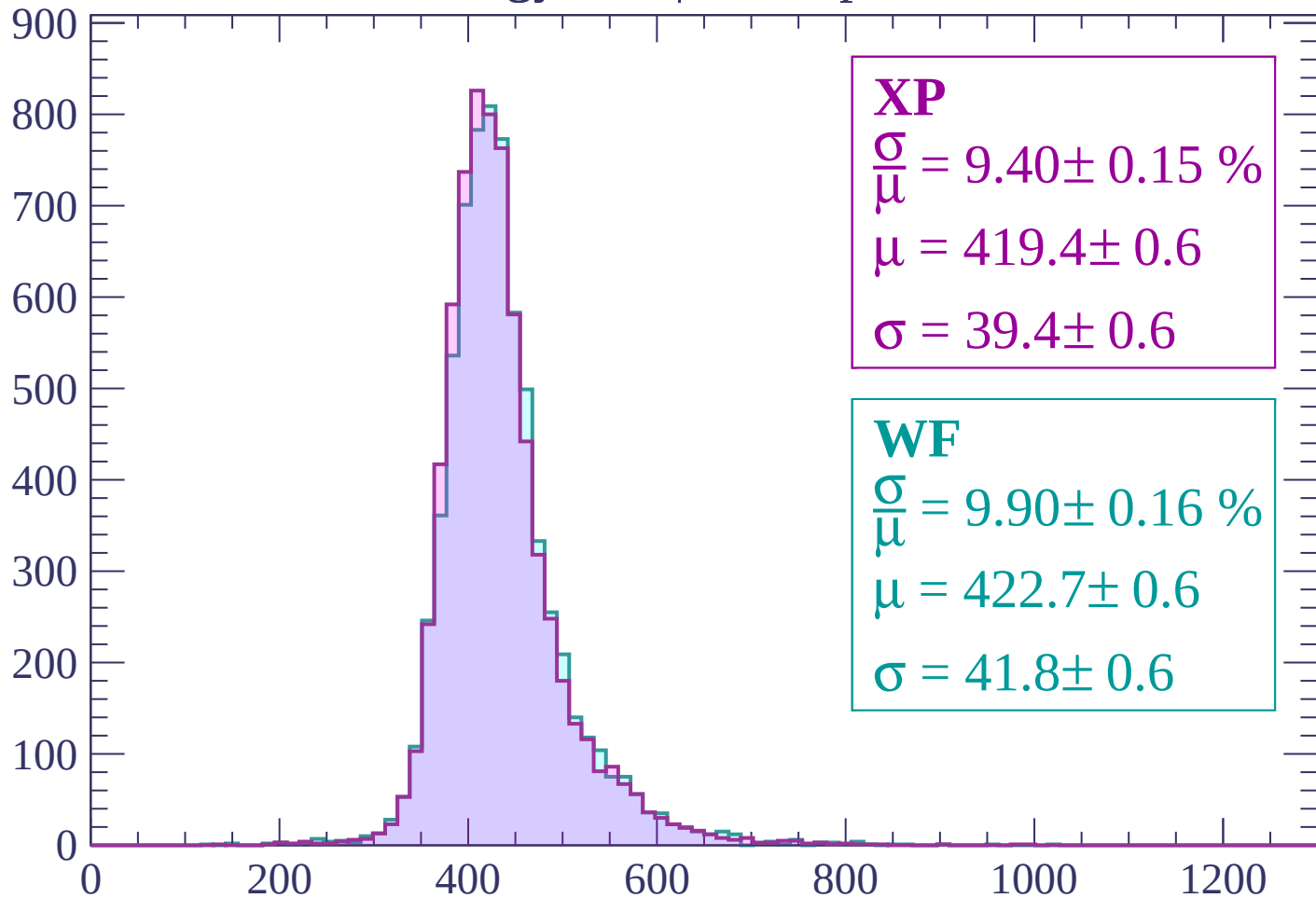
$$\mu = 417.8 \pm 0.6$$

$$\sigma = 39.7 \pm 0.6$$

dE/dx (ADC counts/cm)

Energy loss | $720 < p < 780$

Count



XP

$$\frac{\sigma}{\mu} = 9.40 \pm 0.15 \%$$

$$\mu = 419.4 \pm 0.6$$

$$\sigma = 39.4 \pm 0.6$$

WF

$$\frac{\sigma}{\mu} = 9.90 \pm 0.16 \%$$

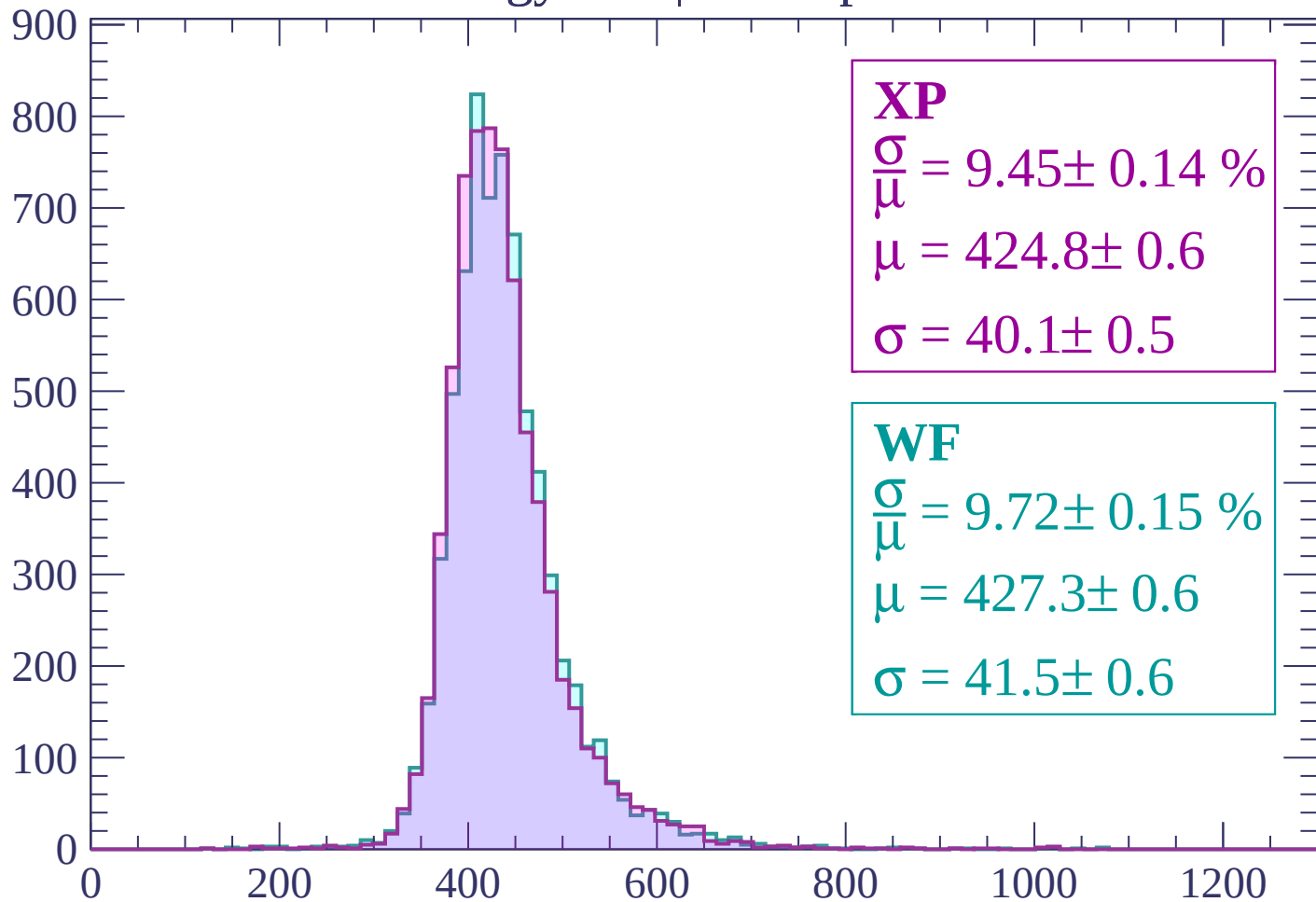
$$\mu = 422.7 \pm 0.6$$

$$\sigma = 41.8 \pm 0.6$$

dE/dx (ADC counts/cm)

Energy loss | $780 < p < 840$

Count



XP

$$\frac{\sigma}{\mu} = 9.45 \pm 0.14 \%$$

$$\mu = 424.8 \pm 0.6$$

$$\sigma = 40.1 \pm 0.5$$

WF

$$\frac{\sigma}{\mu} = 9.72 \pm 0.15 \%$$

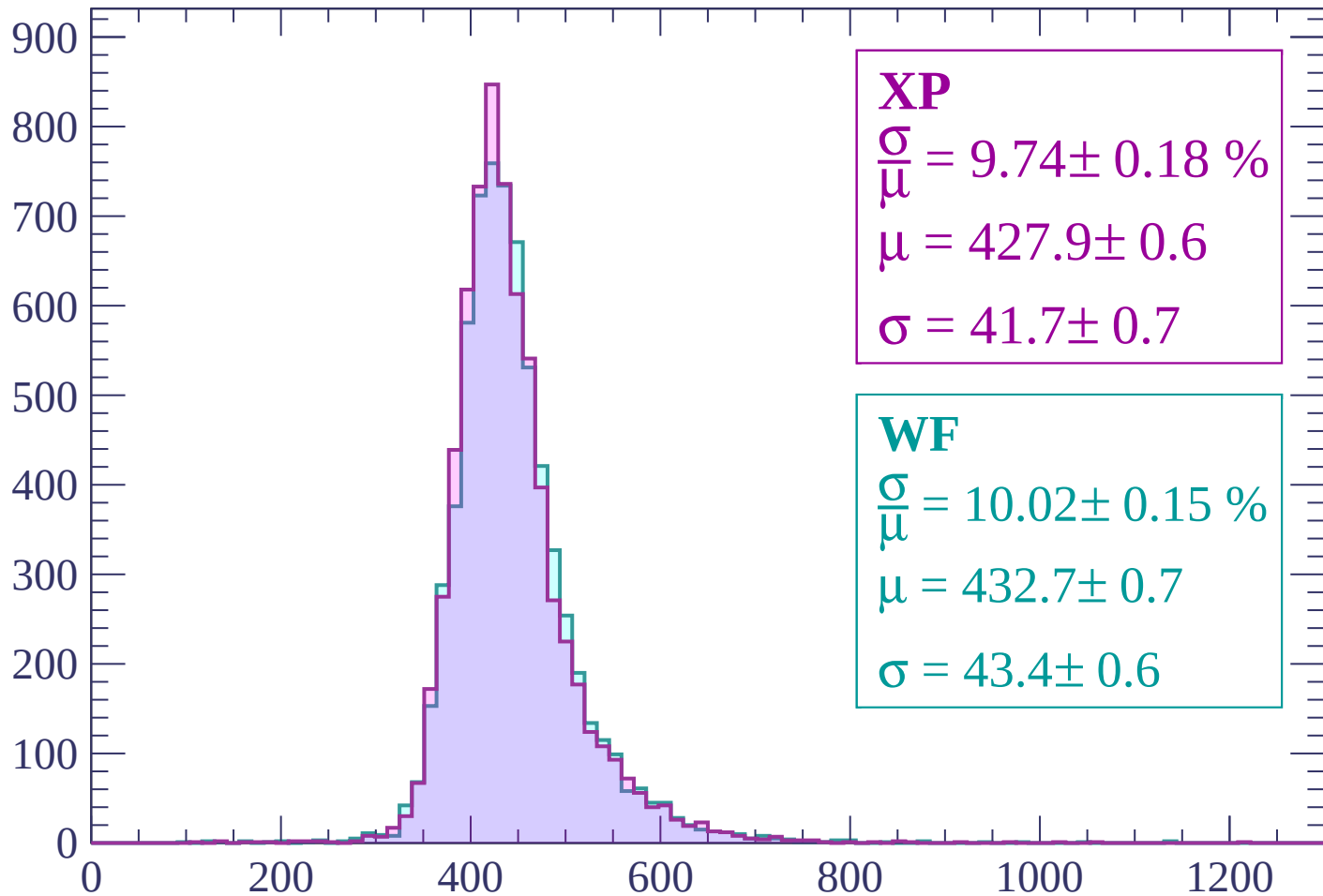
$$\mu = 427.3 \pm 0.6$$

$$\sigma = 41.5 \pm 0.6$$

dE/dx (ADC counts/cm)

Energy loss | $840 < p < 900$

Count



XP

$$\frac{\sigma}{\mu} = 9.74 \pm 0.18 \%$$

$$\mu = 427.9 \pm 0.6$$

$$\sigma = 41.7 \pm 0.7$$

WF

$$\frac{\sigma}{\mu} = 10.02 \pm 0.15 \%$$

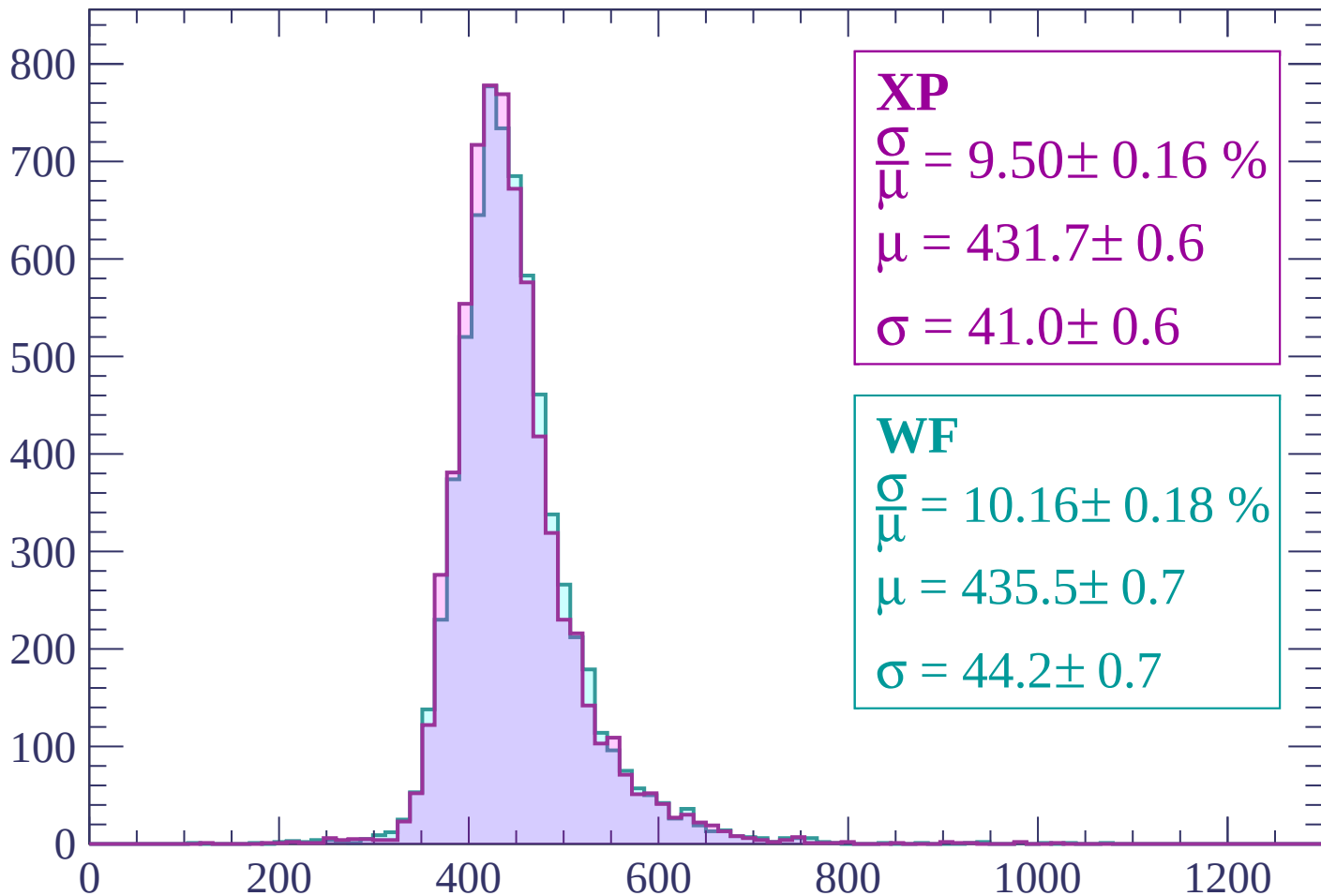
$$\mu = 432.7 \pm 0.7$$

$$\sigma = 43.4 \pm 0.6$$

dE/dx (ADC counts/cm)

Energy loss | $900 < p < 960$

Count



XP

$$\frac{\sigma}{\mu} = 9.50 \pm 0.16 \%$$

$$\mu = 431.7 \pm 0.6$$

$$\sigma = 41.0 \pm 0.6$$

WF

$$\frac{\sigma}{\mu} = 10.16 \pm 0.18 \%$$

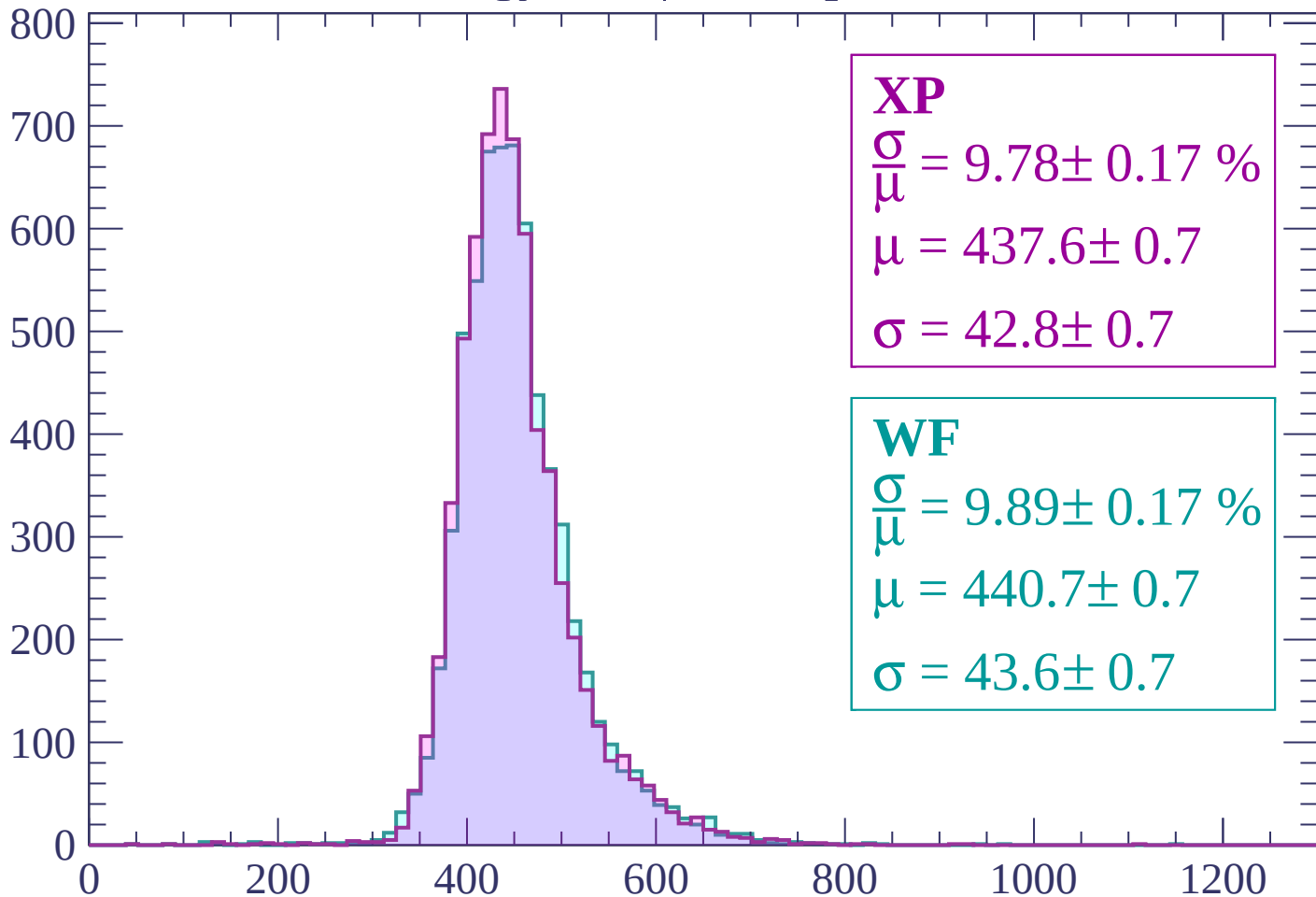
$$\mu = 435.5 \pm 0.7$$

$$\sigma = 44.2 \pm 0.7$$

dE/dx (ADC counts/cm)

Energy loss | $960 < p < 1020$

Count



XP

$$\frac{\sigma}{\mu} = 9.78 \pm 0.17 \%$$

$$\mu = 437.6 \pm 0.7$$

$$\sigma = 42.8 \pm 0.7$$

WF

$$\frac{\sigma}{\mu} = 9.89 \pm 0.17 \%$$

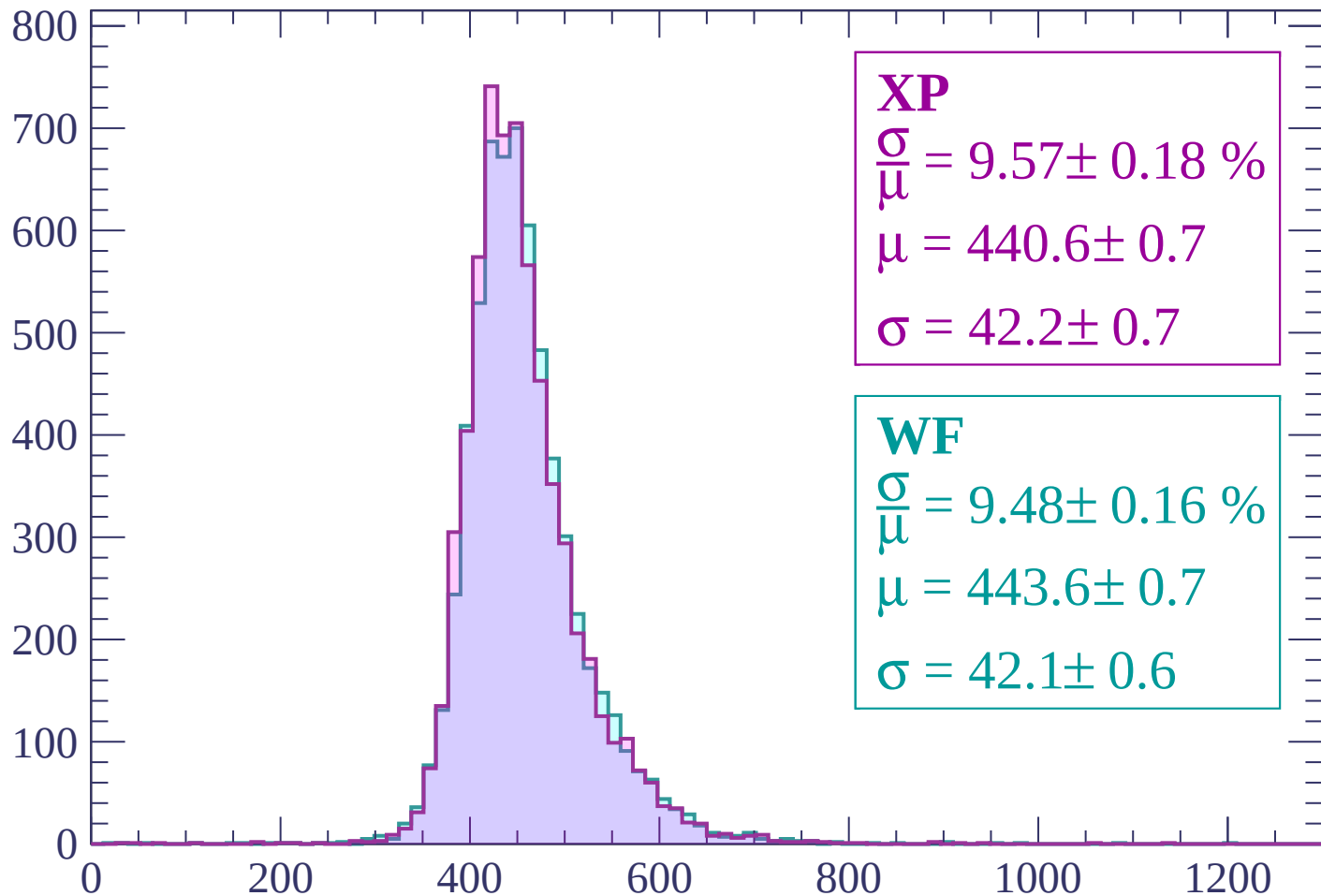
$$\mu = 440.7 \pm 0.7$$

$$\sigma = 43.6 \pm 0.7$$

dE/dx (ADC counts/cm)

Energy loss | $1020 < p < 1080$

Count



XP

$$\frac{\sigma}{\mu} = 9.57 \pm 0.18 \%$$

$$\mu = 440.6 \pm 0.7$$

$$\sigma = 42.2 \pm 0.7$$

WF

$$\frac{\sigma}{\mu} = 9.48 \pm 0.16 \%$$

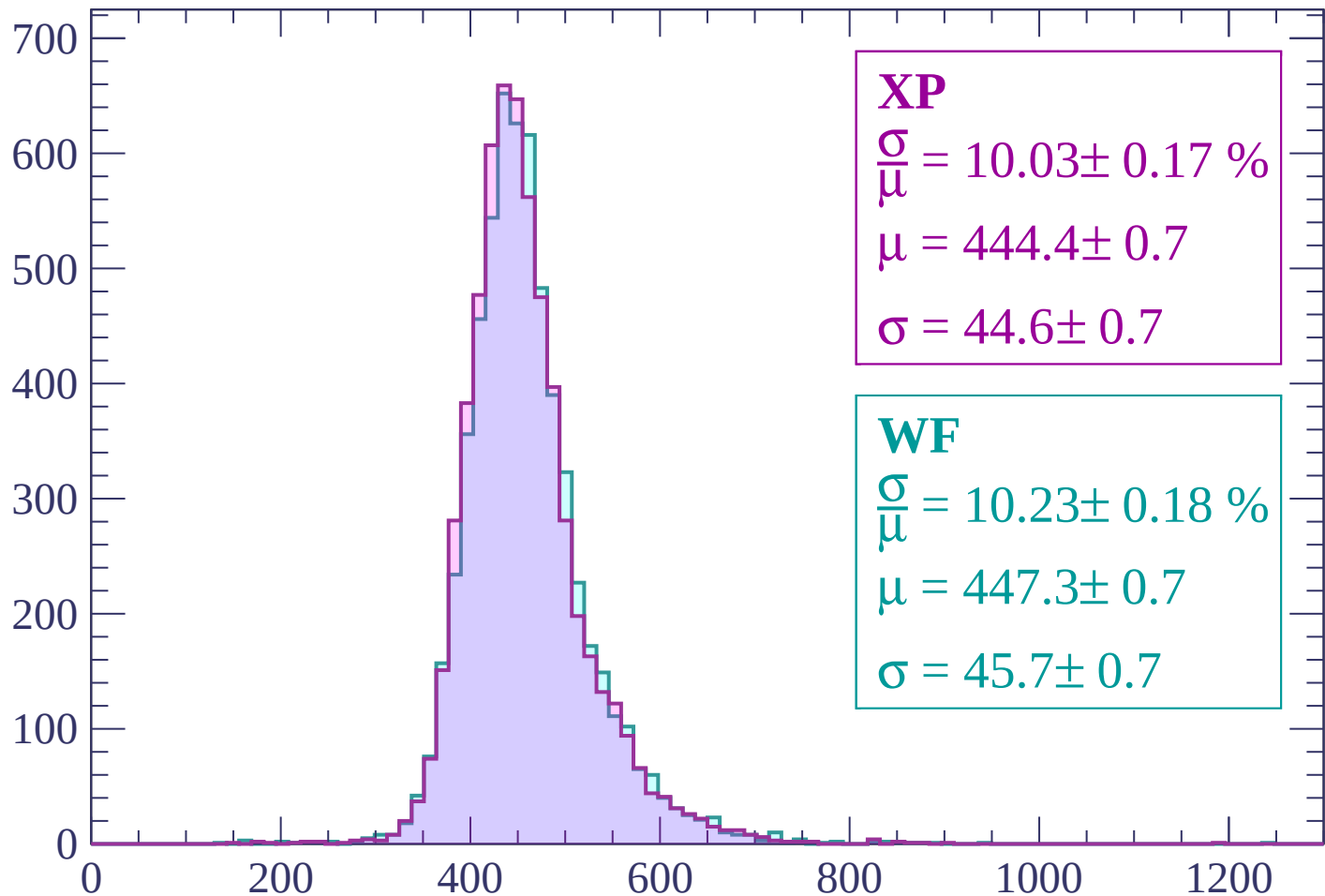
$$\mu = 443.6 \pm 0.7$$

$$\sigma = 42.1 \pm 0.6$$

dE/dx (ADC counts/cm)

Energy loss | $1080 < p < 1140$

Count



XP

$$\frac{\sigma}{\mu} = 10.03 \pm 0.17 \%$$

$$\mu = 444.4 \pm 0.7$$

$$\sigma = 44.6 \pm 0.7$$

WF

$$\frac{\sigma}{\mu} = 10.23 \pm 0.18 \%$$

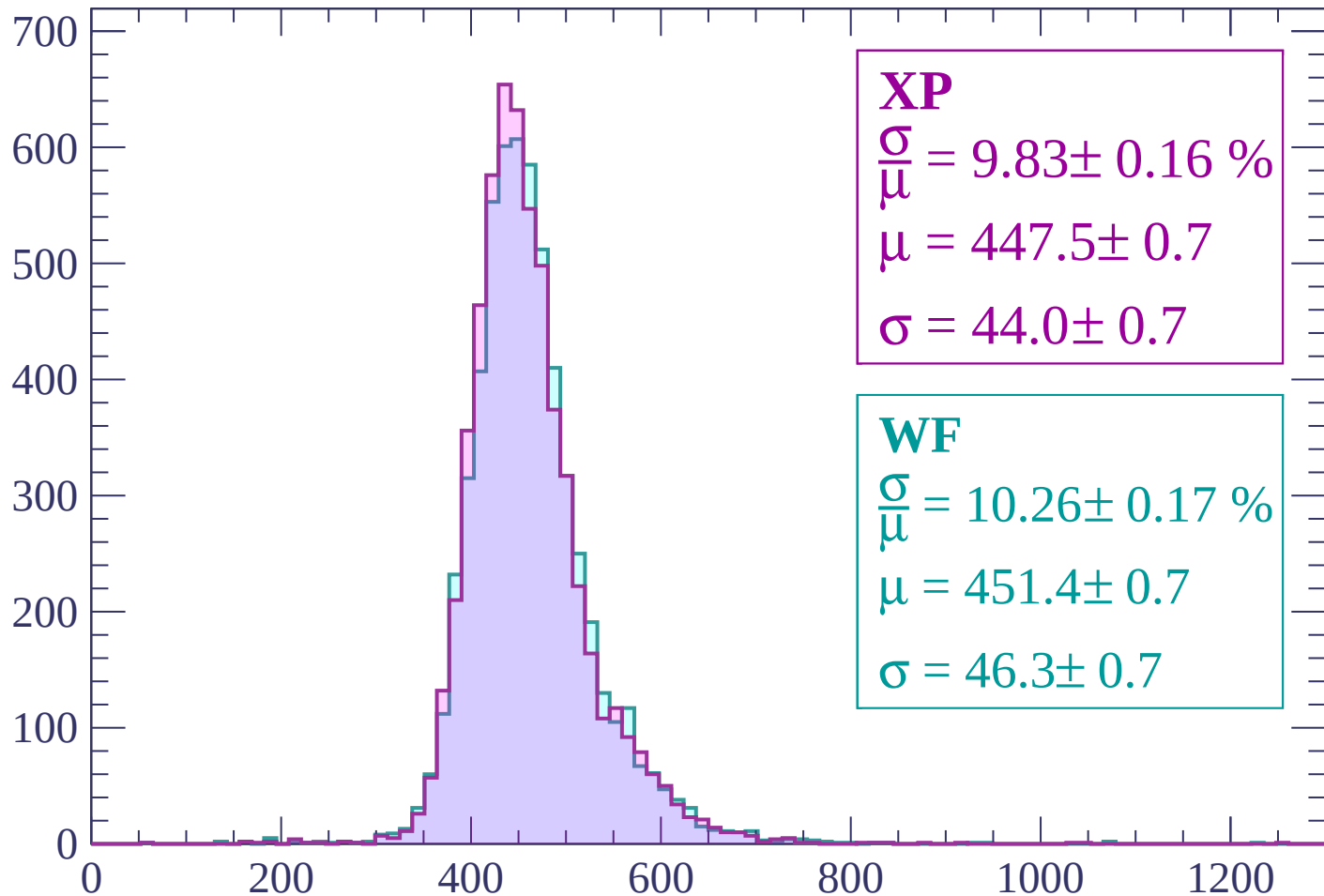
$$\mu = 447.3 \pm 0.7$$

$$\sigma = 45.7 \pm 0.7$$

dE/dx (ADC counts/cm)

Energy loss | $1140 < p < 1200$

Count



XP

$$\frac{\sigma}{\mu} = 9.83 \pm 0.16 \%$$

$$\mu = 447.5 \pm 0.7$$

$$\sigma = 44.0 \pm 0.7$$

WF

$$\frac{\sigma}{\mu} = 10.26 \pm 0.17 \%$$

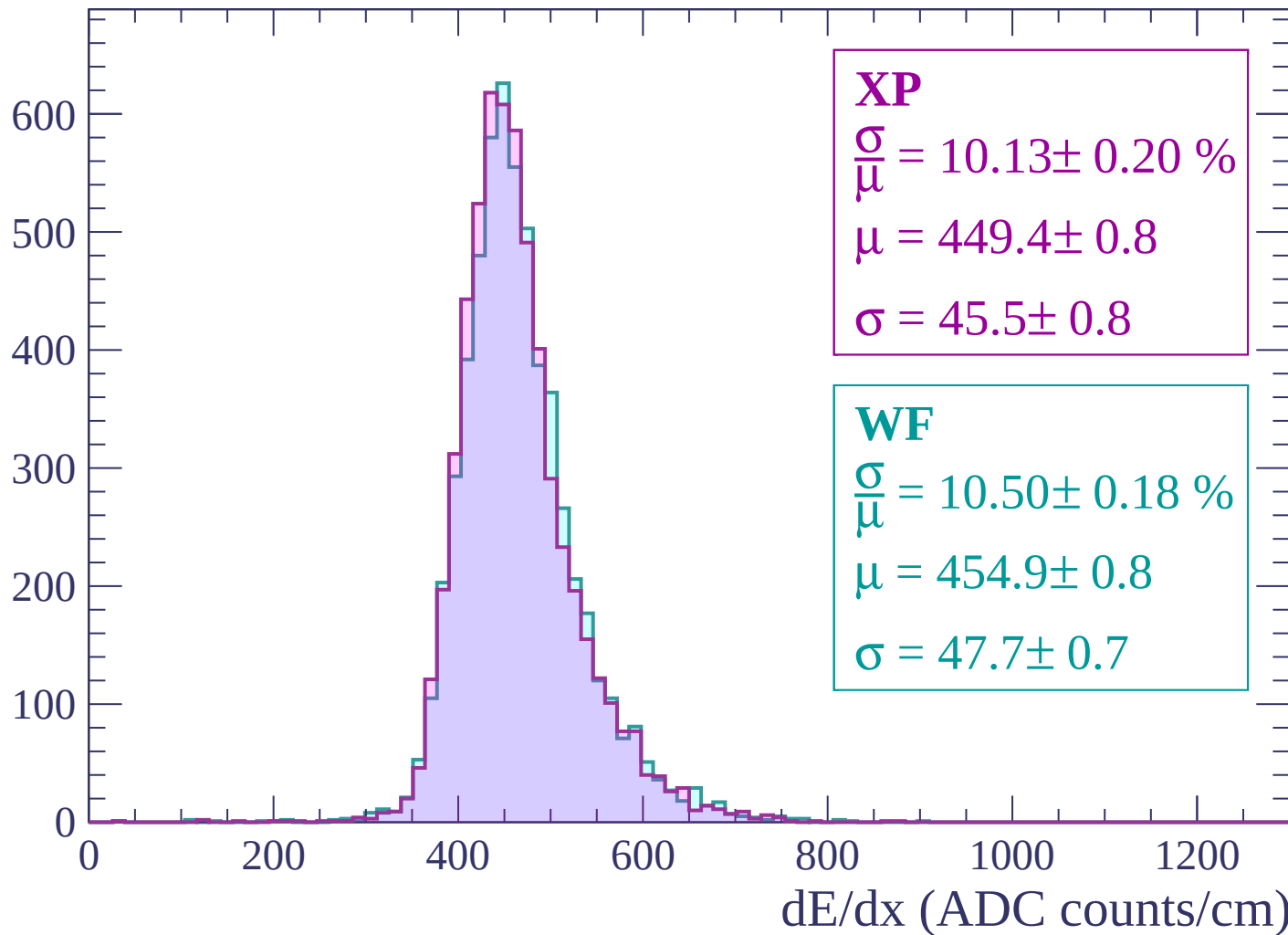
$$\mu = 451.4 \pm 0.7$$

$$\sigma = 46.3 \pm 0.7$$

dE/dx (ADC counts/cm)

Energy loss | $1200 < p < 1260$

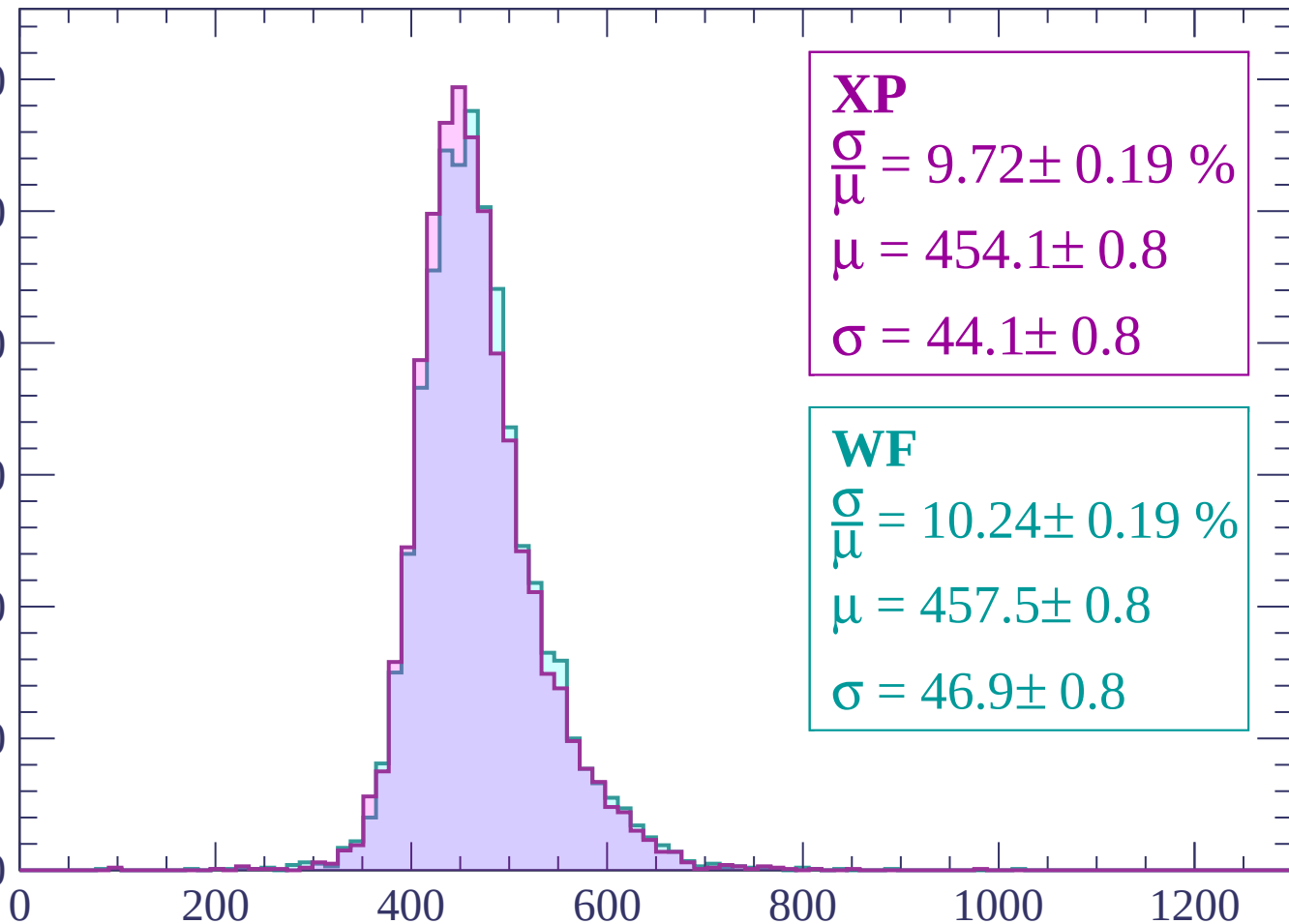
Count



Energy loss | $1260 < p < 1320$

Count

600
500
400
300
200
100
0



XP

$$\frac{\sigma}{\mu} = 9.72 \pm 0.19 \%$$

$$\mu = 454.1 \pm 0.8$$

$$\sigma = 44.1 \pm 0.8$$

WF

$$\frac{\sigma}{\mu} = 10.24 \pm 0.19 \%$$

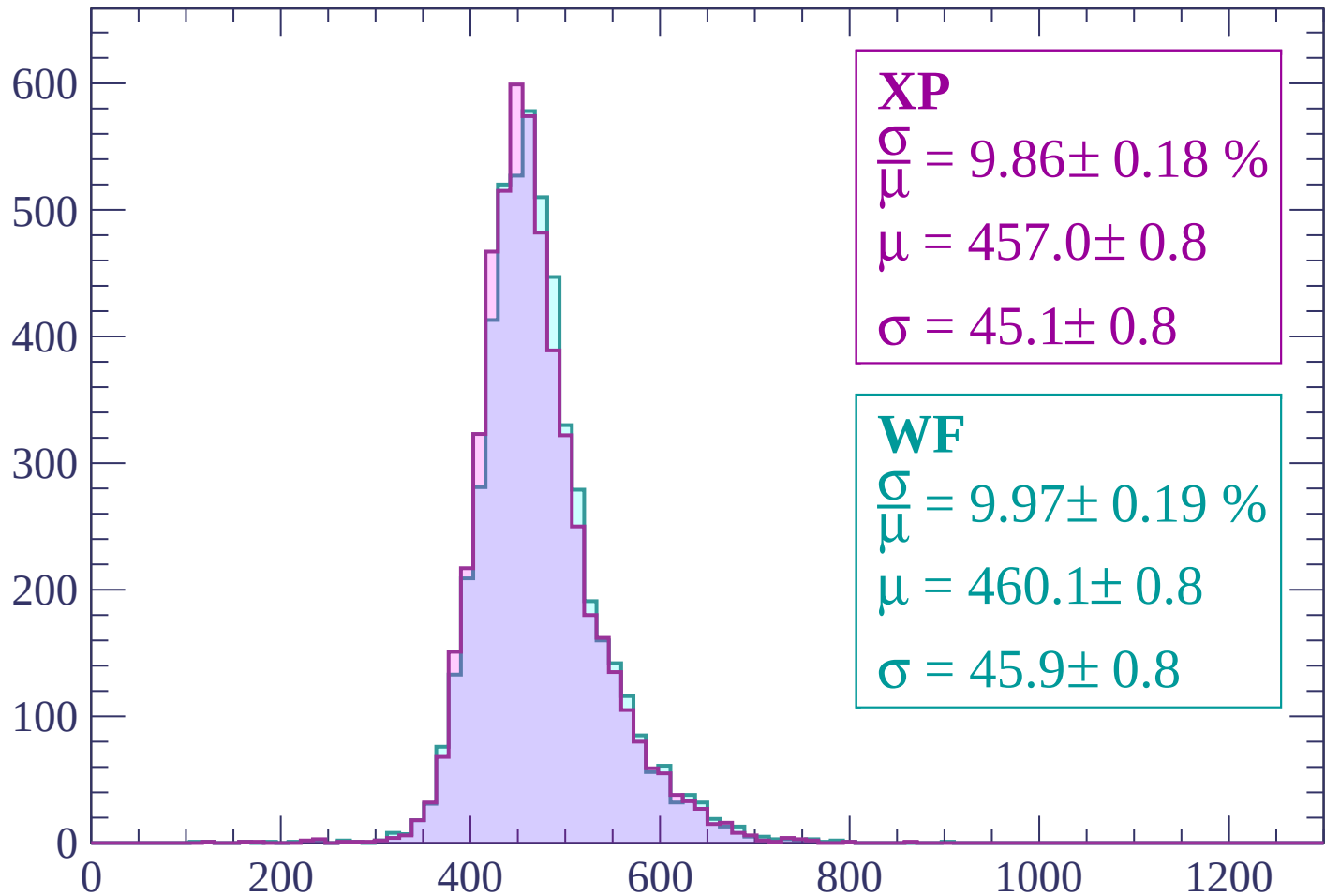
$$\mu = 457.5 \pm 0.8$$

$$\sigma = 46.9 \pm 0.8$$

dE/dx (ADC counts/cm)

Energy loss | $1320 < p < 1380$

Count



XP

$$\frac{\sigma}{\mu} = 9.86 \pm 0.18 \%$$

$$\mu = 457.0 \pm 0.8$$

$$\sigma = 45.1 \pm 0.8$$

WF

$$\frac{\sigma}{\mu} = 9.97 \pm 0.19 \%$$

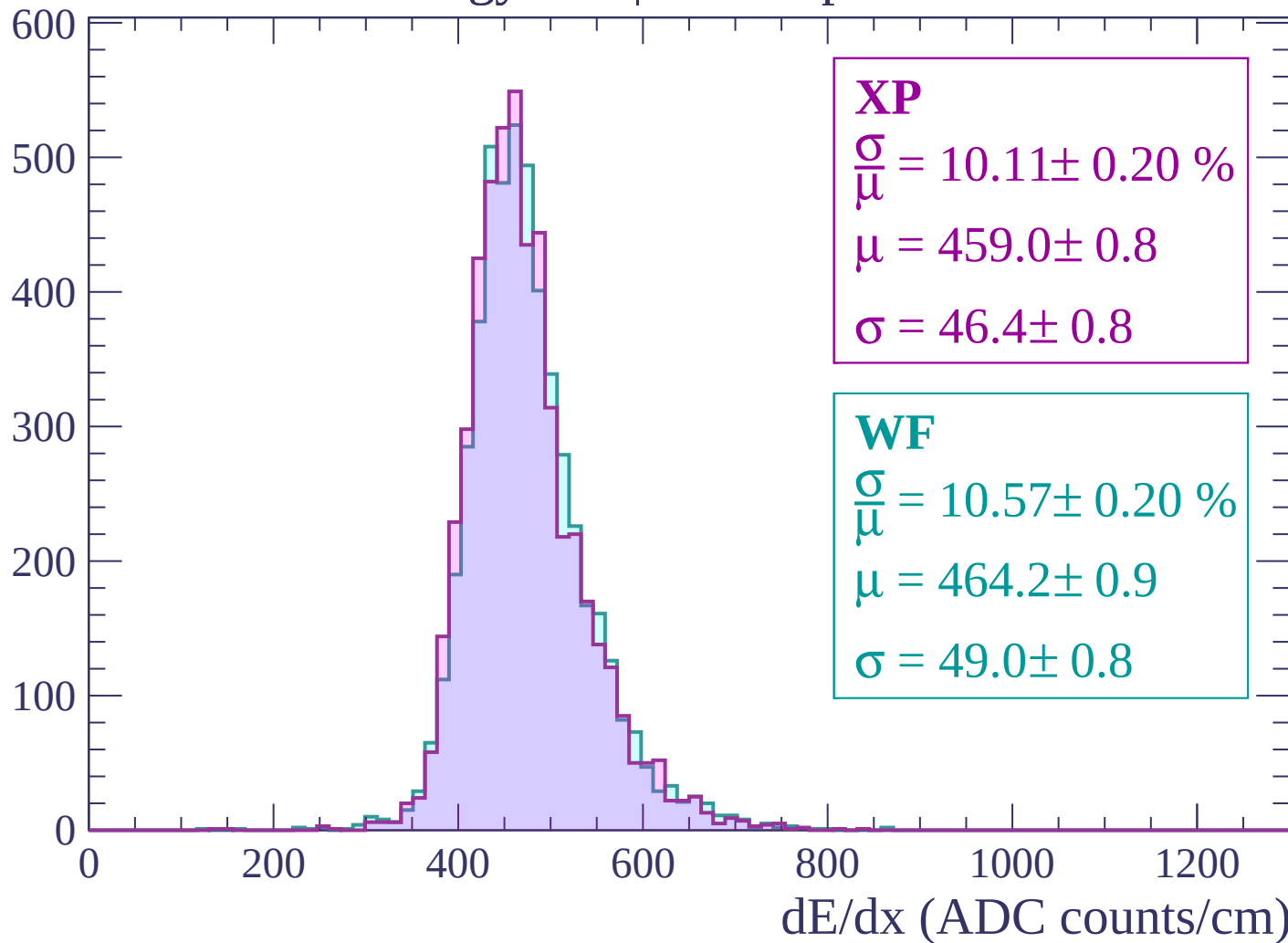
$$\mu = 460.1 \pm 0.8$$

$$\sigma = 45.9 \pm 0.8$$

dE/dx (ADC counts/cm)

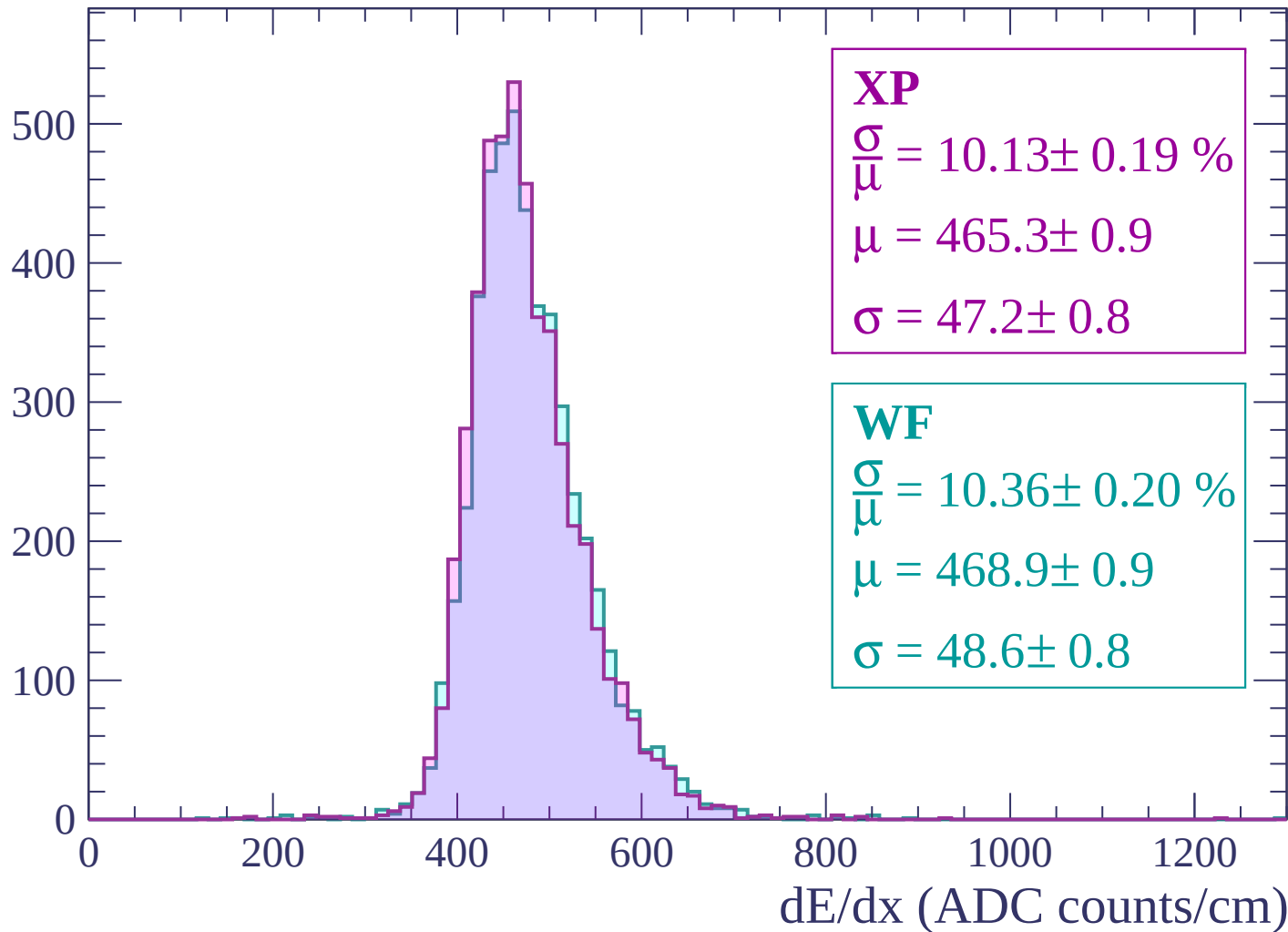
Energy loss | $1380 < p < 1440$

Count



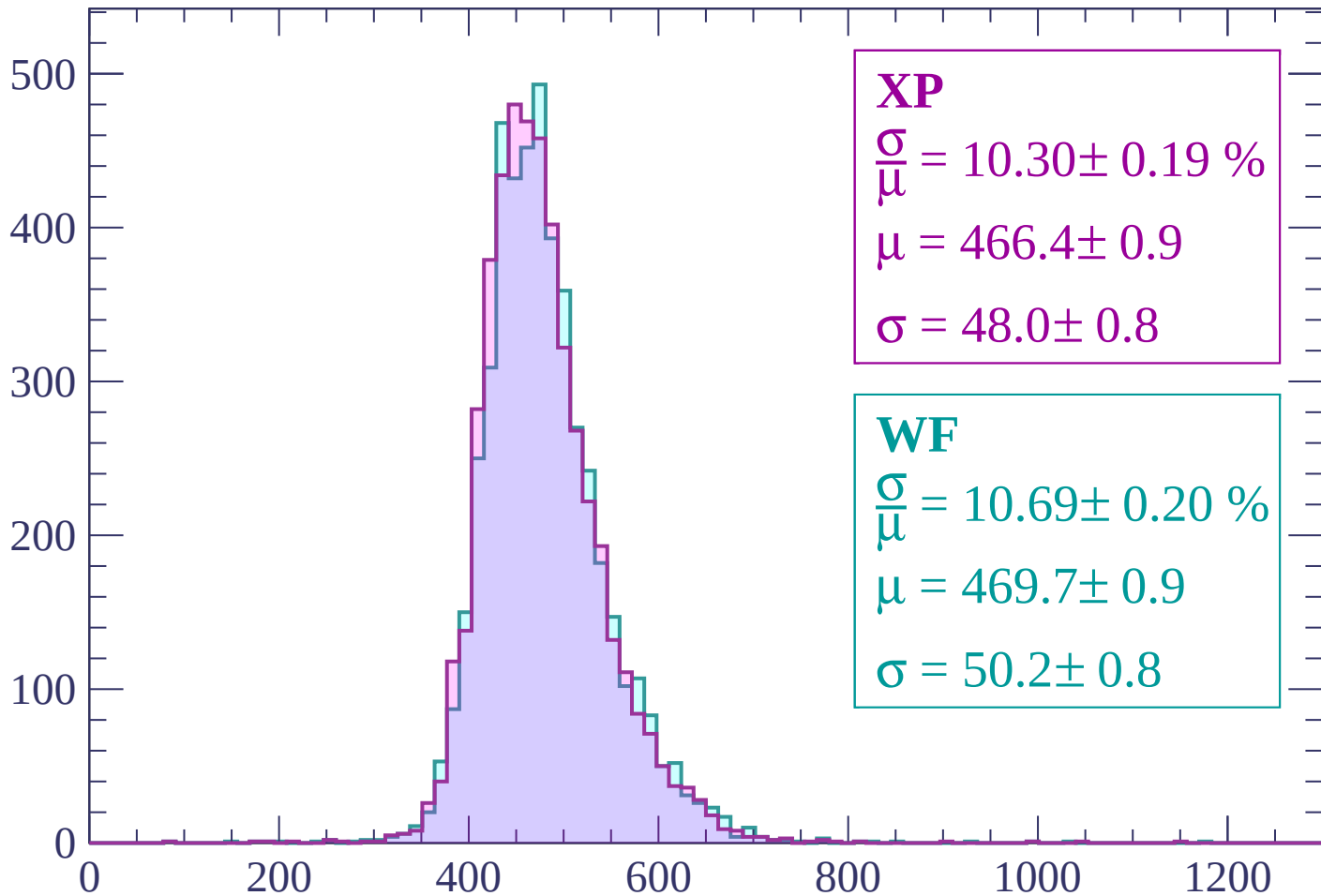
Energy loss | $1440 < p < 1500$

Count



Energy loss | $1500 < p < 1560$

Count



XP

$$\frac{\sigma}{\mu} = 10.30 \pm 0.19 \%$$

$$\mu = 466.4 \pm 0.9$$

$$\sigma = 48.0 \pm 0.8$$

WF

$$\frac{\sigma}{\mu} = 10.69 \pm 0.20 \%$$

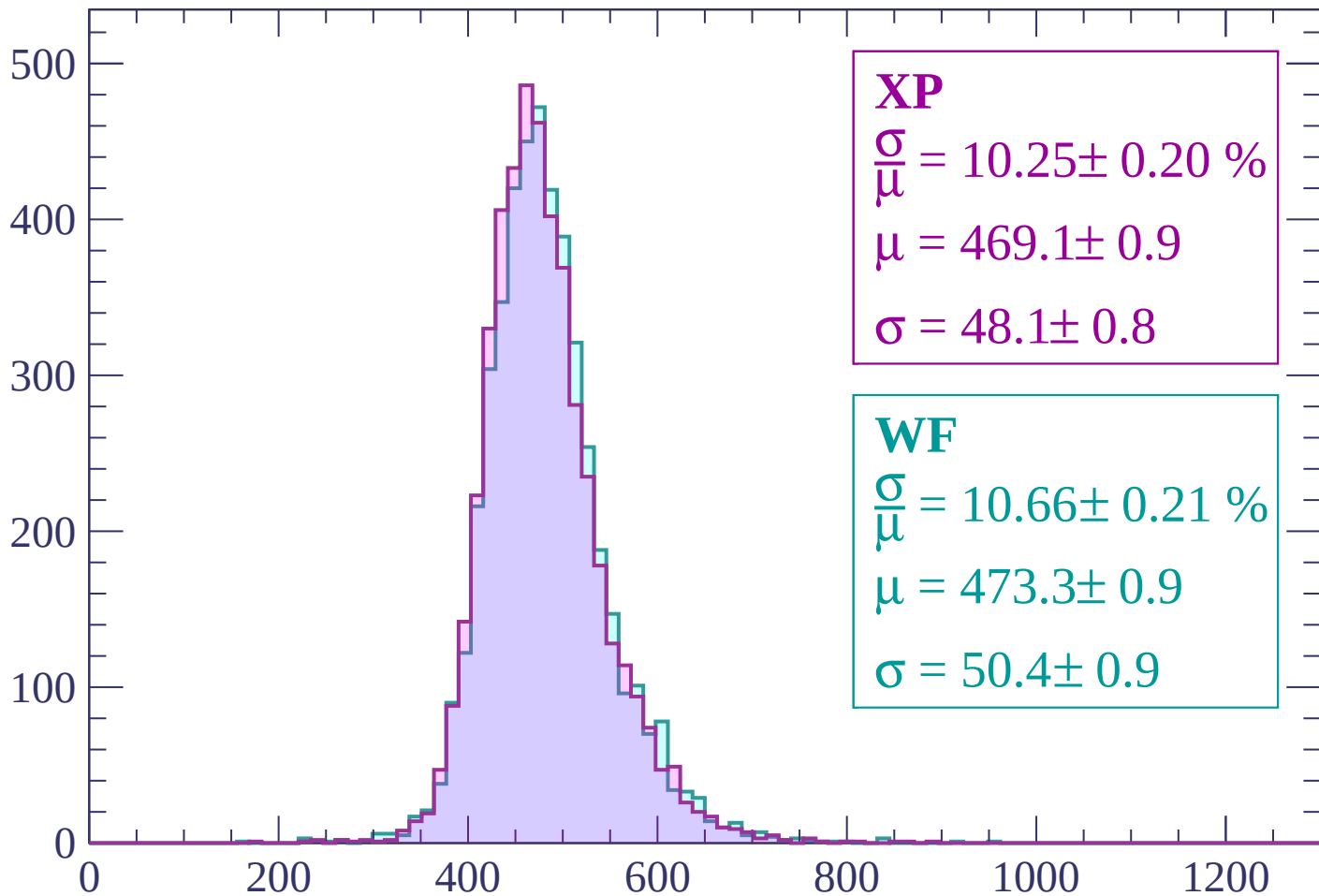
$$\mu = 469.7 \pm 0.9$$

$$\sigma = 50.2 \pm 0.8$$

dE/dx (ADC counts/cm)

Energy loss | $1560 < p < 1620$

Count



XP

$$\frac{\sigma}{\mu} = 10.25 \pm 0.20 \%$$

$$\mu = 469.1 \pm 0.9$$

$$\sigma = 48.1 \pm 0.8$$

WF

$$\frac{\sigma}{\mu} = 10.66 \pm 0.21 \%$$

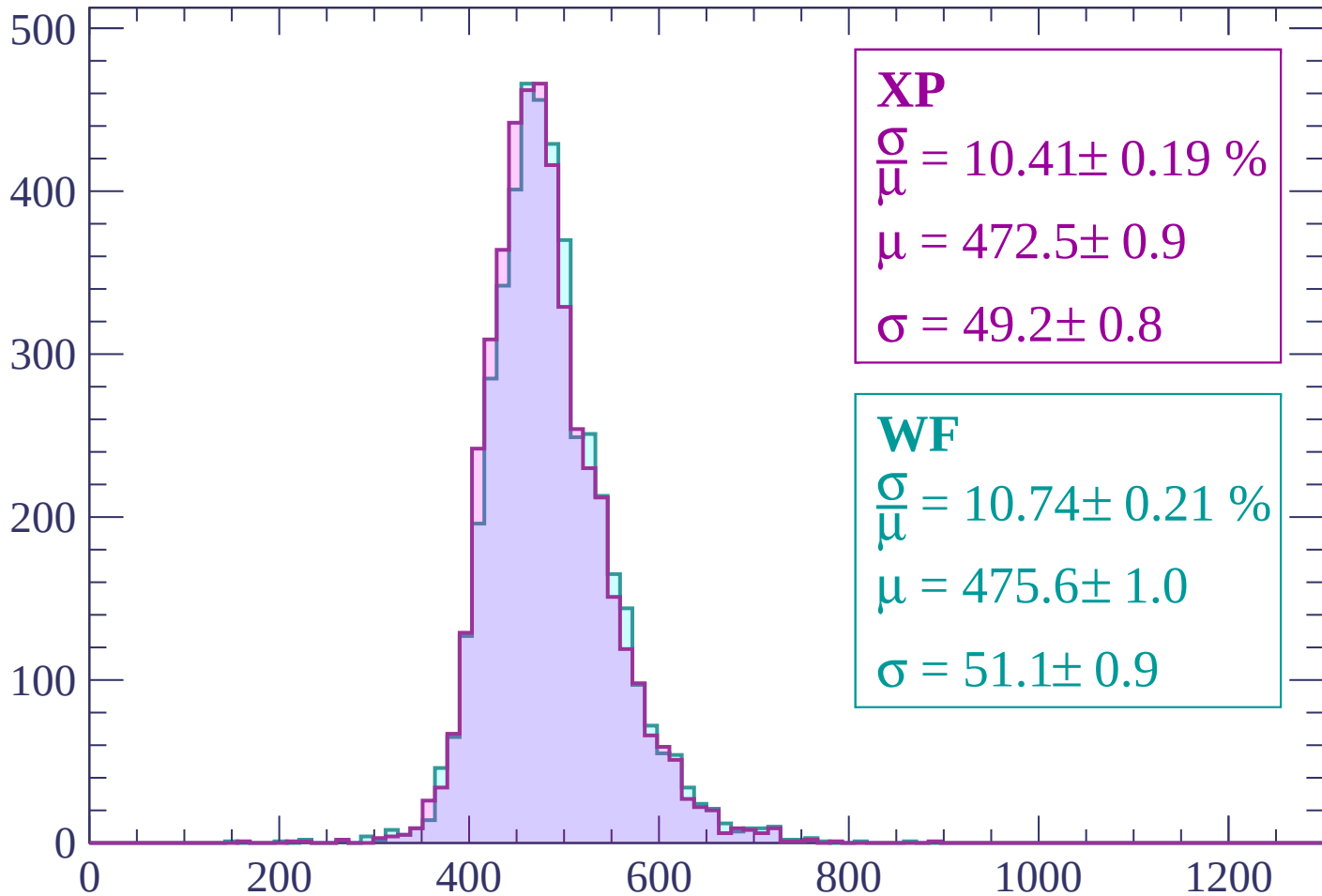
$$\mu = 473.3 \pm 0.9$$

$$\sigma = 50.4 \pm 0.9$$

dE/dx (ADC counts/cm)

Energy loss | $1620 < p < 1680$

Count



XP

$$\frac{\sigma}{\mu} = 10.41 \pm 0.19 \%$$

$$\mu = 472.5 \pm 0.9$$

$$\sigma = 49.2 \pm 0.8$$

WF

$$\frac{\sigma}{\mu} = 10.74 \pm 0.21 \%$$

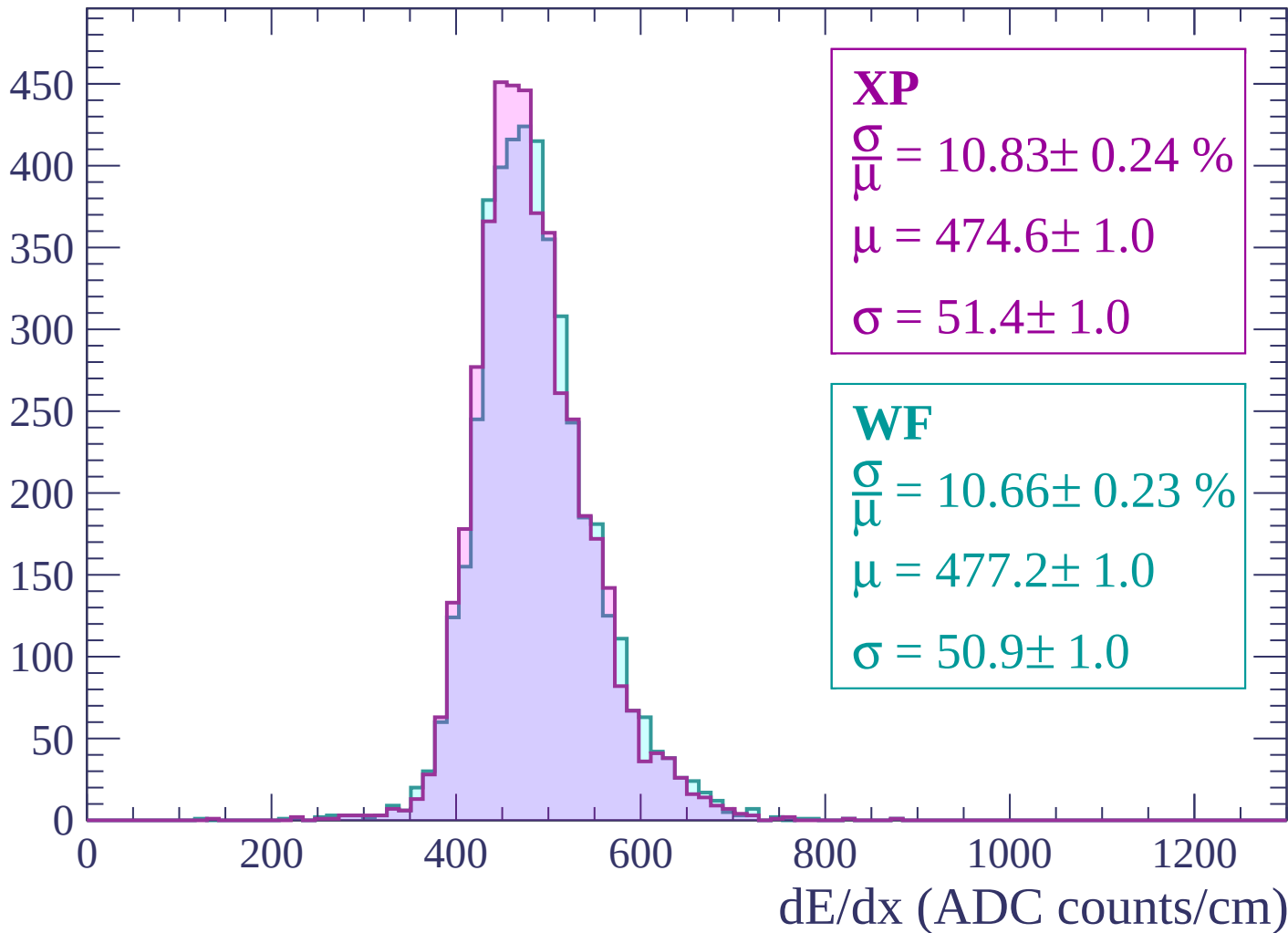
$$\mu = 475.6 \pm 1.0$$

$$\sigma = 51.1 \pm 0.9$$

dE/dx (ADC counts/cm)

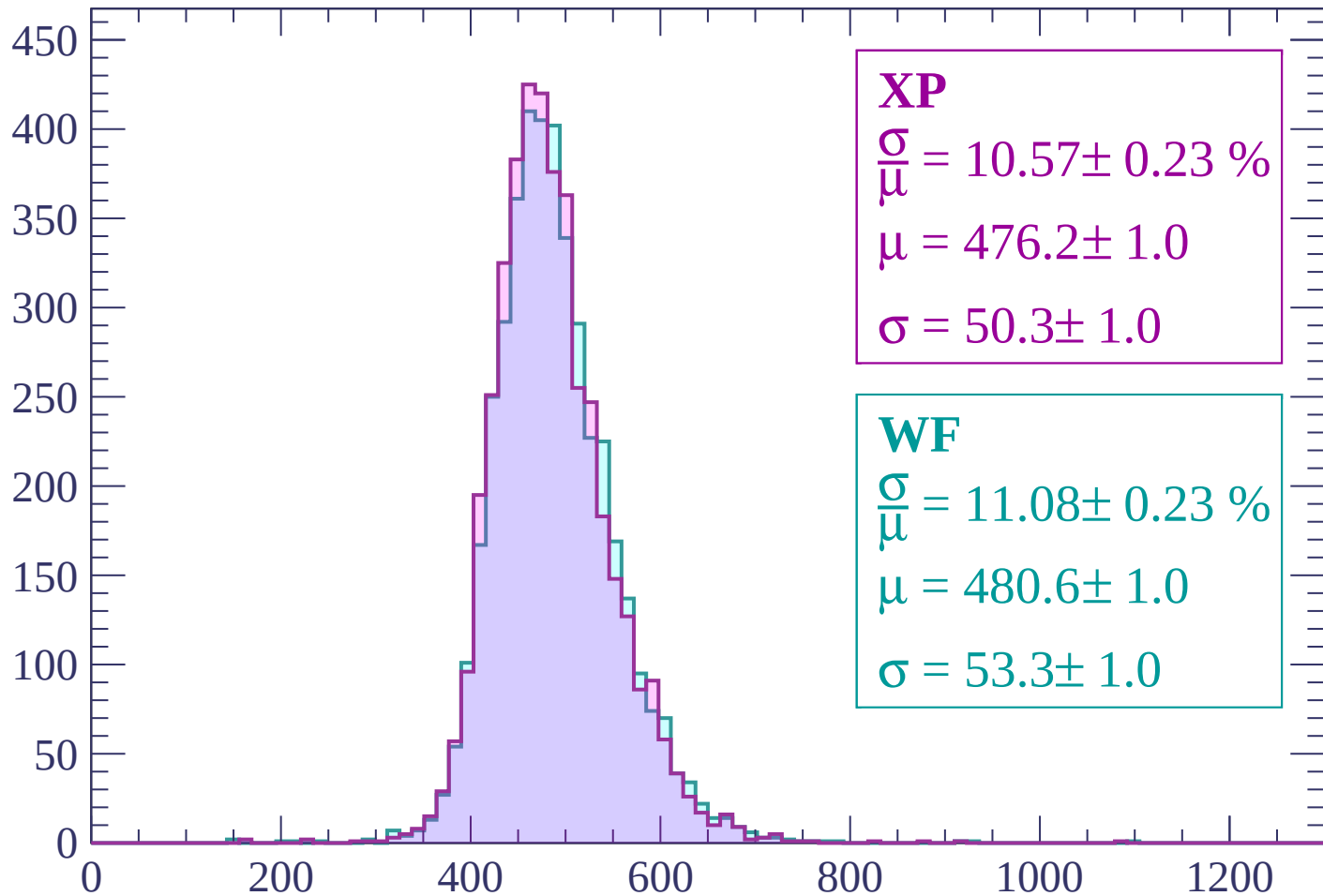
Energy loss | $1680 < p < 1740$

Count



Energy loss | $1740 < p < 1800$

Count



XP

$$\frac{\sigma}{\bar{\mu}} = 10.57 \pm 0.23 \%$$

$$\mu = 476.2 \pm 1.0$$

$$\sigma = 50.3 \pm 1.0$$

WF

$$\frac{\sigma}{\bar{\mu}} = 11.08 \pm 0.23 \%$$

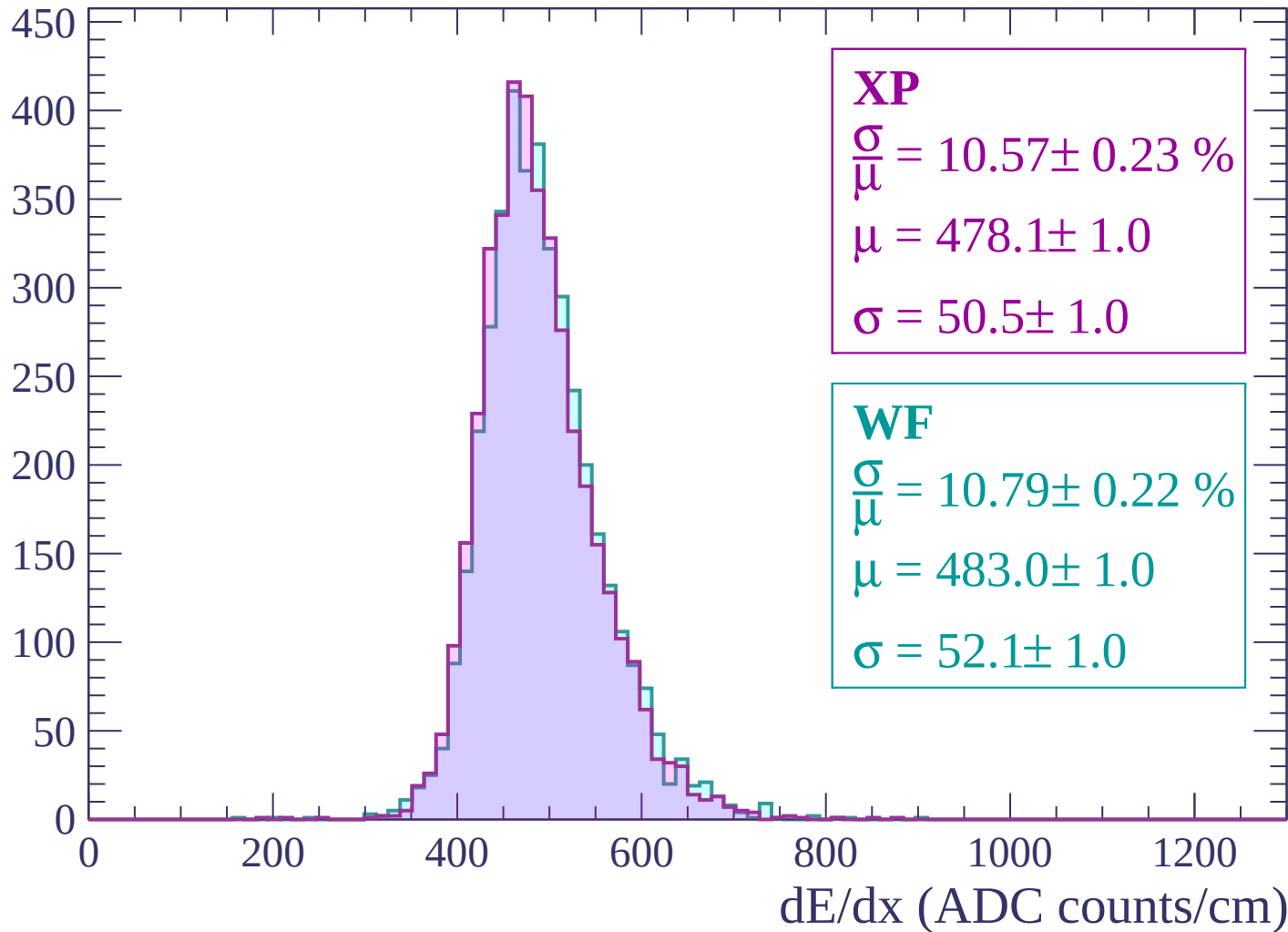
$$\mu = 480.6 \pm 1.0$$

$$\sigma = 53.3 \pm 1.0$$

dE/dx (ADC counts/cm)

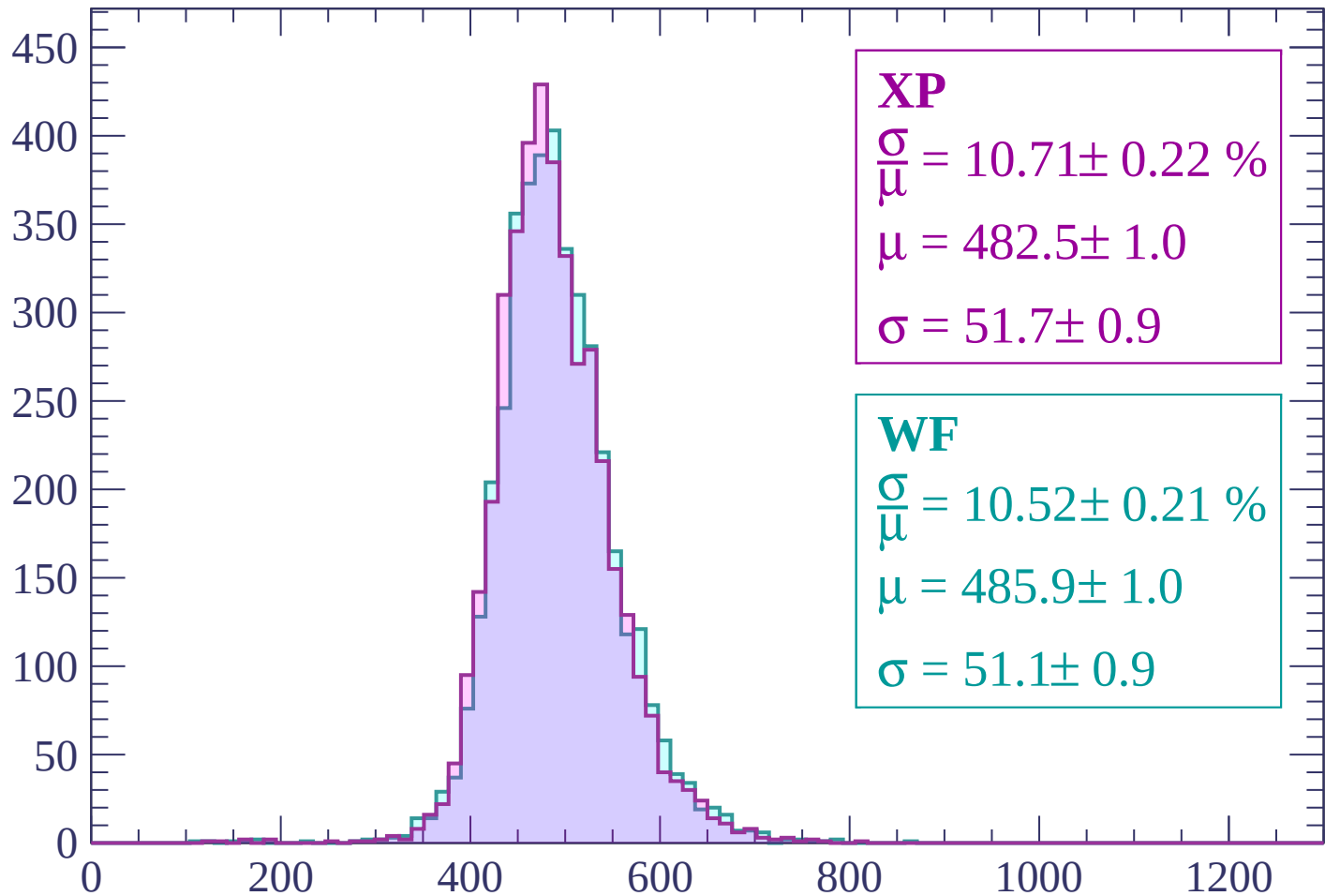
Energy loss | $1800 < p < 1860$

Count



Energy loss | $1860 < p < 1920$

Count



XP

$$\frac{\sigma}{\mu} = 10.71 \pm 0.22 \%$$

$$\mu = 482.5 \pm 1.0$$

$$\sigma = 51.7 \pm 0.9$$

WF

$$\frac{\sigma}{\mu} = 10.52 \pm 0.21 \%$$

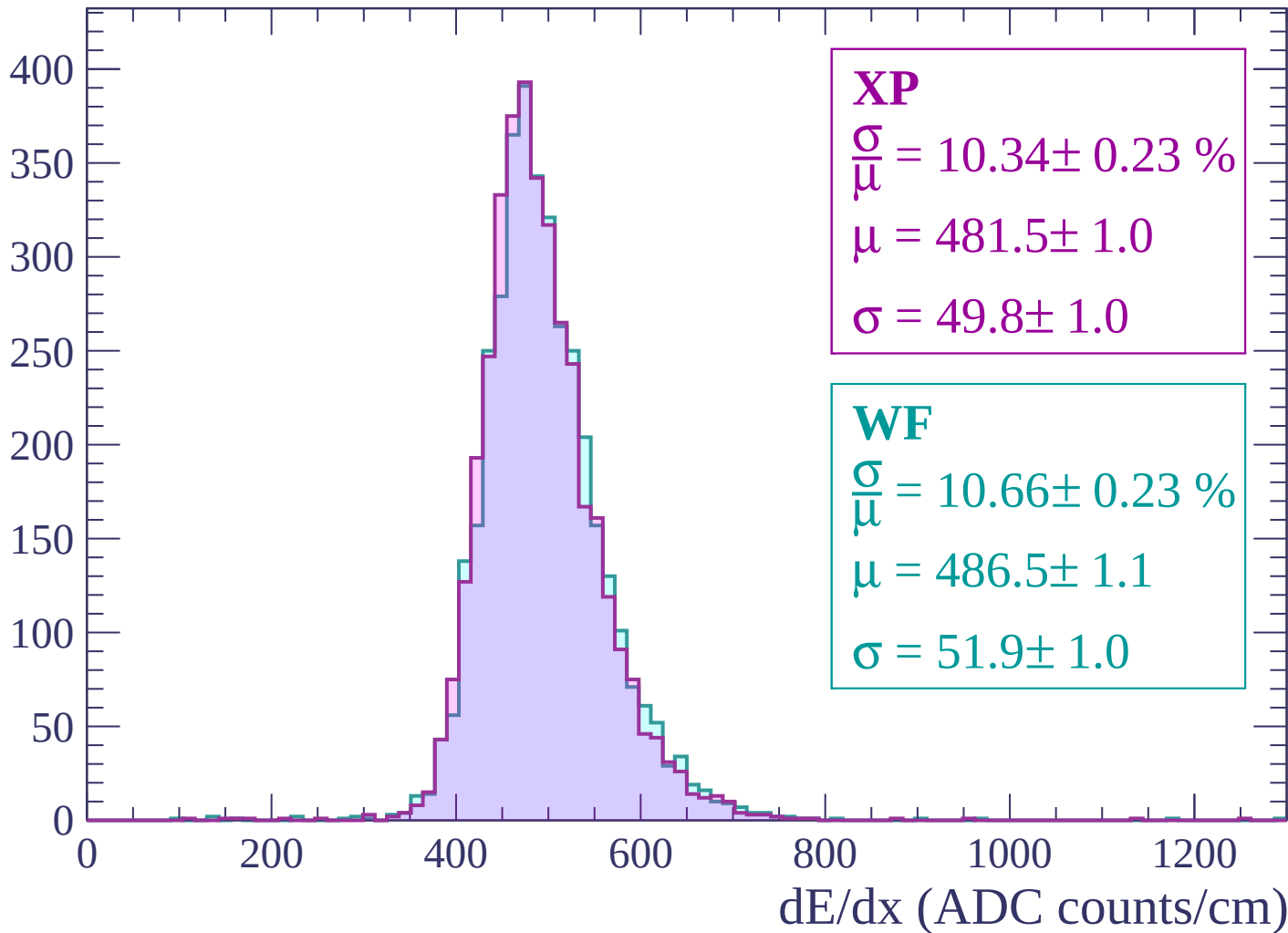
$$\mu = 485.9 \pm 1.0$$

$$\sigma = 51.1 \pm 0.9$$

dE/dx (ADC counts/cm)

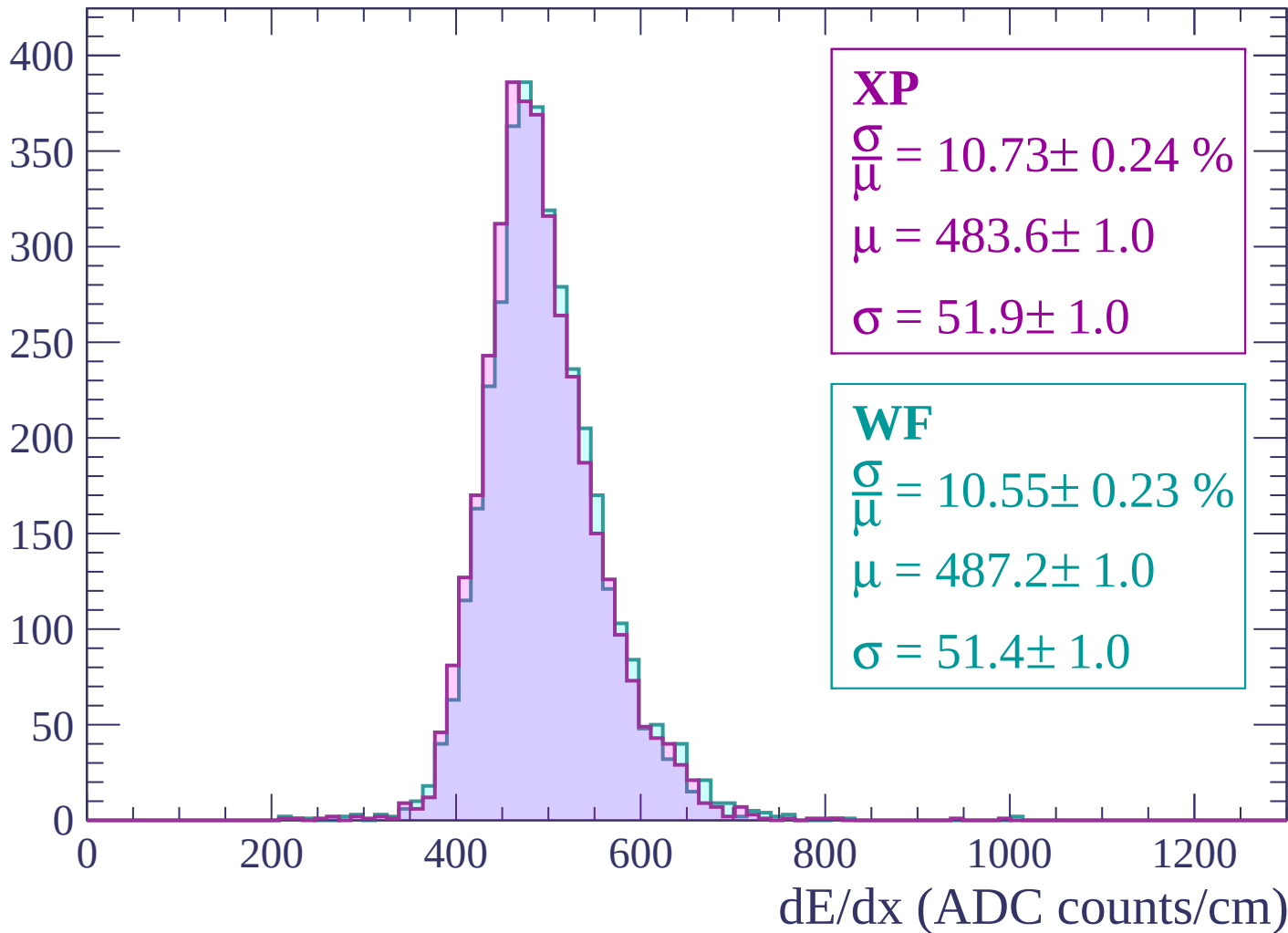
Energy loss | $1920 < p < 1980$

Count

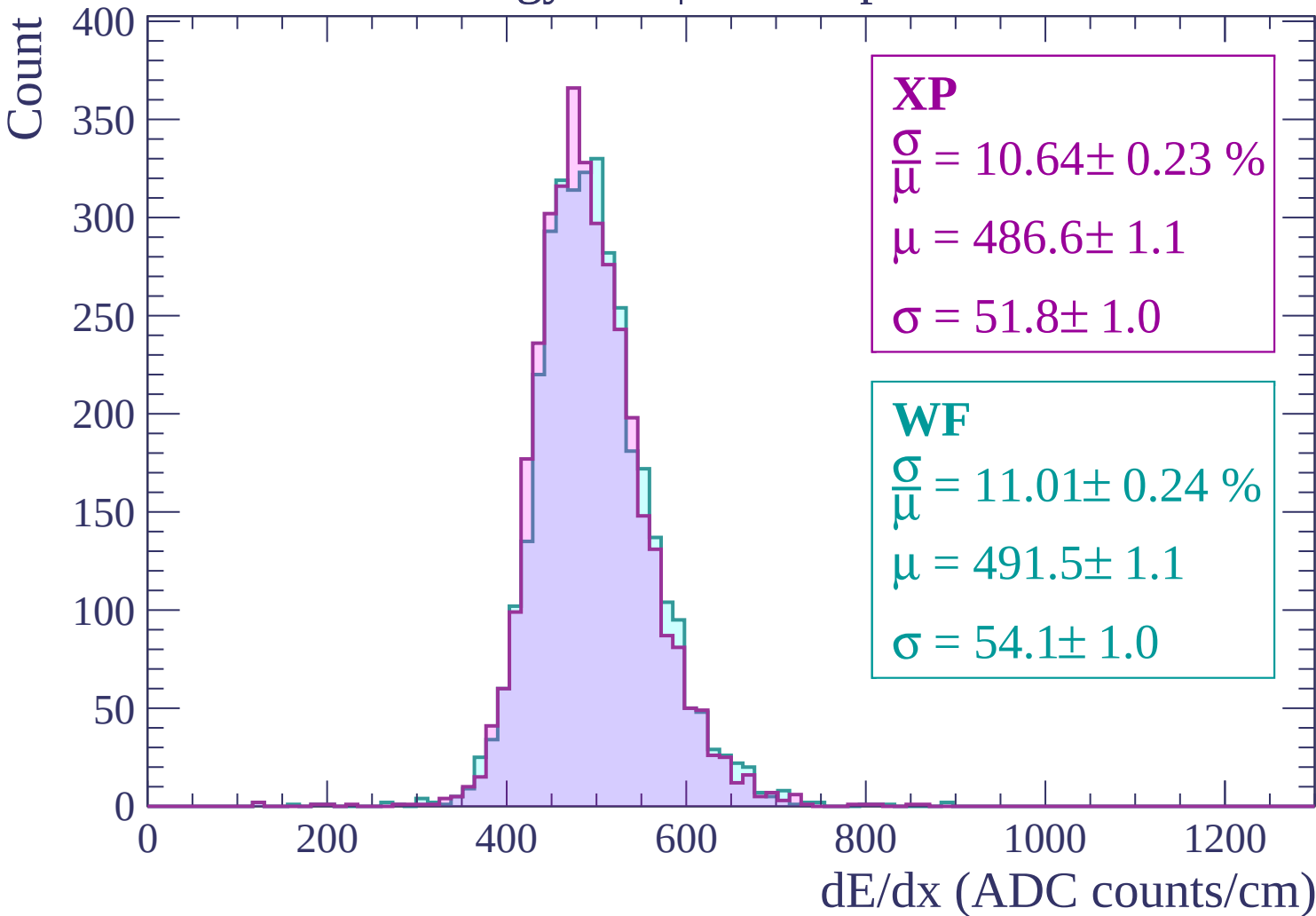


Energy loss | $1980 < p < 2040$

Count

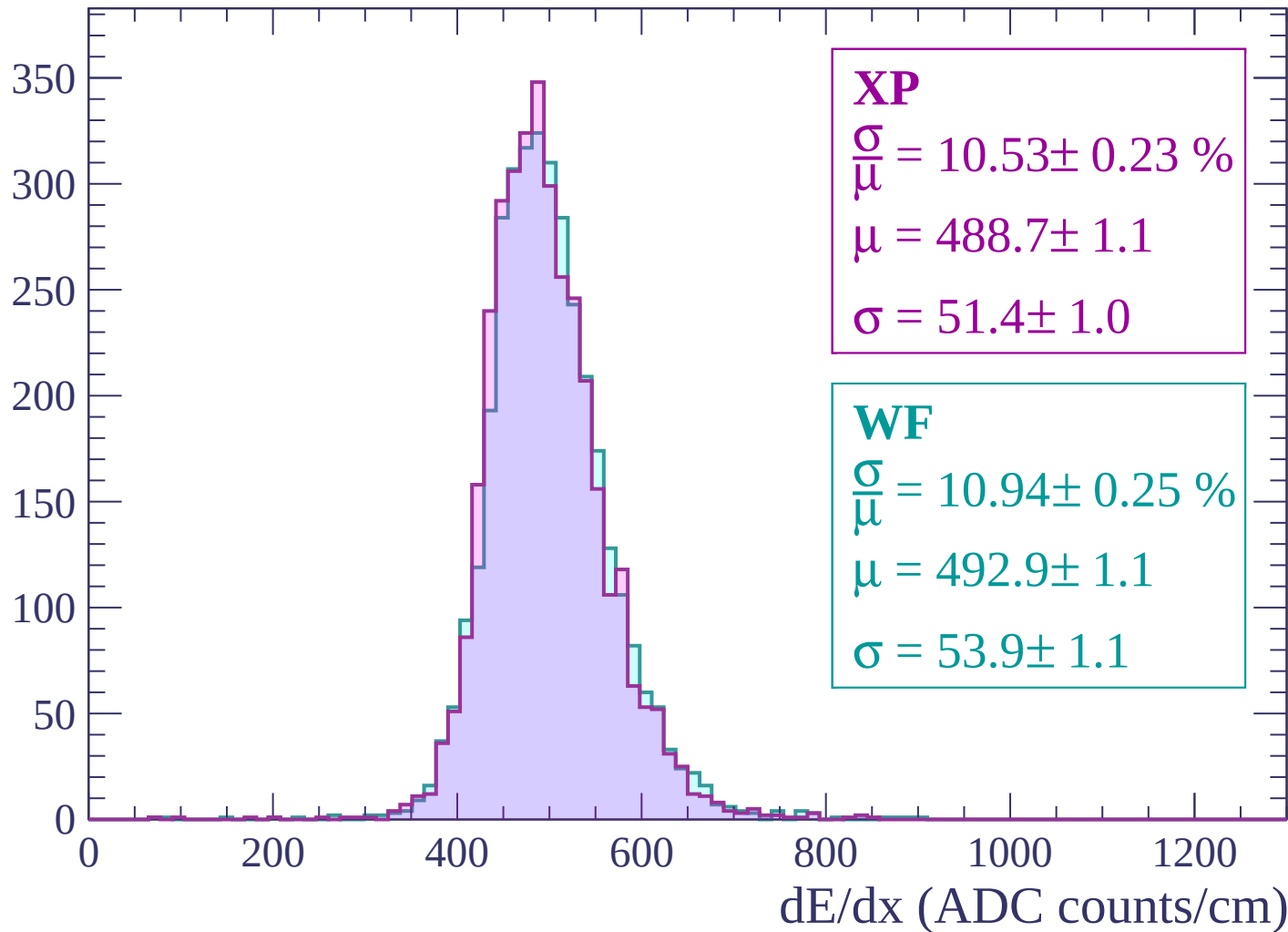


Energy loss | $2040 < p < 2100$



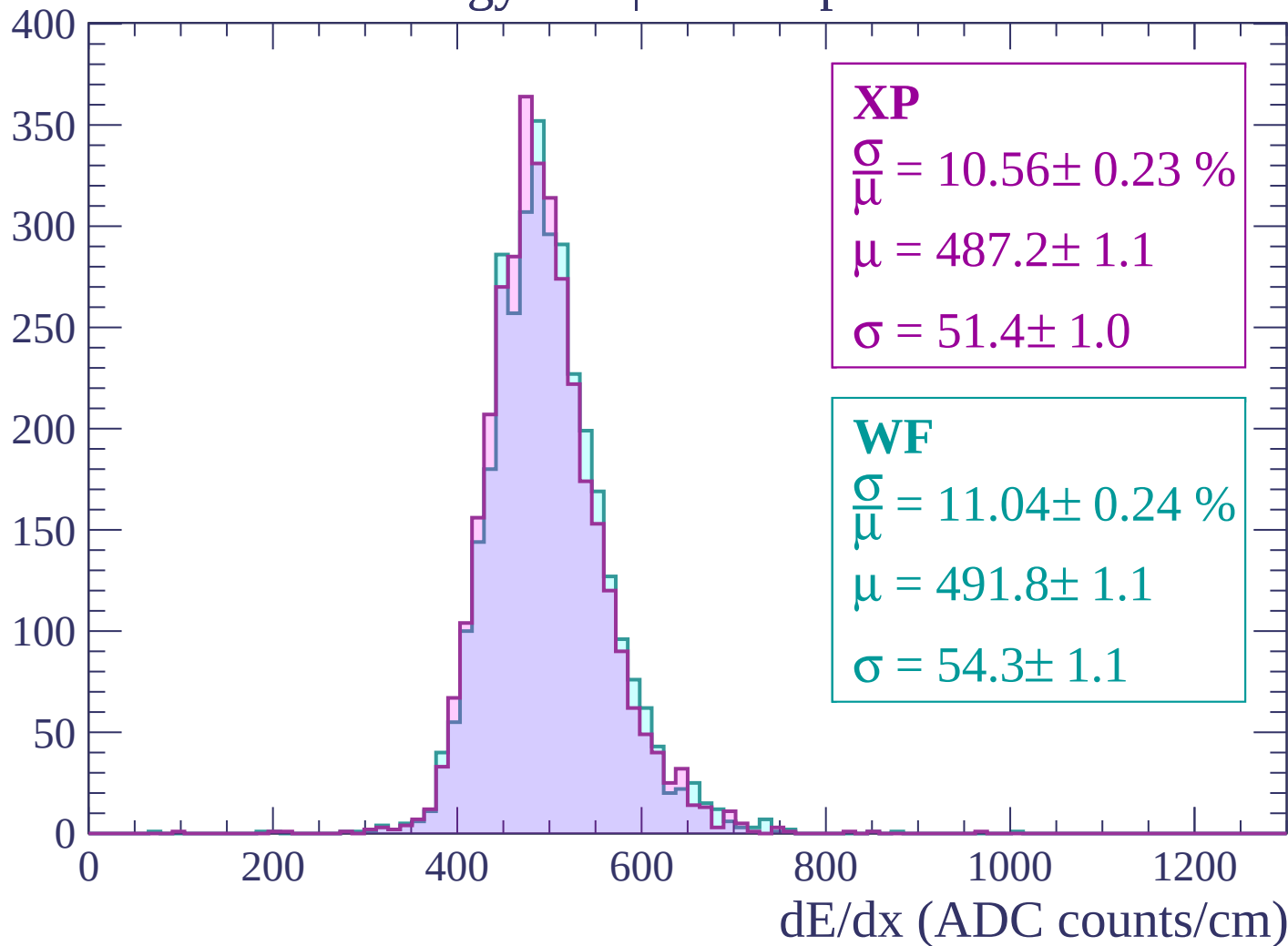
Energy loss | $2100 < p < 2160$

Count



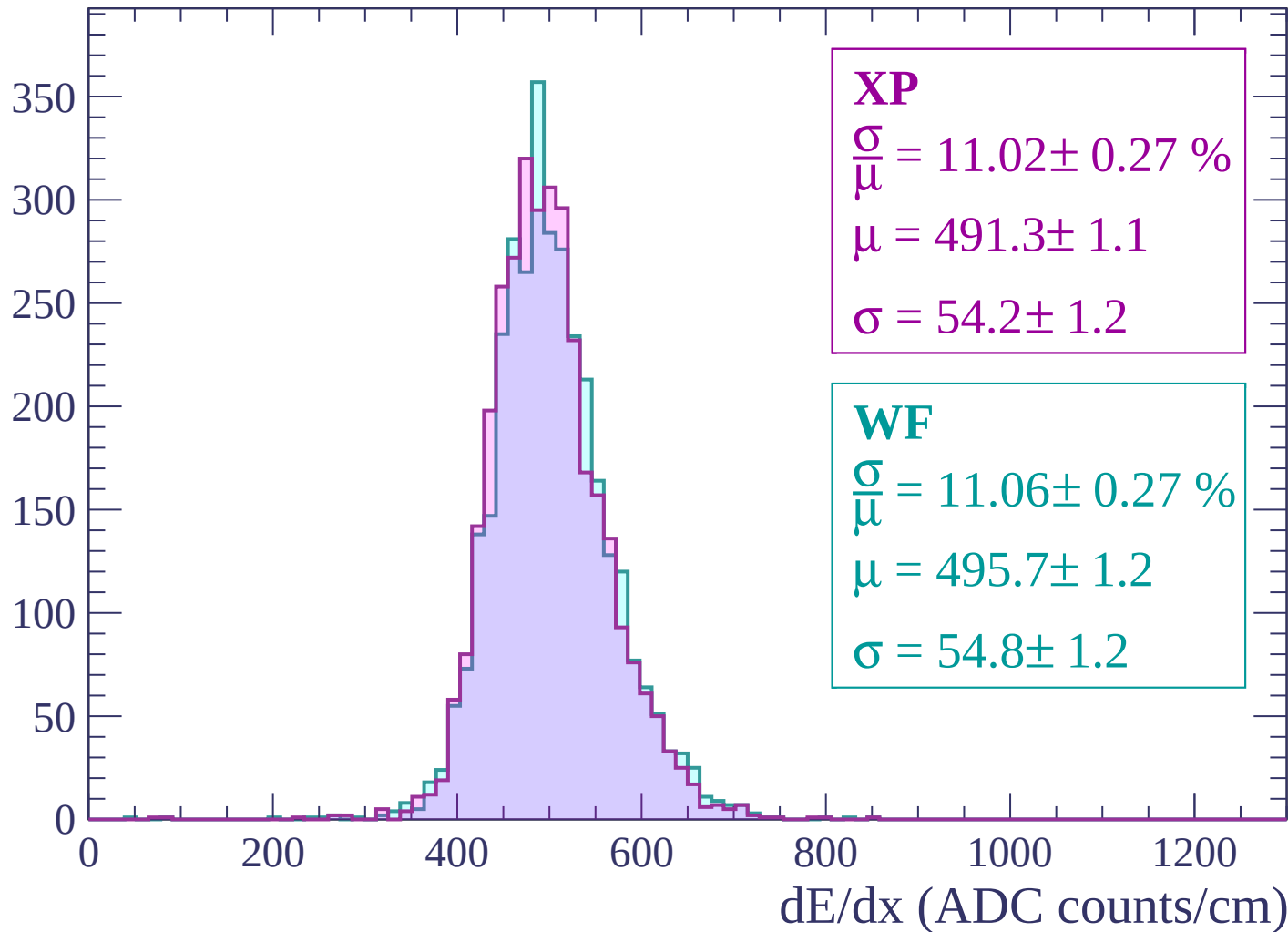
Energy loss | $2160 < p < 2220$

Count



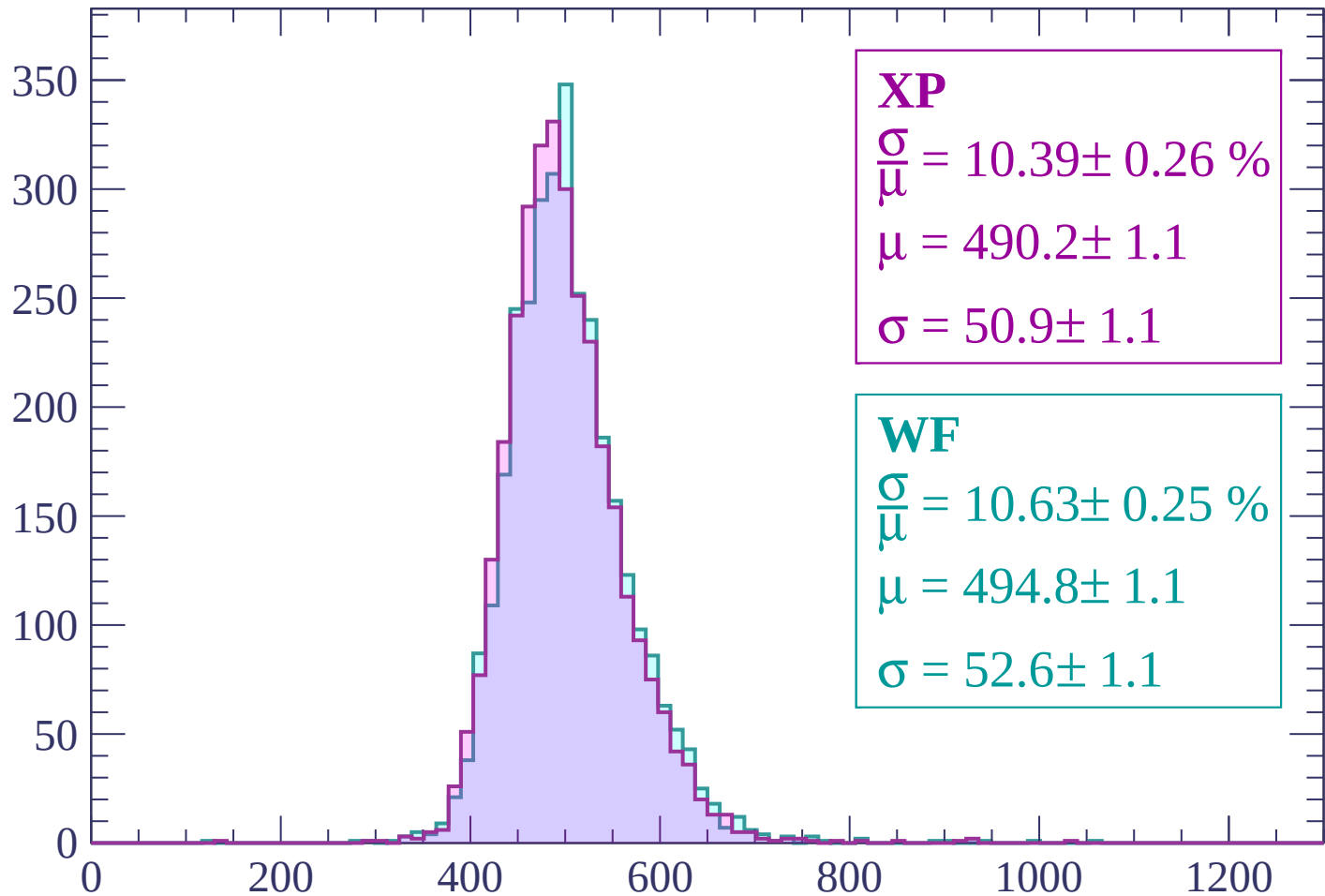
Energy loss | $2220 < p < 2280$

Count



Energy loss | $2280 < p < 2340$

Count



XP

$$\frac{\sigma}{\mu} = 10.39 \pm 0.26 \%$$

$$\mu = 490.2 \pm 1.1$$

$$\sigma = 50.9 \pm 1.1$$

WF

$$\frac{\sigma}{\mu} = 10.63 \pm 0.25 \%$$

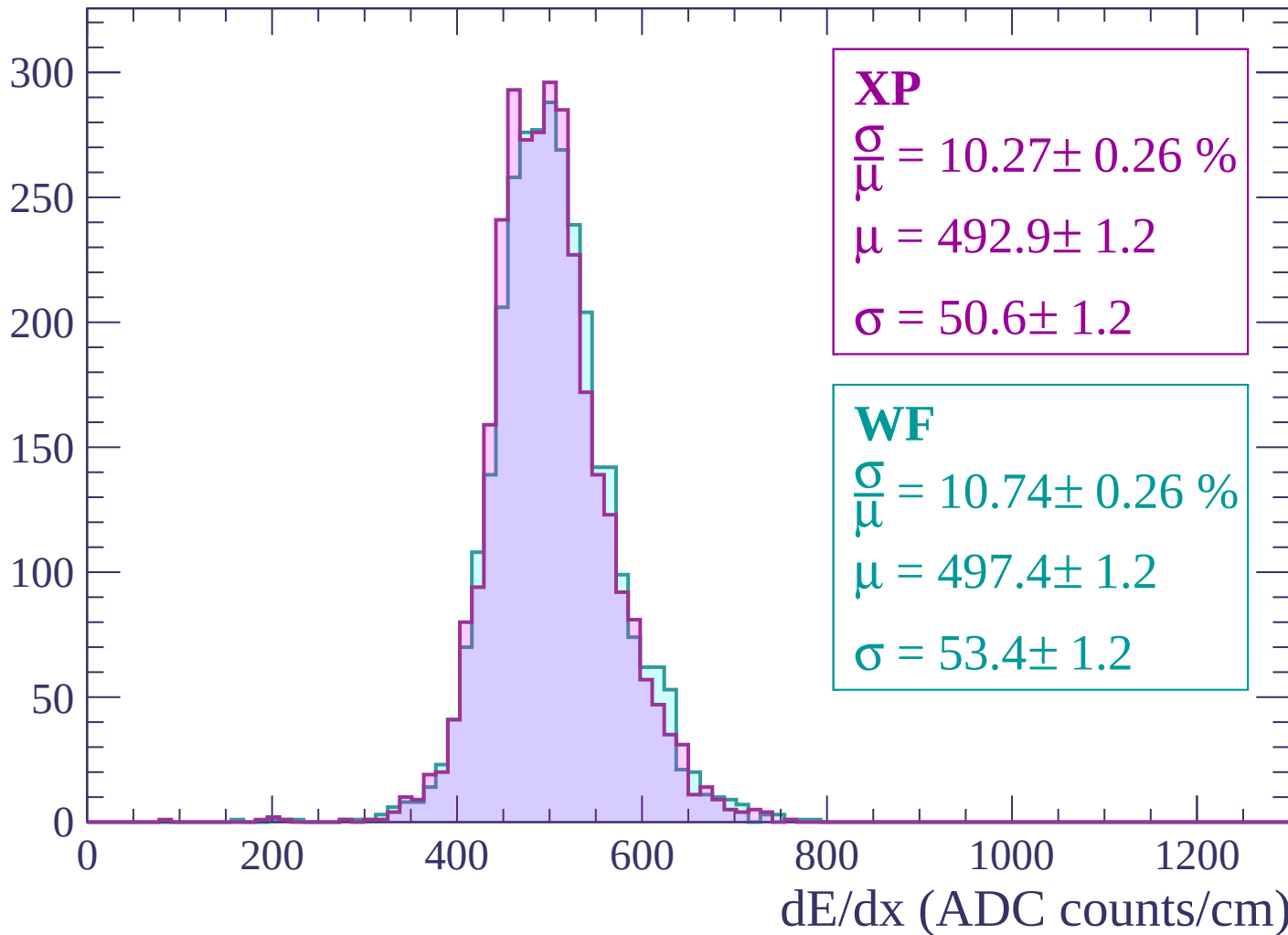
$$\mu = 494.8 \pm 1.1$$

$$\sigma = 52.6 \pm 1.1$$

dE/dx (ADC counts/cm)

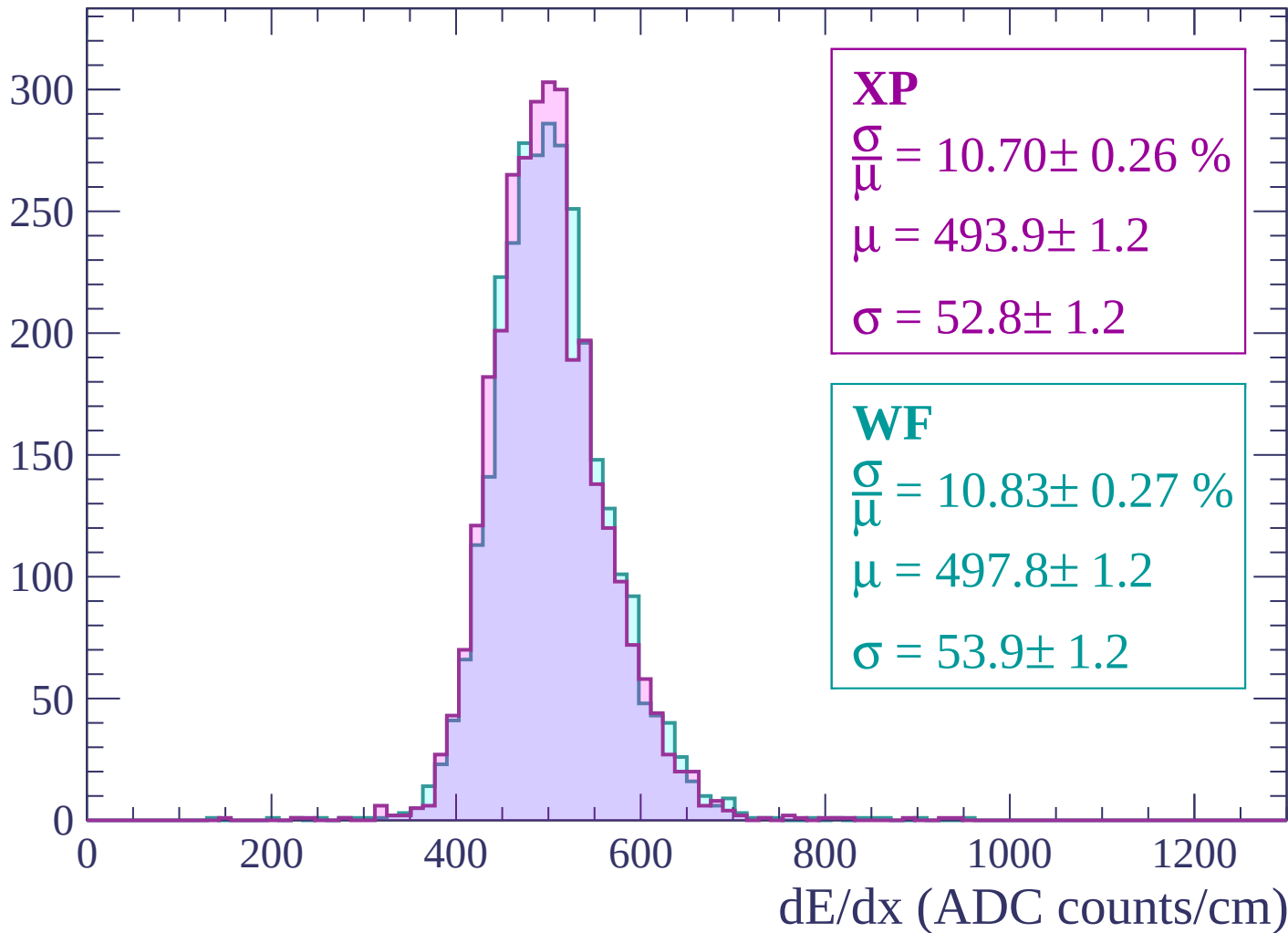
Energy loss | $2340 < p < 2400$

Count



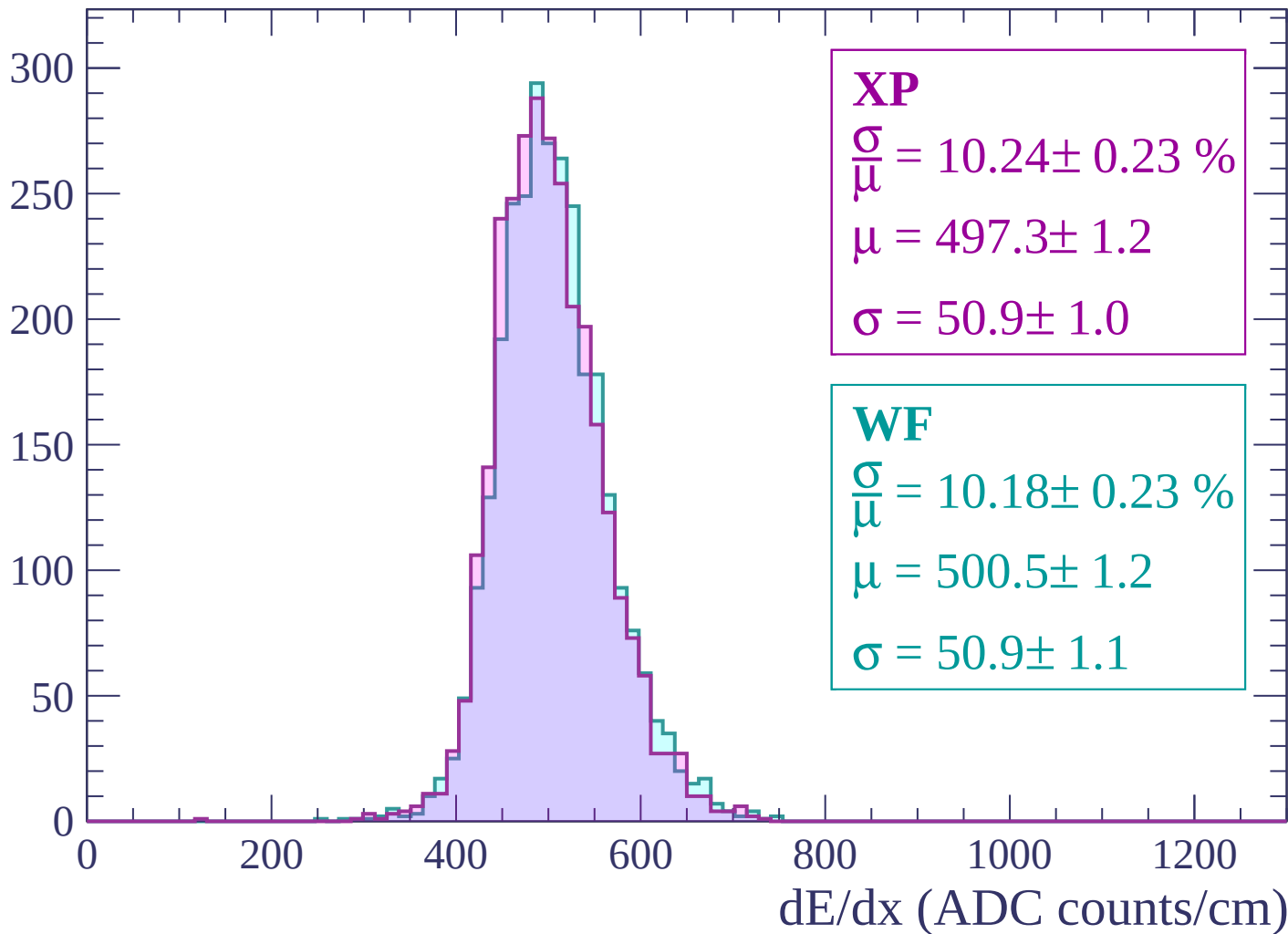
Energy loss | $2400 < p < 2460$

Count



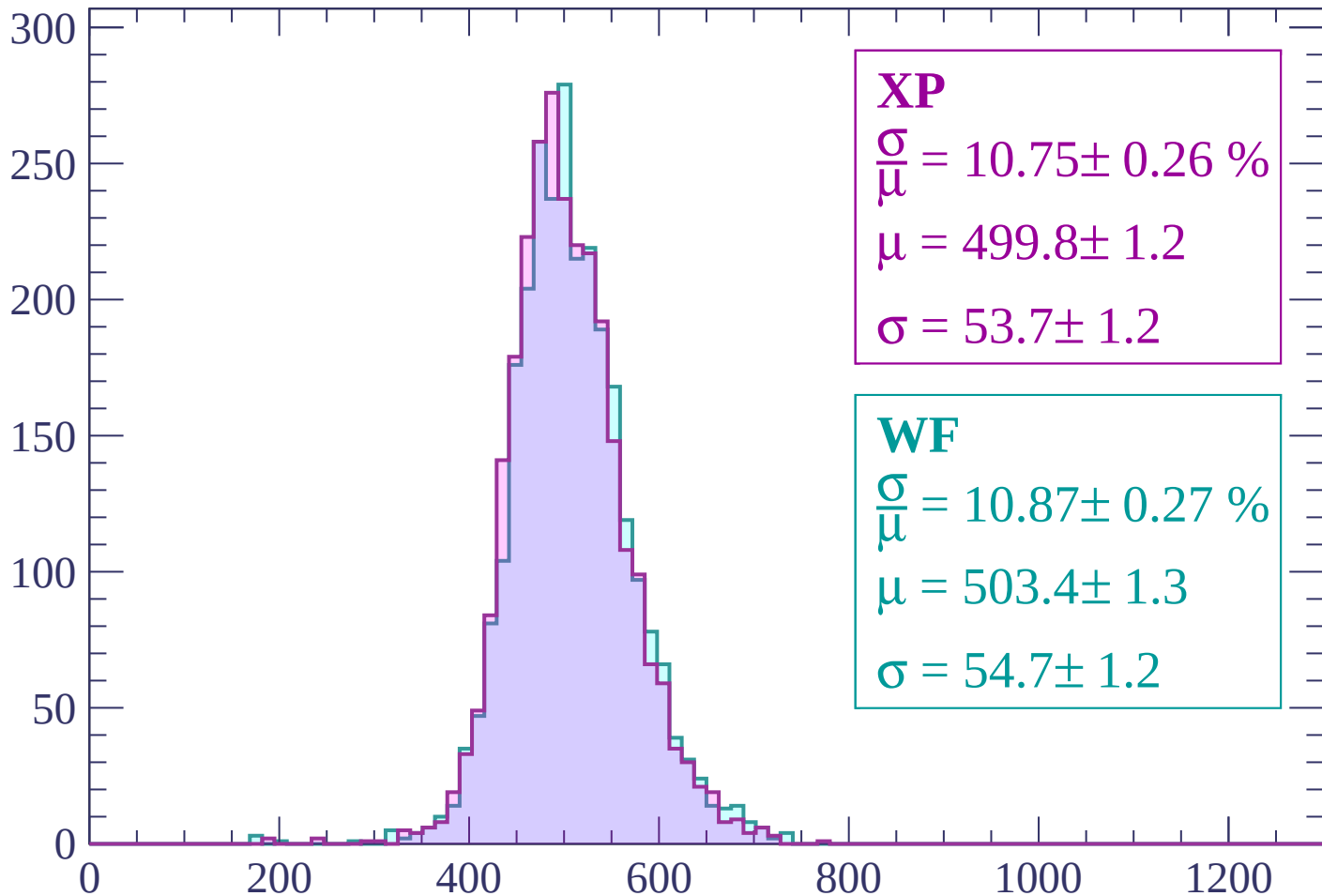
Energy loss | $2460 < p < 2520$

Count



Energy loss | $2520 < p < 2580$

Count



XP

$$\frac{\sigma}{\mu} = 10.75 \pm 0.26 \%$$

$$\mu = 499.8 \pm 1.2$$

$$\sigma = 53.7 \pm 1.2$$

WF

$$\frac{\sigma}{\mu} = 10.87 \pm 0.27 \%$$

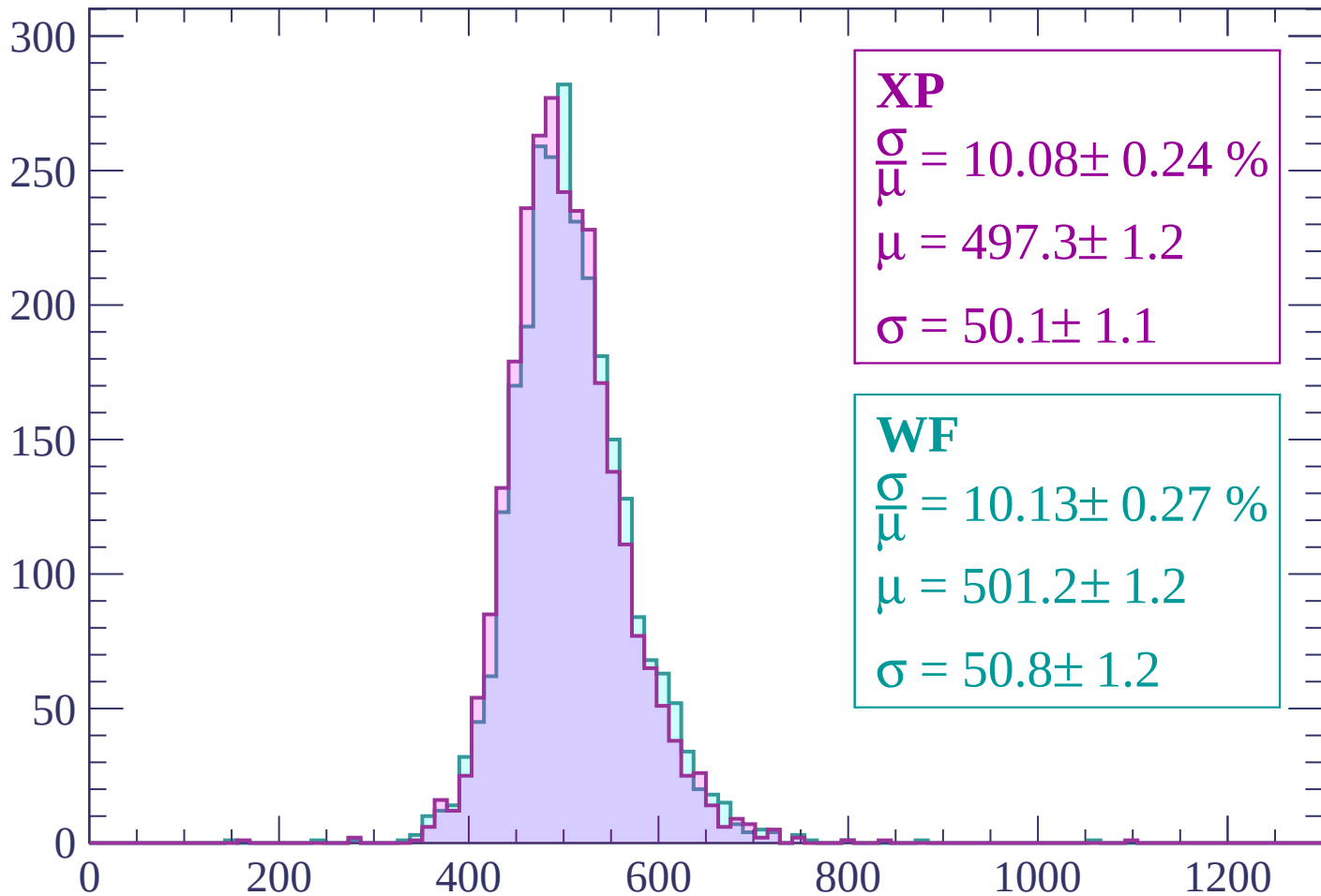
$$\mu = 503.4 \pm 1.3$$

$$\sigma = 54.7 \pm 1.2$$

dE/dx (ADC counts/cm)

Energy loss | $2580 < p < 2640$

Count



XP

$$\frac{\sigma}{\mu} = 10.08 \pm 0.24 \%$$

$$\mu = 497.3 \pm 1.2$$

$$\sigma = 50.1 \pm 1.1$$

WF

$$\frac{\sigma}{\mu} = 10.13 \pm 0.27 \%$$

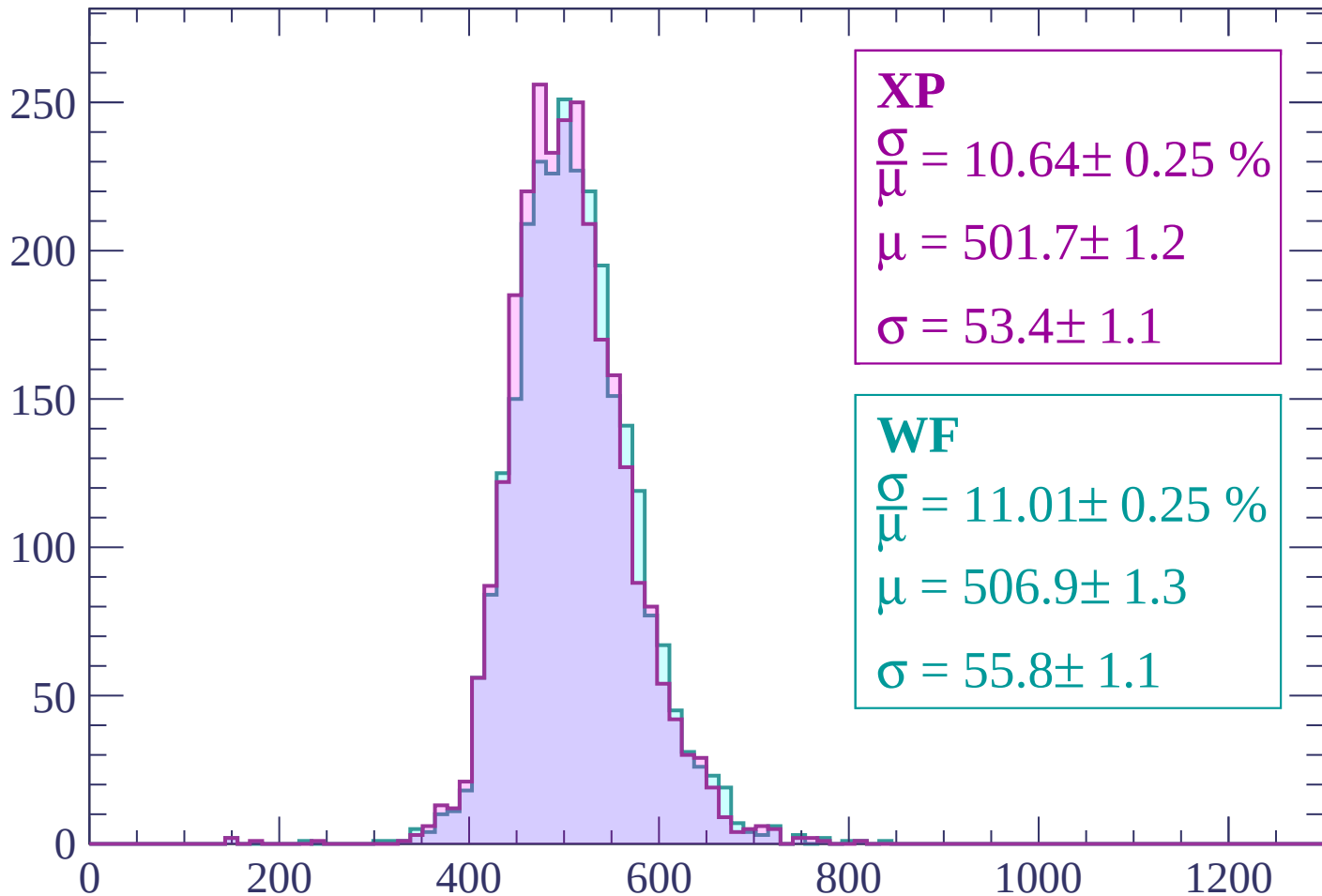
$$\mu = 501.2 \pm 1.2$$

$$\sigma = 50.8 \pm 1.2$$

dE/dx (ADC counts/cm)

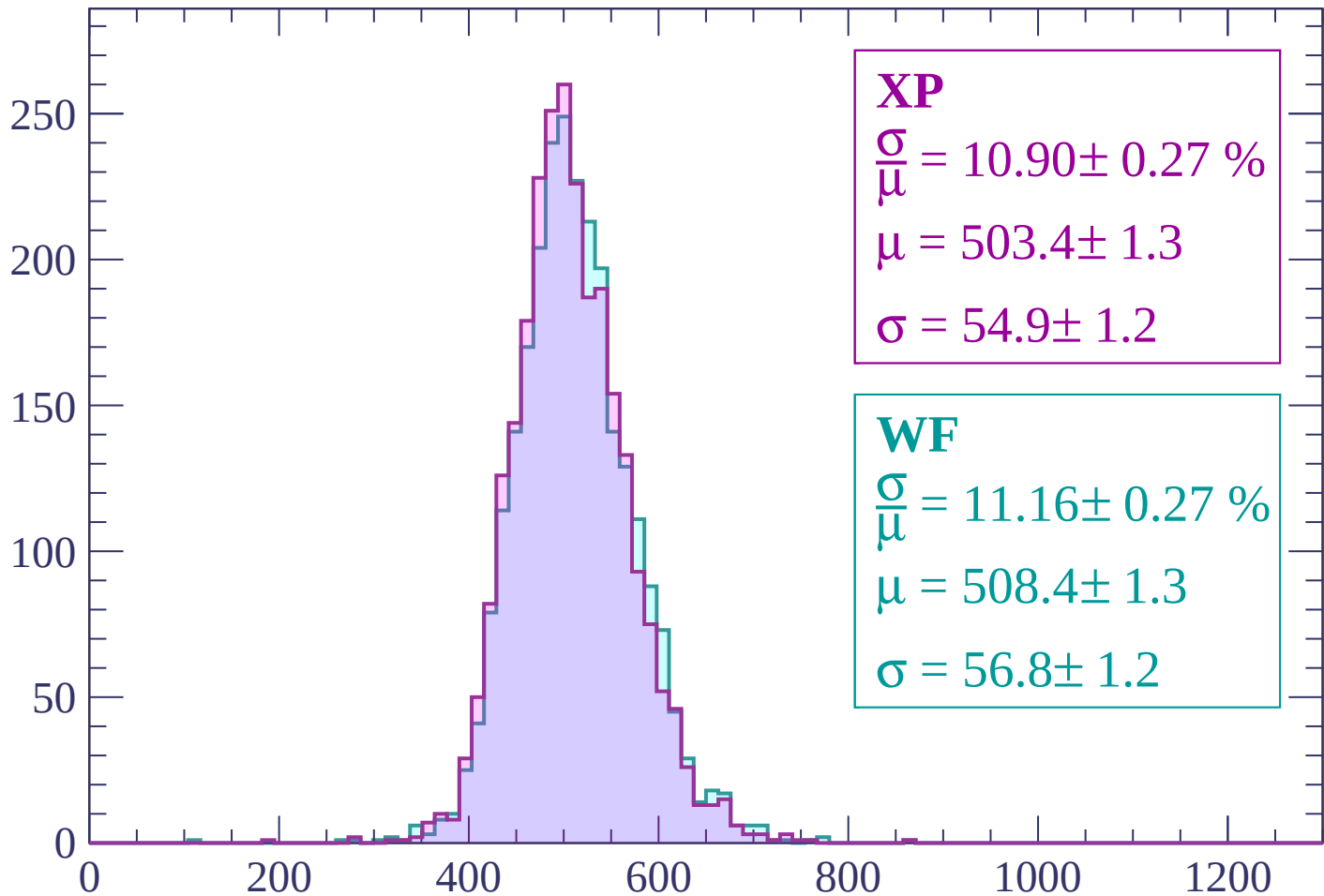
Energy loss | $2640 < p < 2700$

Count



Energy loss | $2700 < p < 2760$

Count



XP

$$\frac{\sigma}{\mu} = 10.90 \pm 0.27 \%$$

$$\mu = 503.4 \pm 1.3$$

$$\sigma = 54.9 \pm 1.2$$

WF

$$\frac{\sigma}{\mu} = 11.16 \pm 0.27 \%$$

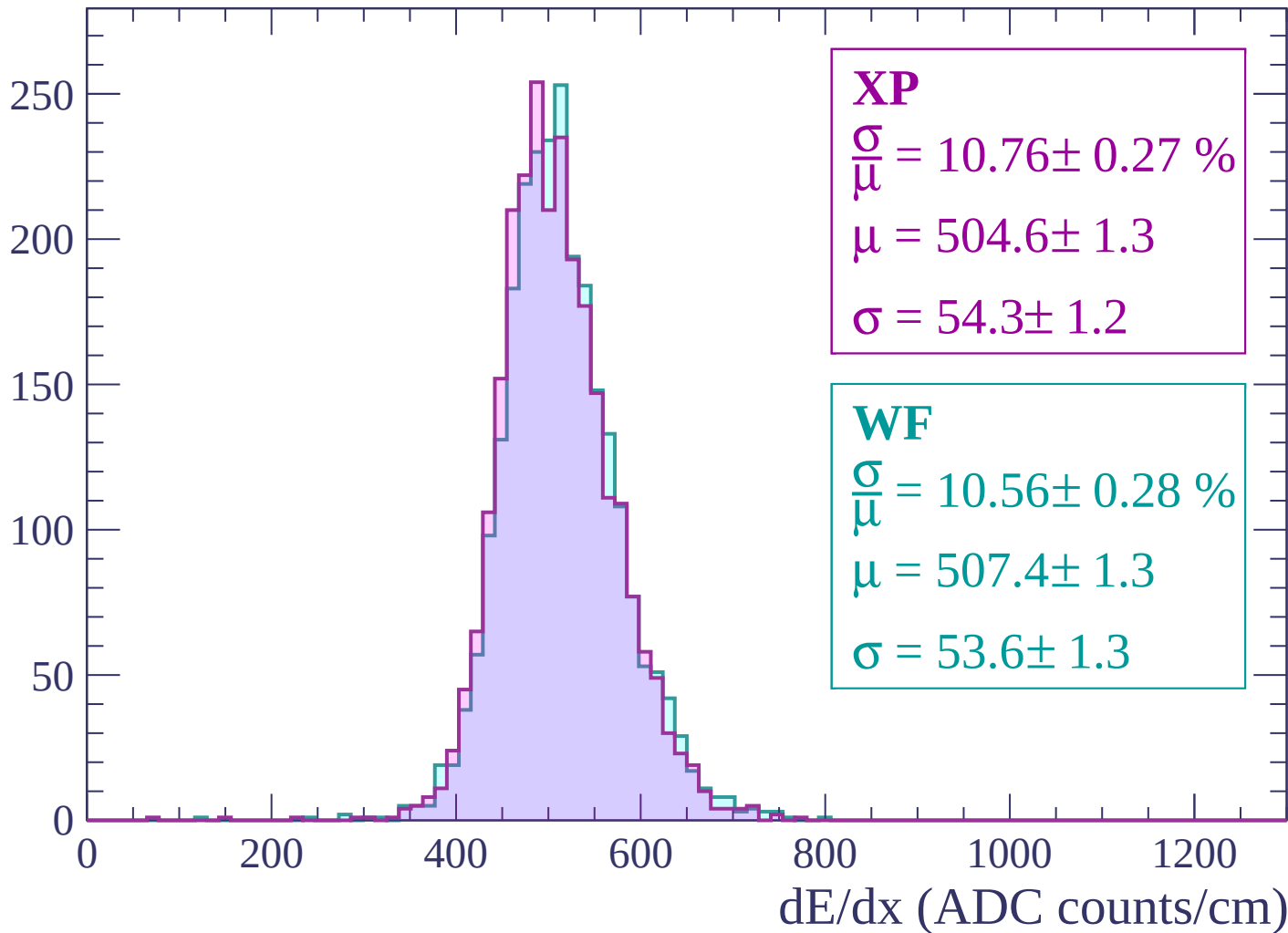
$$\mu = 508.4 \pm 1.3$$

$$\sigma = 56.8 \pm 1.2$$

dE/dx (ADC counts/cm)

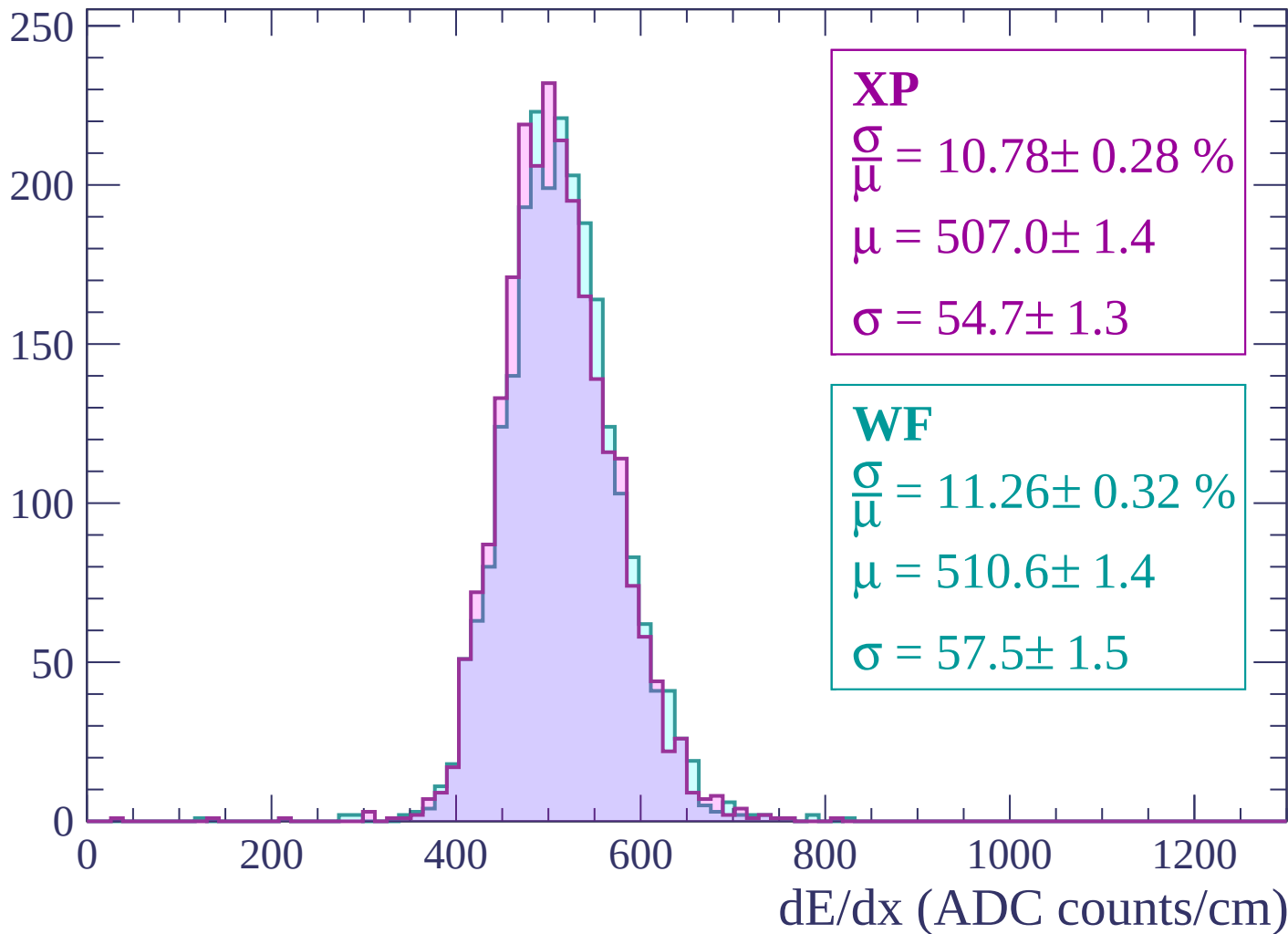
Energy loss | $2760 < p < 2820$

Count



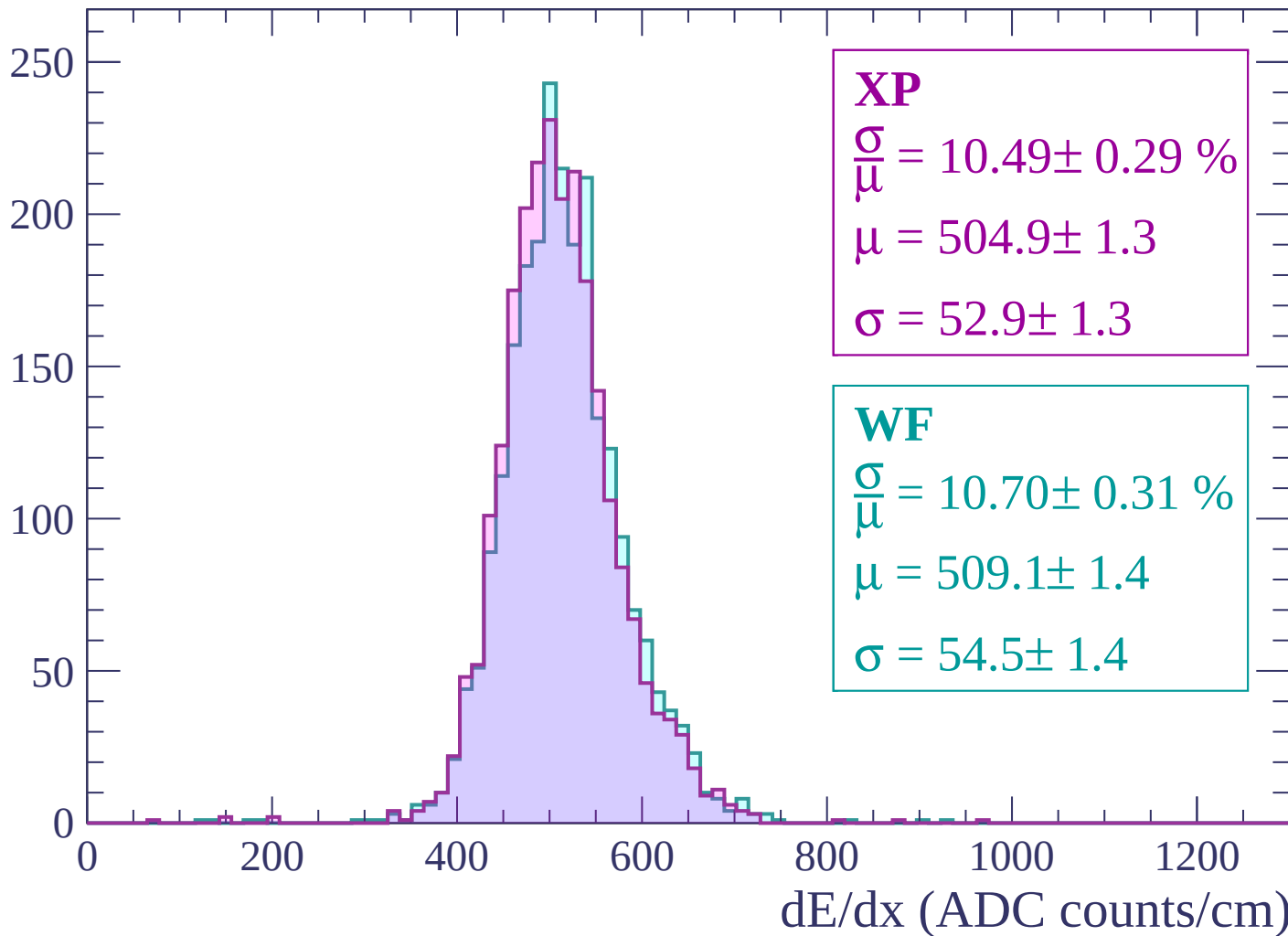
Energy loss | $2820 < p < 2880$

Count



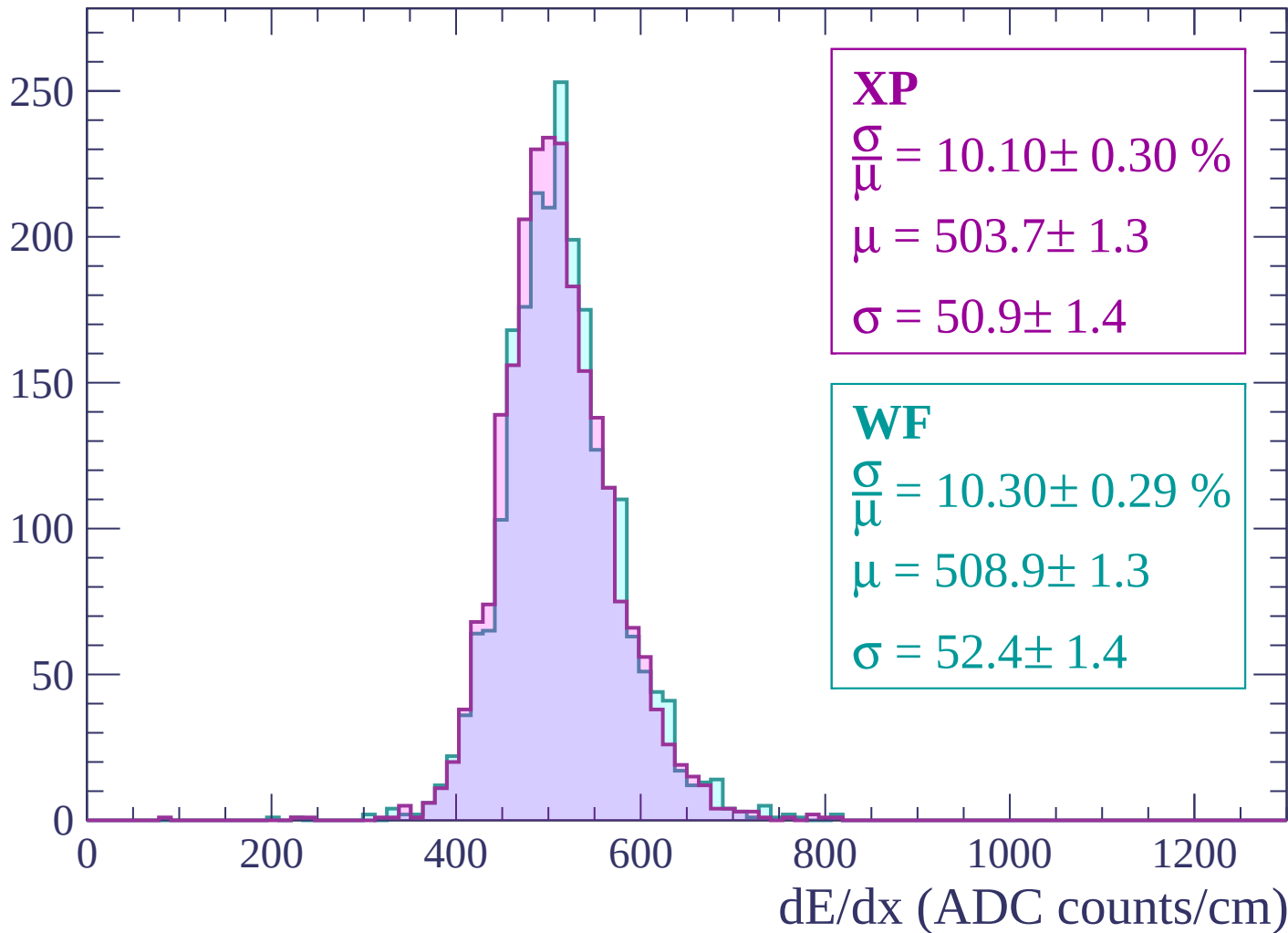
Energy loss | $2880 < p < 2940$

Count



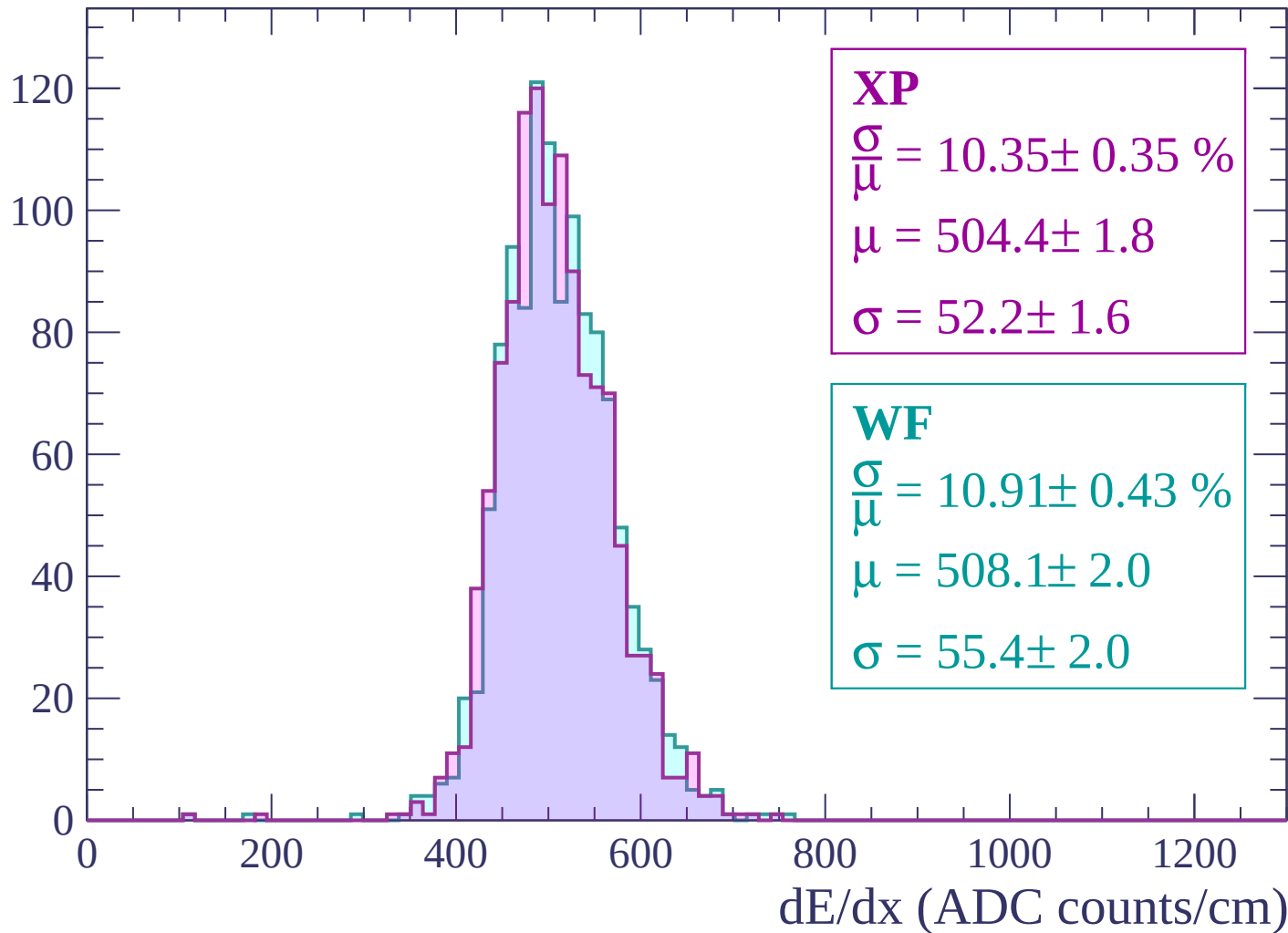
Energy loss | $2940 < p < 3000$

Count



Energy loss | $3000 < p < 3060$

Count



Energy loss | $-90 < \theta < -88$

Count

1.0
0.9
0.8
0.7
0.6
0.5
0.4
0.3
0.2
0.1
0.0

0

200

400

600

800

1000

1200

dE/dx (ADC counts/cm)

XP

$\frac{\sigma}{\mu} = \text{-nan} \pm \text{-nan} \%$

$\mu = 0.0 \pm 0.0$

$\sigma = 0.0 \pm 0.0$

WF

$\frac{\sigma}{\mu} = \text{-nan} \pm \text{-nan} \%$

$\mu = 0.0 \pm 0.0$

$\sigma = 0.0 \pm 0.0$

Energy loss | $-88 < \theta < -86$

Count

1.0
0.9
0.8
0.7
0.6
0.5
0.4
0.3
0.2
0.1
0.0

0

200

400

600

800

1000

1200

dE/dx (ADC counts/cm)

XP

$\frac{\sigma}{\mu} = \text{-nan} \pm \text{-nan} \%$

$\mu = 0.0 \pm 0.0$

$\sigma = 0.0 \pm 0.0$

WF

$\frac{\sigma}{\mu} = \text{-nan} \pm \text{-nan} \%$

$\mu = 0.0 \pm 0.0$

$\sigma = 0.0 \pm 0.0$

Energy loss | $-86 < \theta < -84$

Count

1.0
0.9
0.8
0.7
0.6
0.5
0.4
0.3
0.2
0.1
0.0

0

200

400

600

800

1000

1200

dE/dx (ADC counts/cm)

XP

$\frac{\sigma}{\mu} = -\text{nan} \pm -\text{nan} \%$

$\mu = 0.0 \pm 0.0$

$\sigma = 0.0 \pm 0.0$

WF

$\frac{\sigma}{\mu} = -\text{nan} \pm -\text{nan} \%$

$\mu = 0.0 \pm 0.0$

$\sigma = 0.0 \pm 0.0$

Energy loss | $-84 < \theta < -82$

Count

1.0
0.9
0.8
0.7
0.6
0.5
0.4
0.3
0.2
0.1
0.0

0

200

400

600

800

1000

1200

dE/dx (ADC counts/cm)

XP

$\frac{\sigma}{\mu} = -\text{nan} \pm -\text{nan} \%$

$\mu = 0.0 \pm 0.0$

$\sigma = 0.0 \pm 0.0$

WF

$\frac{\sigma}{\mu} = -\text{nan} \pm -\text{nan} \%$

$\mu = 0.0 \pm 0.0$

$\sigma = 0.0 \pm 0.0$

Energy loss | $-82 < \theta < -80$

Count

1.0
0.9
0.8
0.7
0.6
0.5
0.4
0.3
0.2
0.1
0.0

0

200

400

600

800

1000

1200

dE/dx (ADC counts/cm)

XP

$\frac{\sigma}{\mu} = \text{-nan} \pm \text{-nan} \%$

$\mu = 0.0 \pm 0.0$

$\sigma = 0.0 \pm 0.0$

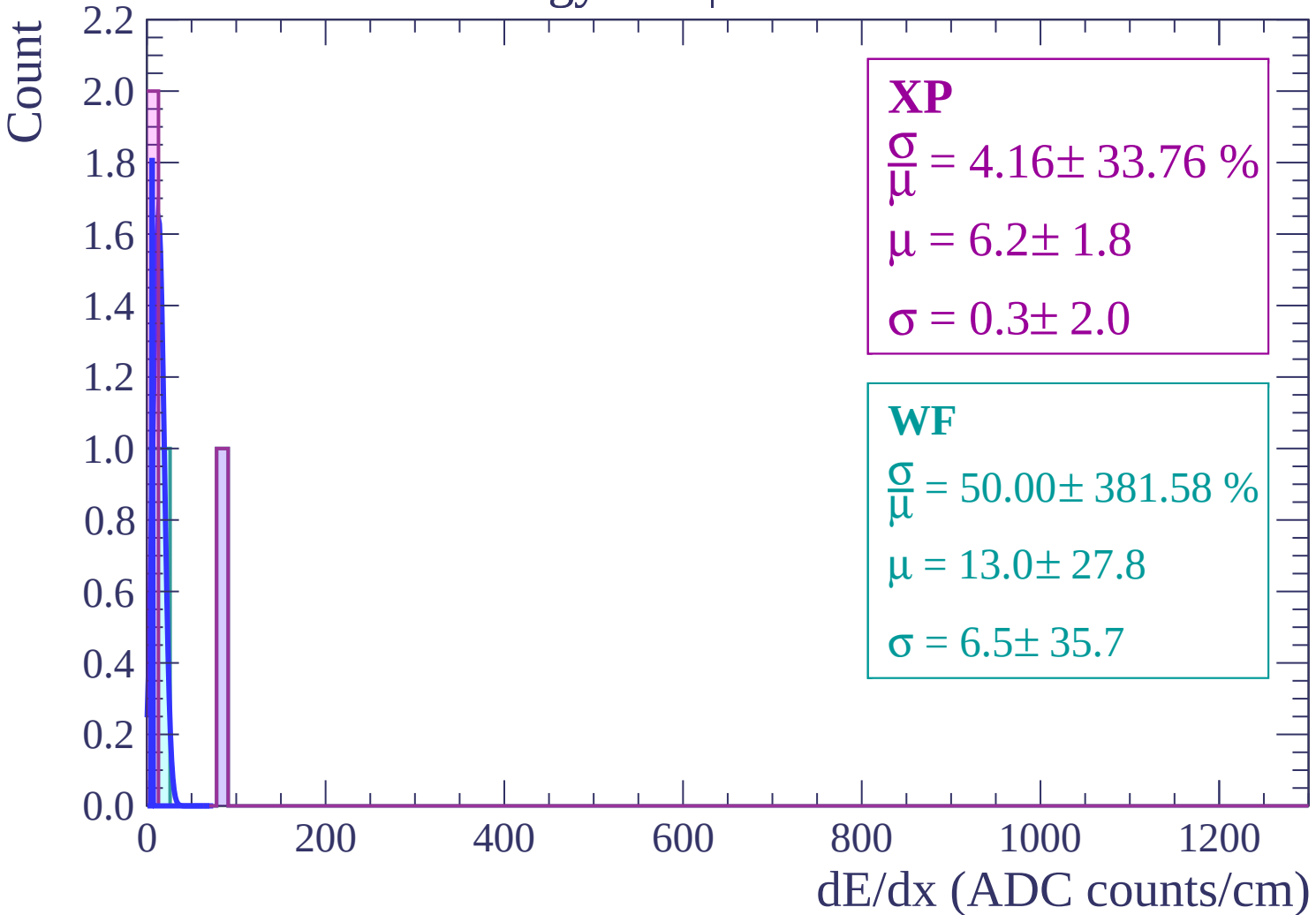
WF

$\frac{\sigma}{\mu} = \text{-nan} \pm \text{-nan} \%$

$\mu = 0.0 \pm 0.0$

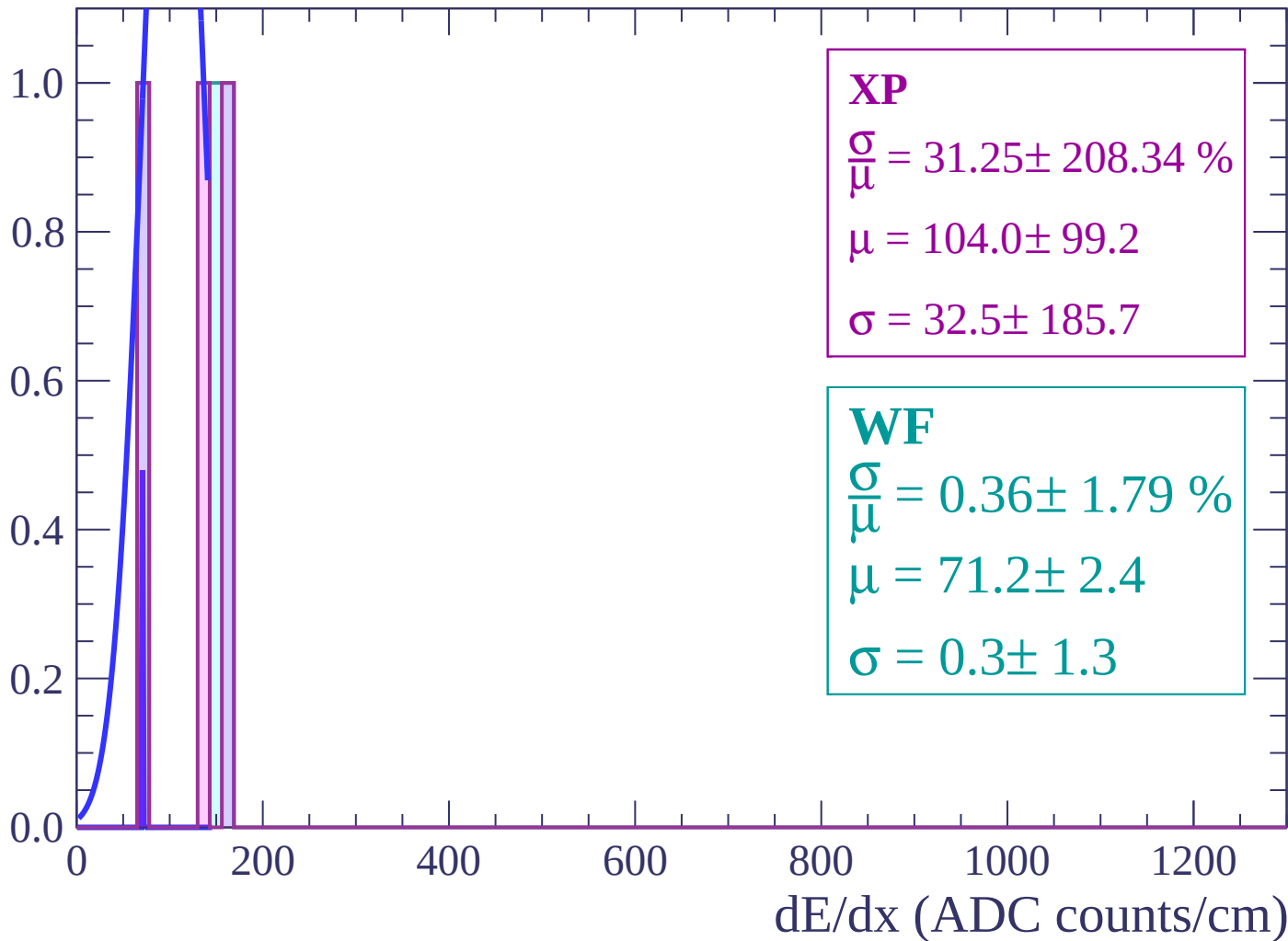
$\sigma = 0.0 \pm 0.0$

Energy loss | $-80 < \theta < -78$



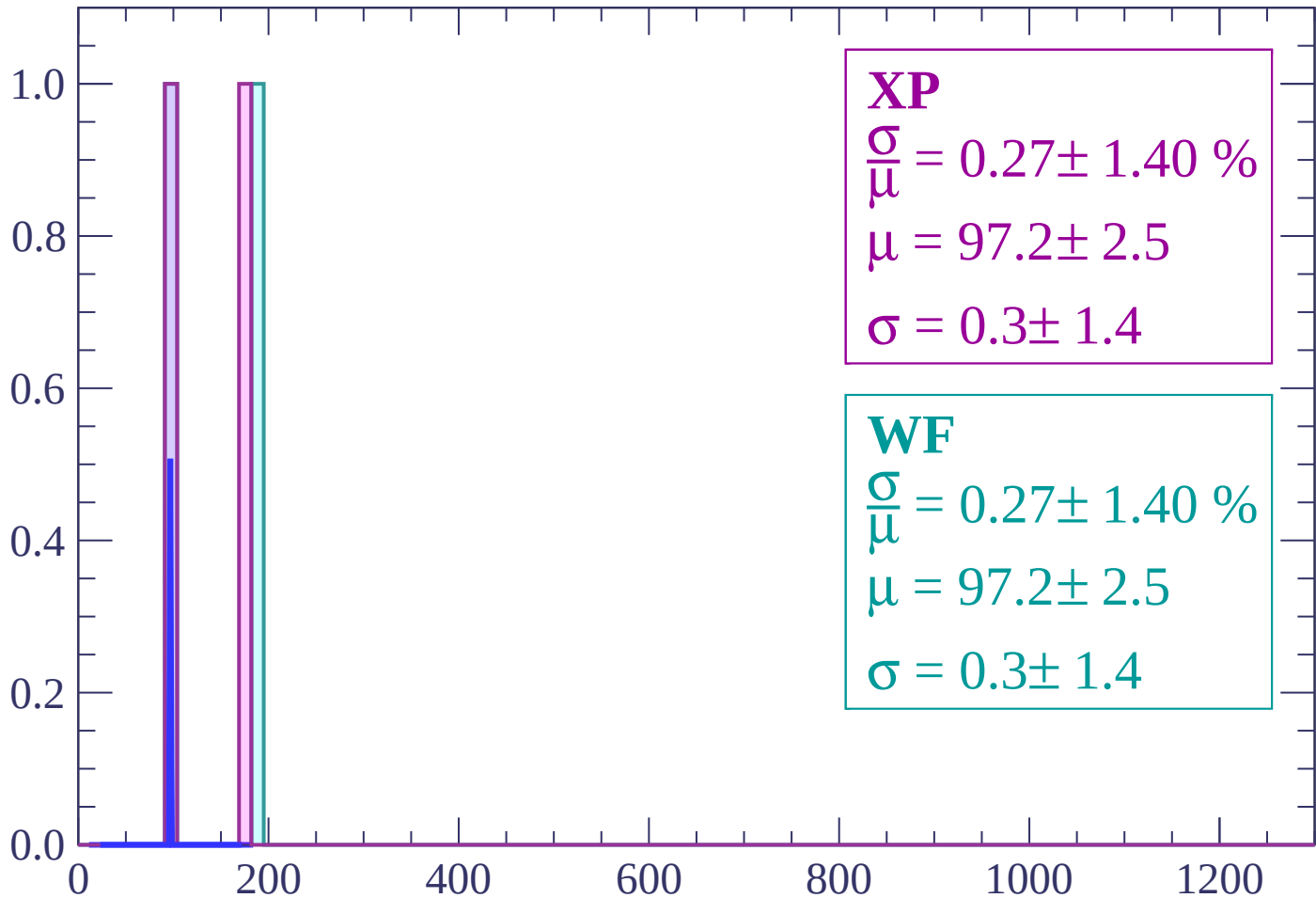
Energy loss | $-78 < \theta < -76$

Count



Energy loss | $-76 < \theta < -74$

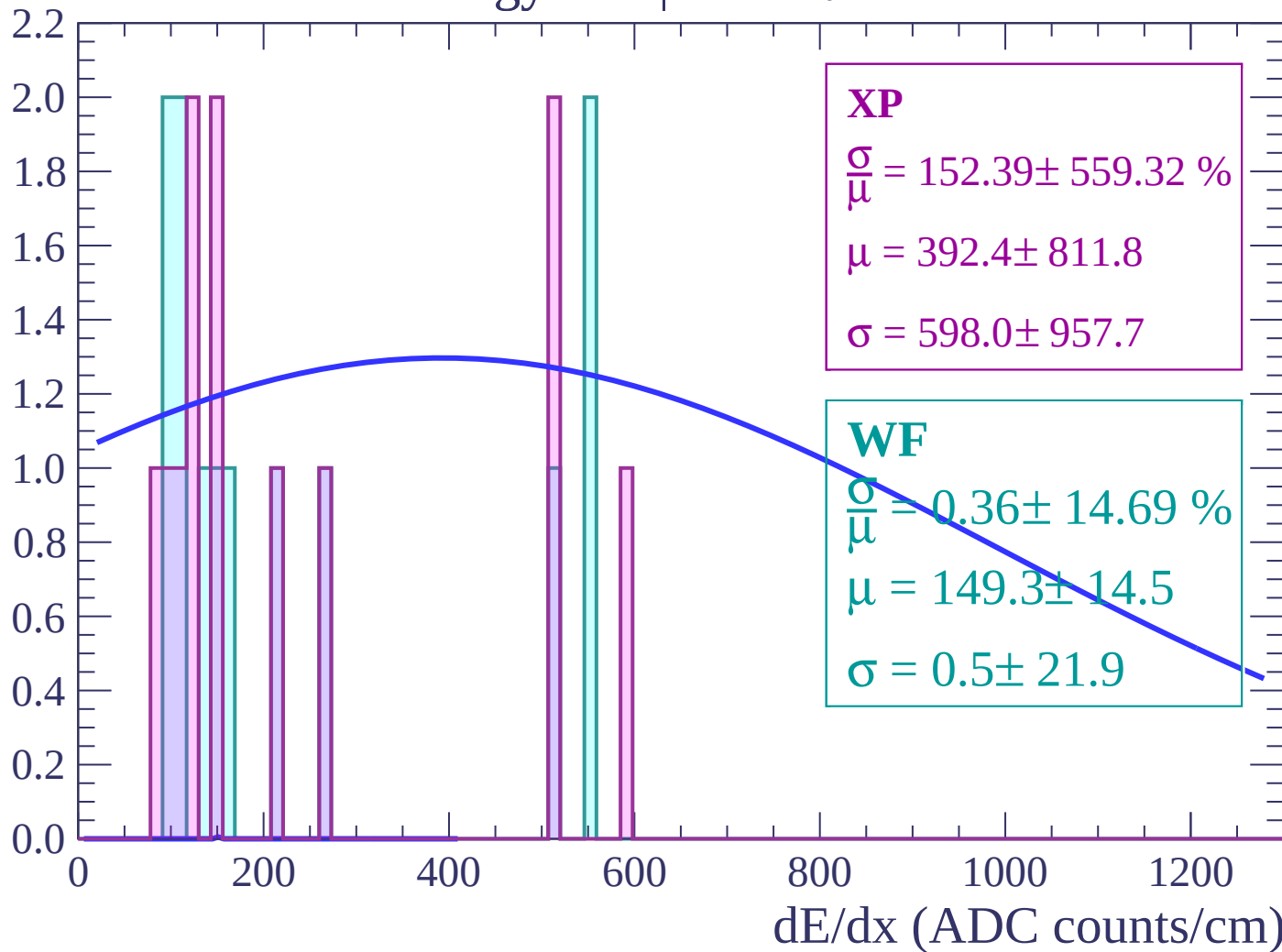
Count



dE/dx (ADC counts/cm)

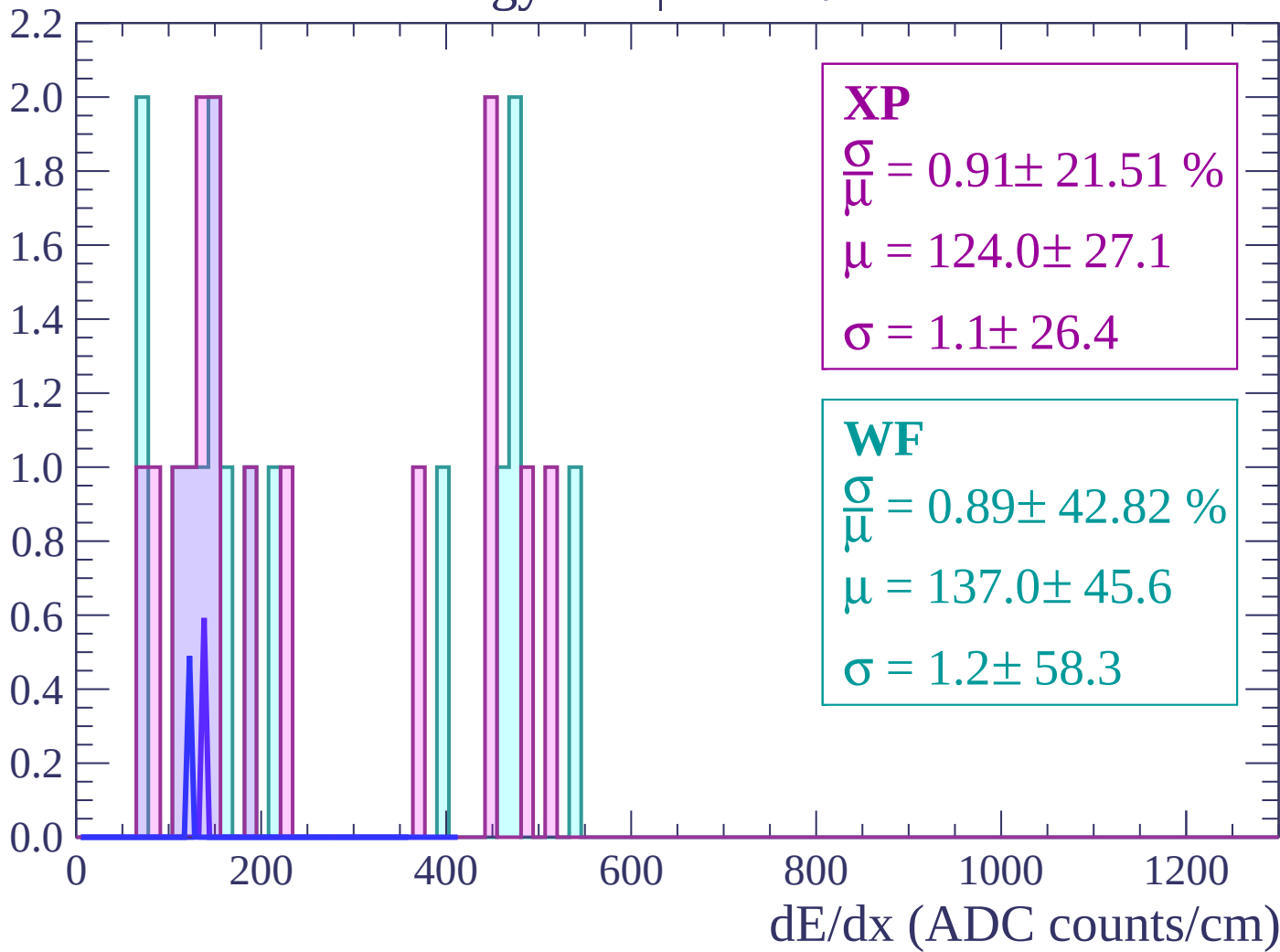
Energy loss | $-74 < \theta < -72$

Count



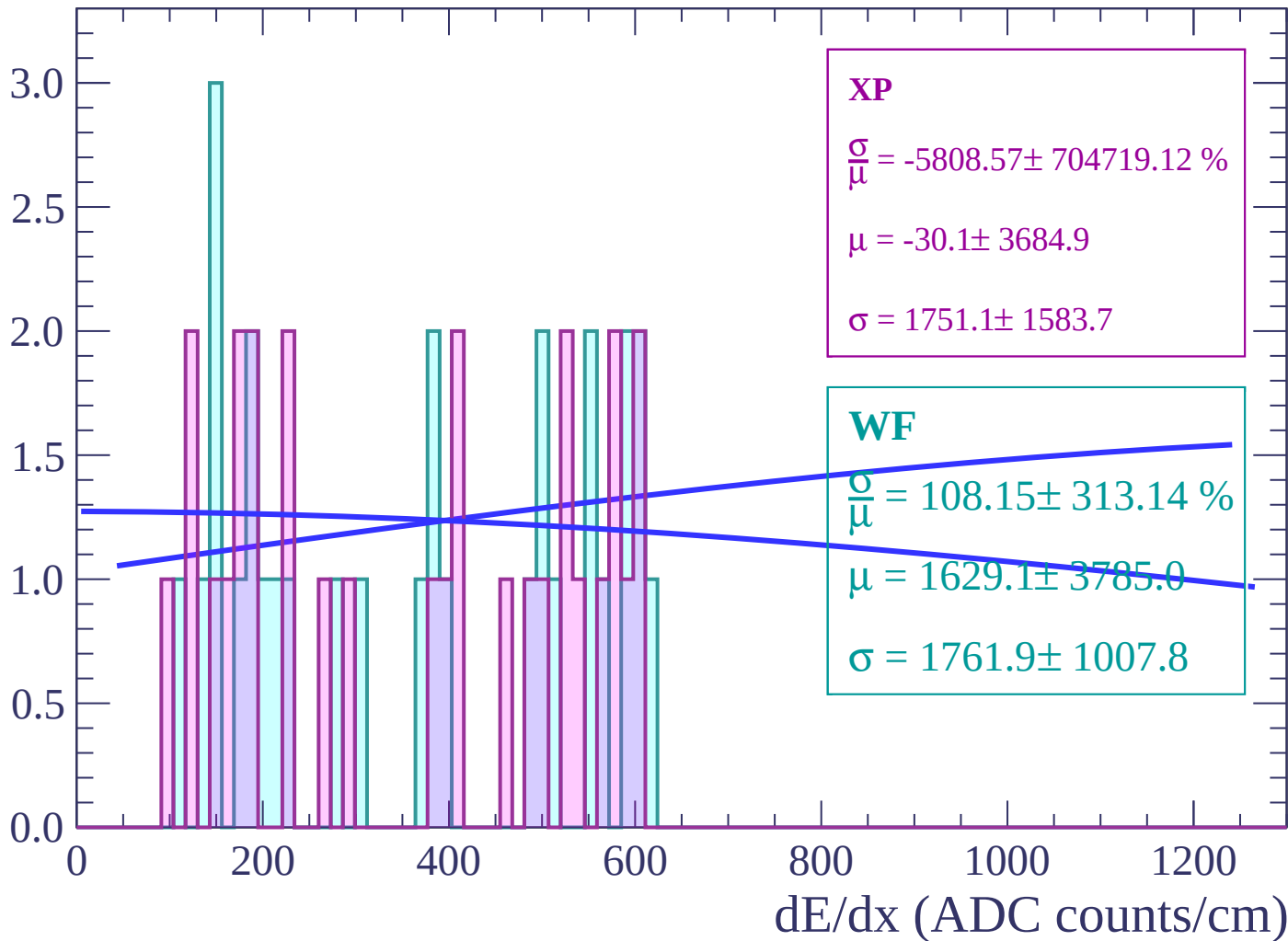
Energy loss | $-72 < \theta < -70$

Count



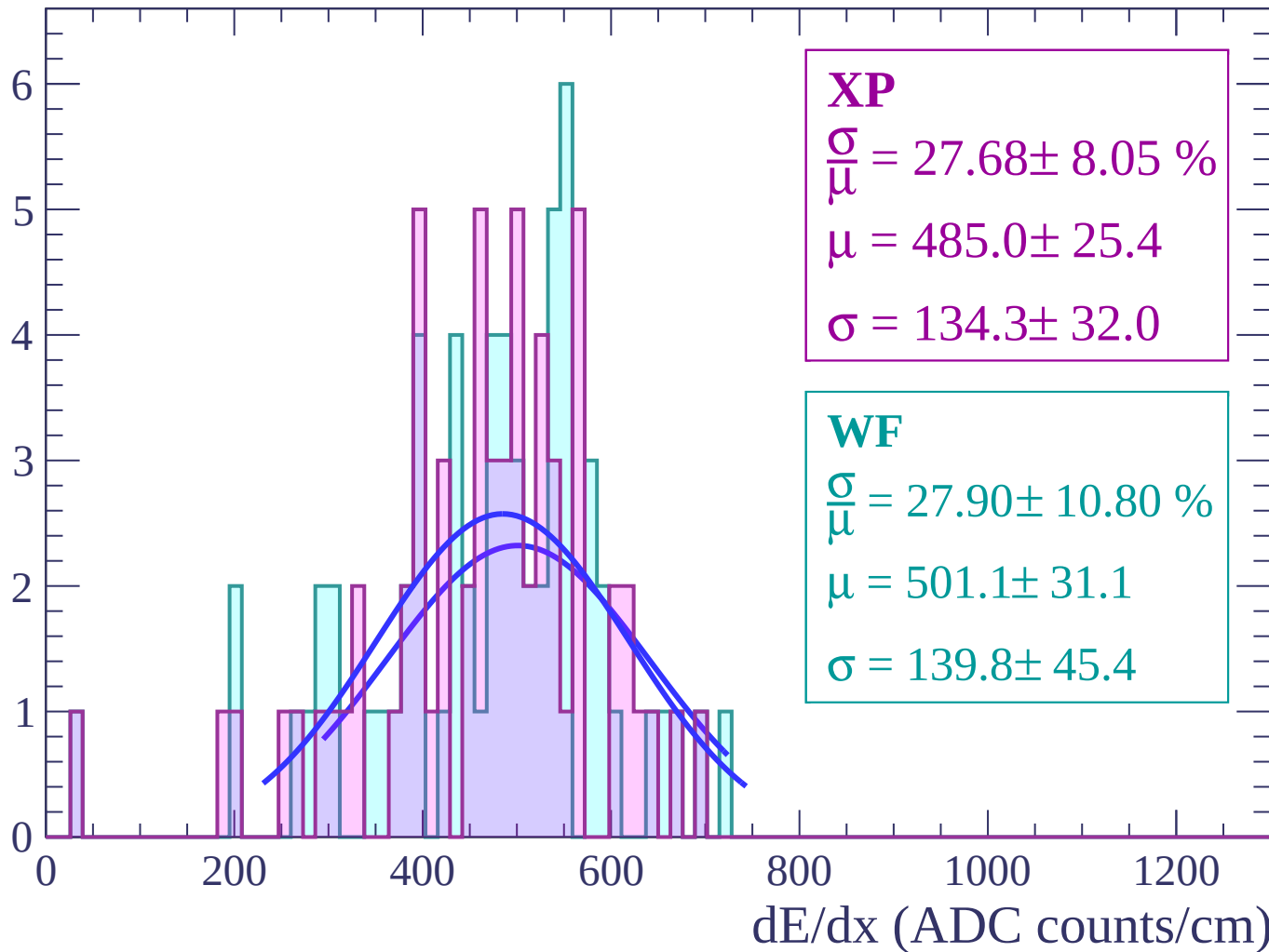
Energy loss | $-70 < \theta < -68$

Count



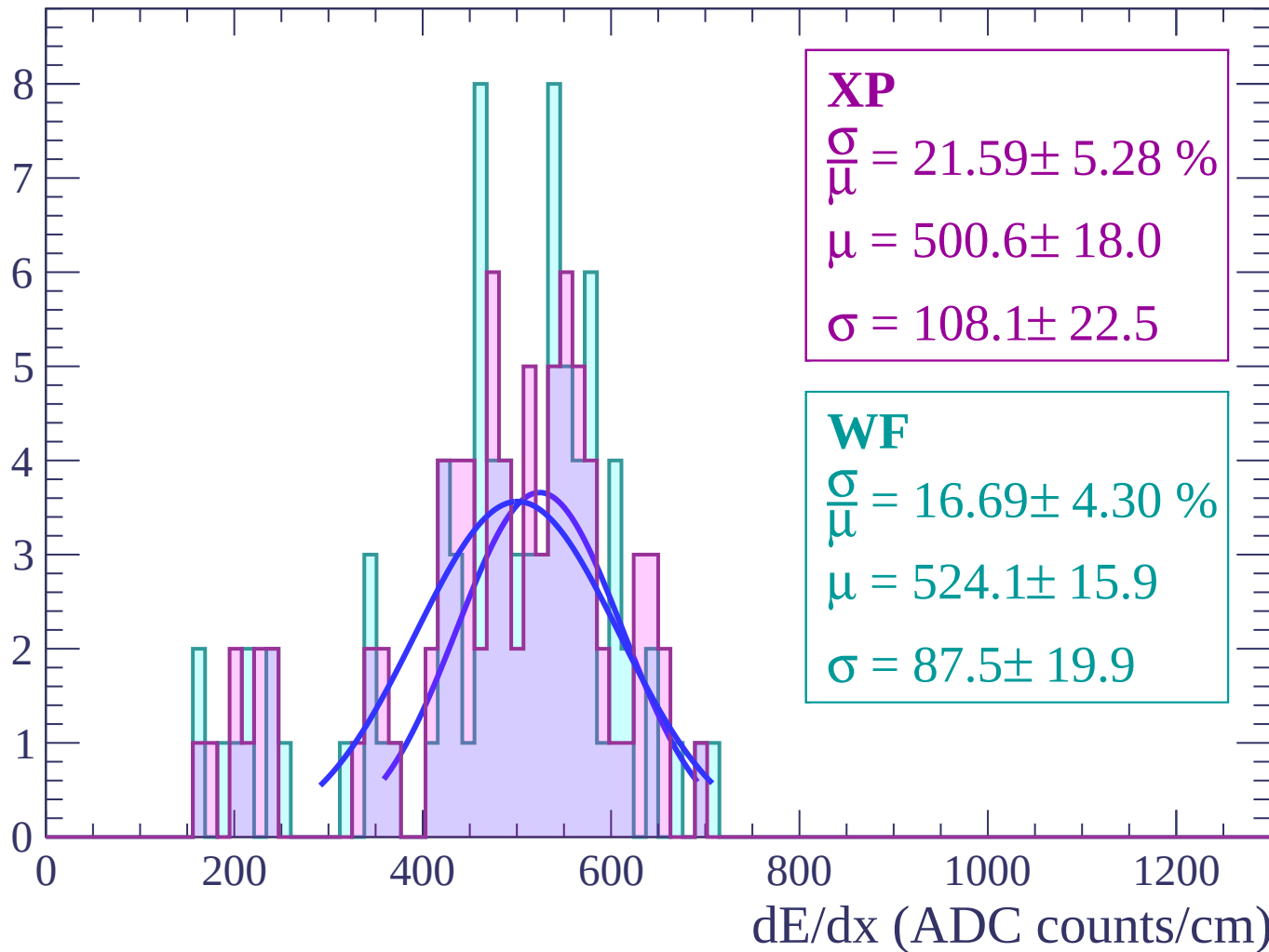
Energy loss | $-68 < \theta < -66$

Count



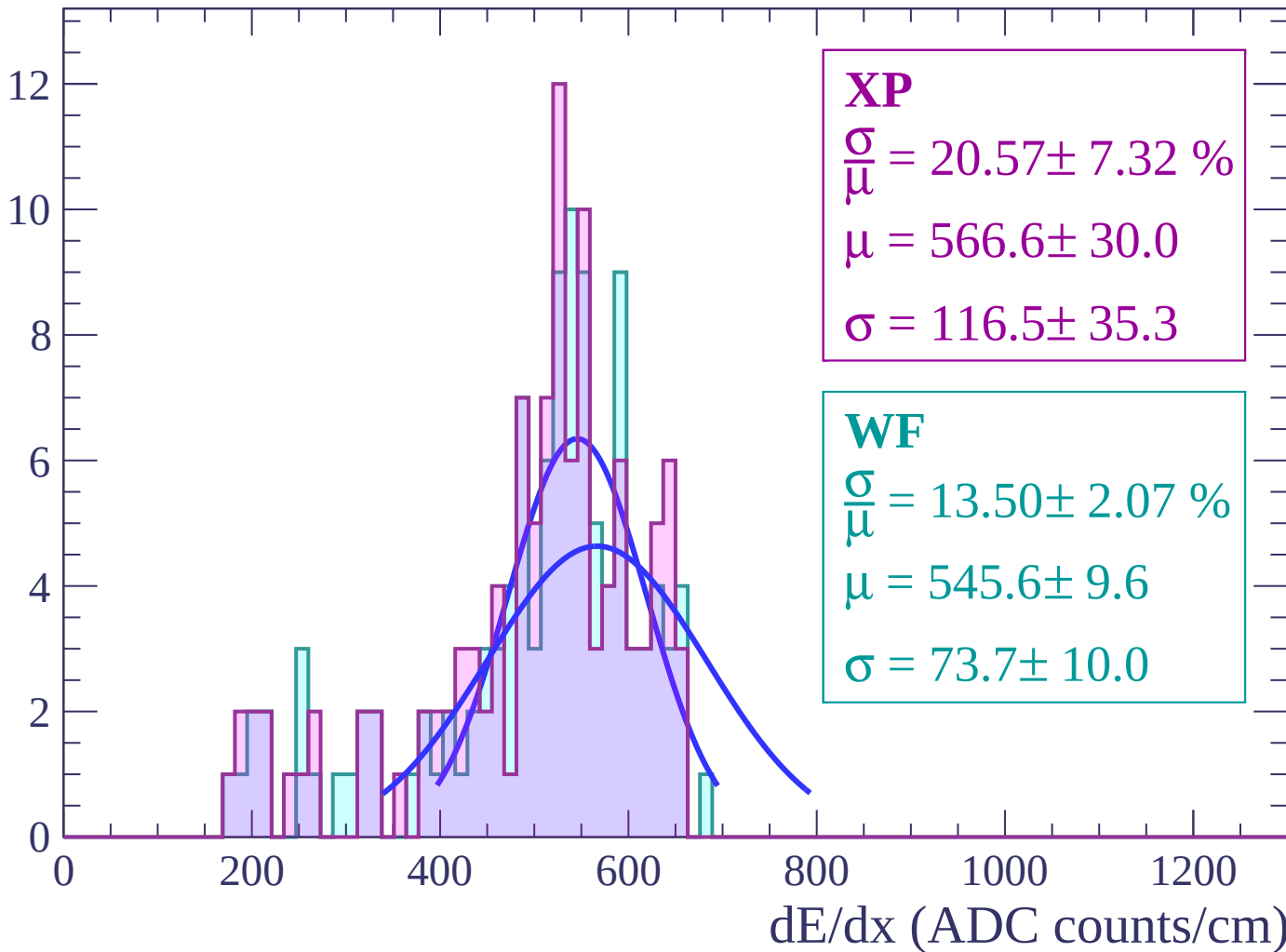
Energy loss | $-66 < \theta < -64$

Count



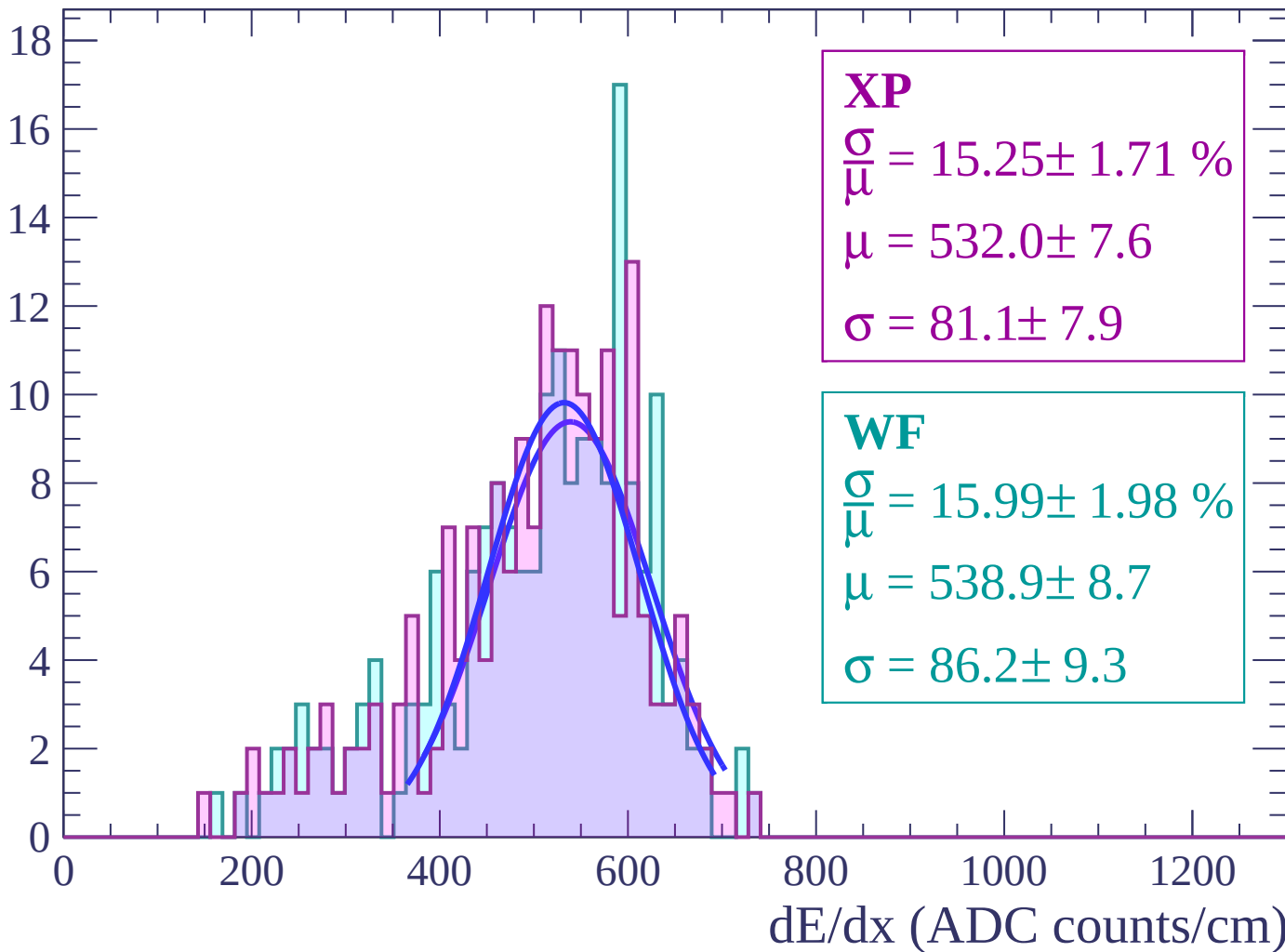
Energy loss | $-64 < \theta < -62$

Count



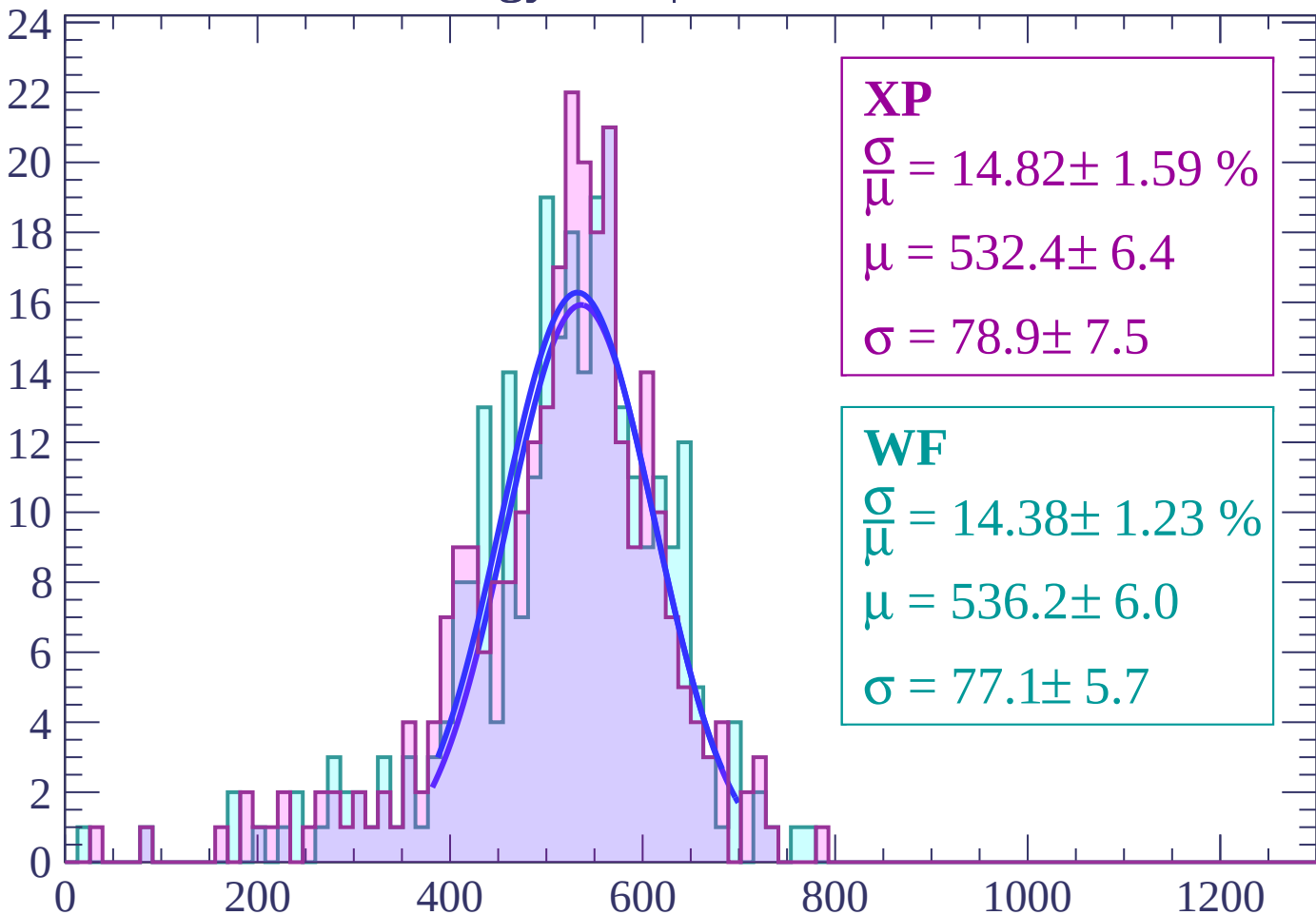
Energy loss | $-62 < \theta < -60$

Count



Energy loss | $-60 < \theta < -58$

Count



XP

$$\frac{\sigma}{\mu} = 14.82 \pm 1.59 \%$$

$$\mu = 532.4 \pm 6.4$$

$$\sigma = 78.9 \pm 7.5$$

WF

$$\frac{\sigma}{\mu} = 14.38 \pm 1.23 \%$$

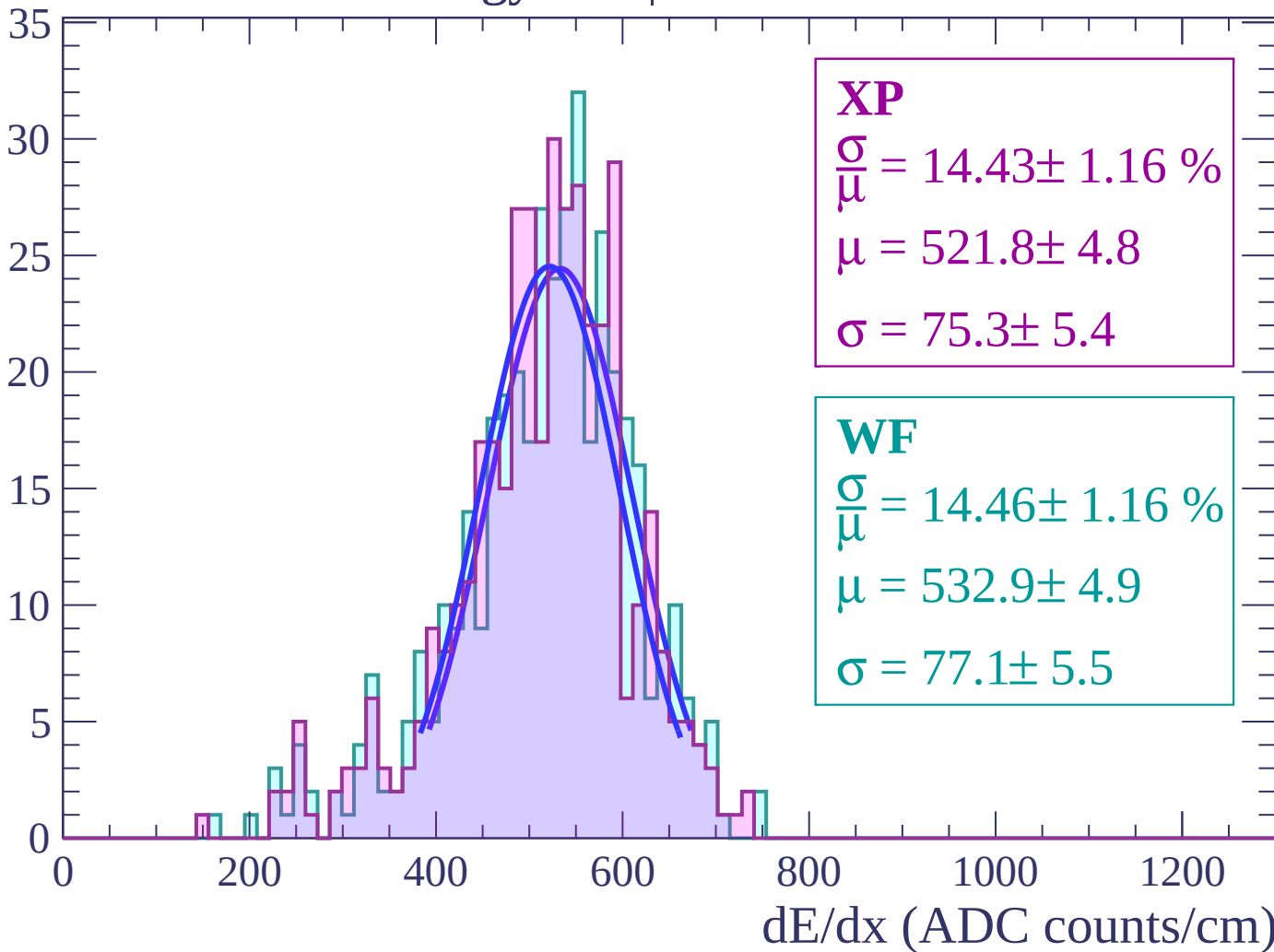
$$\mu = 536.2 \pm 6.0$$

$$\sigma = 77.1 \pm 5.7$$

dE/dx (ADC counts/cm)

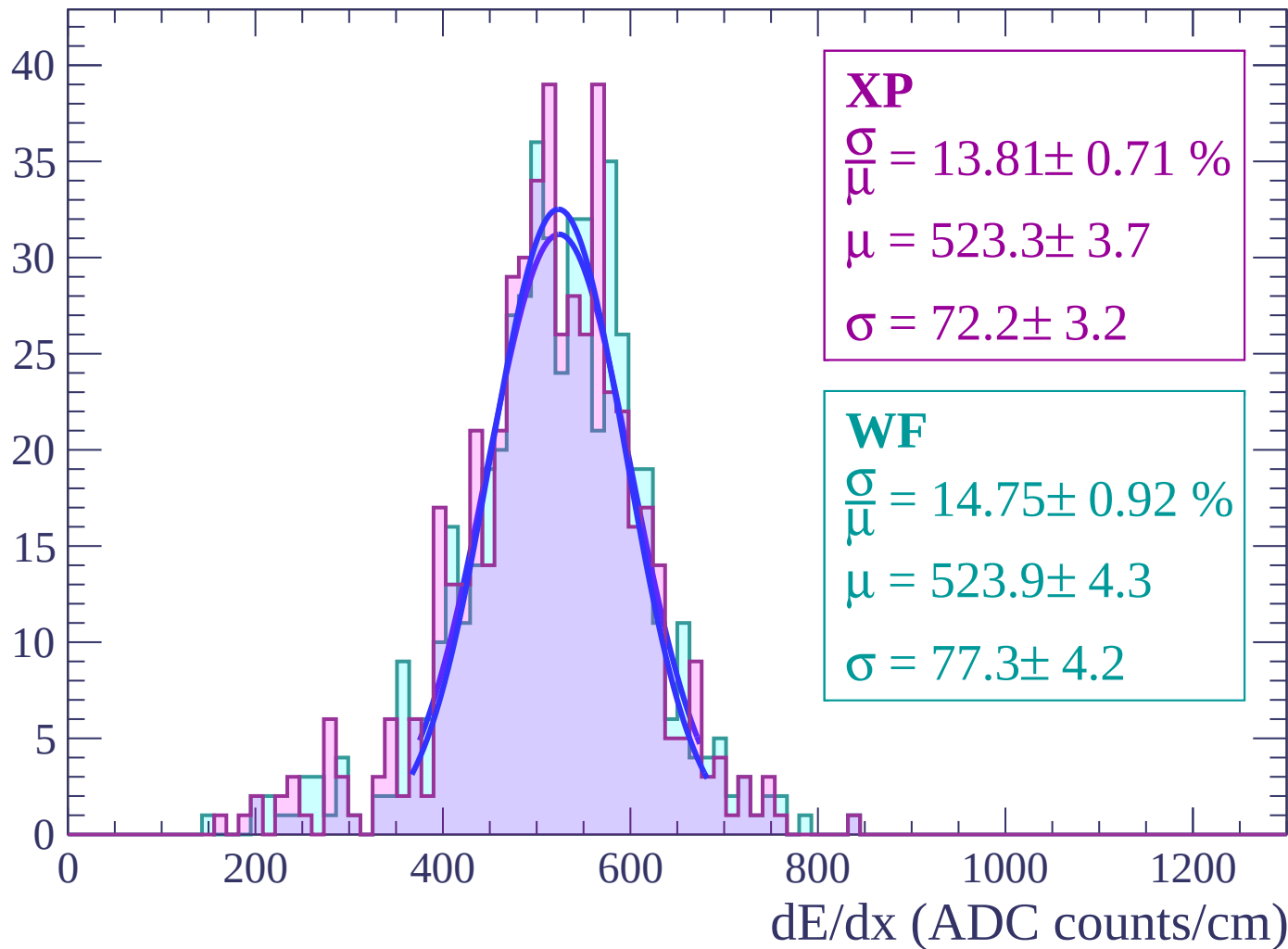
Energy loss | $-58 < \theta < -56$

Count



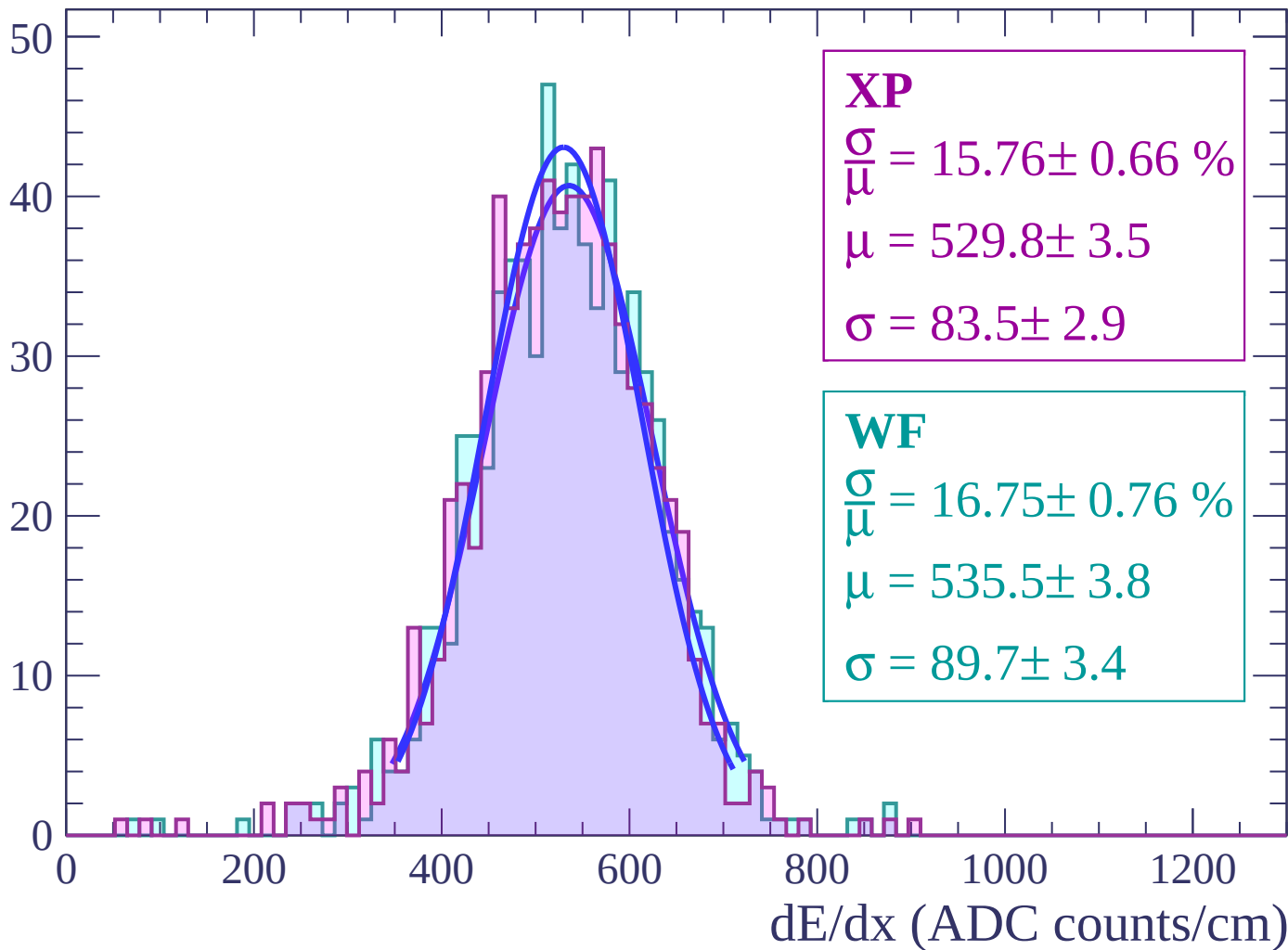
Energy loss | $-56 < \theta < -54$

Count



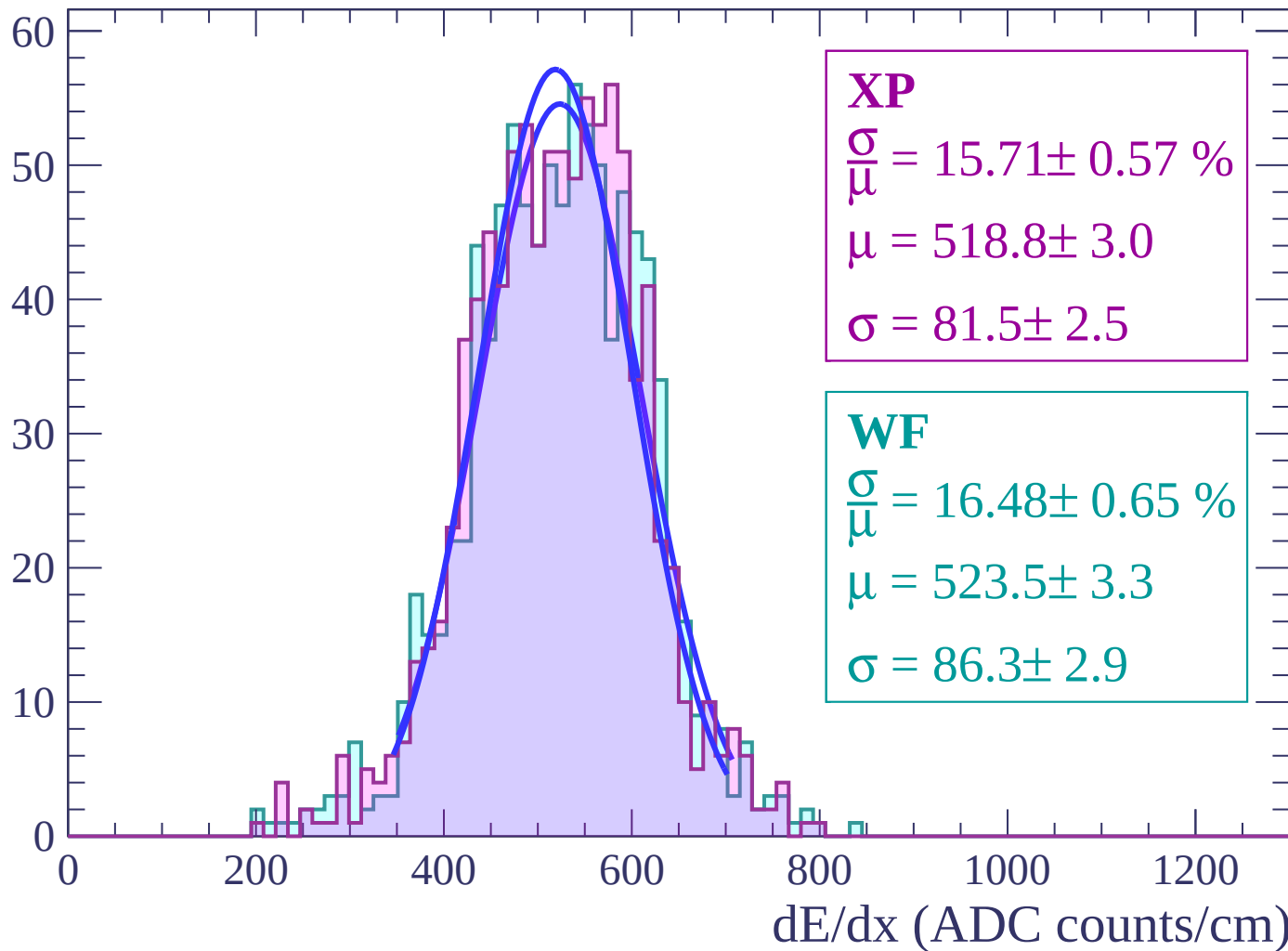
Energy loss | $-54 < \theta < -52$

Count



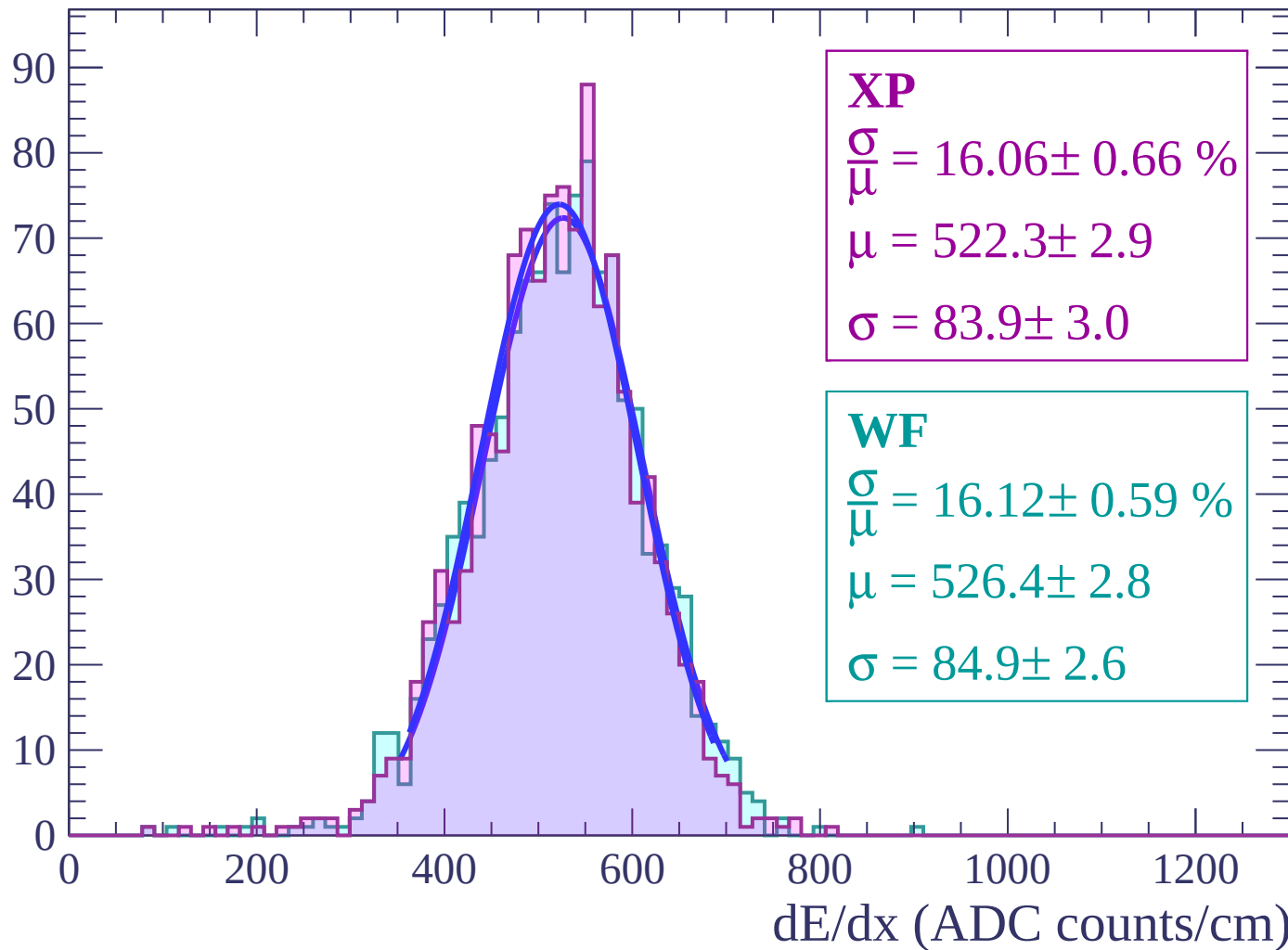
Energy loss | $-52 < \theta < -50$

Count



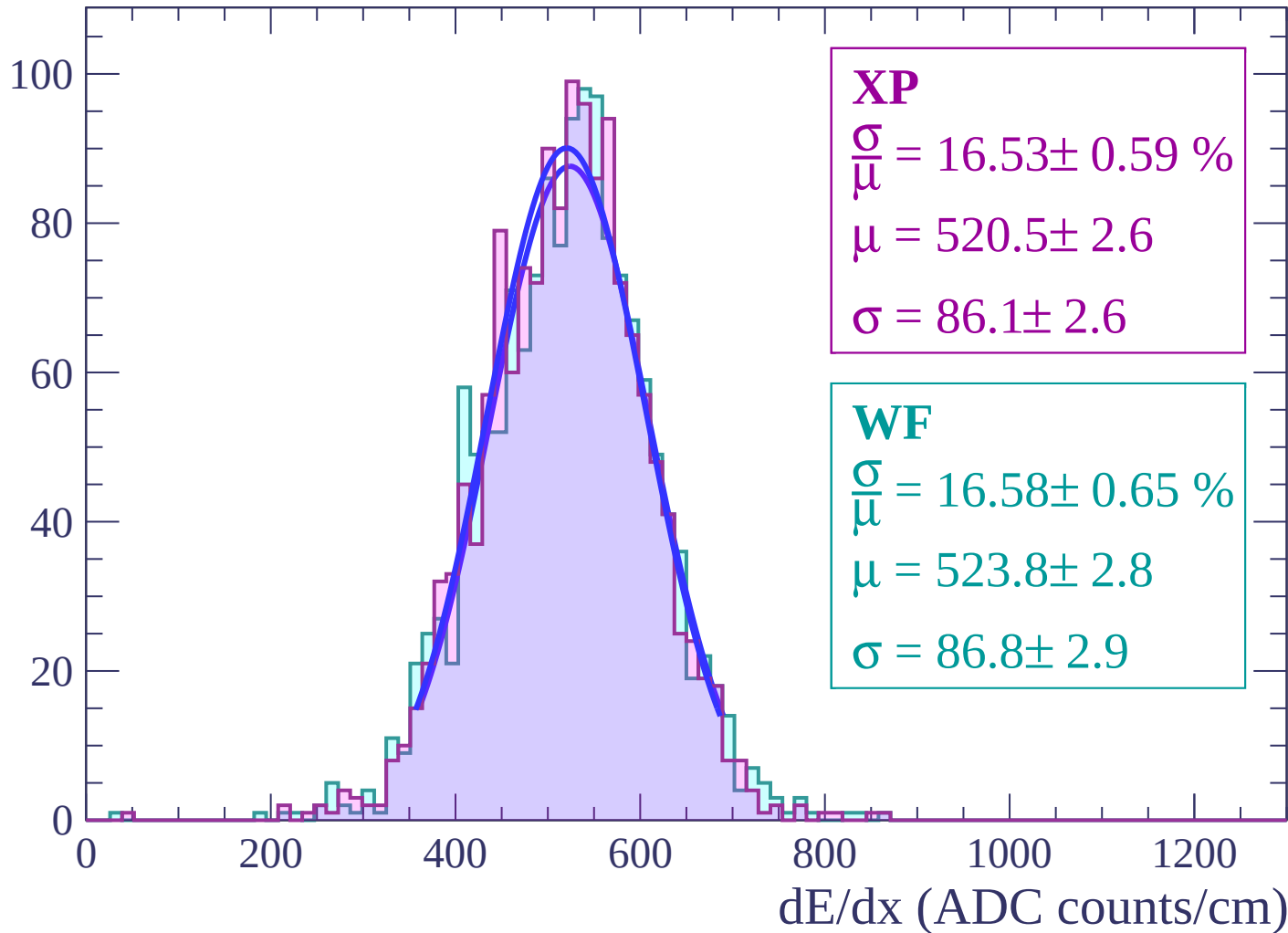
Energy loss | $-50 < \theta < -48$

Count



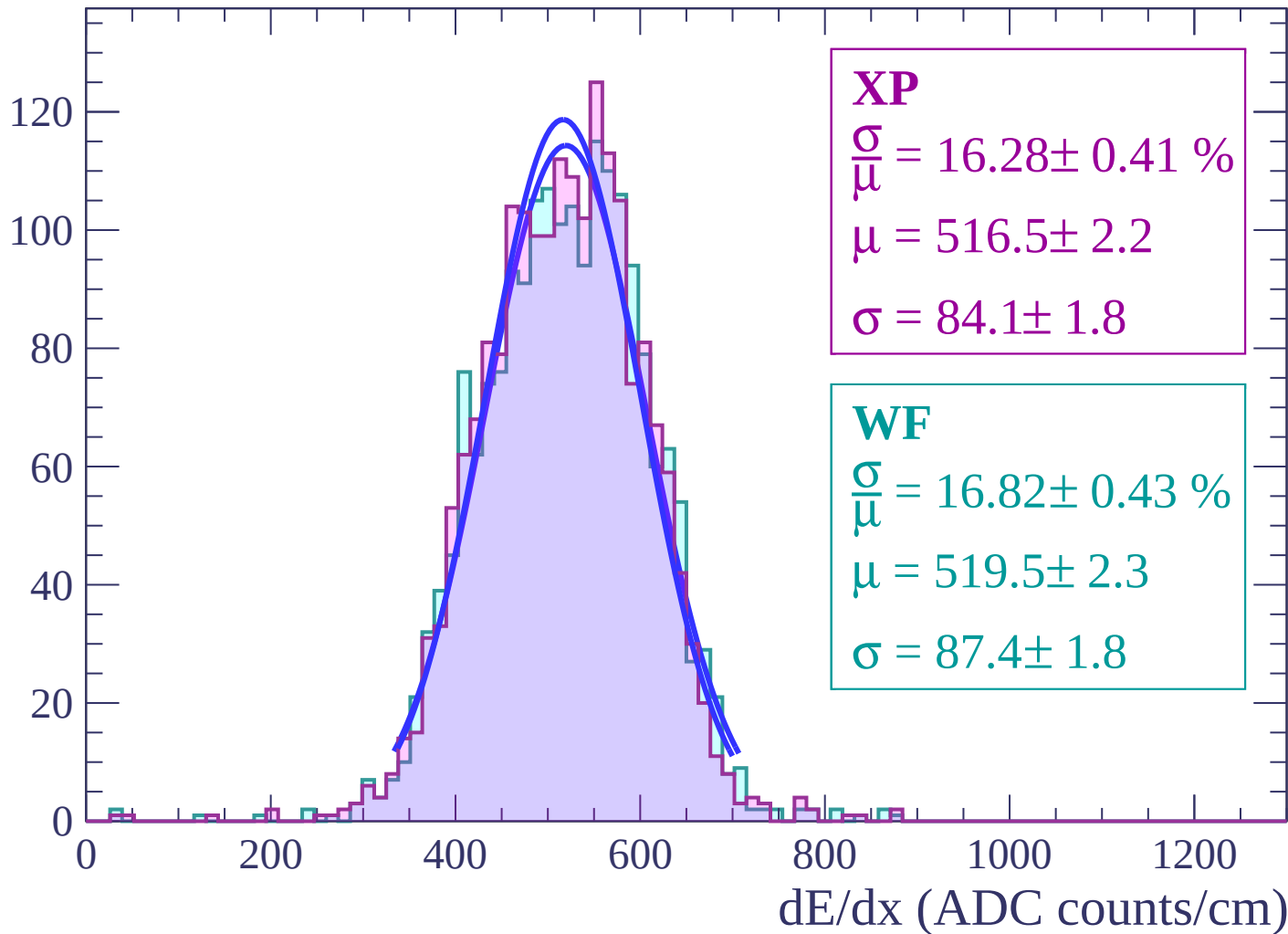
Energy loss | $-48 < \theta < -46$

Count



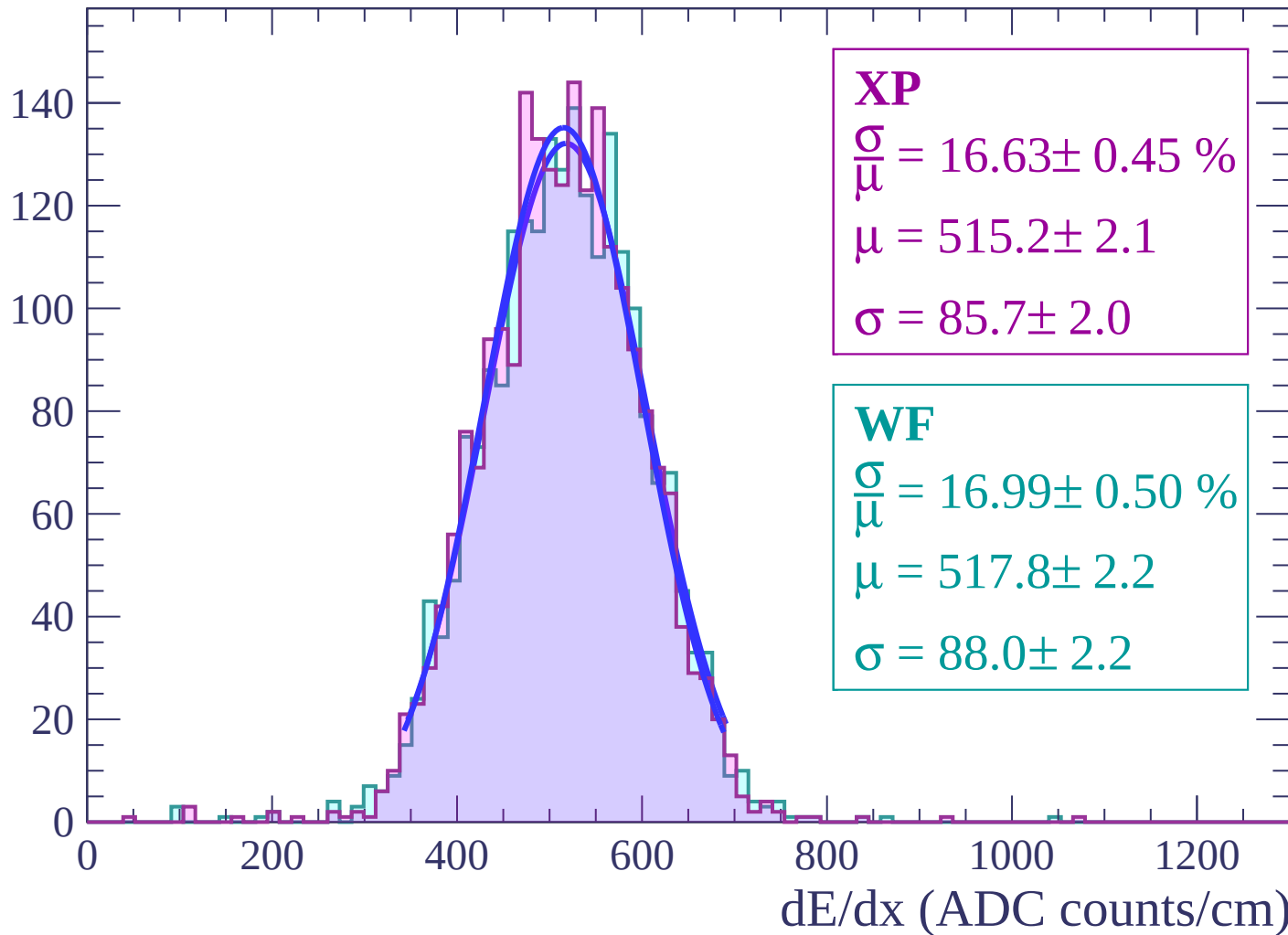
Energy loss | $-46 < \theta < -44$

Count



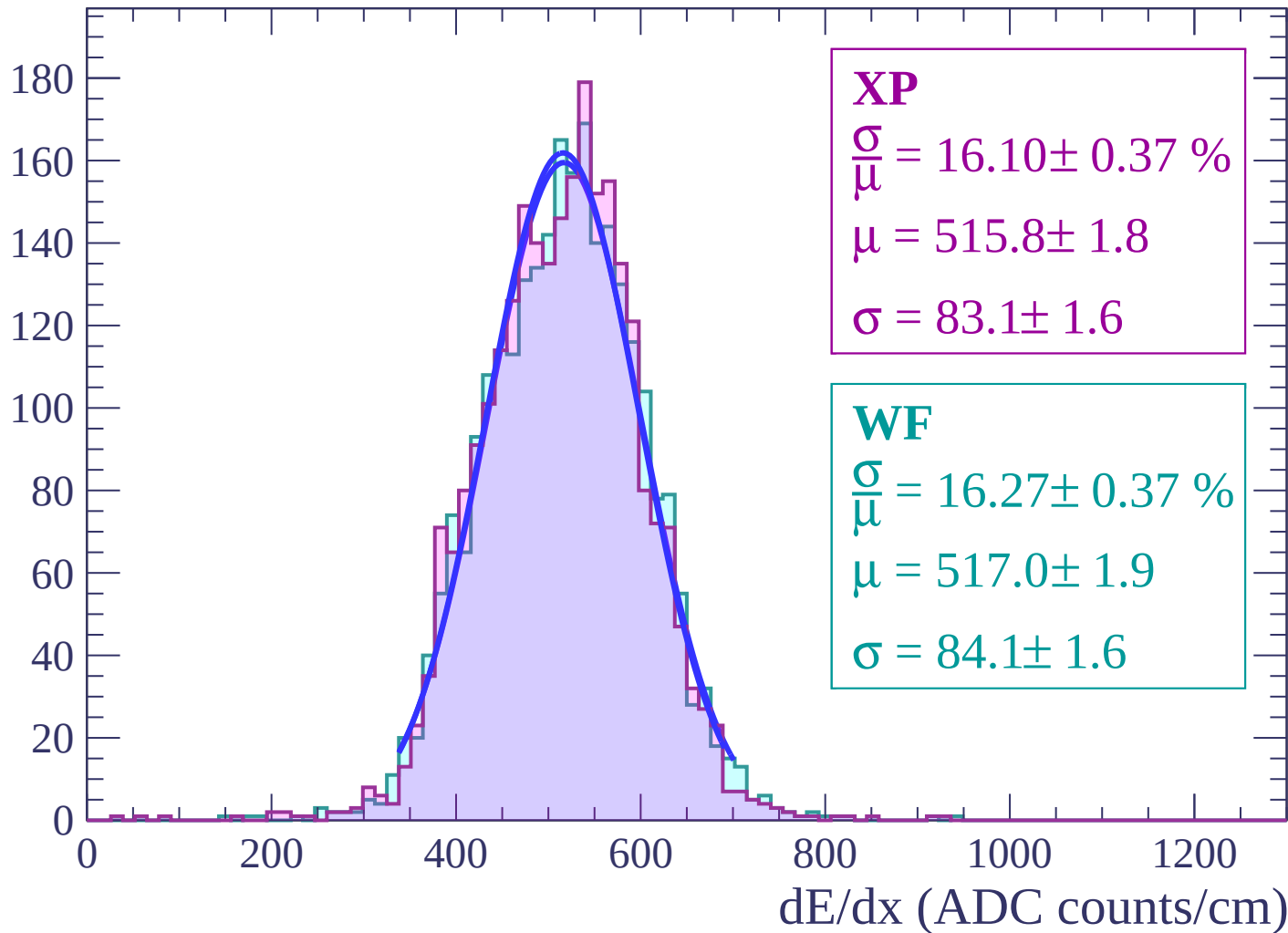
Energy loss | $-44 < \theta < -42$

Count



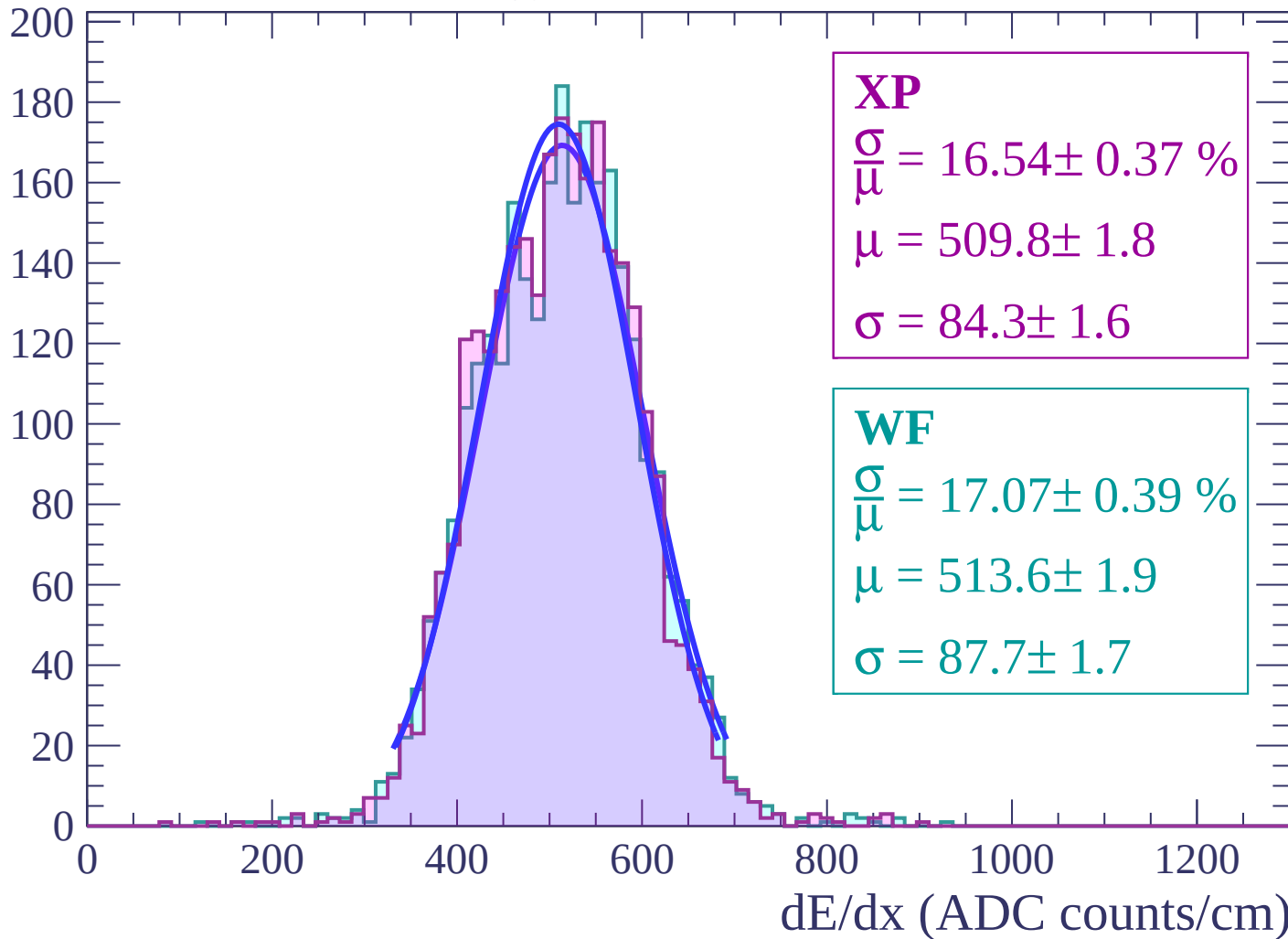
Energy loss | $-42 < \theta < -40$

Count



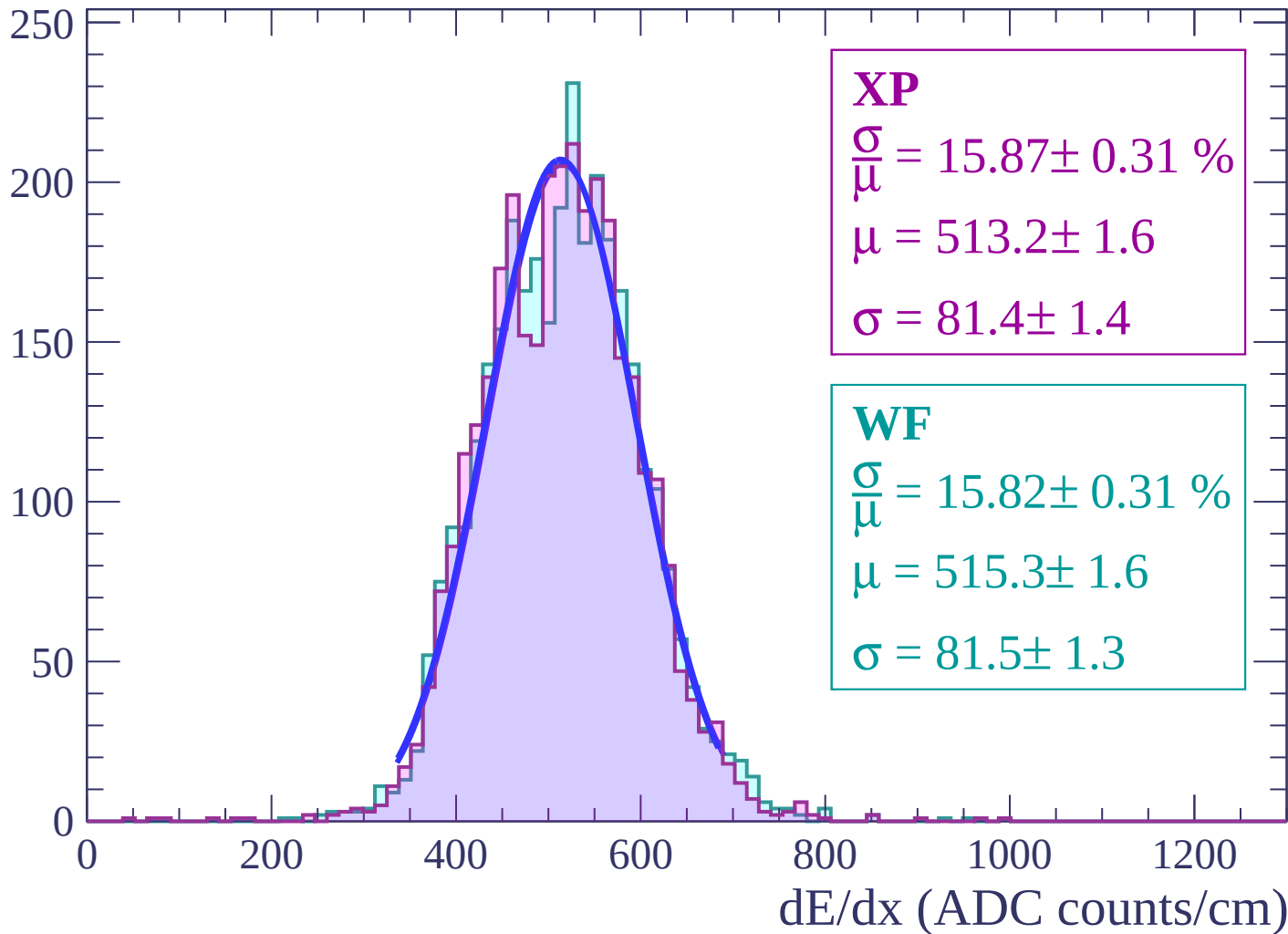
Energy loss | $-40 < \theta < -38$

Count



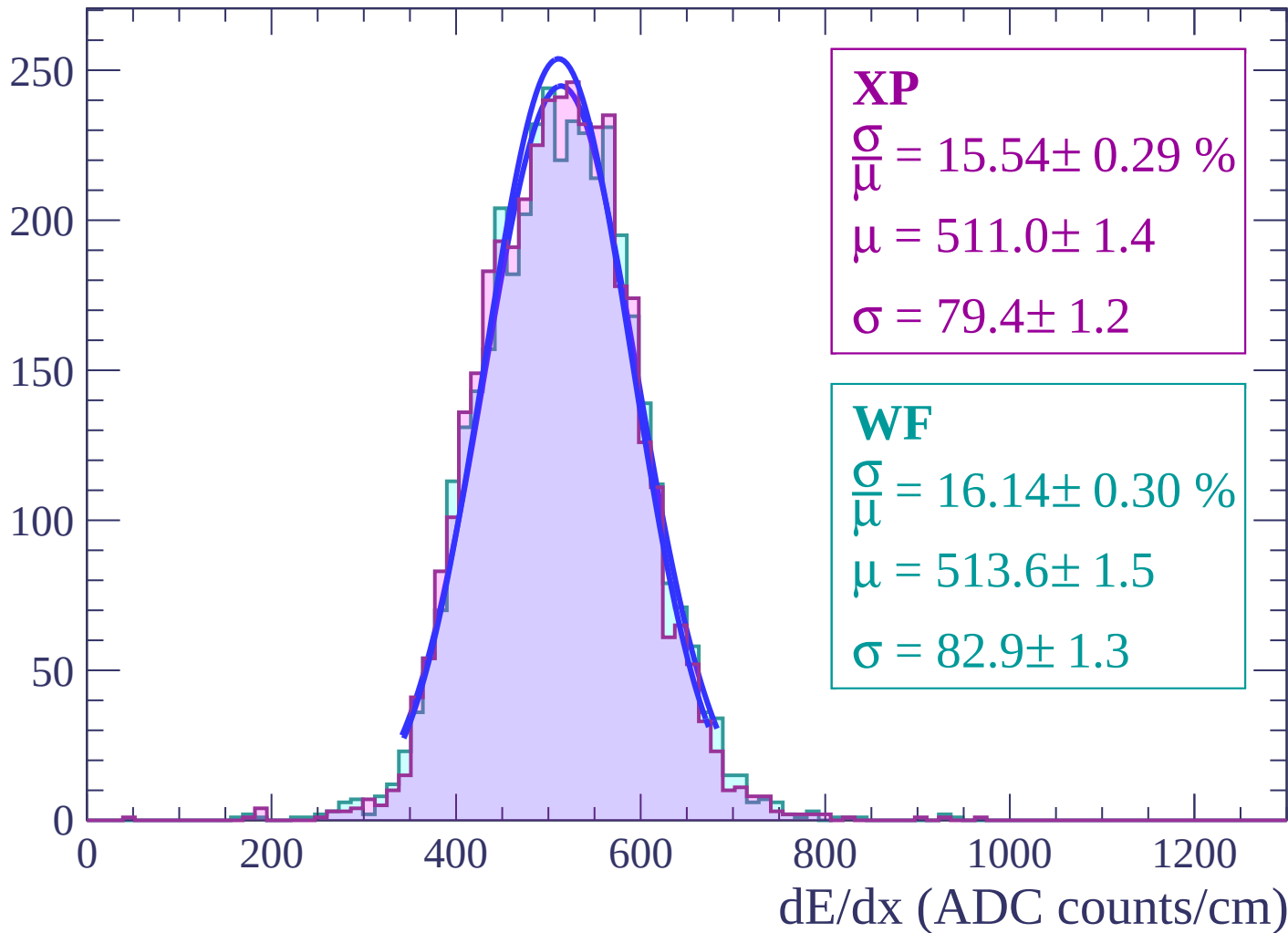
Energy loss | $-38 < \theta < -36$

Count



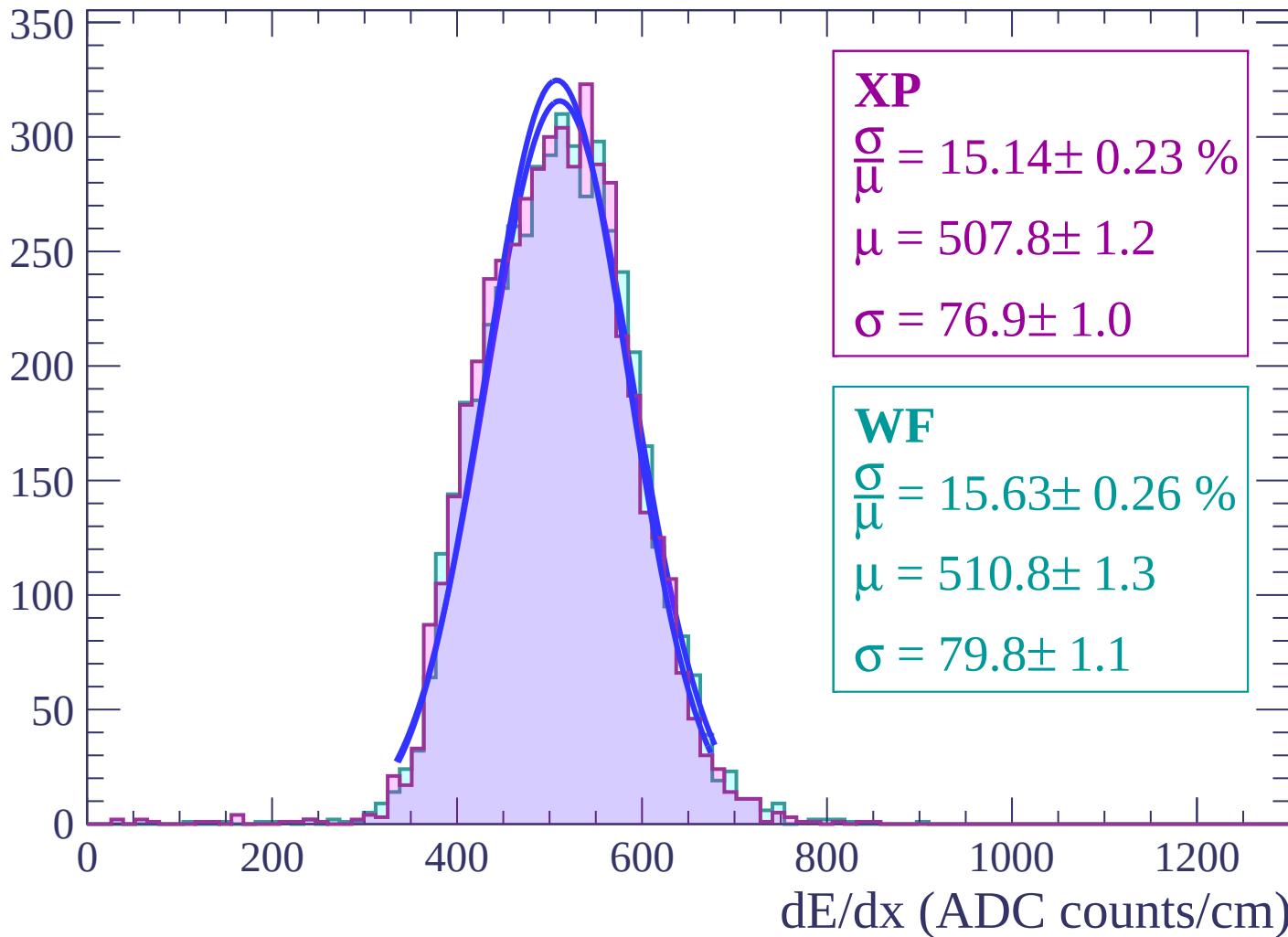
Energy loss | $-36 < \theta < -34$

Count



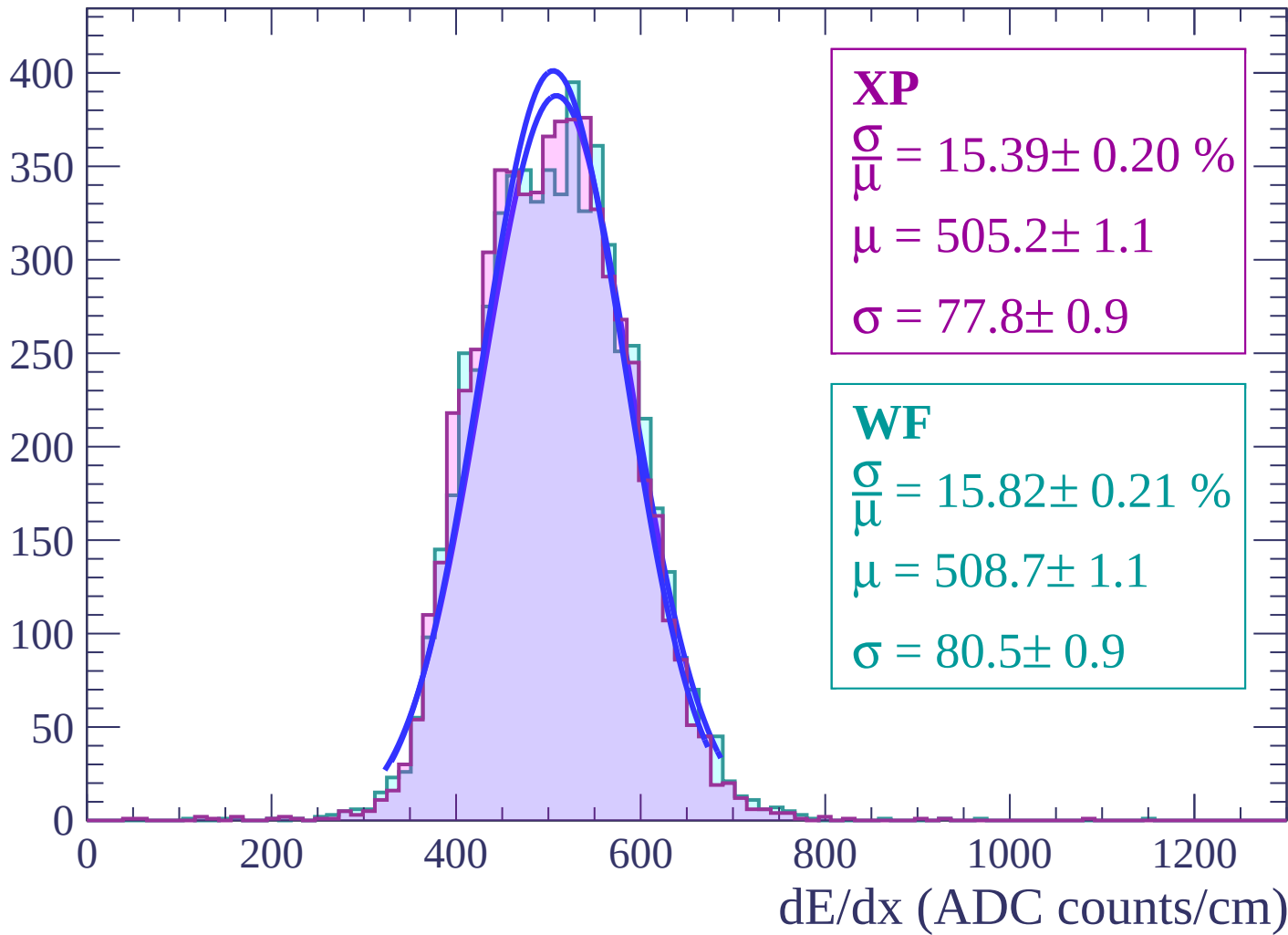
Energy loss | $-34 < \theta < -32$

Count



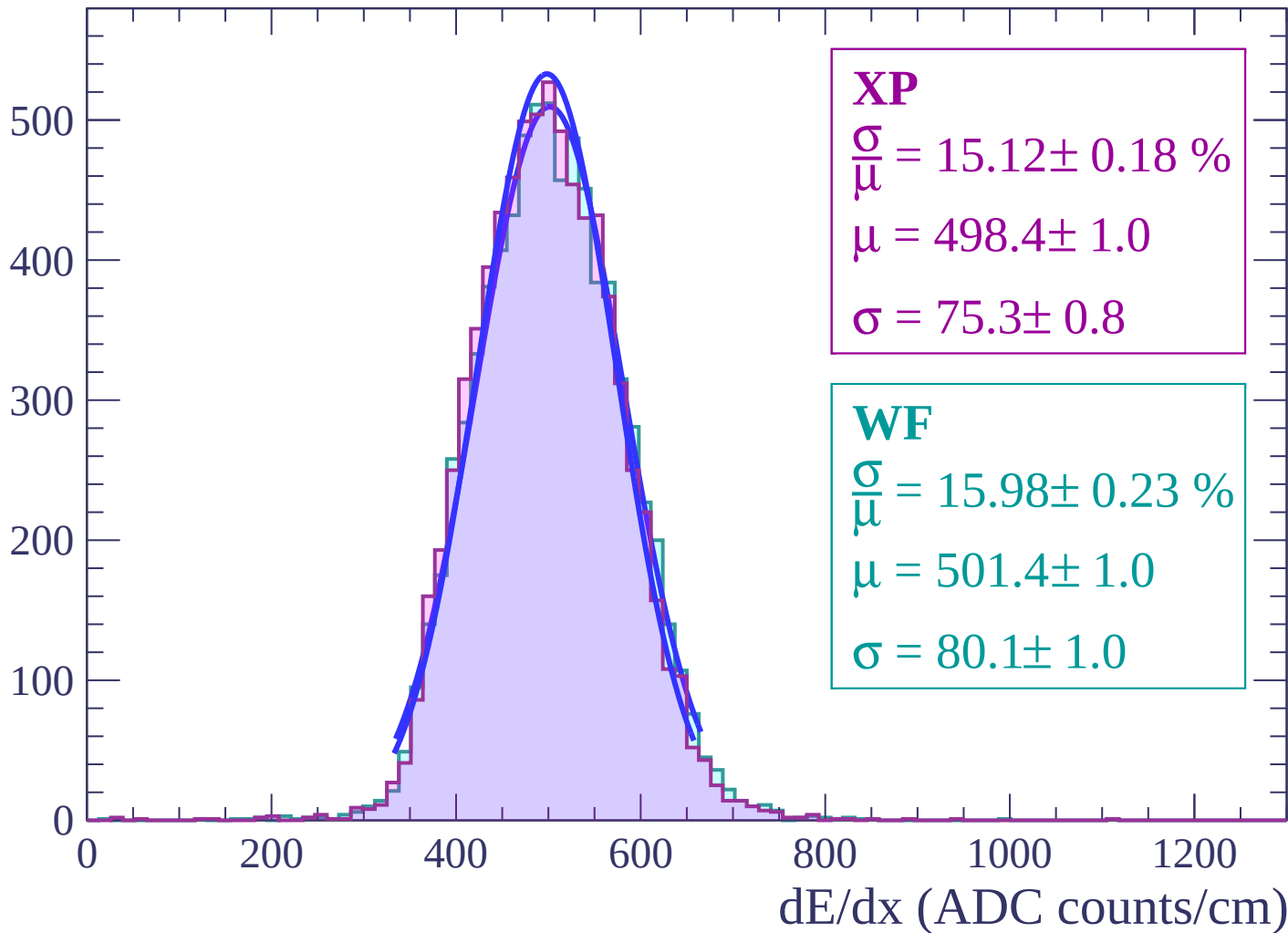
Energy loss | $-32 < \theta < -30$

Count



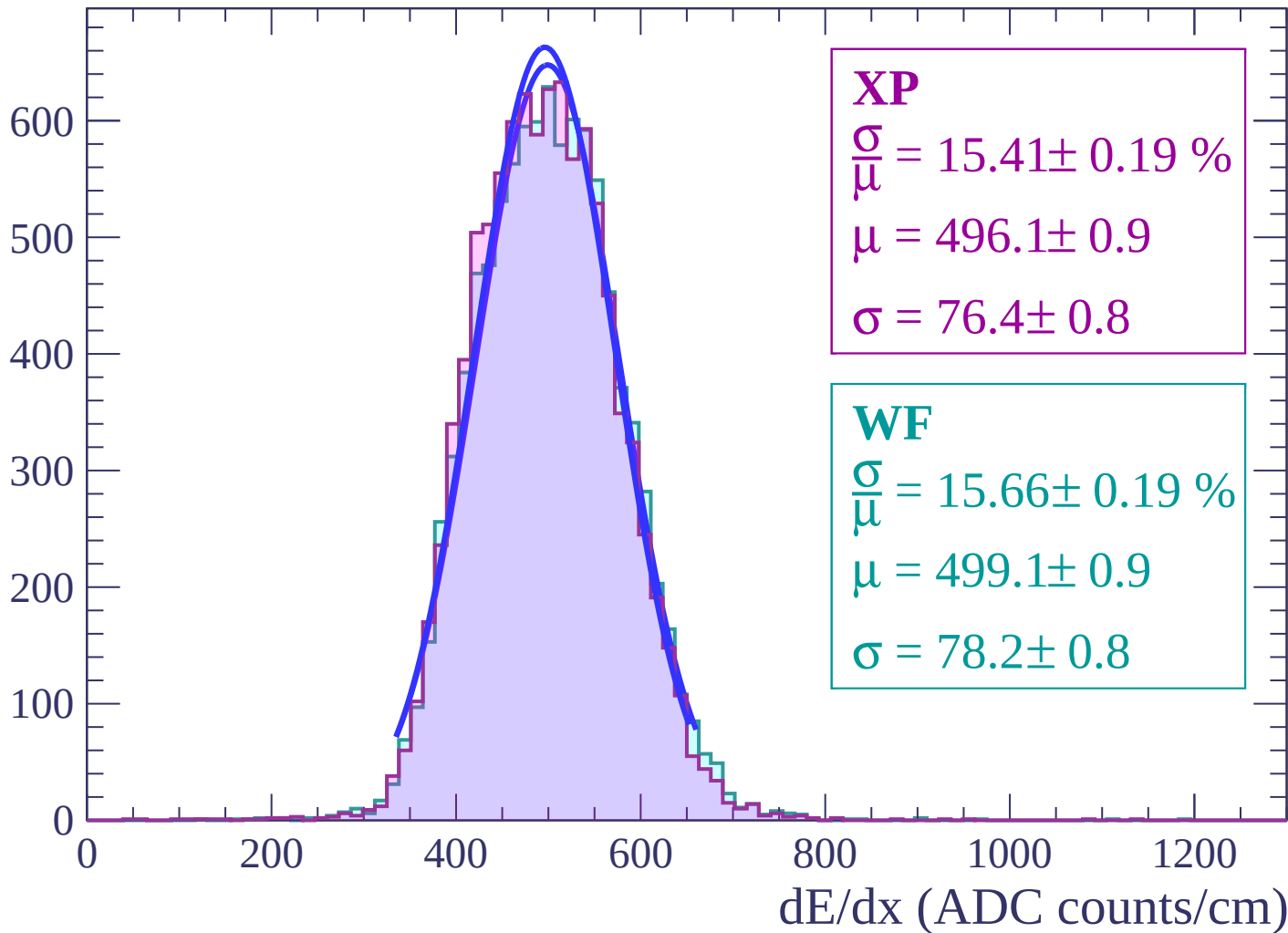
Energy loss | $-30 < \theta < -28$

Count



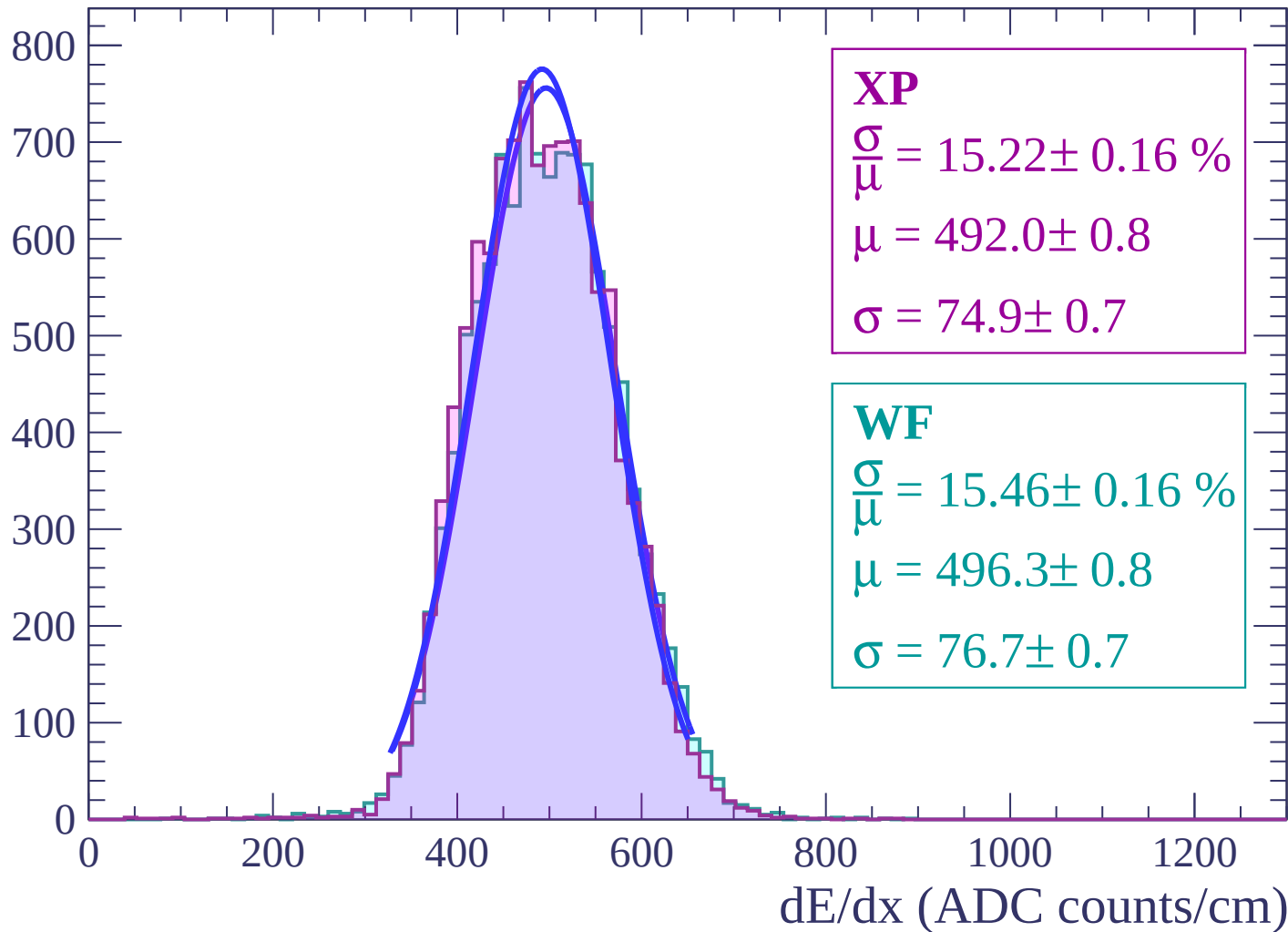
Energy loss | $-28 < \theta < -26$

Count

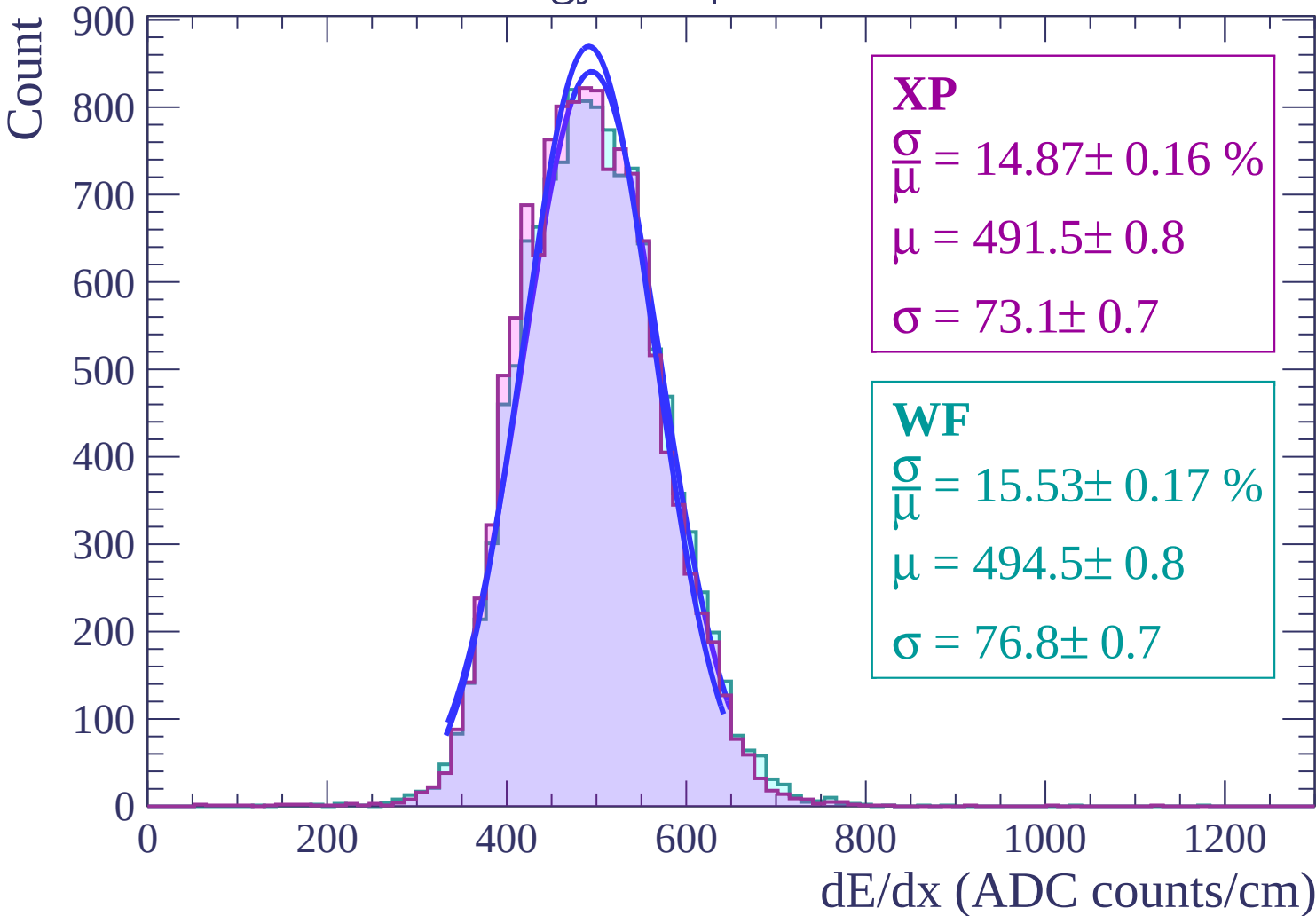


Energy loss | $-26 < \theta < -24$

Count

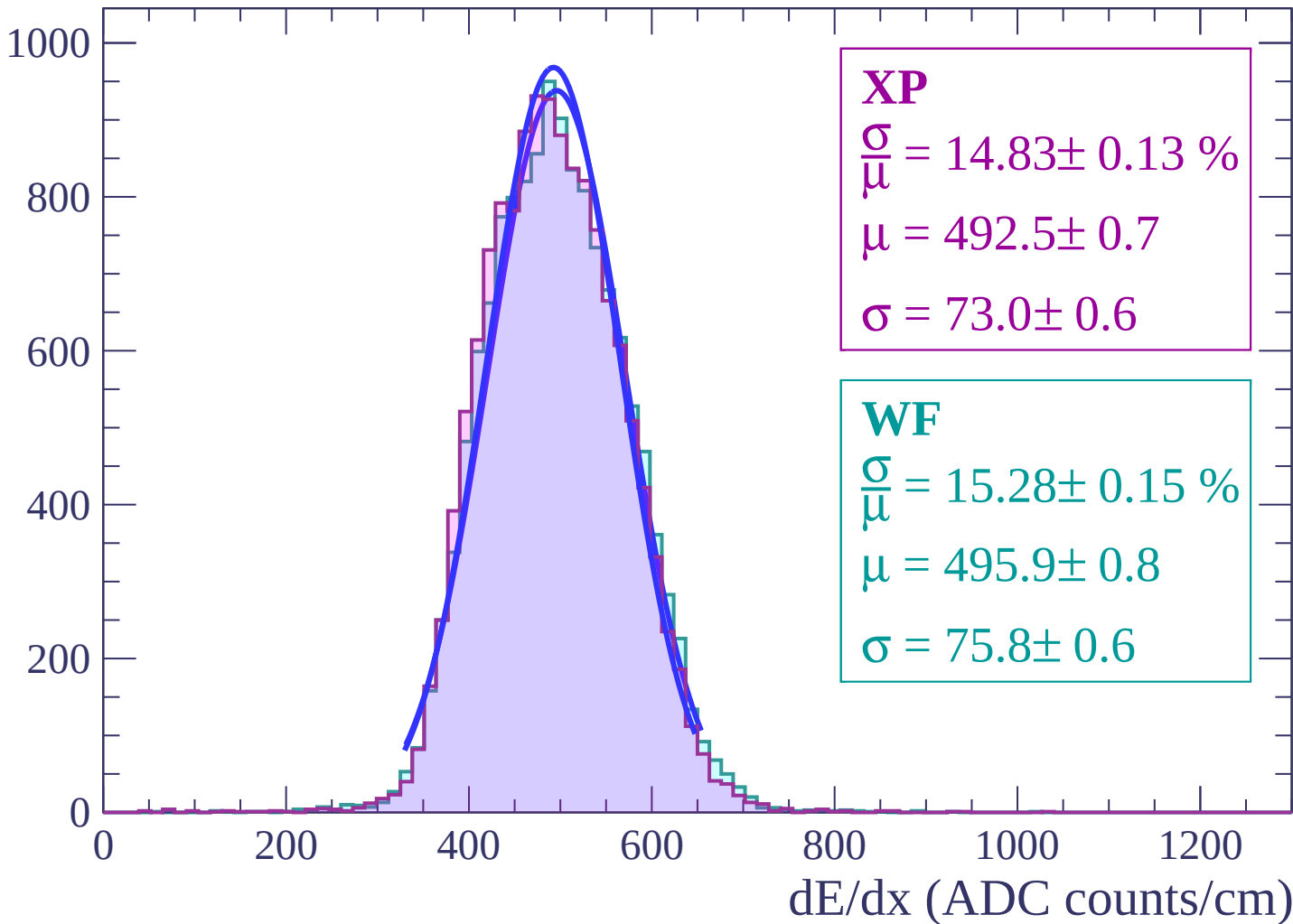


Energy loss | $-24 < \theta < -22$



Energy loss | $-22 < \theta < -20$

Count



Energy loss | $-20 < \theta < -18$

Count

1000

800

600

400

200

0

200

400

600

800

1000

1200

dE/dx (ADC counts/cm)

XP

$$\frac{\sigma}{\mu} = 14.92 \pm 0.13 \%$$

$$\mu = 491.2 \pm 0.7$$

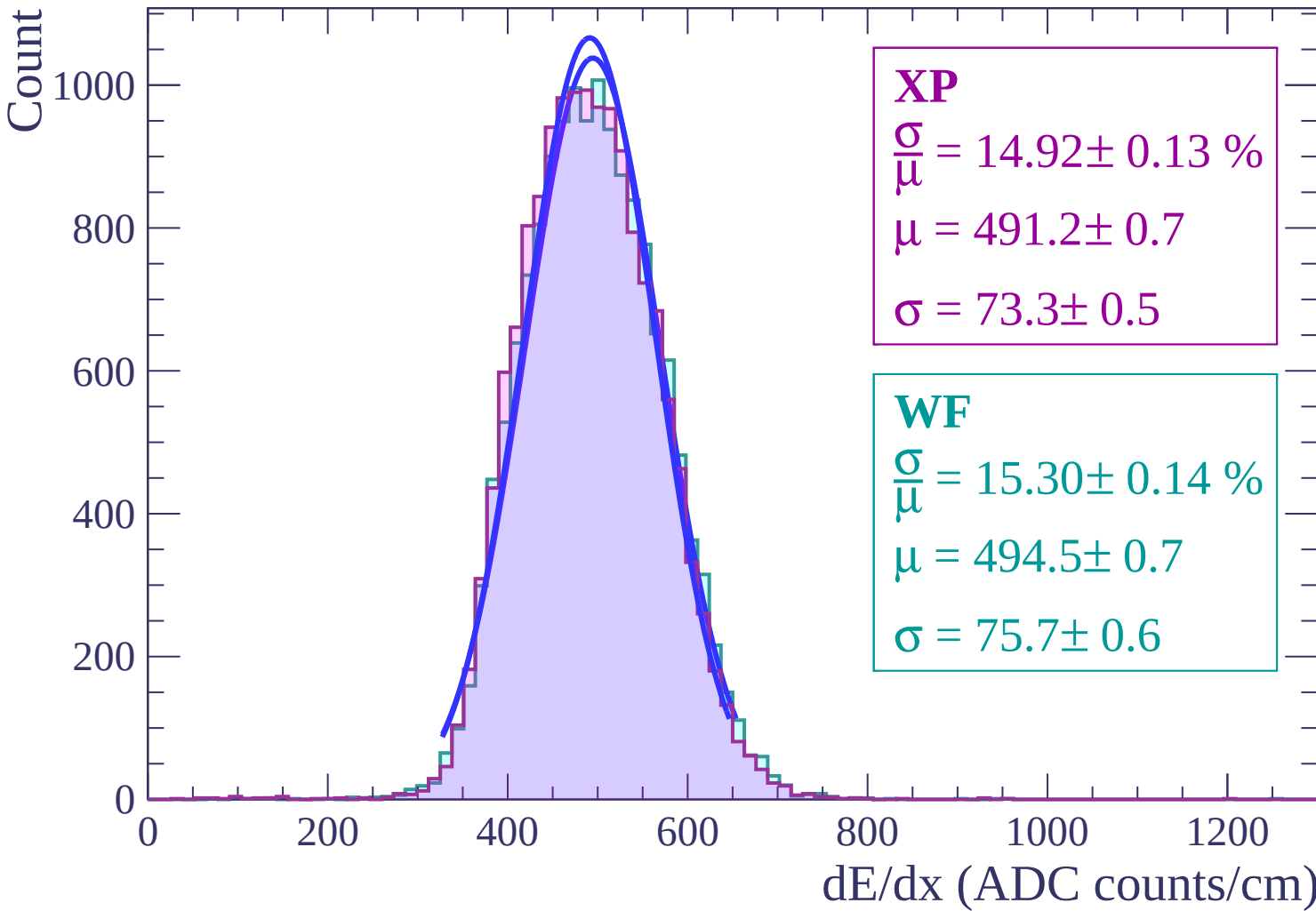
$$\sigma = 73.3 \pm 0.5$$

WF

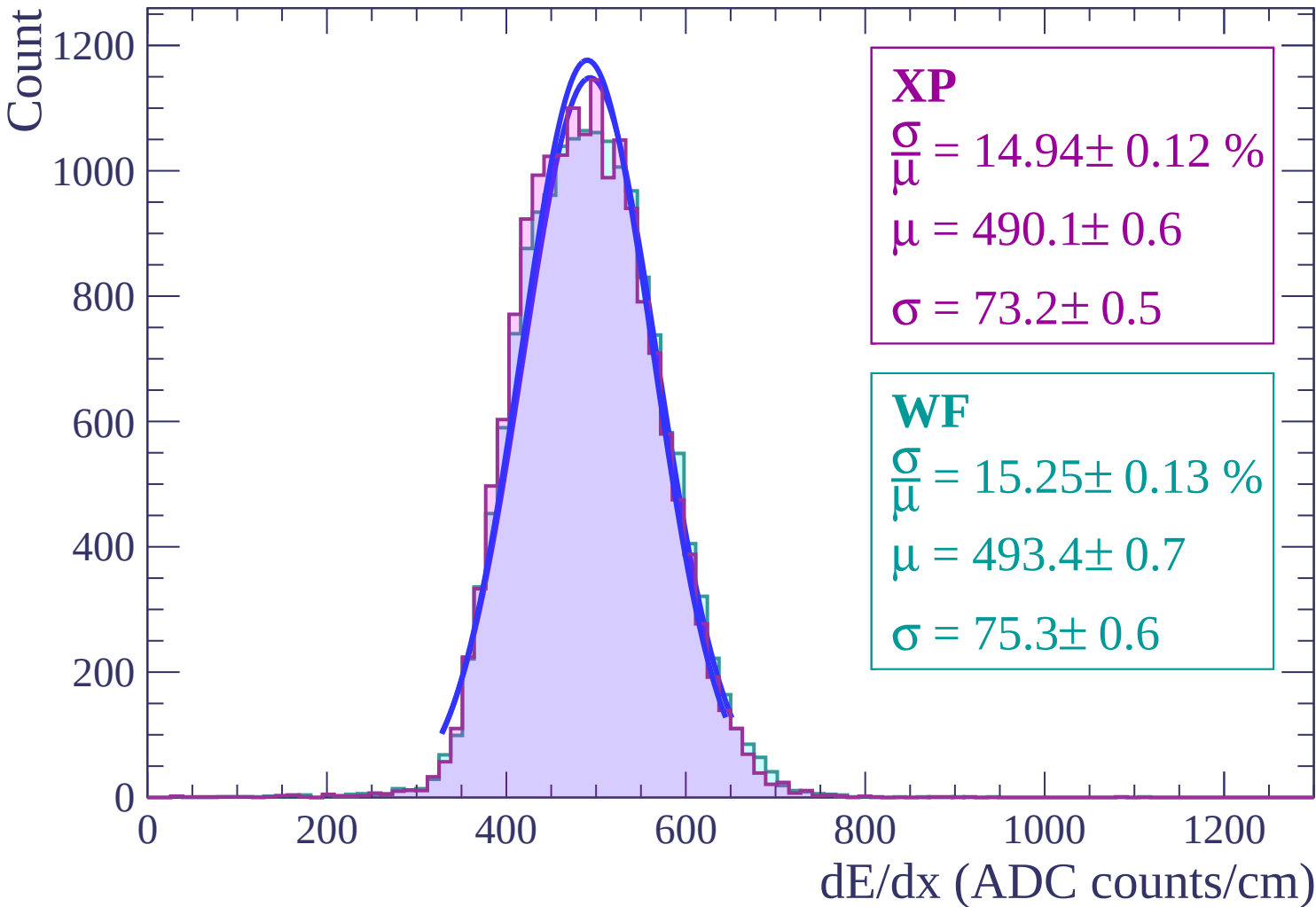
$$\frac{\sigma}{\mu} = 15.30 \pm 0.14 \%$$

$$\mu = 494.5 \pm 0.7$$

$$\sigma = 75.7 \pm 0.6$$

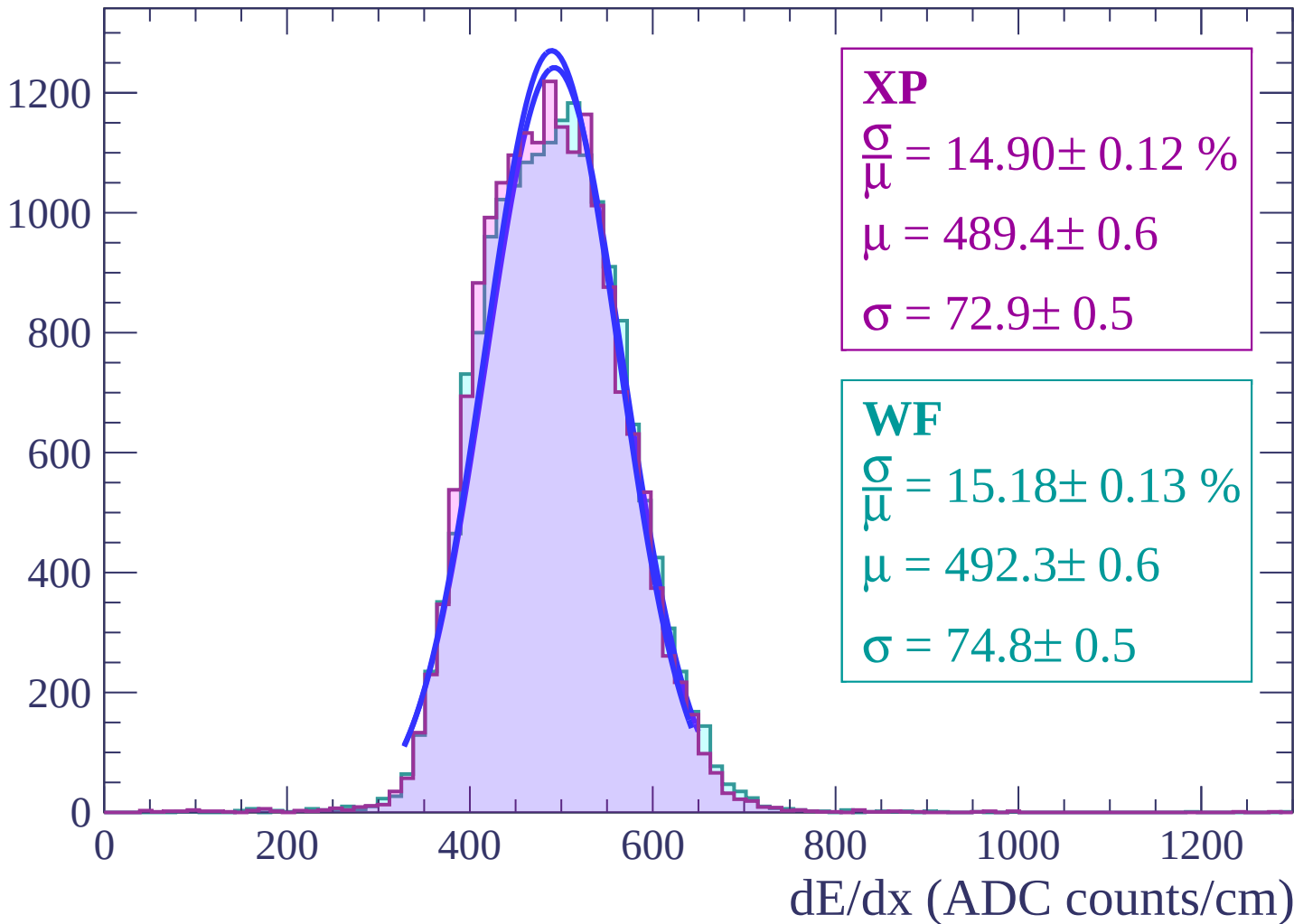


Energy loss | $-18 < \theta < -16$

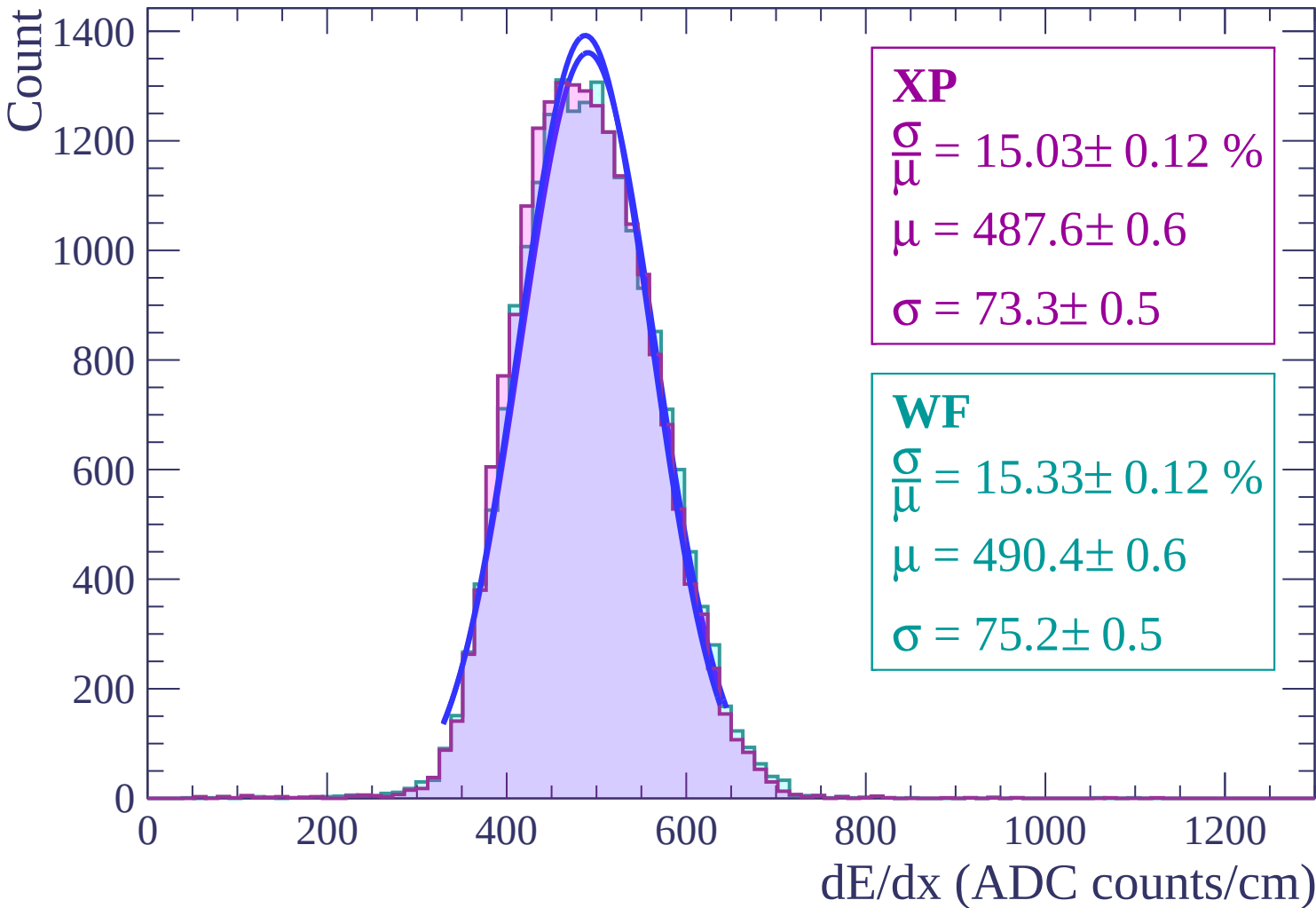


Energy loss | $-16 < \theta < -14$

Count



Energy loss | $-14 < \theta < -12$



Energy loss | $-12 < \theta < -10$

Count

1400
1200
1000
800
600
400
200
0

XP

$$\frac{\sigma}{\mu} = 15.10 \pm 0.12 \%$$

$$\mu = 486.9 \pm 0.6$$

$$\sigma = 73.5 \pm 0.5$$

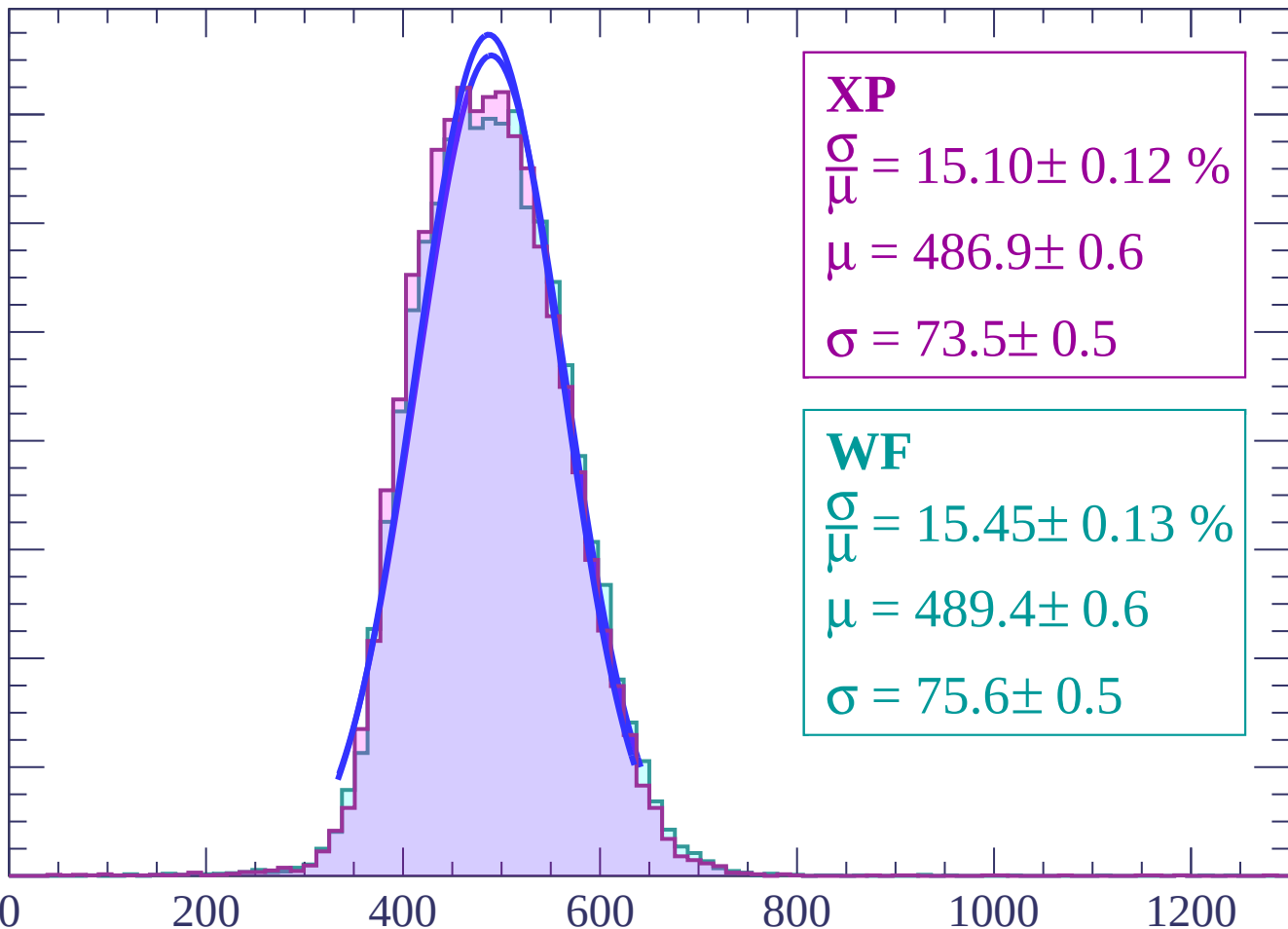
WF

$$\frac{\sigma}{\mu} = 15.45 \pm 0.13 \%$$

$$\mu = 489.4 \pm 0.6$$

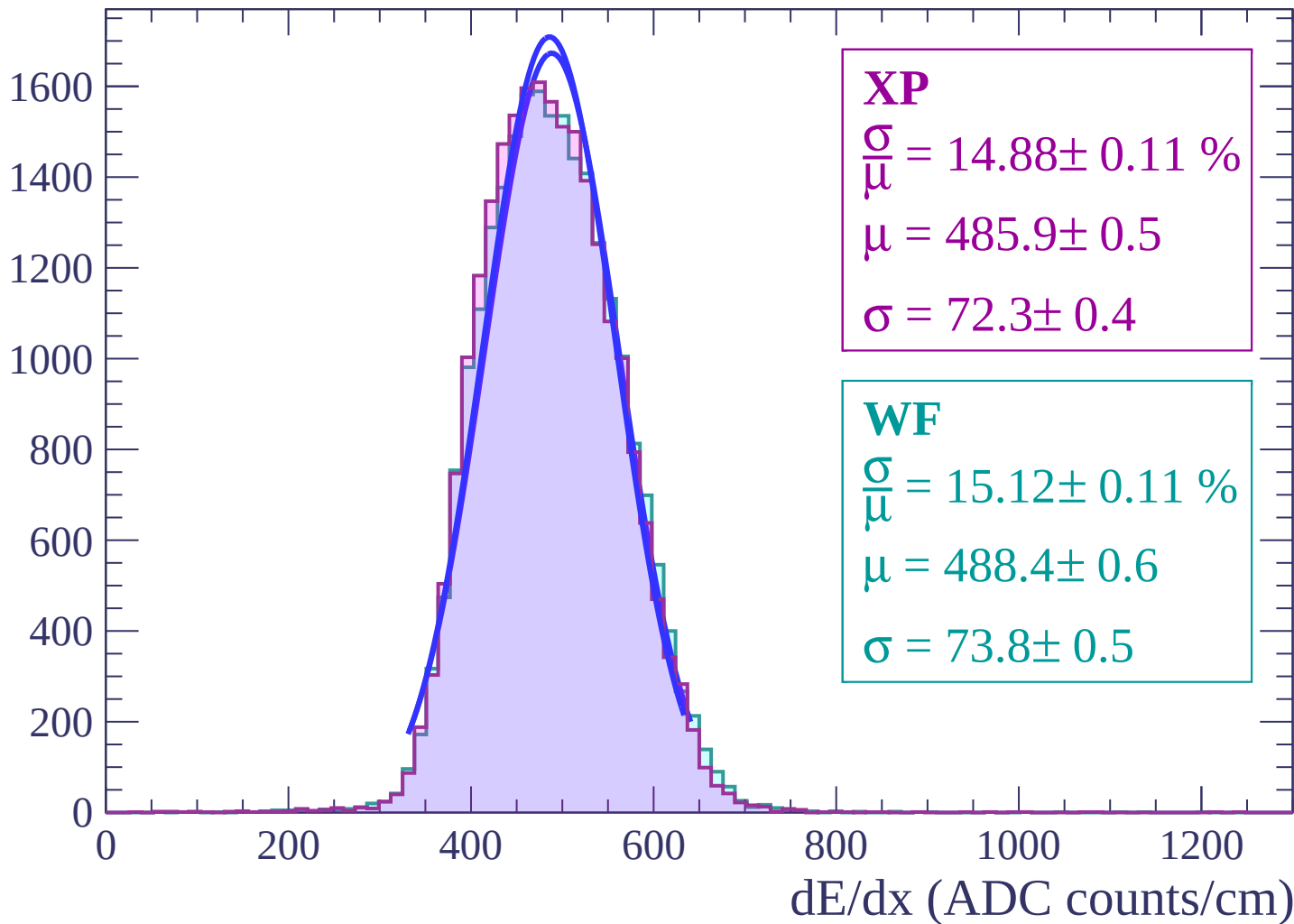
$$\sigma = 75.6 \pm 0.5$$

dE/dx (ADC counts/cm)

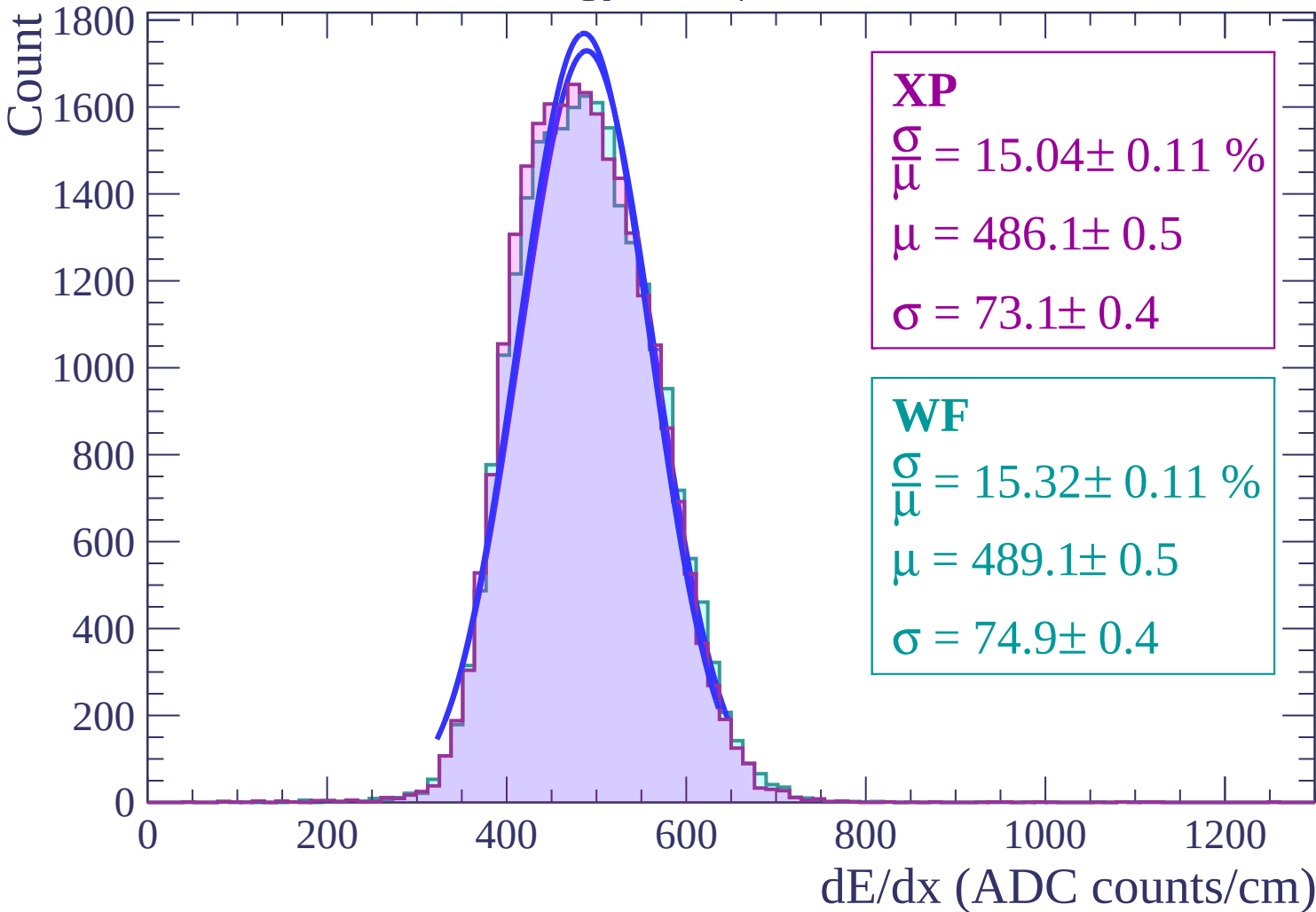


Energy loss | $-10 < \theta < -8$

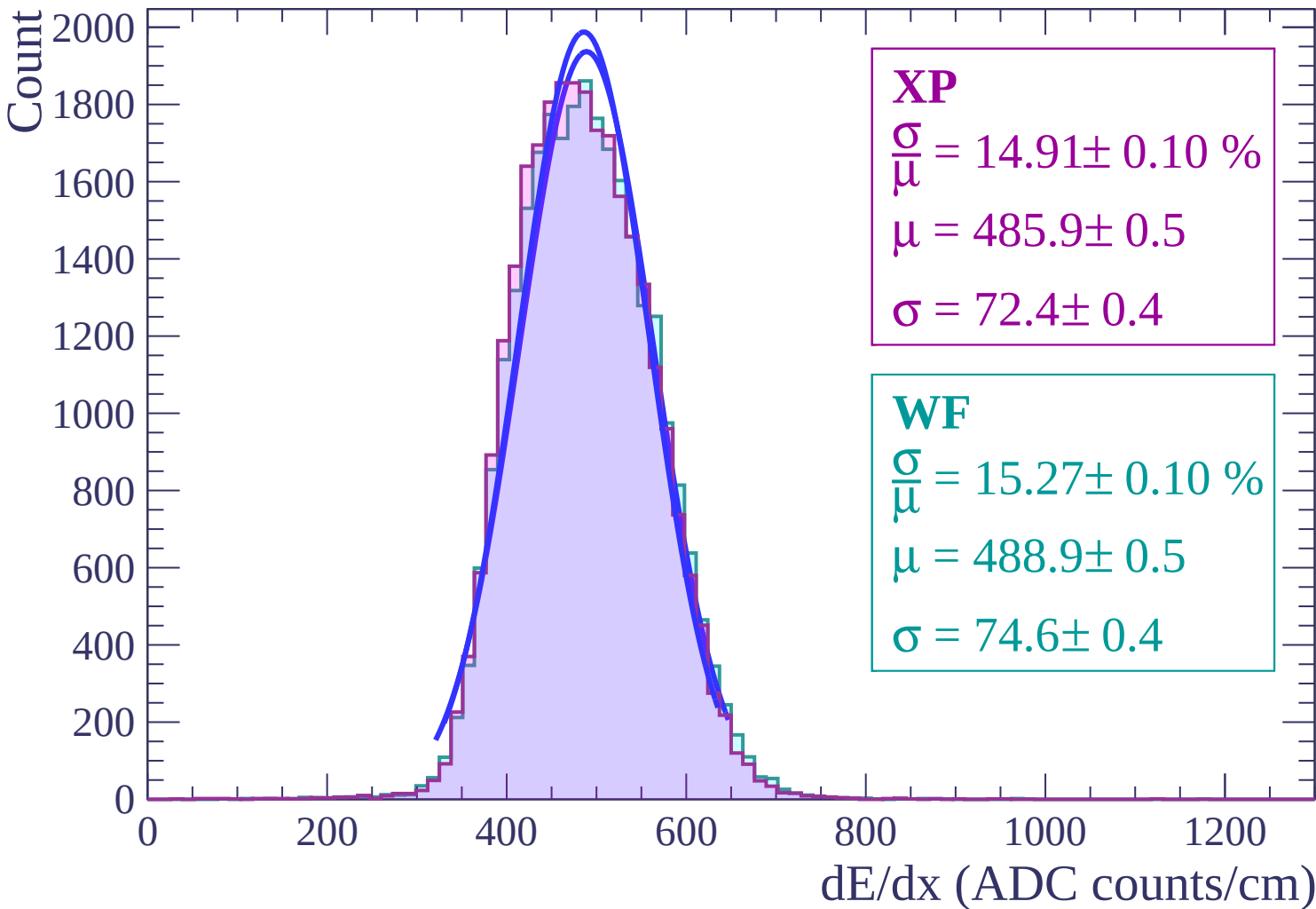
Count



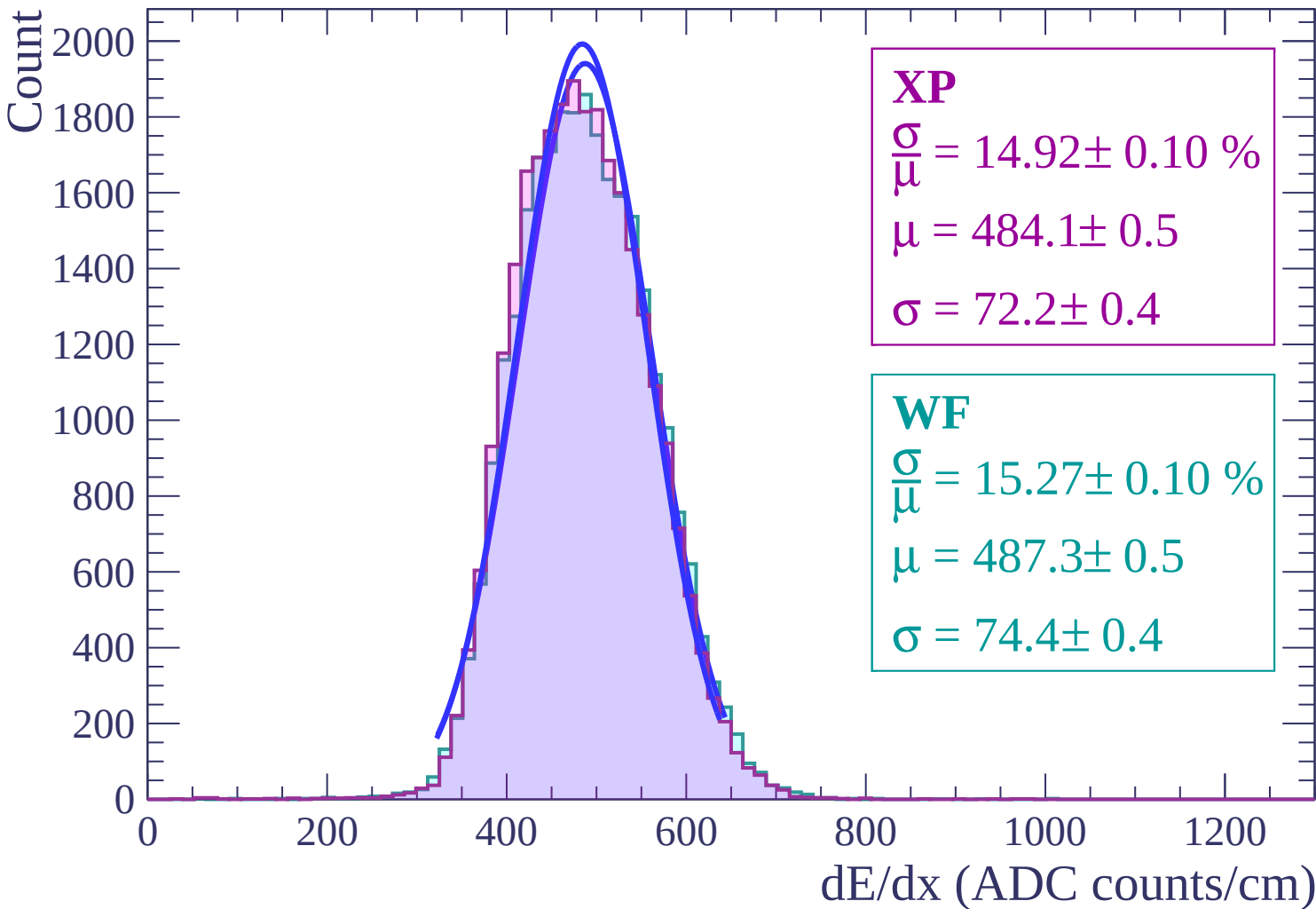
Energy loss | $-8 < \theta < -6$



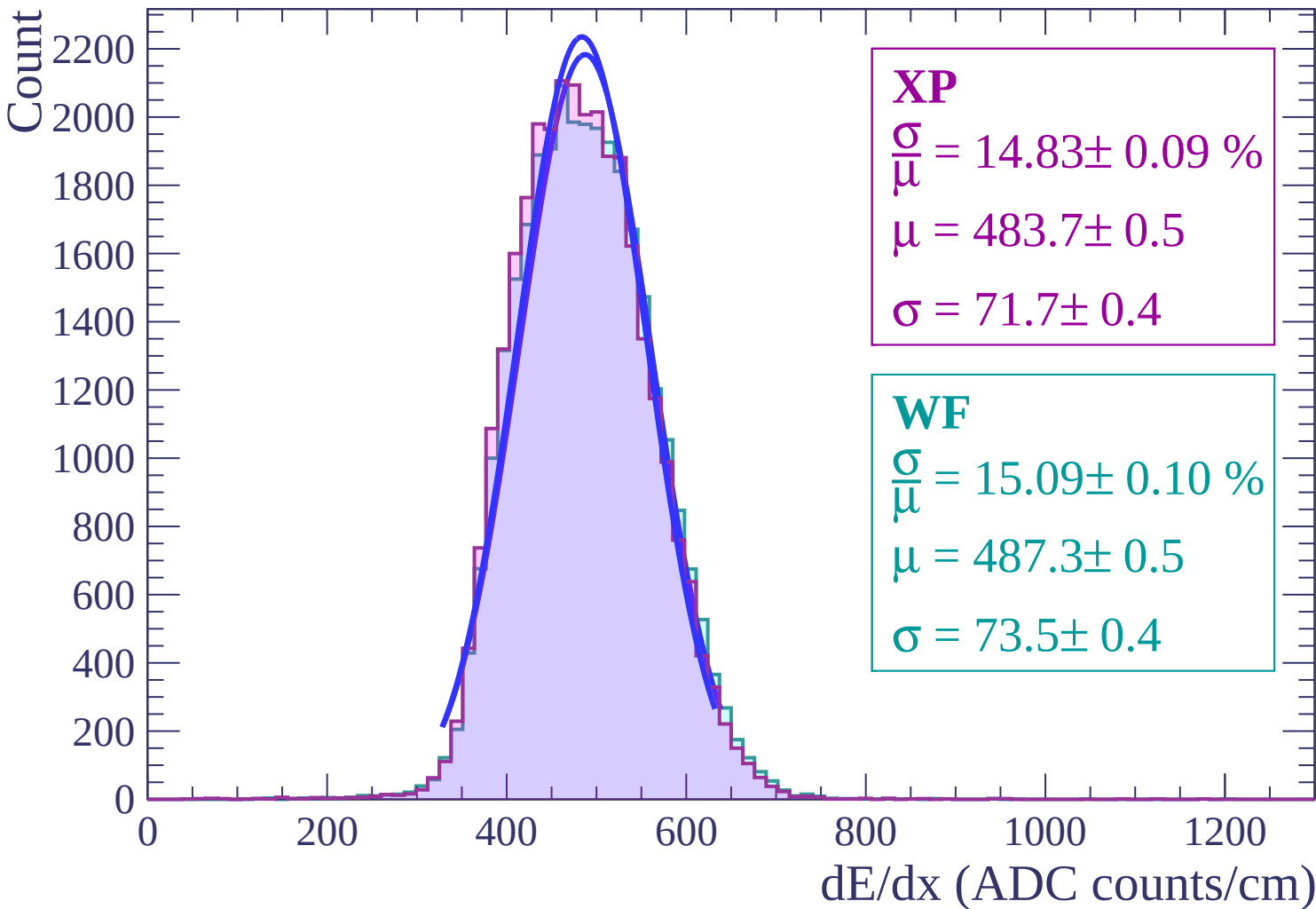
Energy loss | $-6 < \theta < -4$



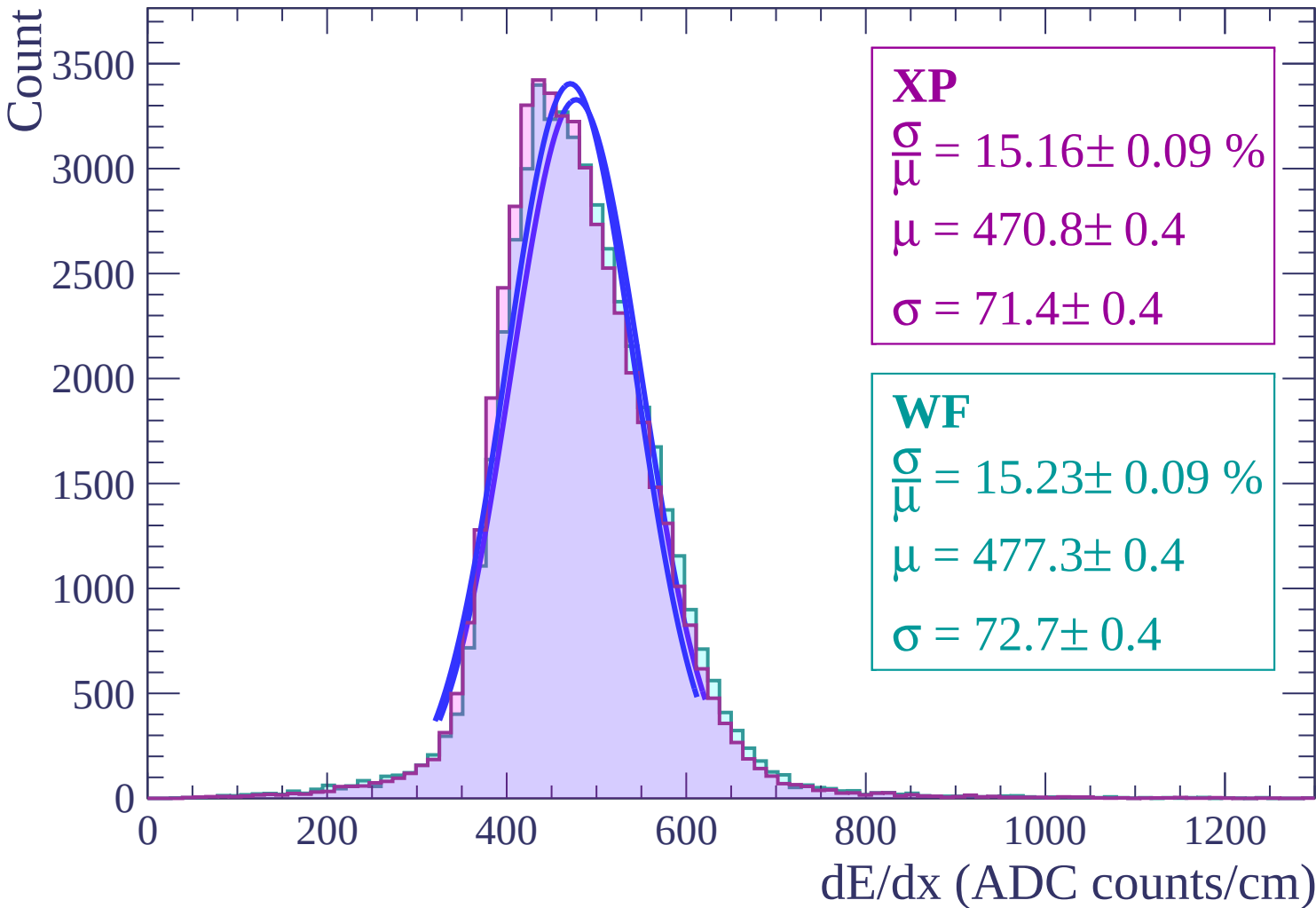
Energy loss | $-4 < \theta < -2$



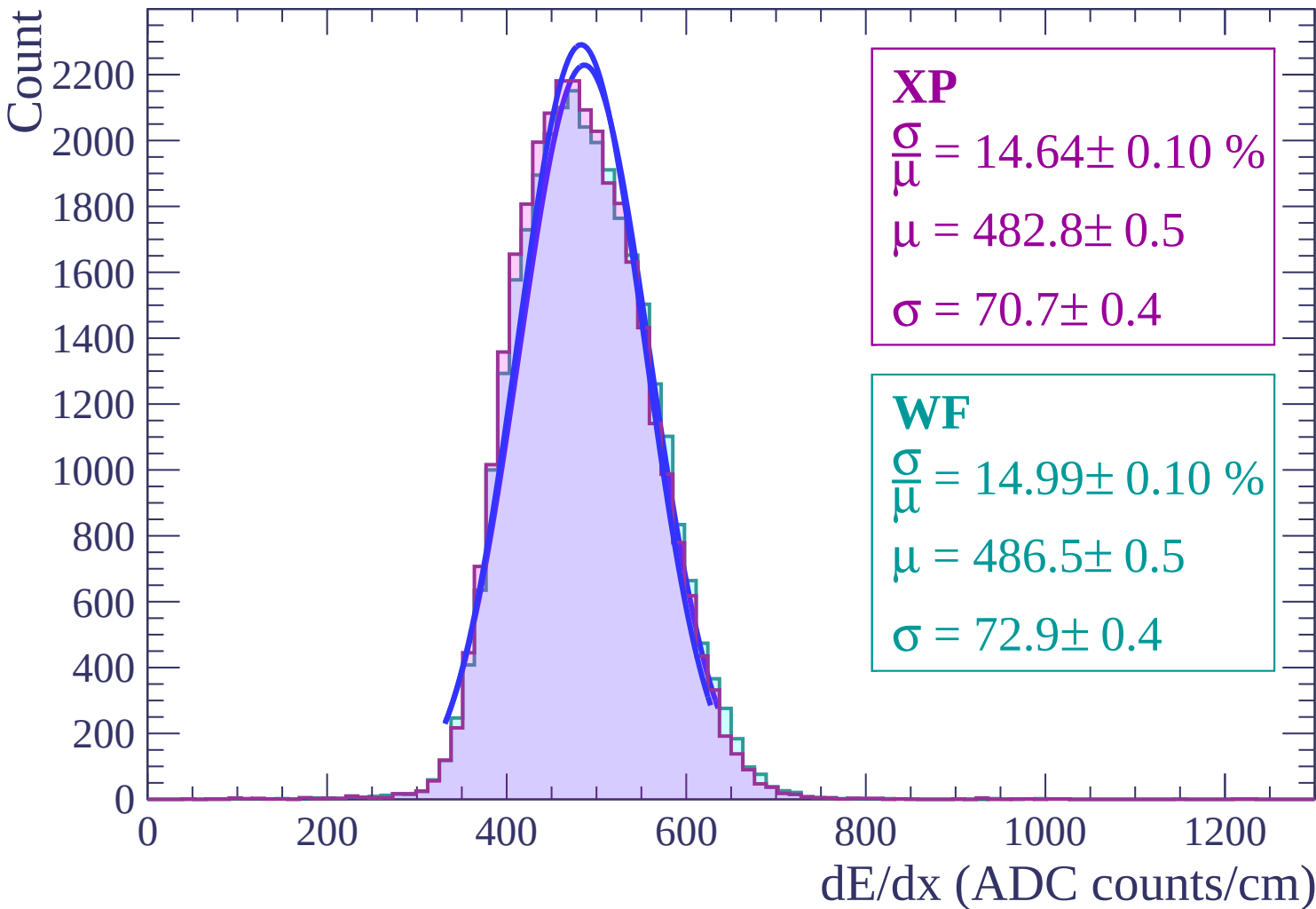
Energy loss $-2 < \theta < 0$



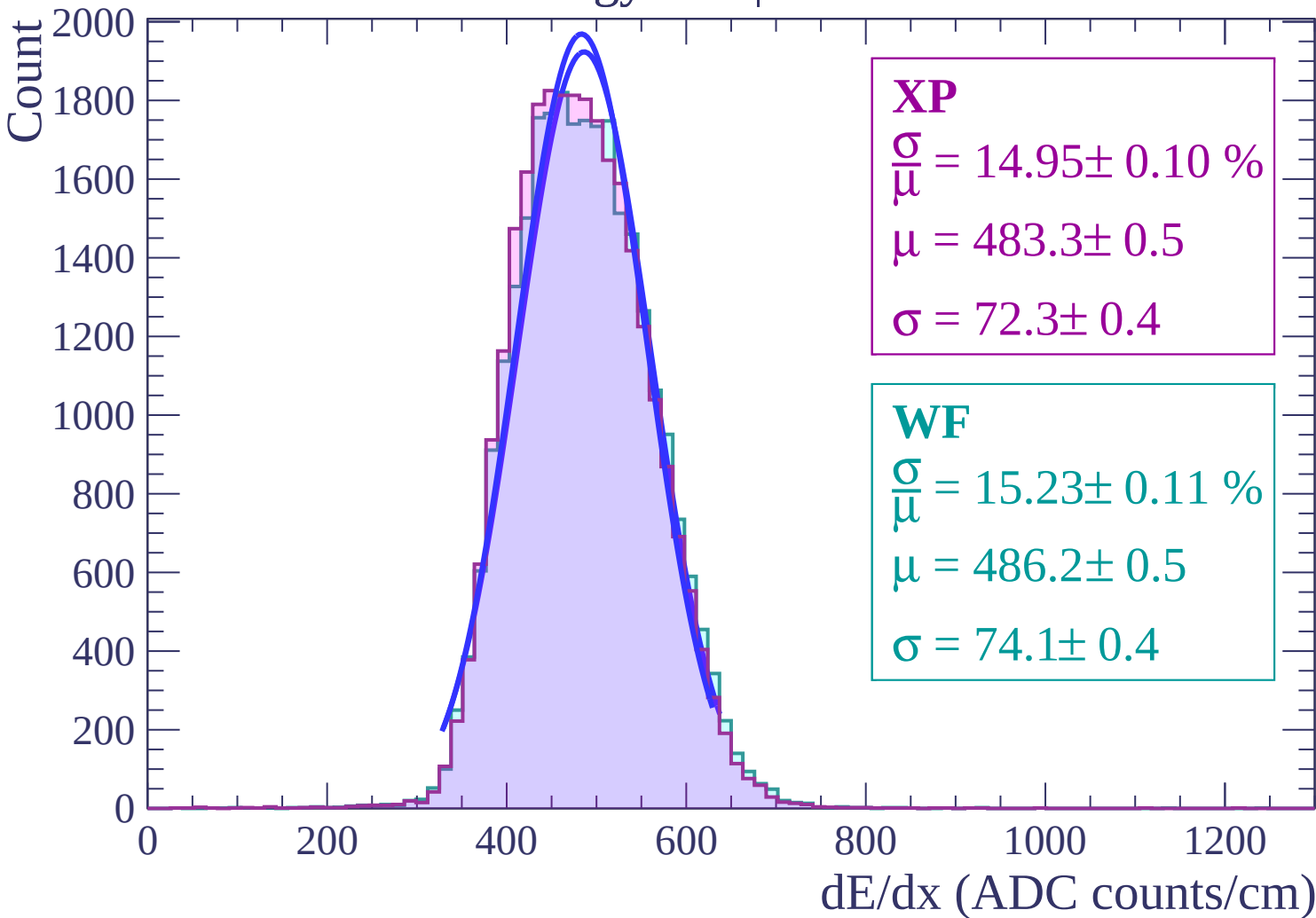
Energy loss | $0 < \theta < 2$



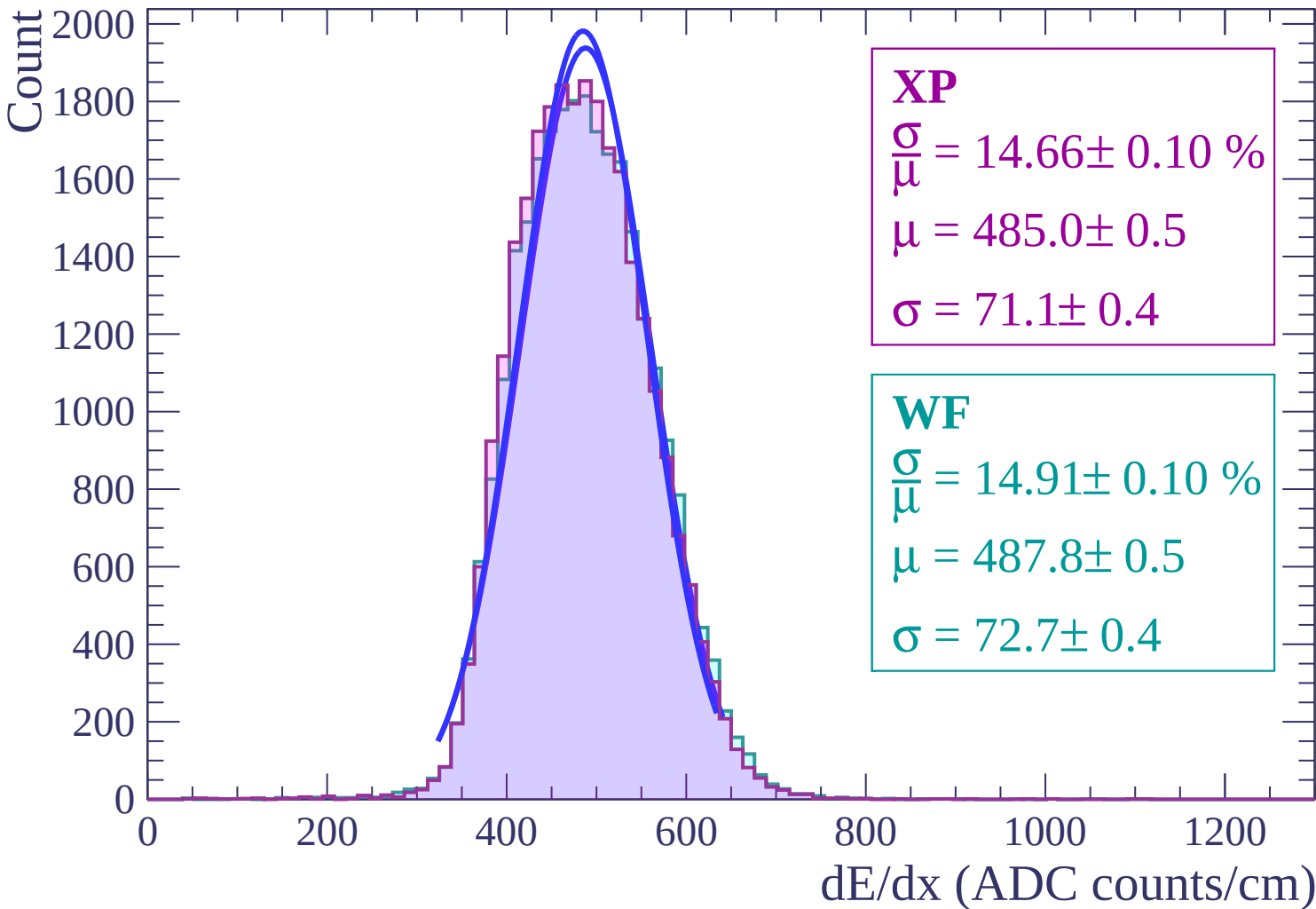
Energy loss | $2 < \theta < 4$



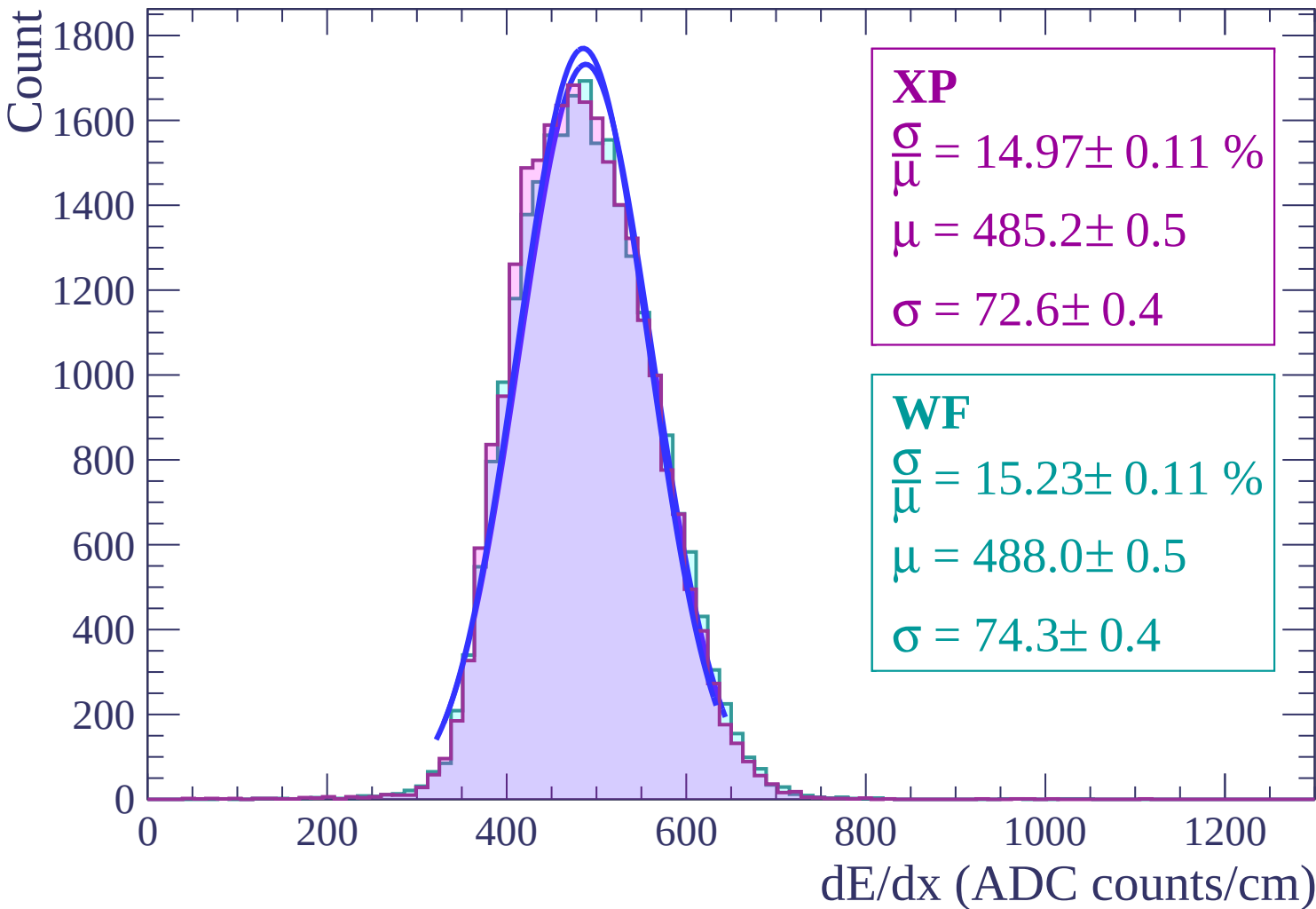
Energy loss | $4 < \theta < 6$



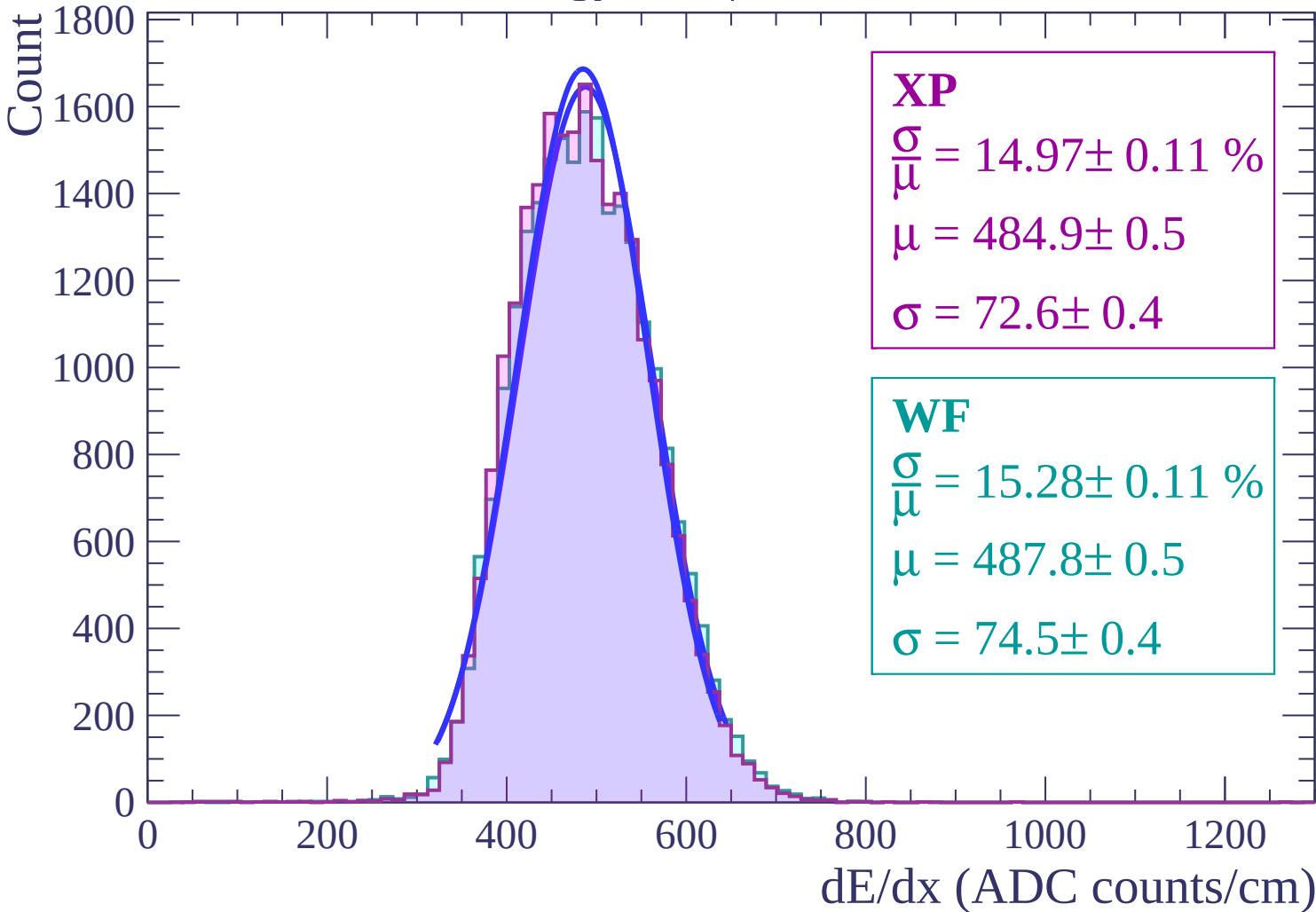
Energy loss | $6 < \theta < 8$



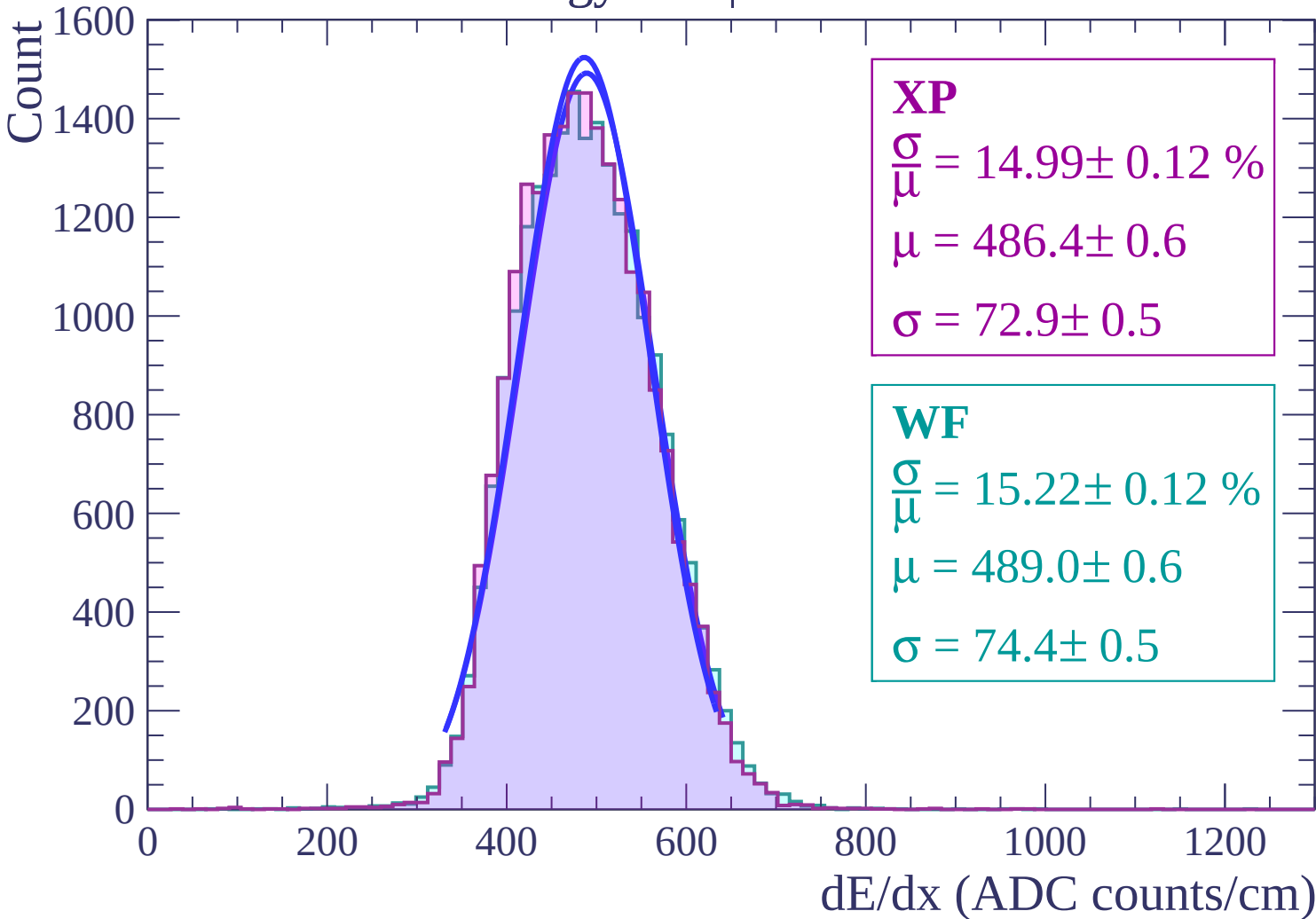
Energy loss | $8 < \theta < 10$



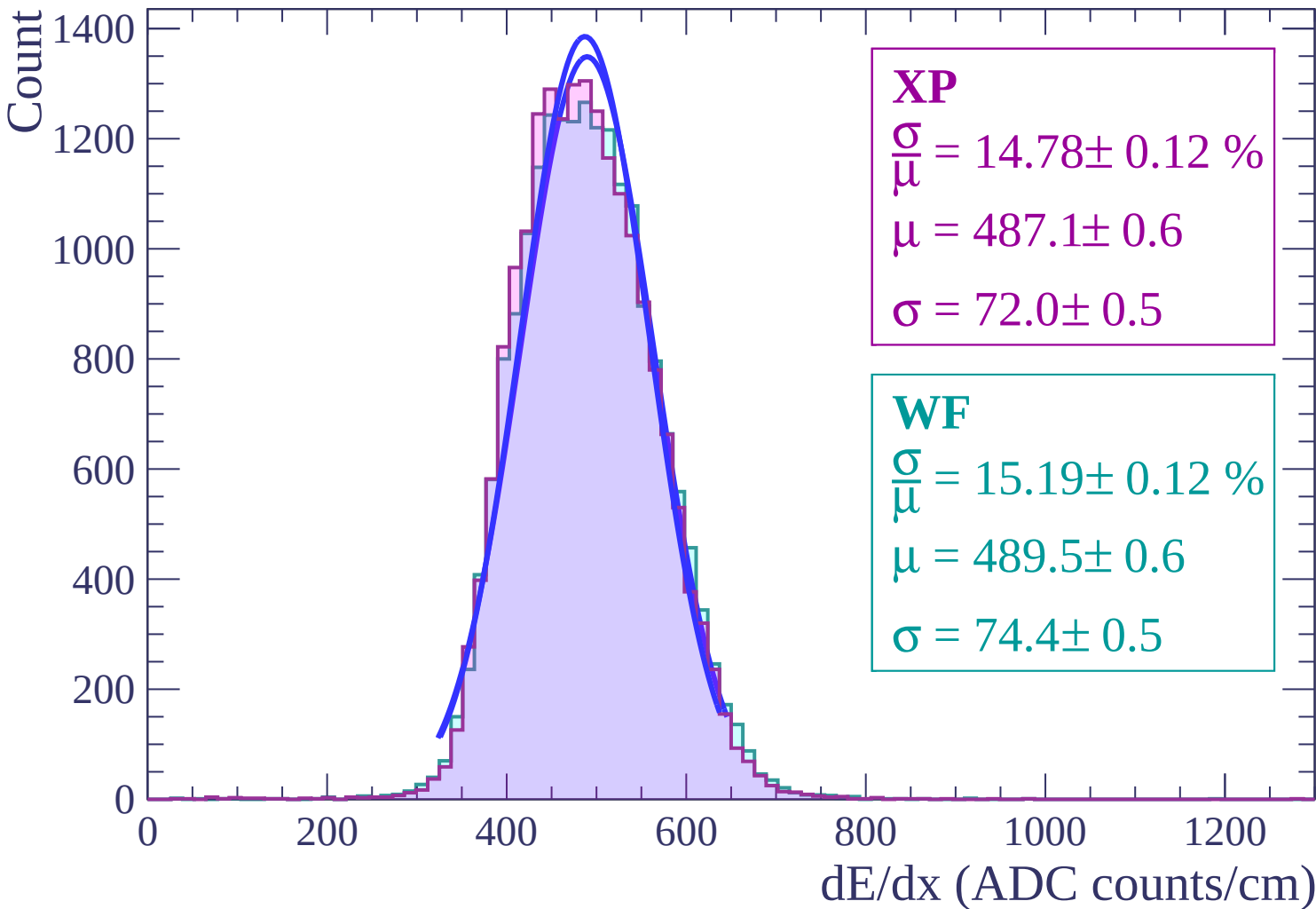
Energy loss | $10 < \theta < 12$



Energy loss | $12 < \theta < 14$



Energy loss | $14 < \theta < 16$



Energy loss | $16 < \theta < 18$

Count

1200

1000

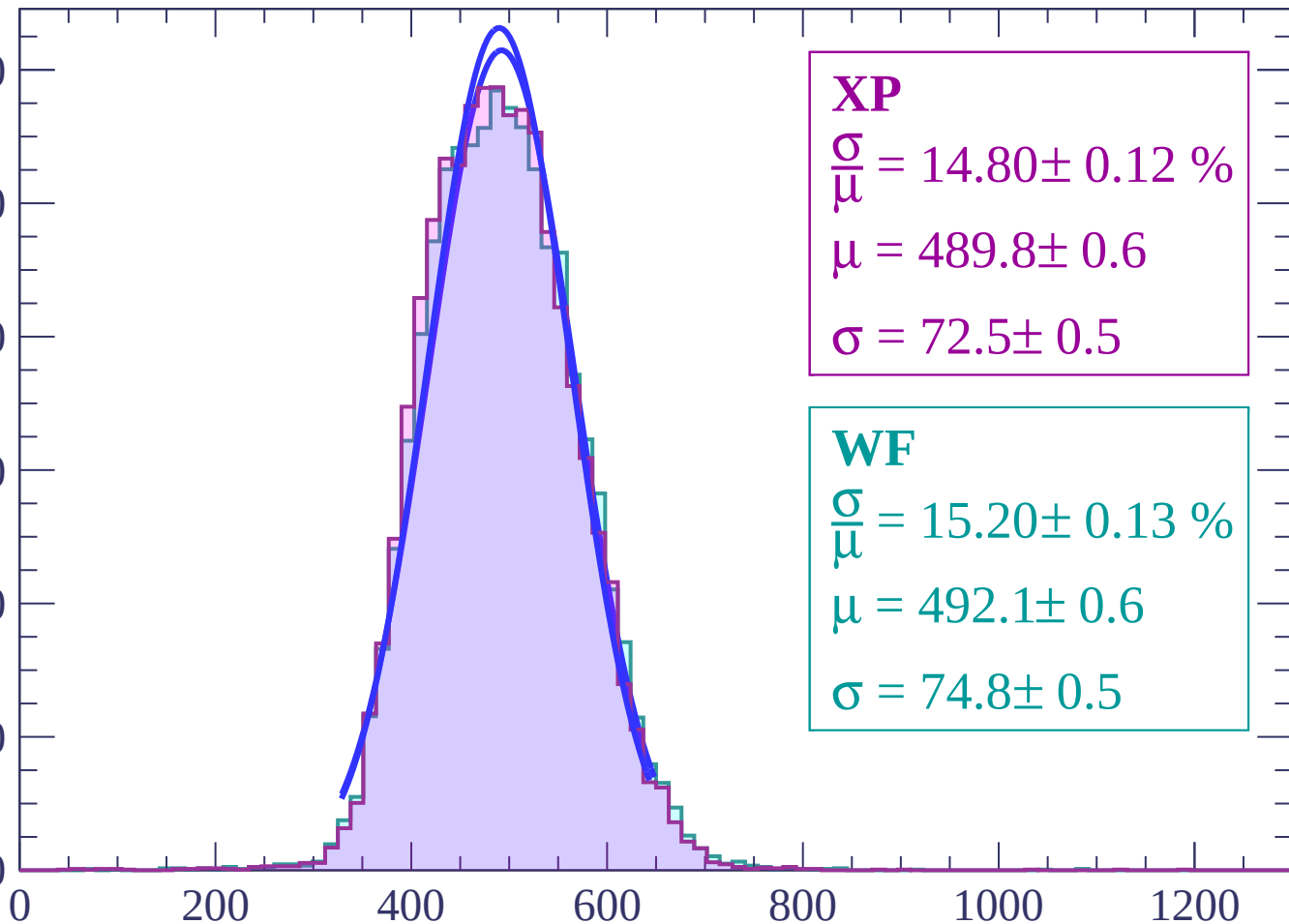
800

600

400

200

0



XP

$$\frac{\sigma}{\mu} = 14.80 \pm 0.12 \%$$

$$\mu = 489.8 \pm 0.6$$

$$\sigma = 72.5 \pm 0.5$$

WF

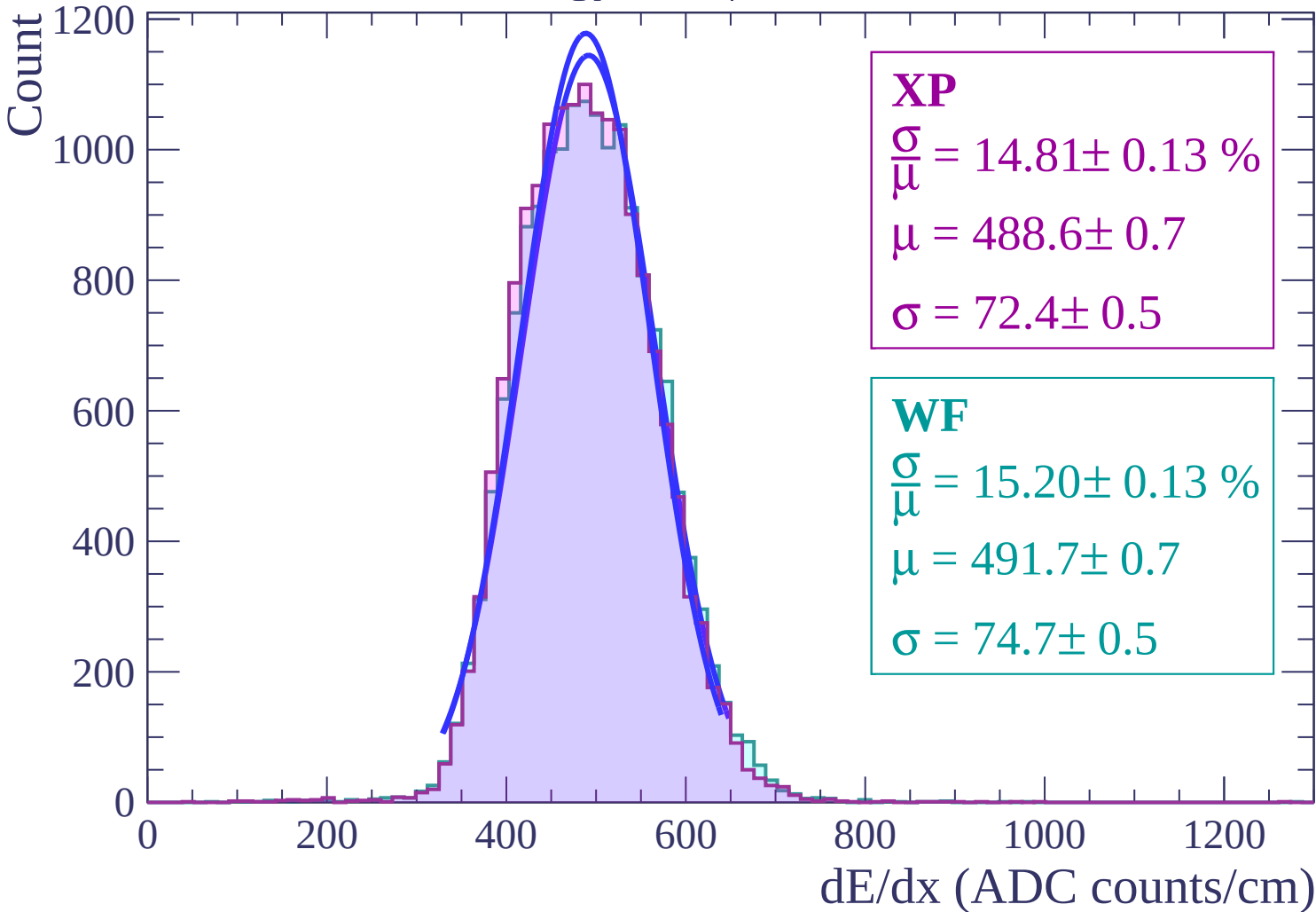
$$\frac{\sigma}{\mu} = 15.20 \pm 0.13 \%$$

$$\mu = 492.1 \pm 0.6$$

$$\sigma = 74.8 \pm 0.5$$

dE/dx (ADC counts/cm)

Energy loss | $18 < \theta < 20$



Energy loss | $20 < \theta < 22$

Count

1000

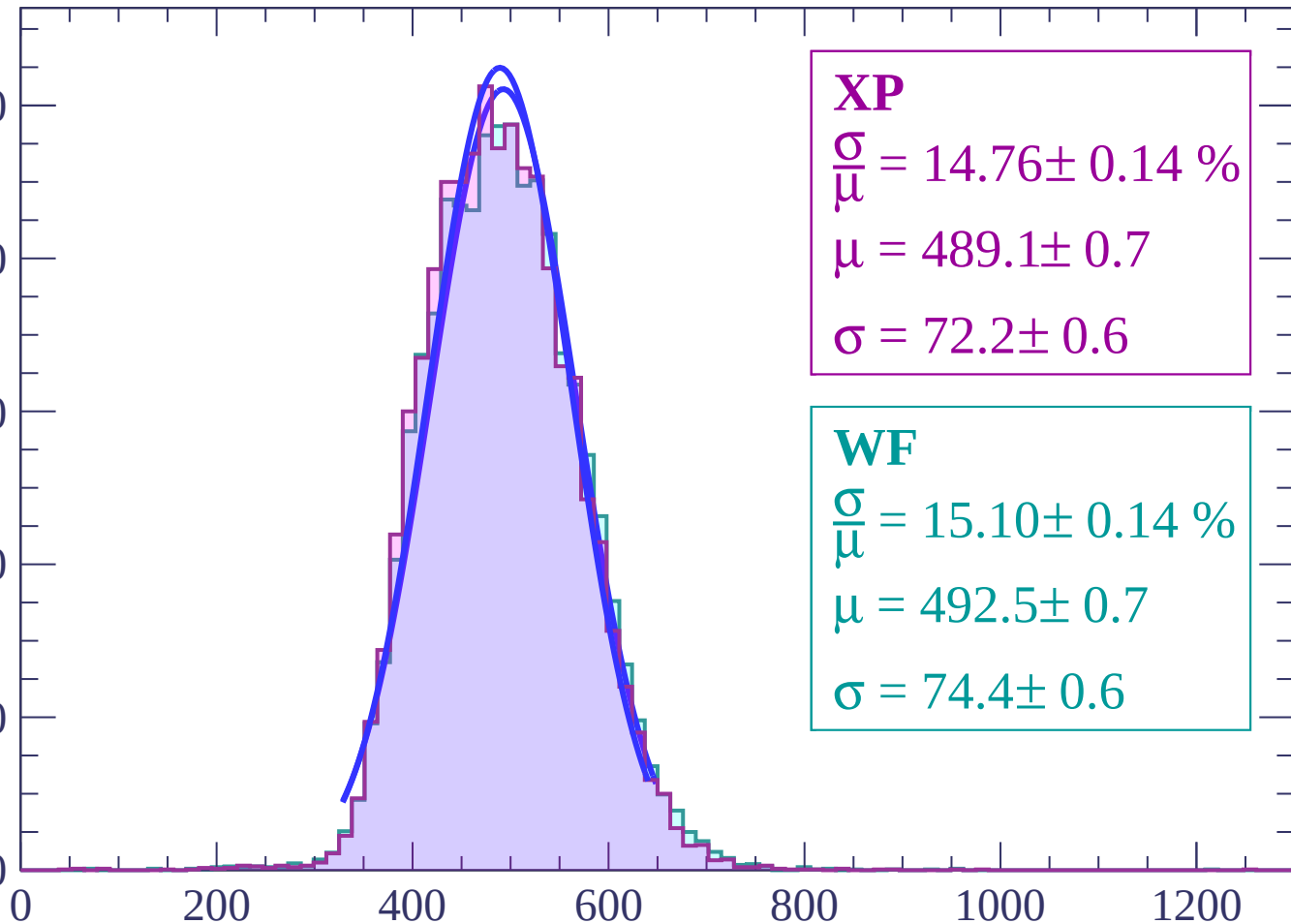
800

600

400

200

0



XP

$$\frac{\sigma}{\mu} = 14.76 \pm 0.14 \%$$

$$\mu = 489.1 \pm 0.7$$

$$\sigma = 72.2 \pm 0.6$$

WF

$$\frac{\sigma}{\mu} = 15.10 \pm 0.14 \%$$

$$\mu = 492.5 \pm 0.7$$

$$\sigma = 74.4 \pm 0.6$$

dE/dx (ADC counts/cm)

Energy loss | $22 < \theta < 24$

Count

1000

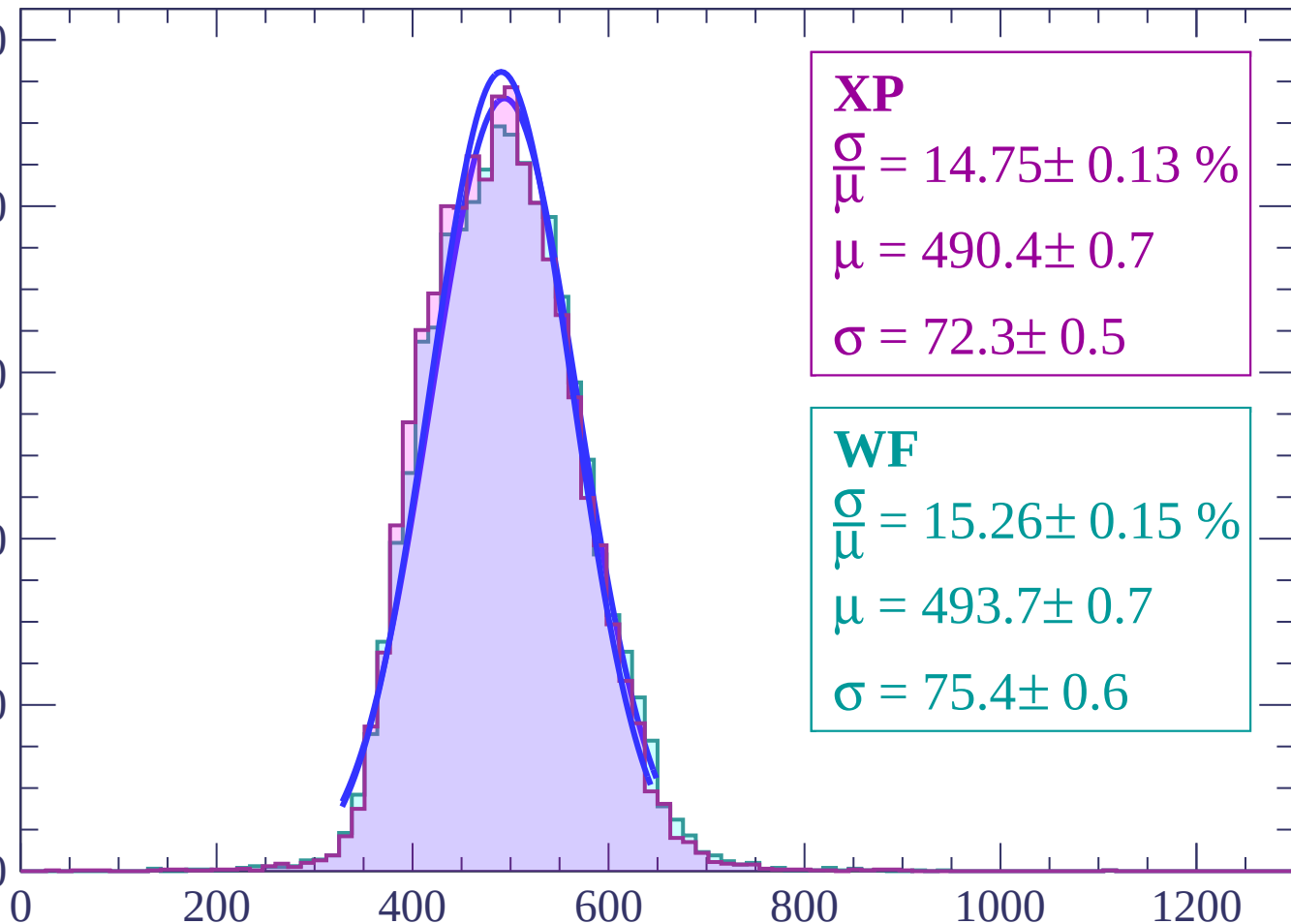
800

600

400

200

0



XP

$$\frac{\sigma}{\mu} = 14.75 \pm 0.13 \%$$

$$\mu = 490.4 \pm 0.7$$

$$\sigma = 72.3 \pm 0.5$$

WF

$$\frac{\sigma}{\mu} = 15.26 \pm 0.15 \%$$

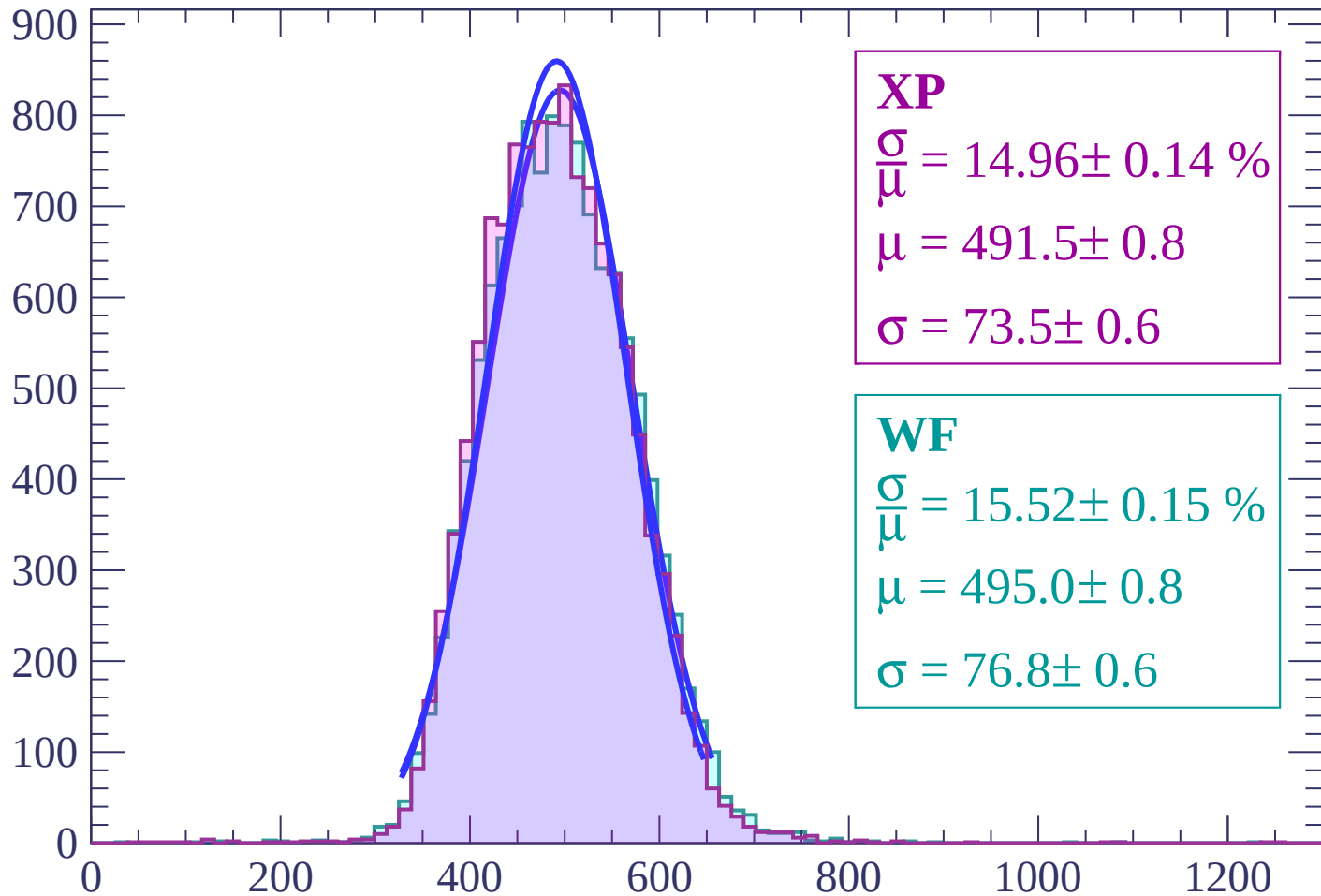
$$\mu = 493.7 \pm 0.7$$

$$\sigma = 75.4 \pm 0.6$$

dE/dx (ADC counts/cm)

Energy loss | $24 < \theta < 26$

Count



XP

$$\frac{\sigma}{\mu} = 14.96 \pm 0.14 \%$$

$$\mu = 491.5 \pm 0.8$$

$$\sigma = 73.5 \pm 0.6$$

WF

$$\frac{\sigma}{\mu} = 15.52 \pm 0.15 \%$$

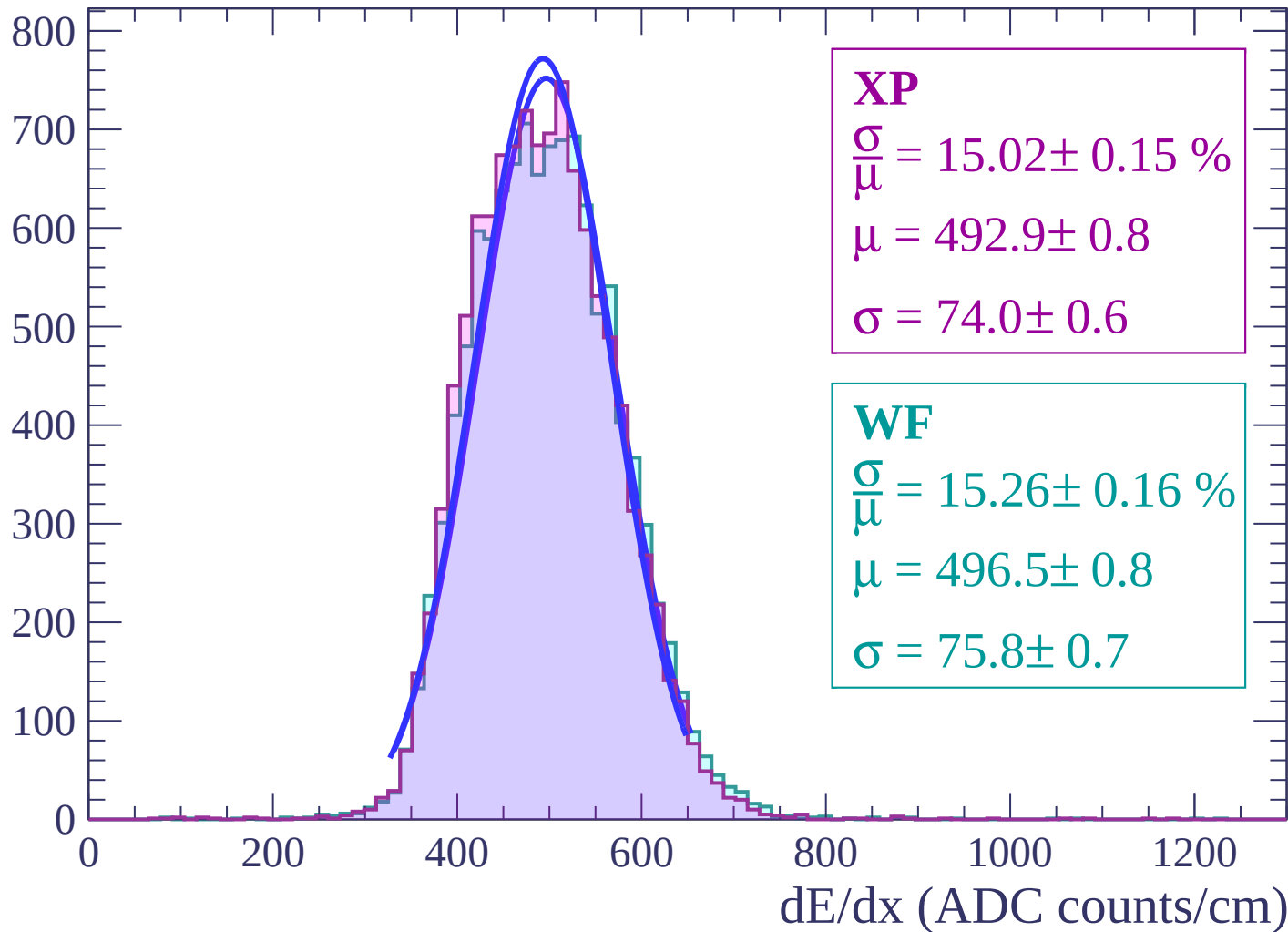
$$\mu = 495.0 \pm 0.8$$

$$\sigma = 76.8 \pm 0.6$$

dE/dx (ADC counts/cm)

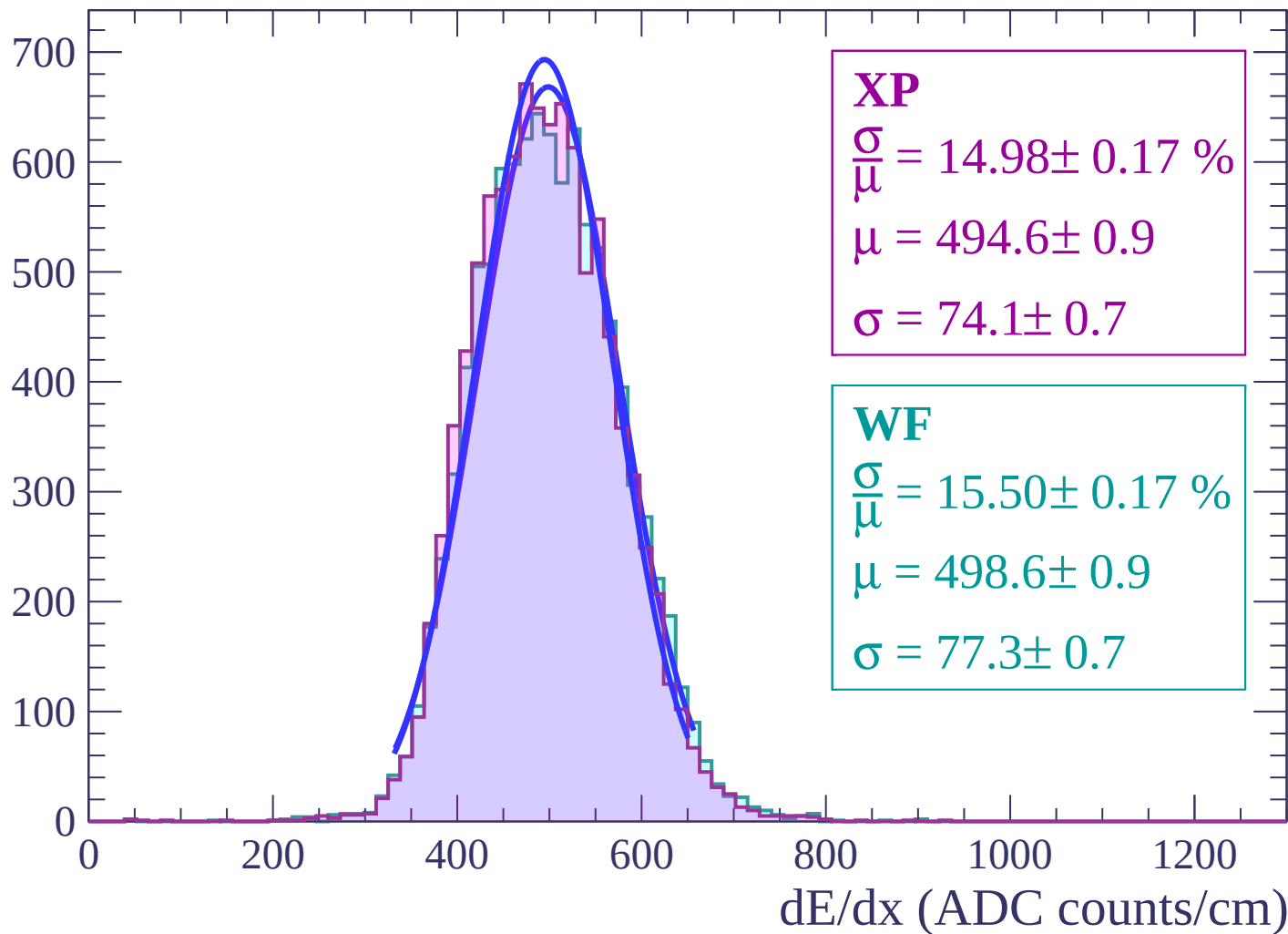
Energy loss | $26 < \theta < 28$

Count



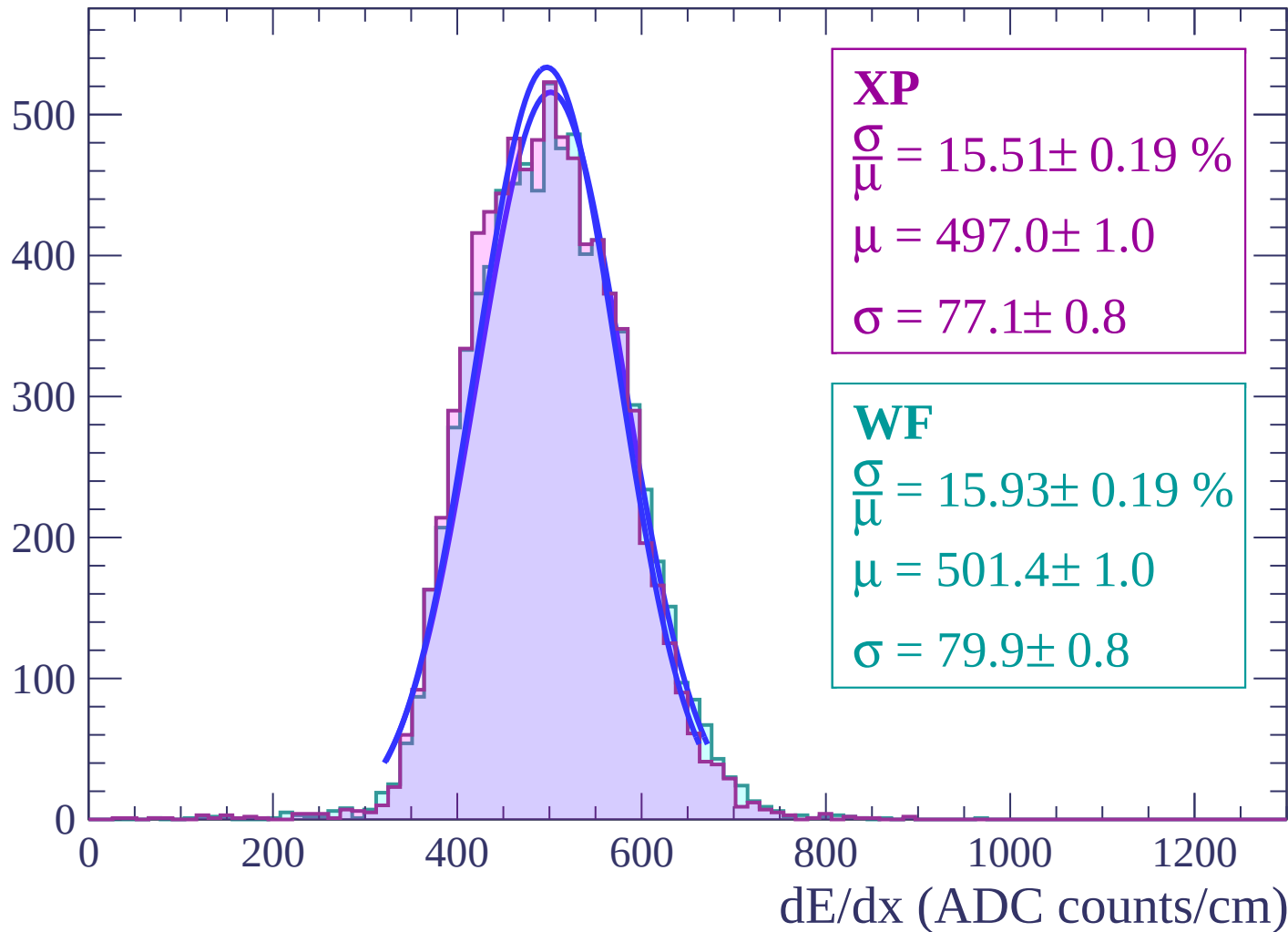
Energy loss | $28 < \theta < 30$

Count



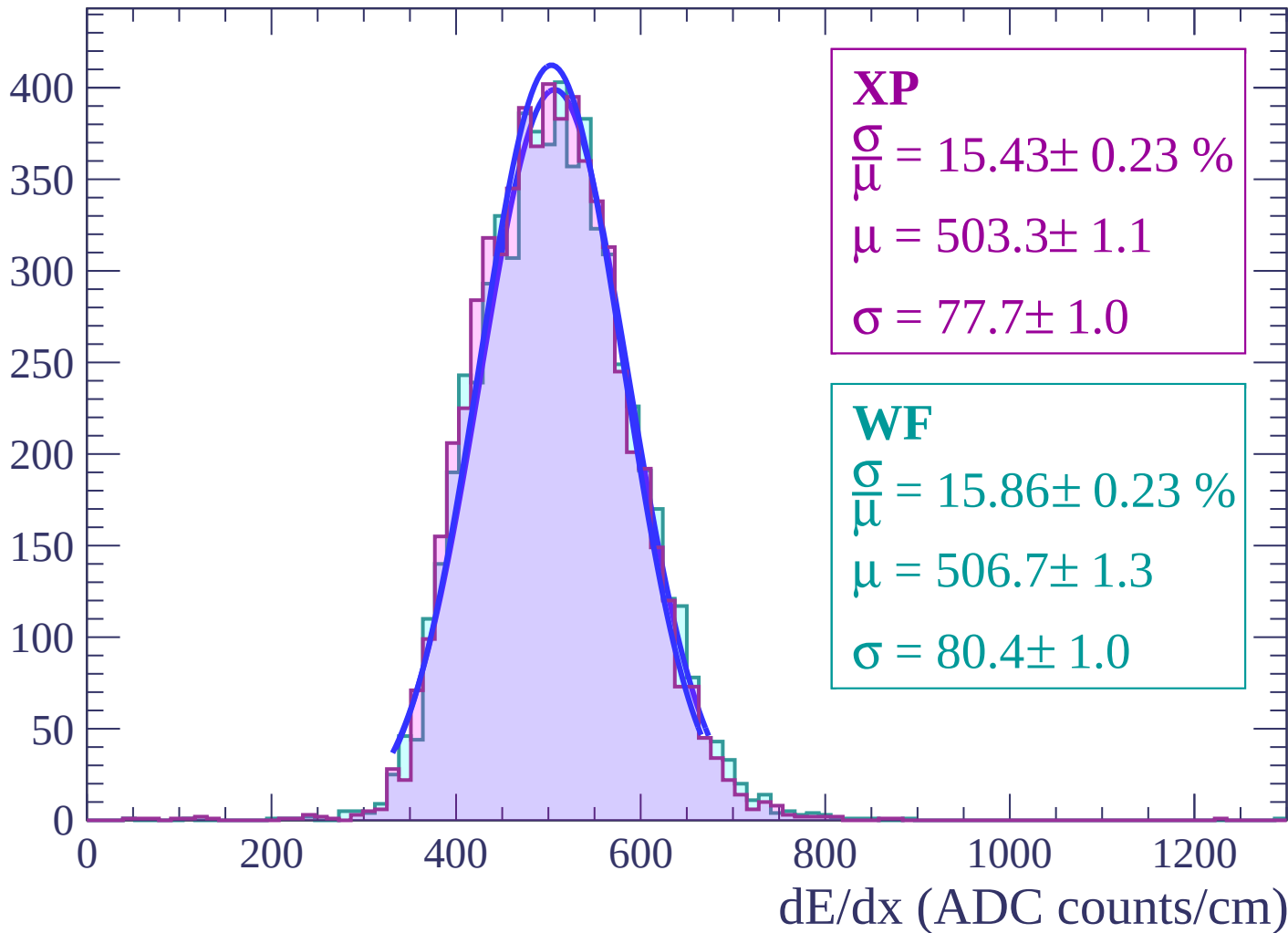
Energy loss | $30 < \theta < 32$

Count



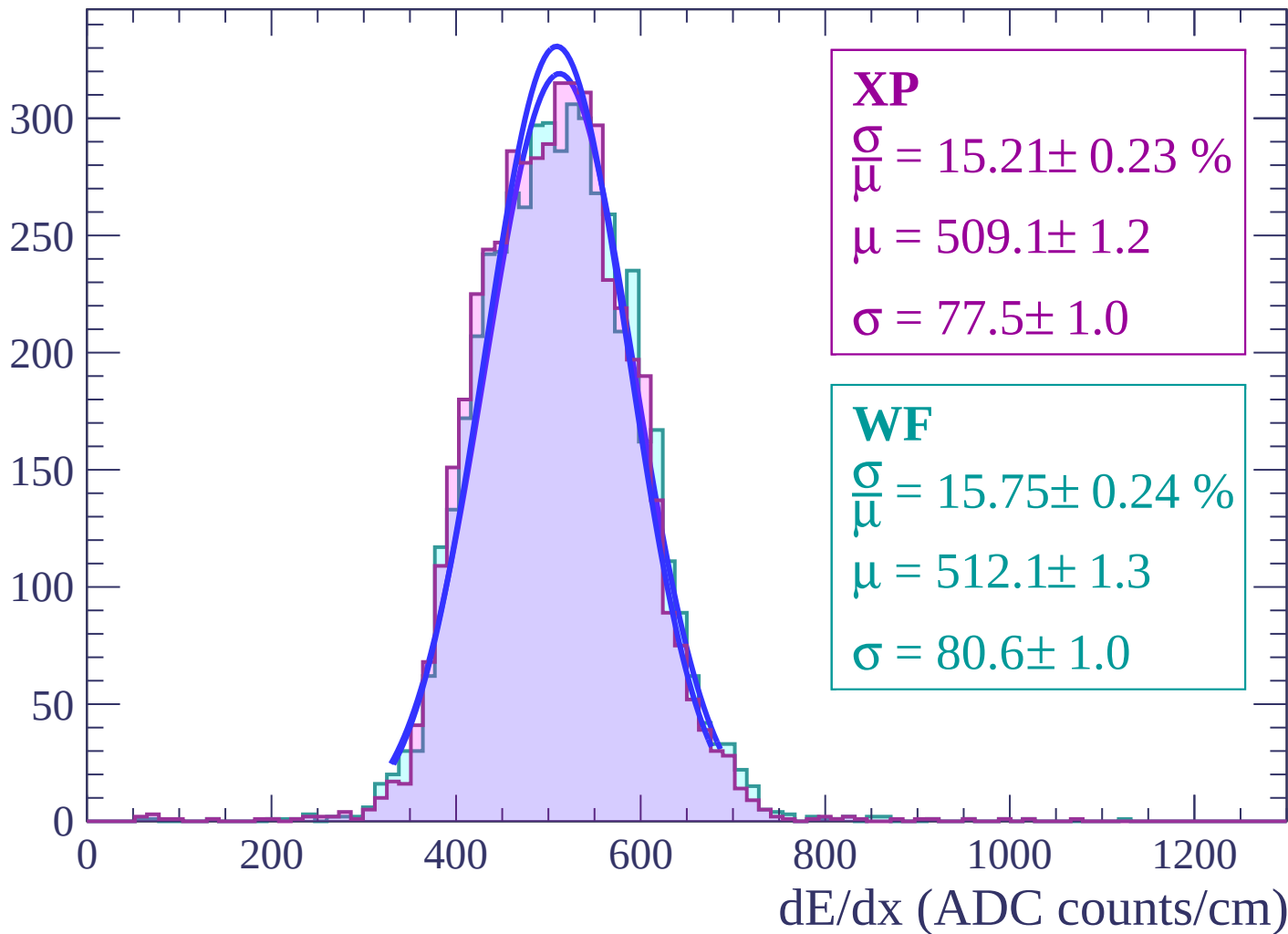
Energy loss | $32 < \theta < 34$

Count



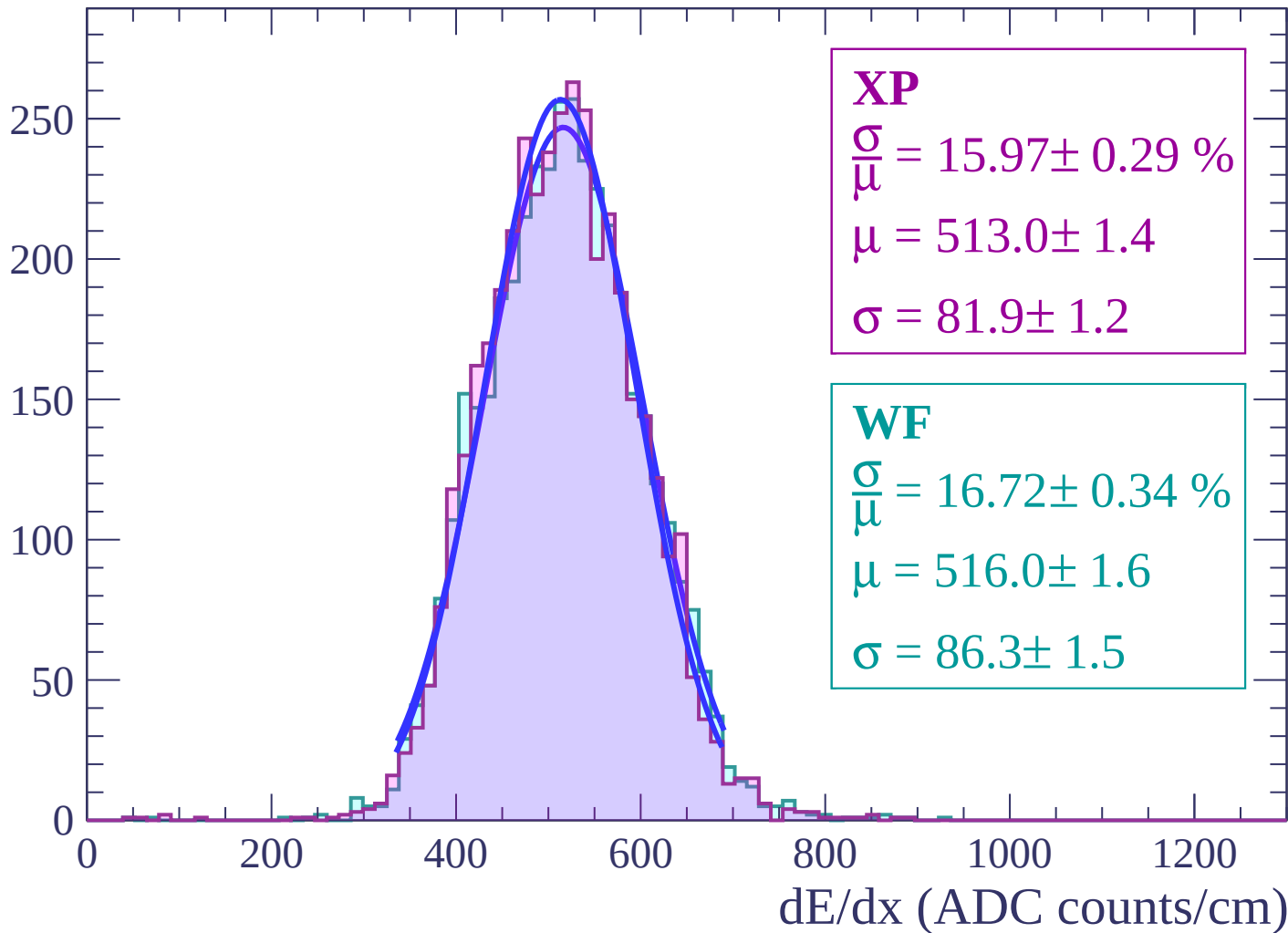
Energy loss | $34 < \theta < 36$

Count



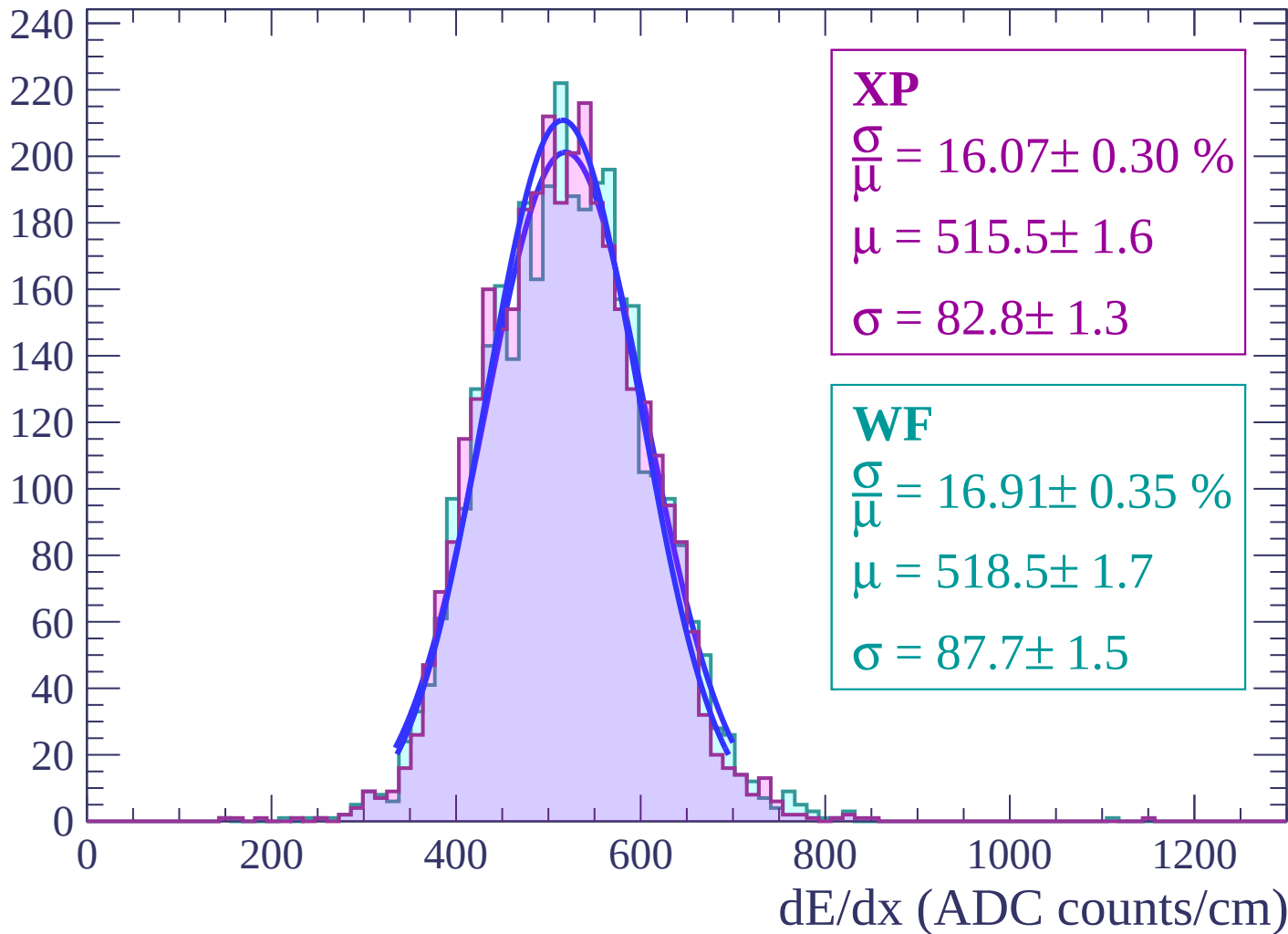
Energy loss | $36 < \theta < 38$

Count



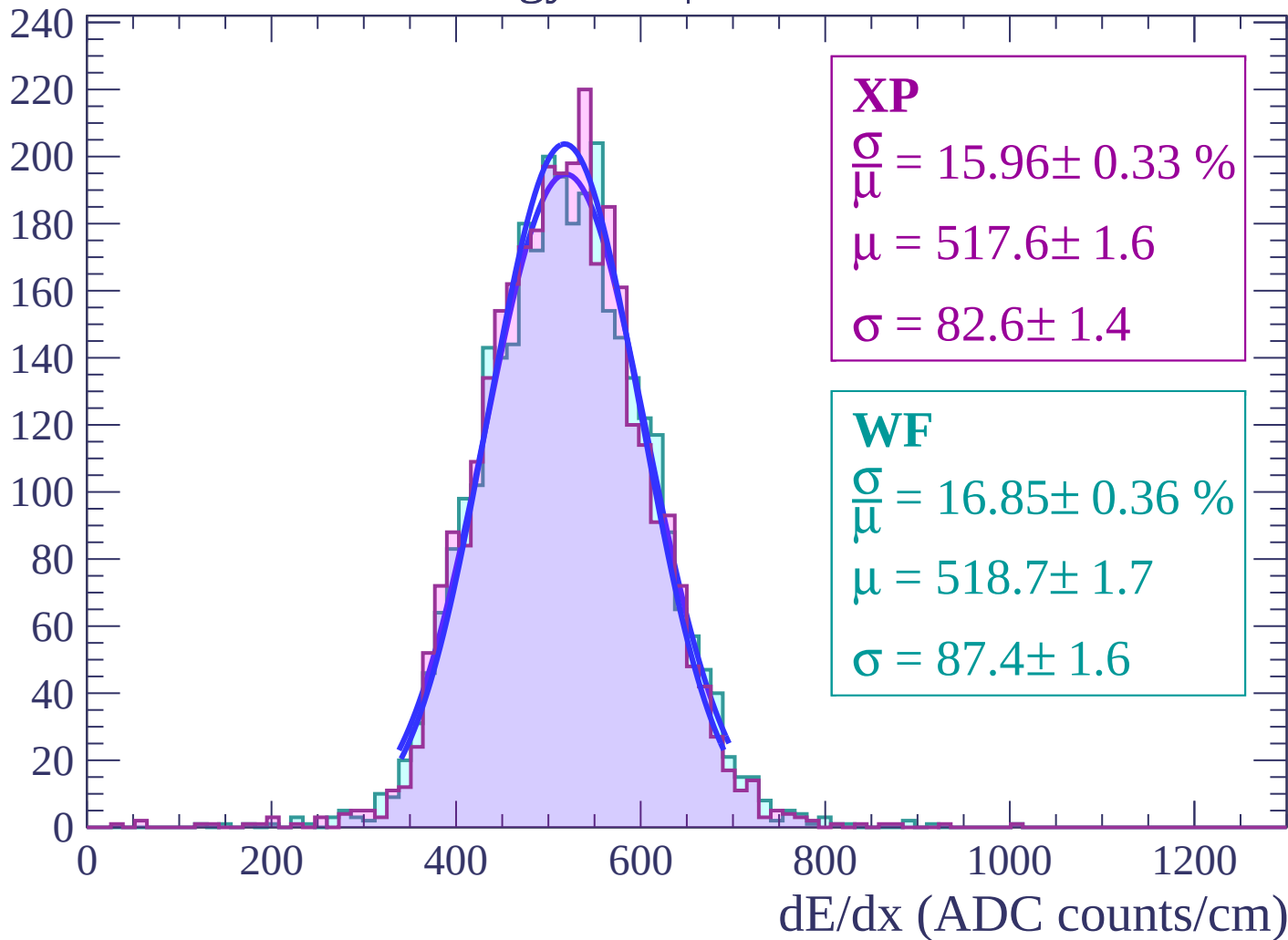
Energy loss | $38 < \theta < 40$

Count



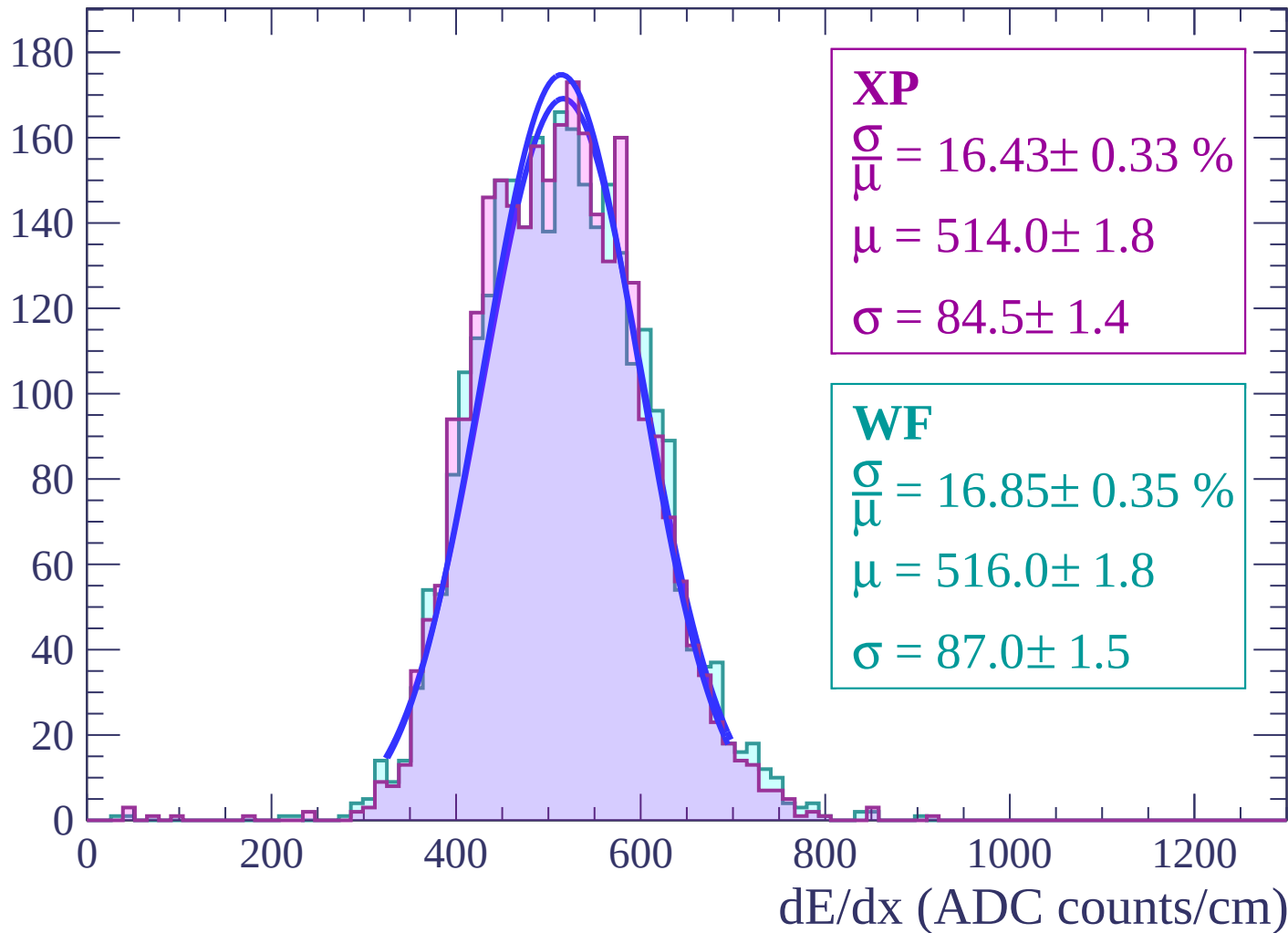
Energy loss | $40 < \theta < 42$

Count



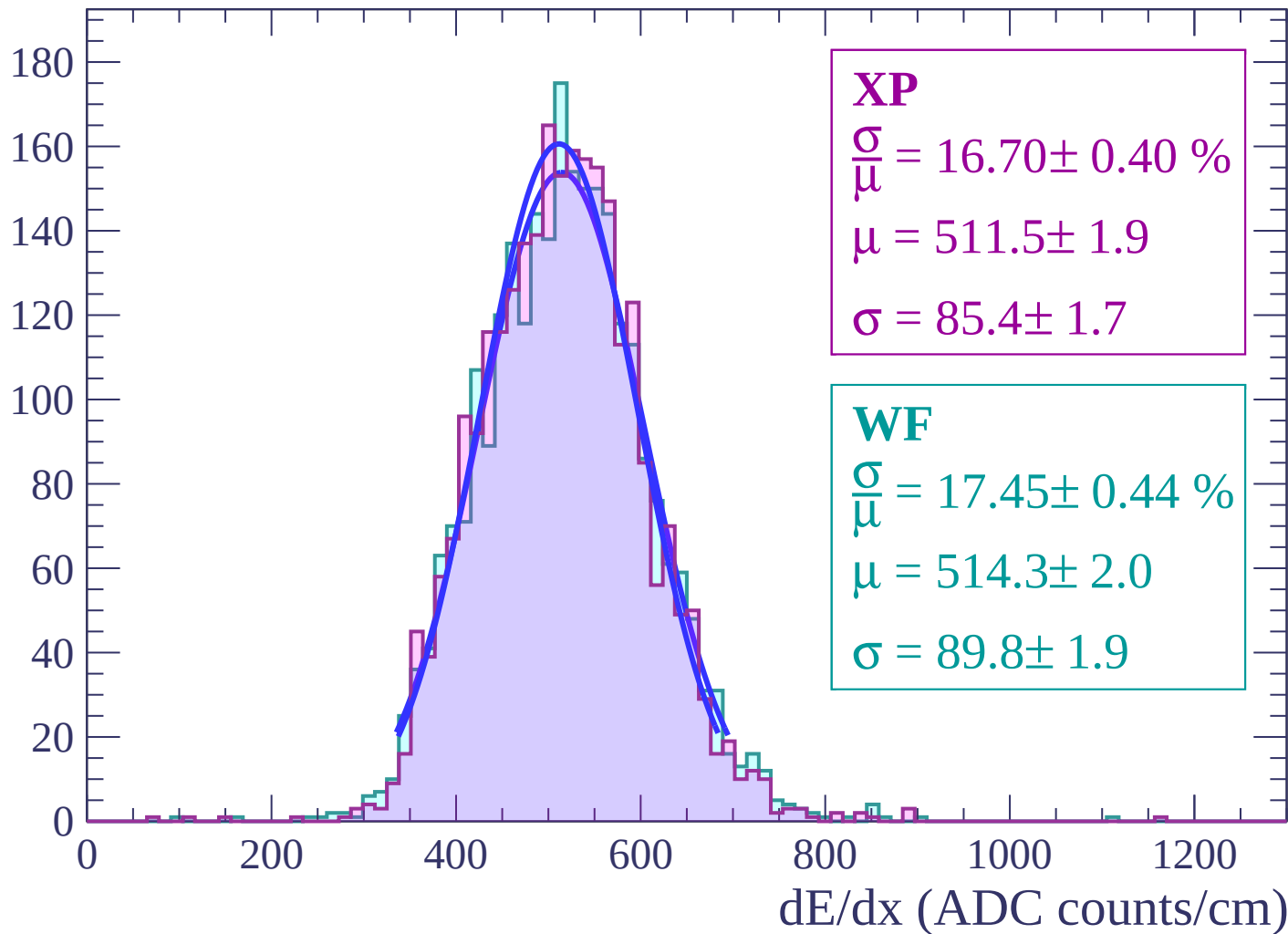
Energy loss | $42 < \theta < 44$

Count



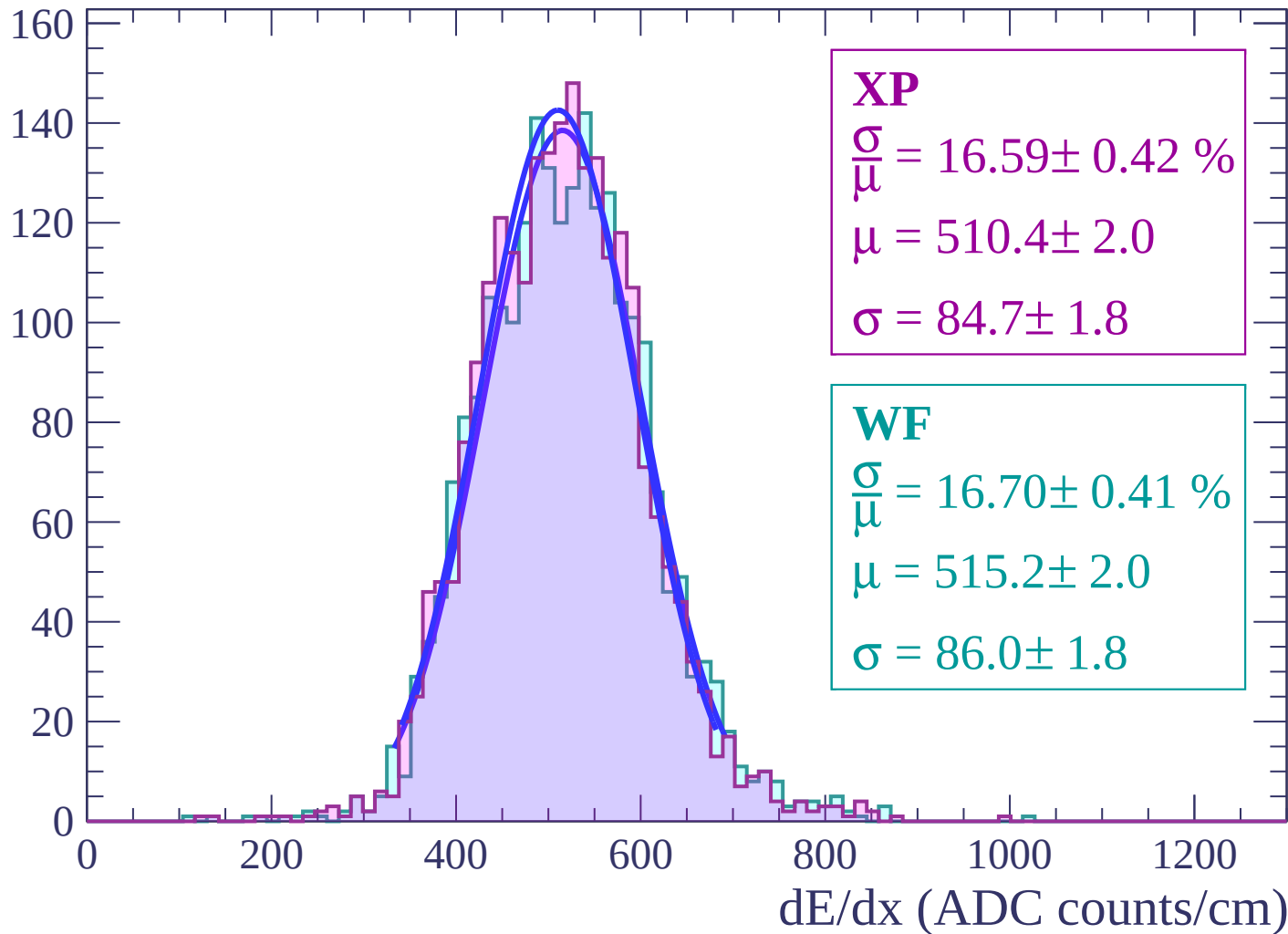
Energy loss | $44 < \theta < 46$

Count



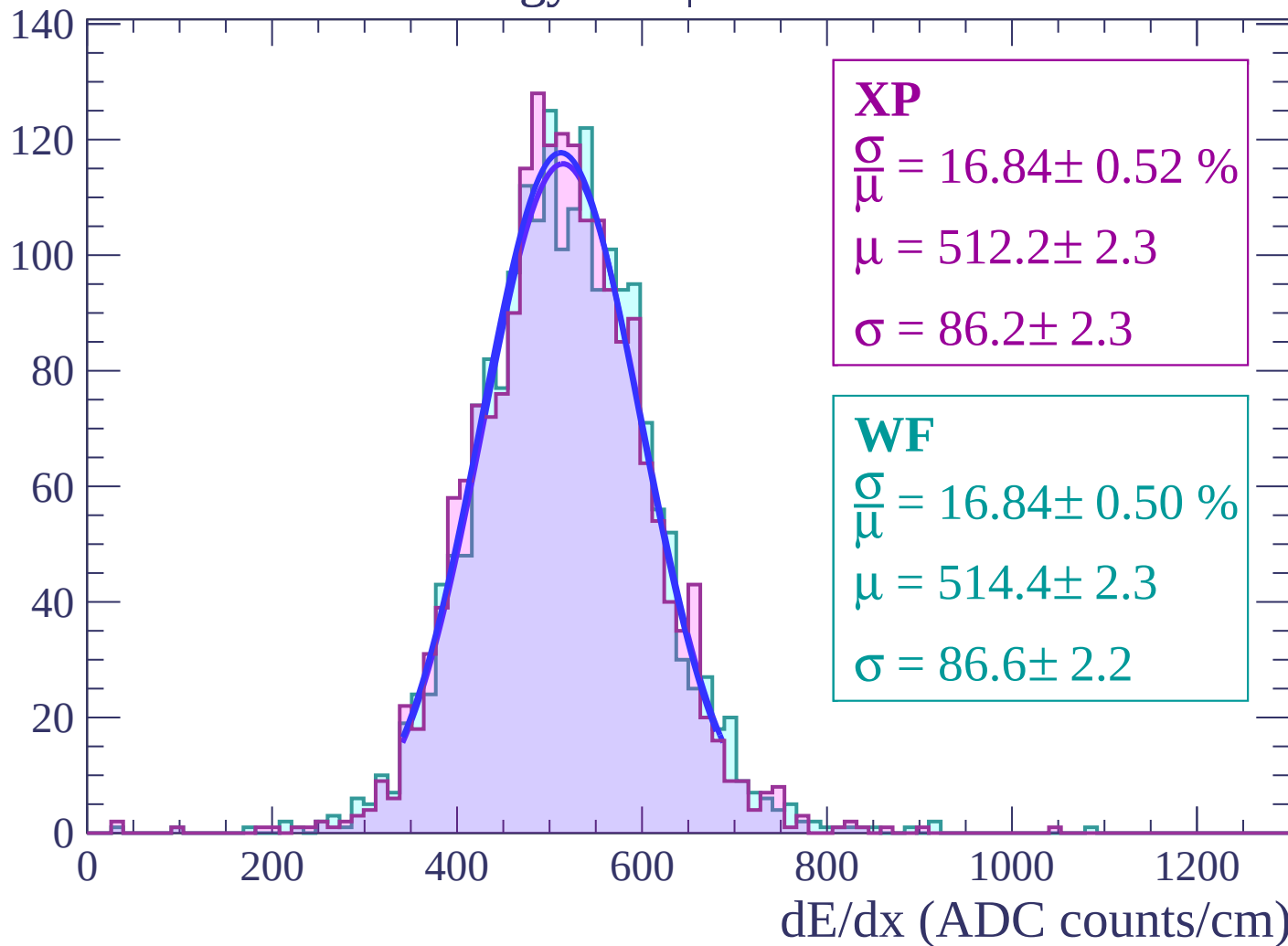
Energy loss | $46 < \theta < 48$

Count



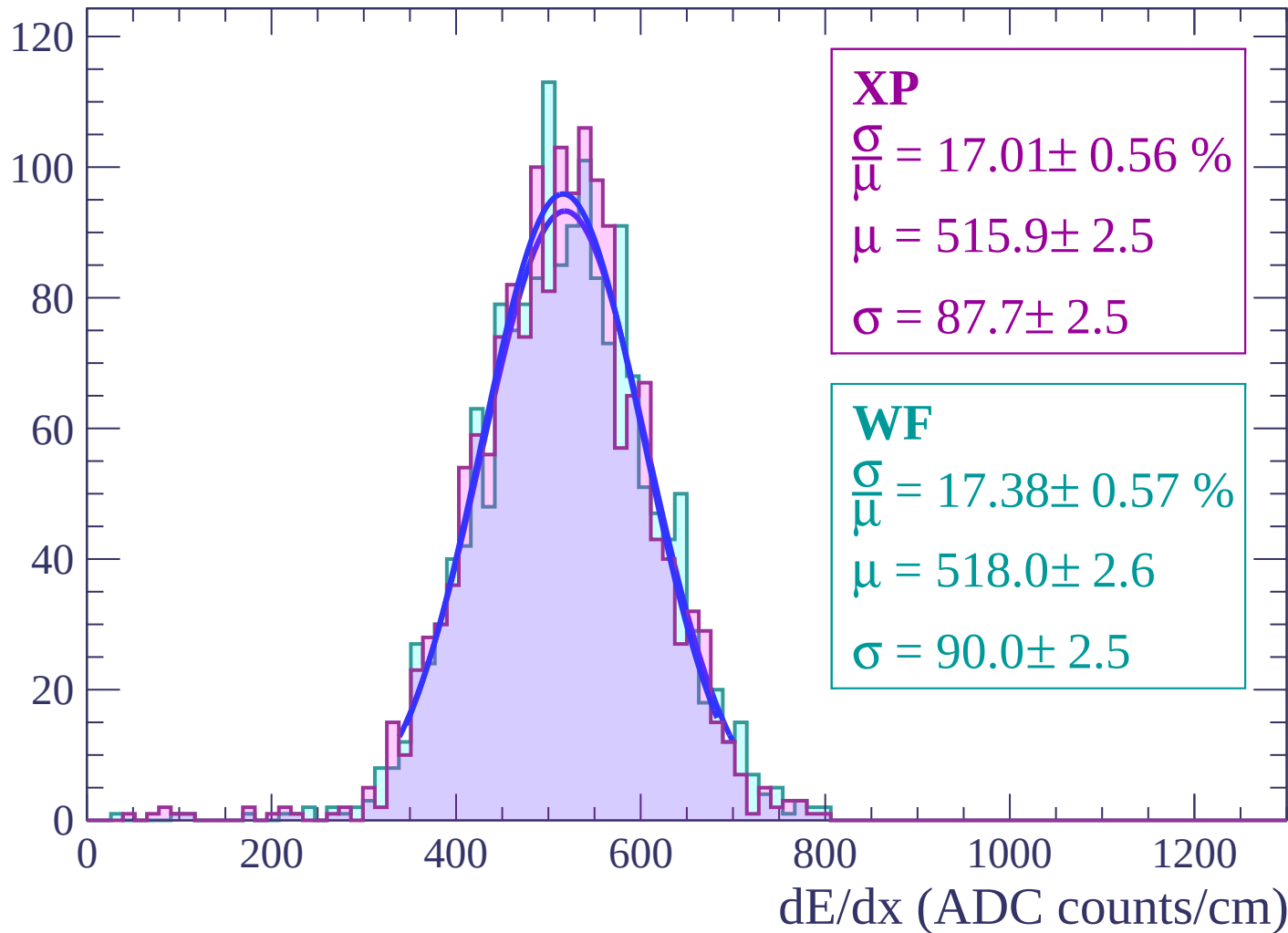
Energy loss | $48 < \theta < 50$

Count



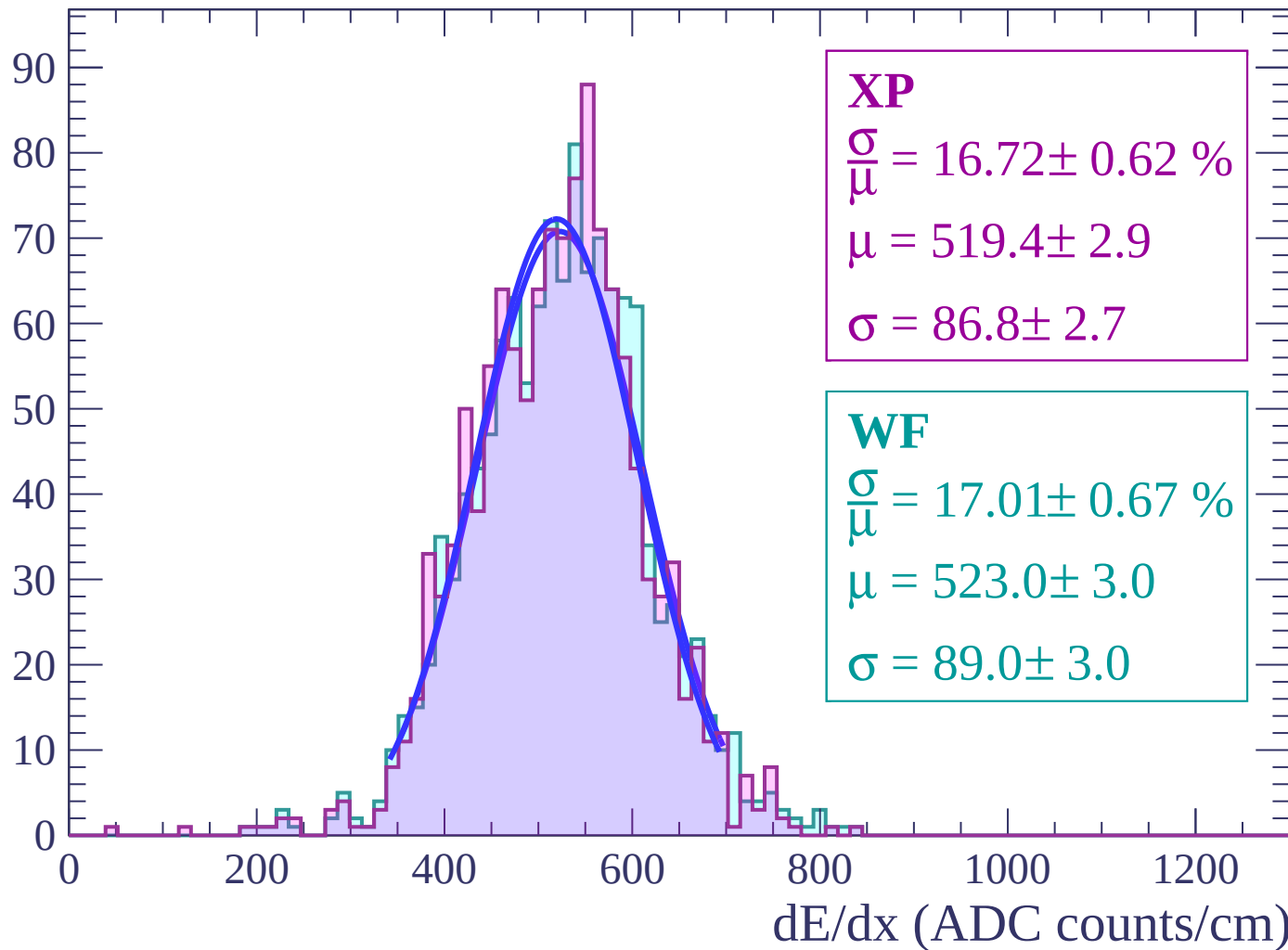
Energy loss | $50 < \theta < 52$

Count



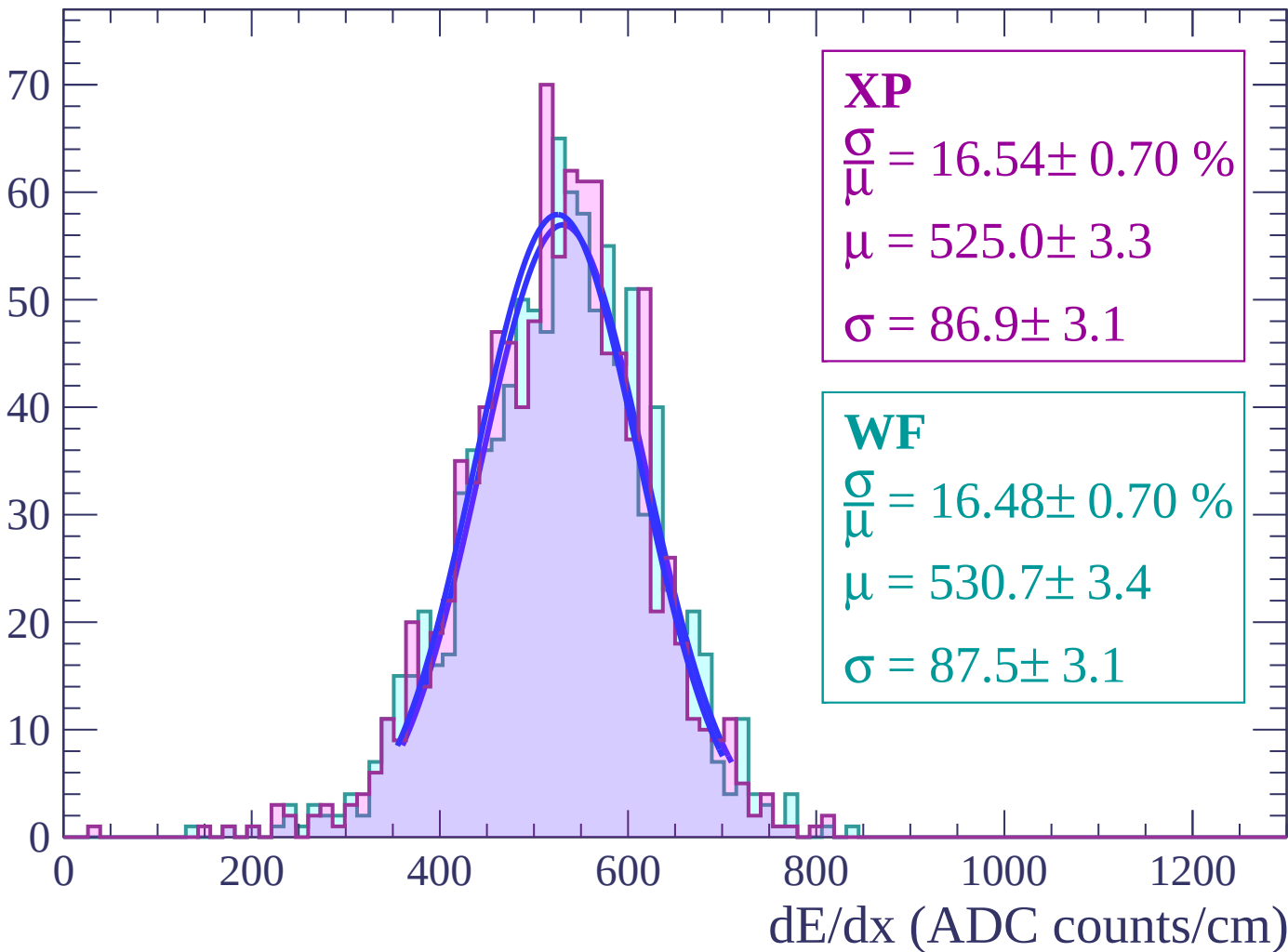
Energy loss | $52 < \theta < 54$

Count



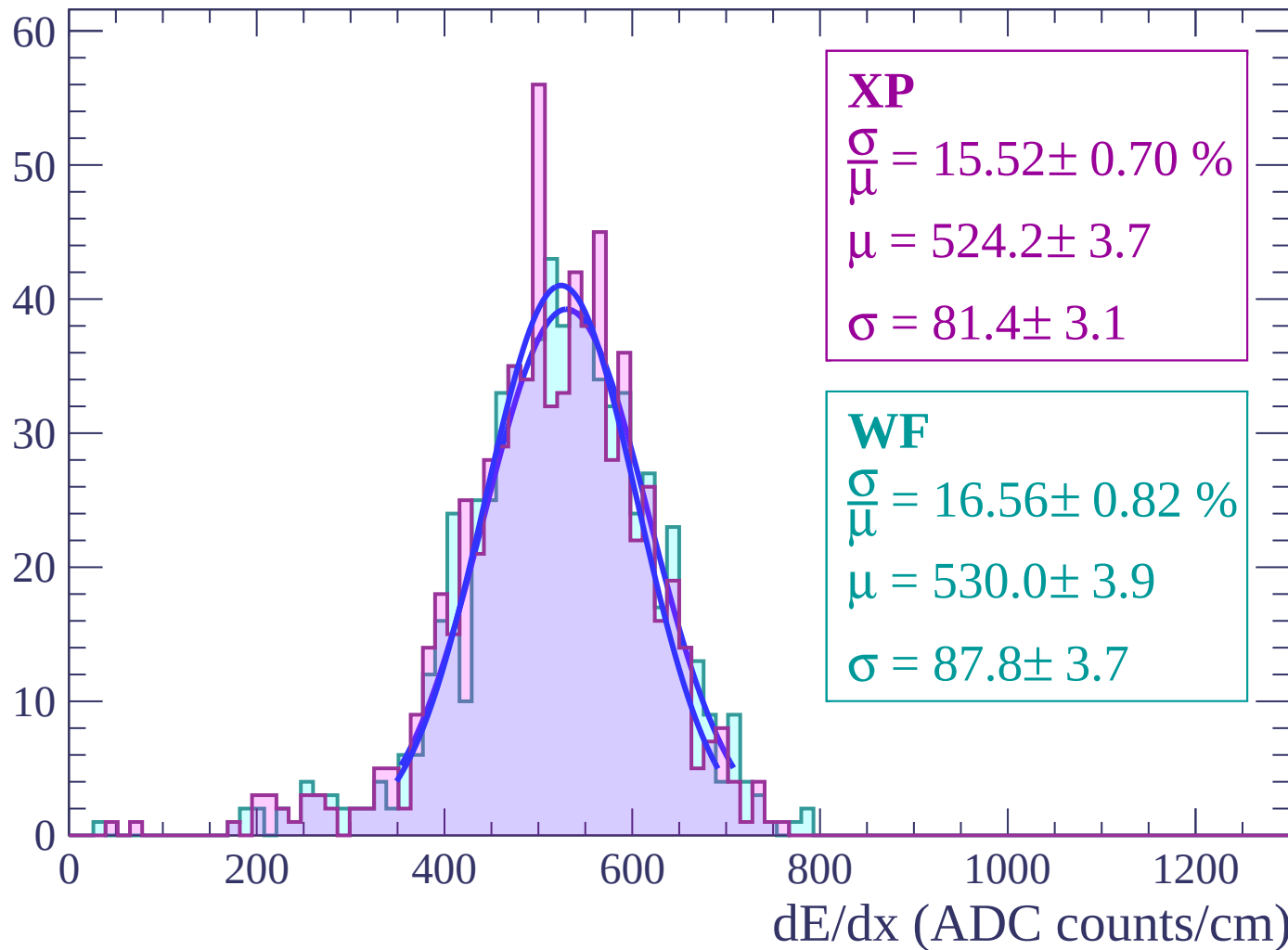
Energy loss | $54 < \theta < 56$

Count



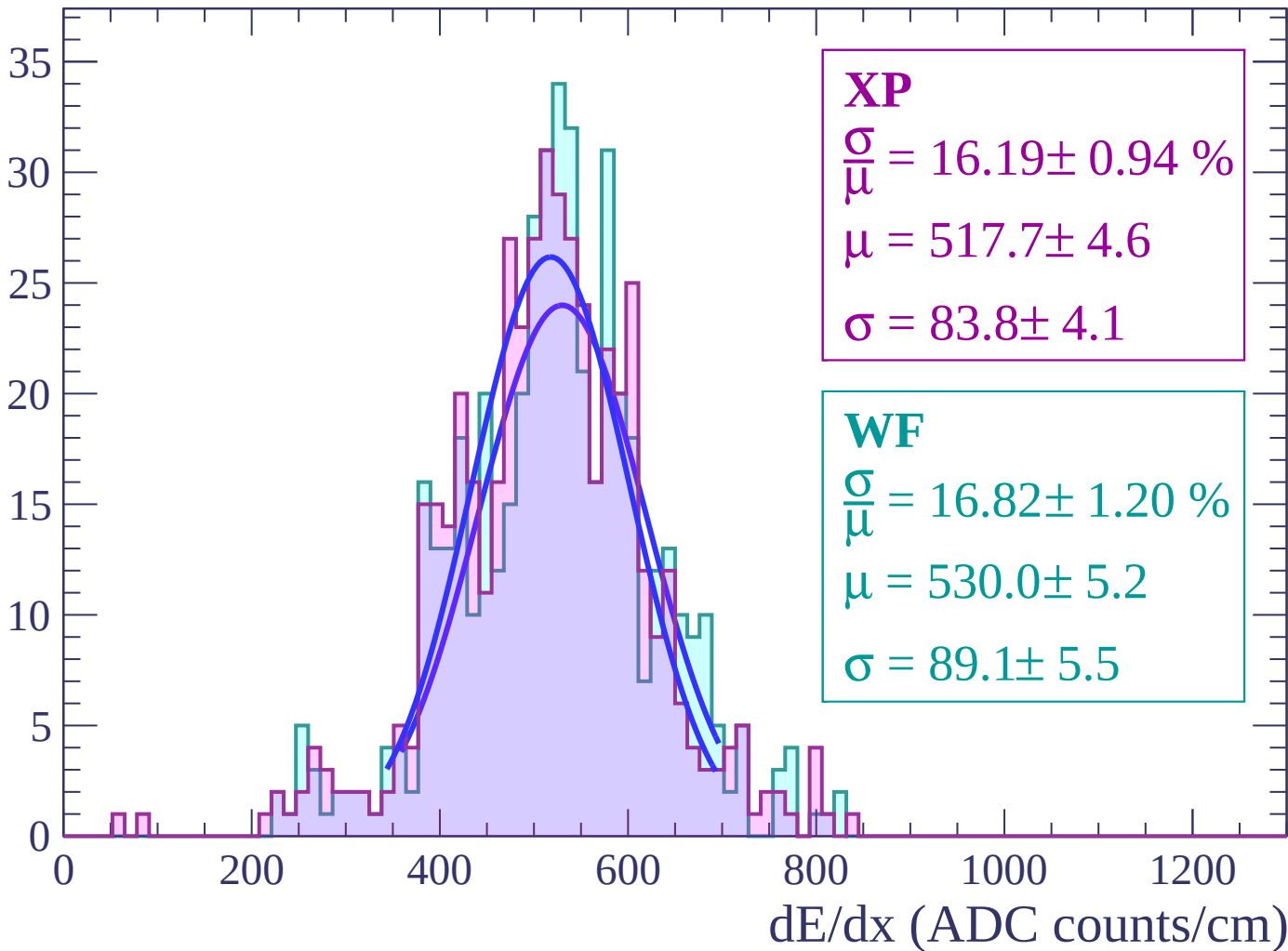
Energy loss | $56 < \theta < 58$

Count



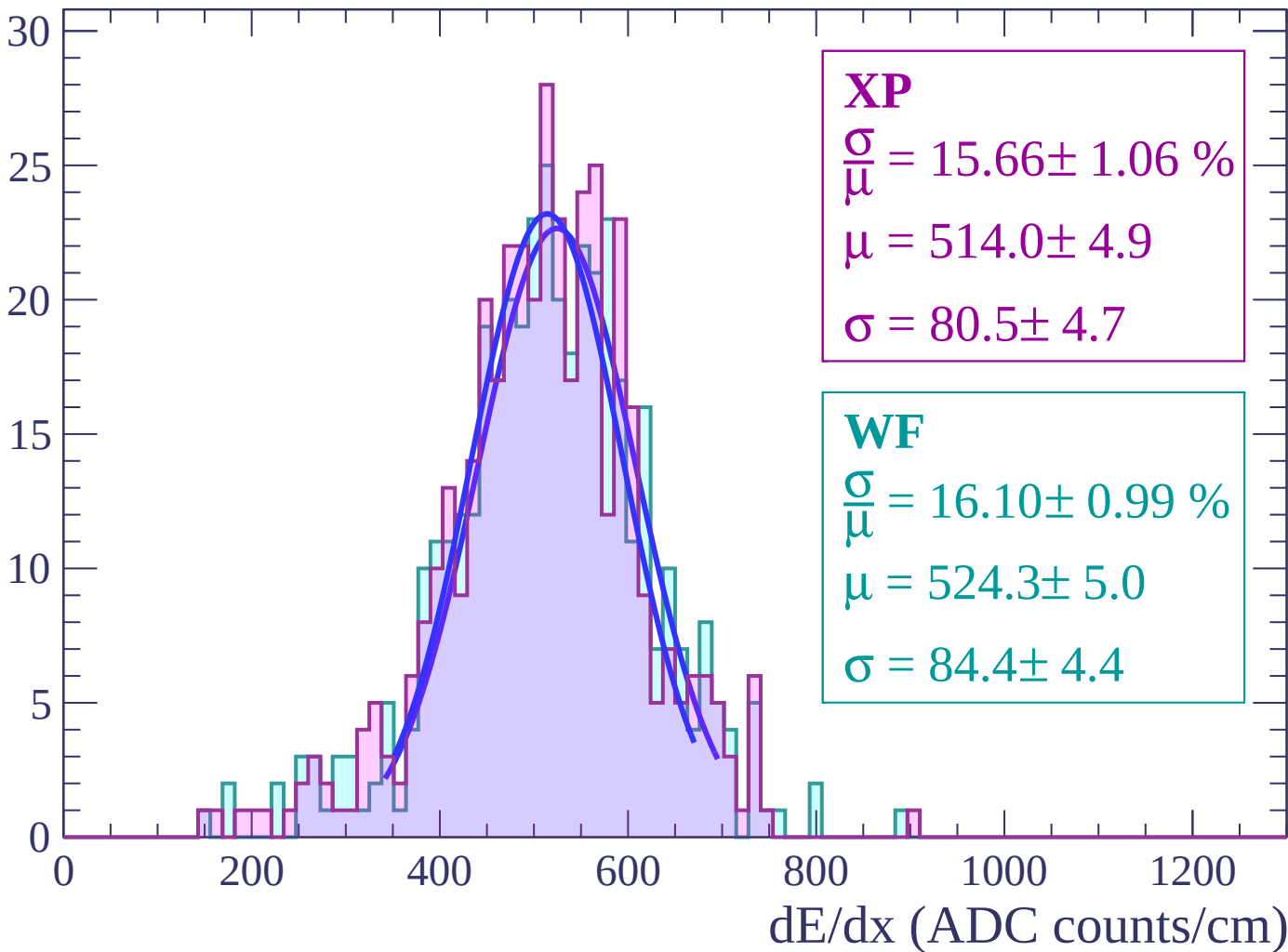
Energy loss | $58 < \theta < 60$

Count



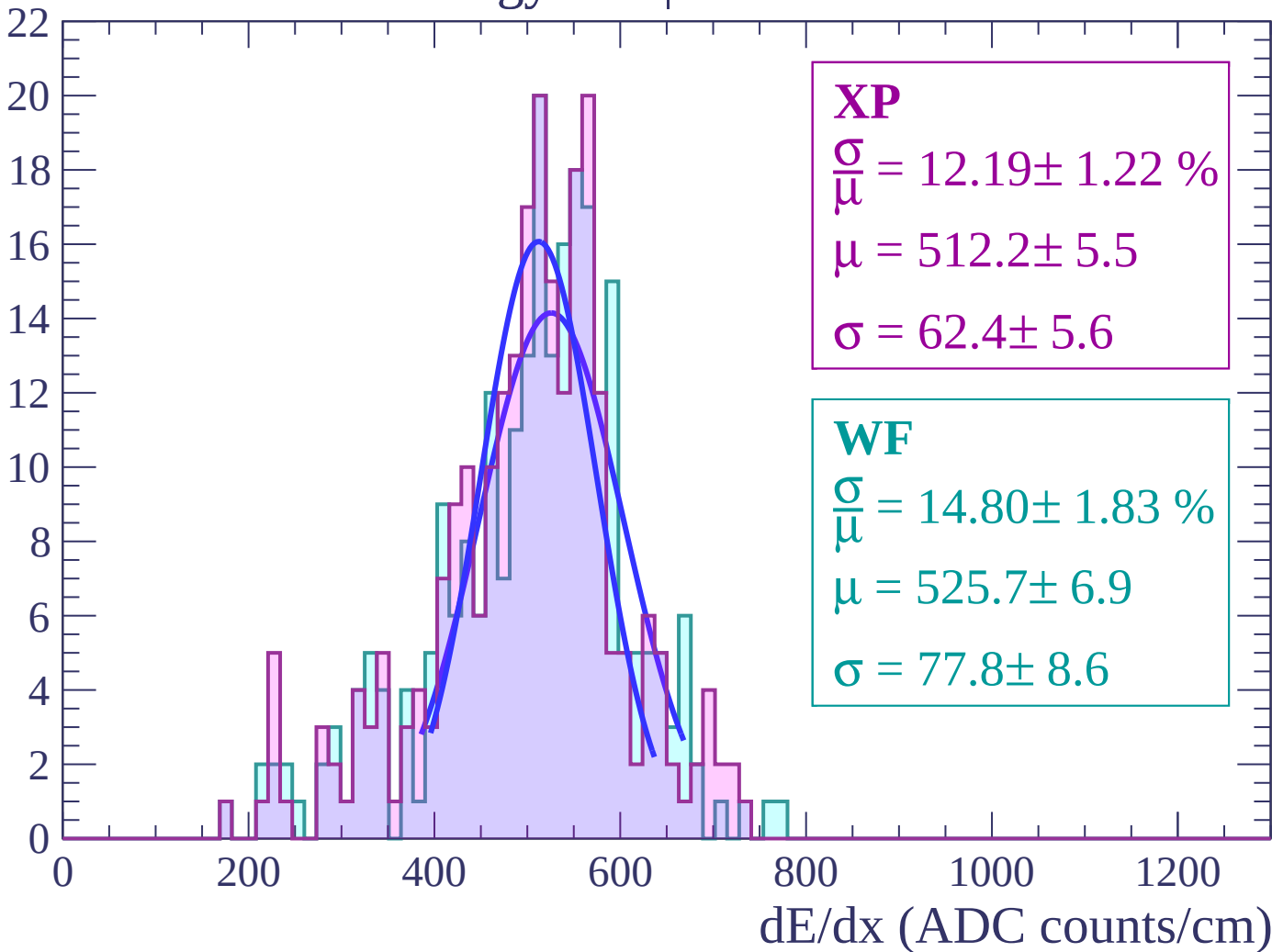
Energy loss | $60 < \theta < 62$

Count



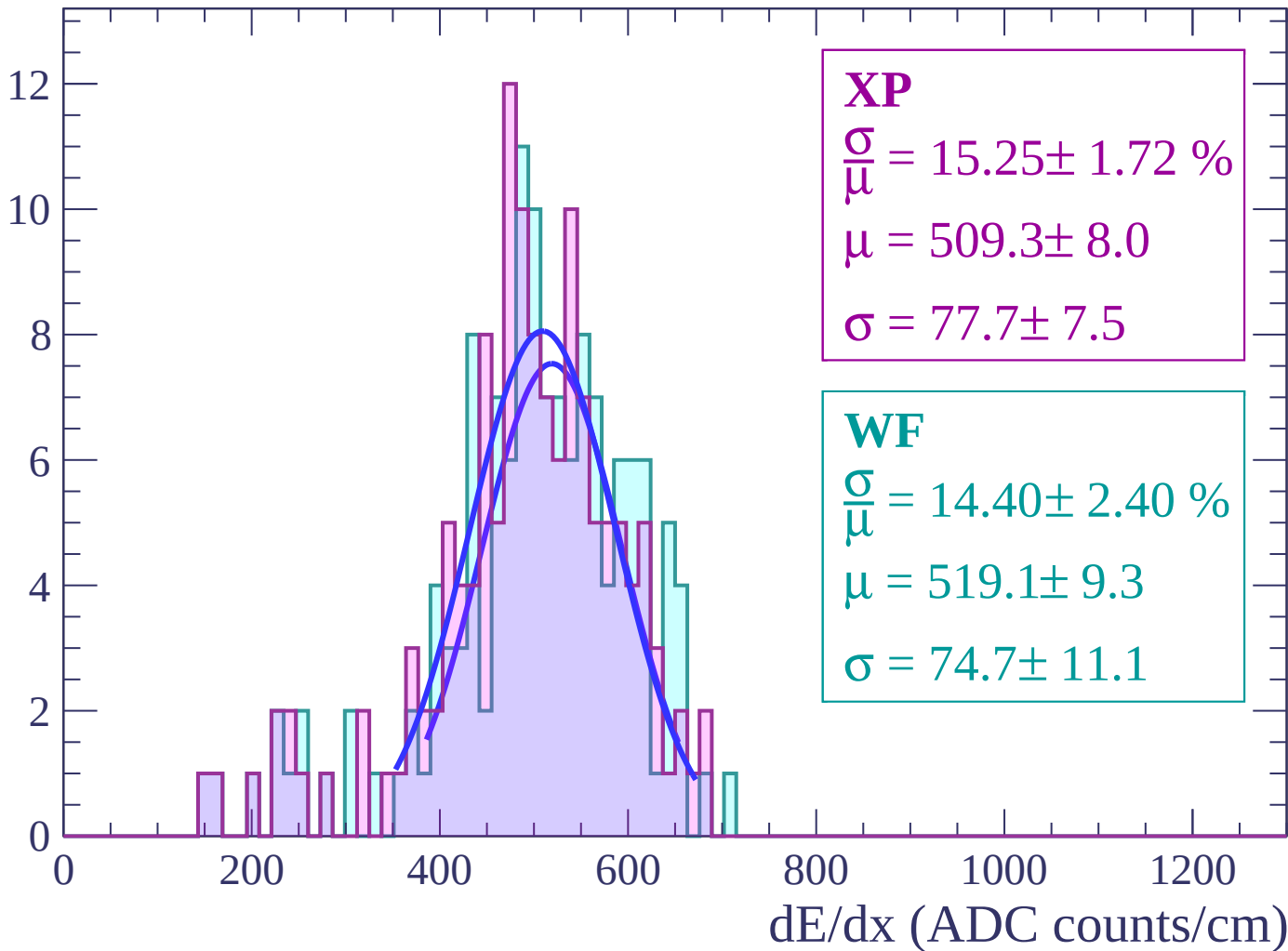
Energy loss | $62 < \theta < 64$

Count



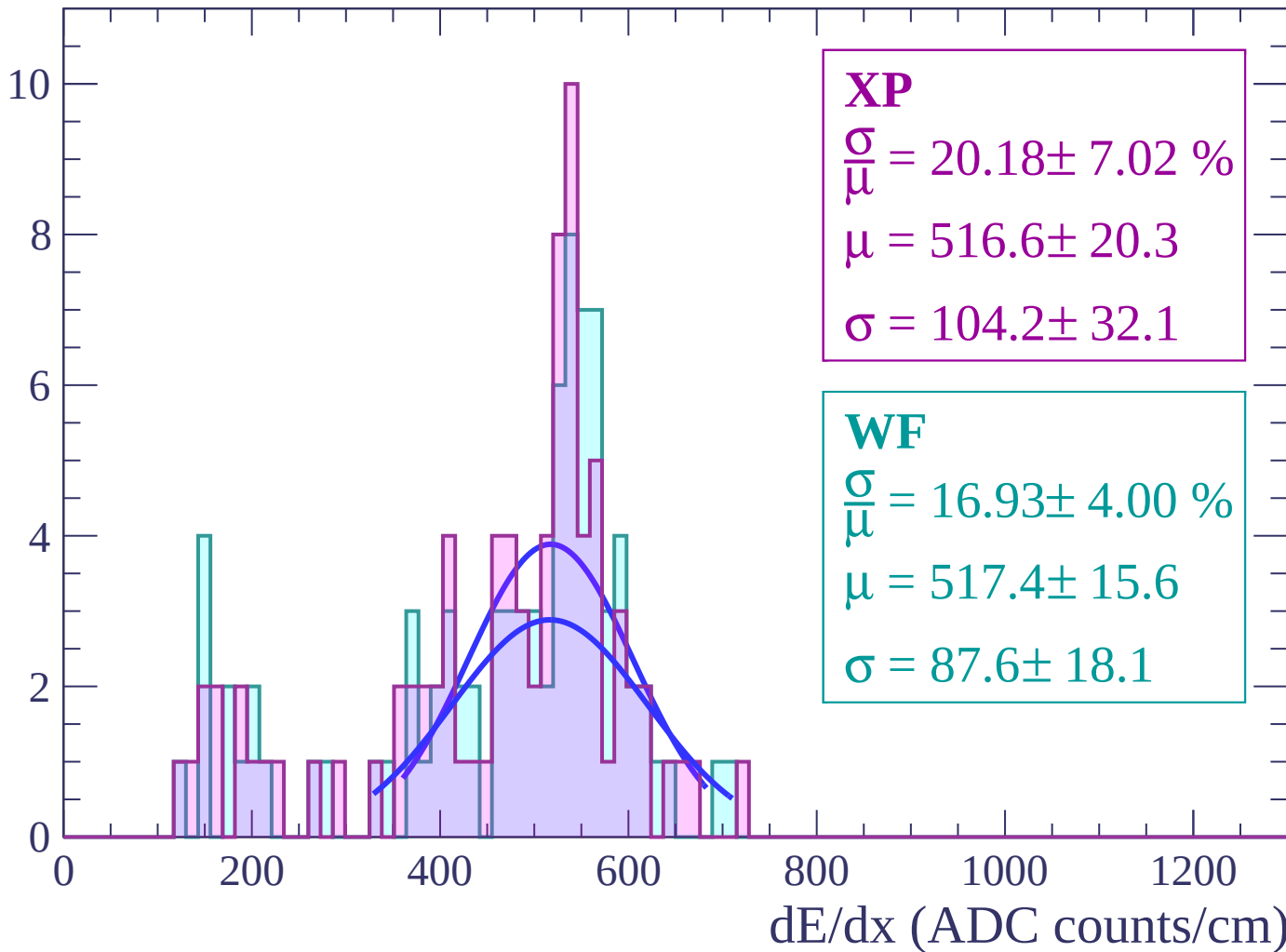
Energy loss | $64 < \theta < 66$

Count



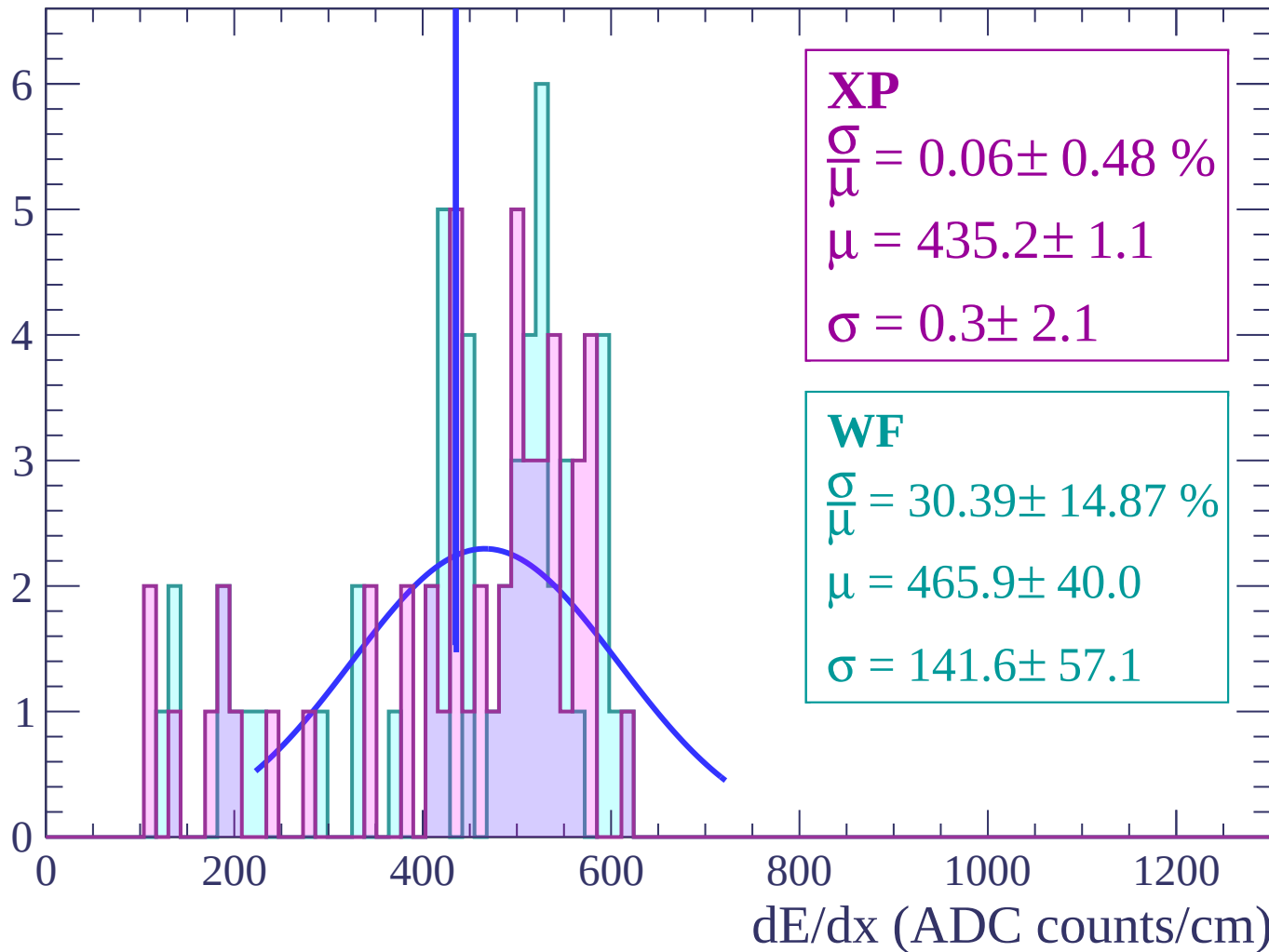
Energy loss | $66 < \theta < 68$

Count



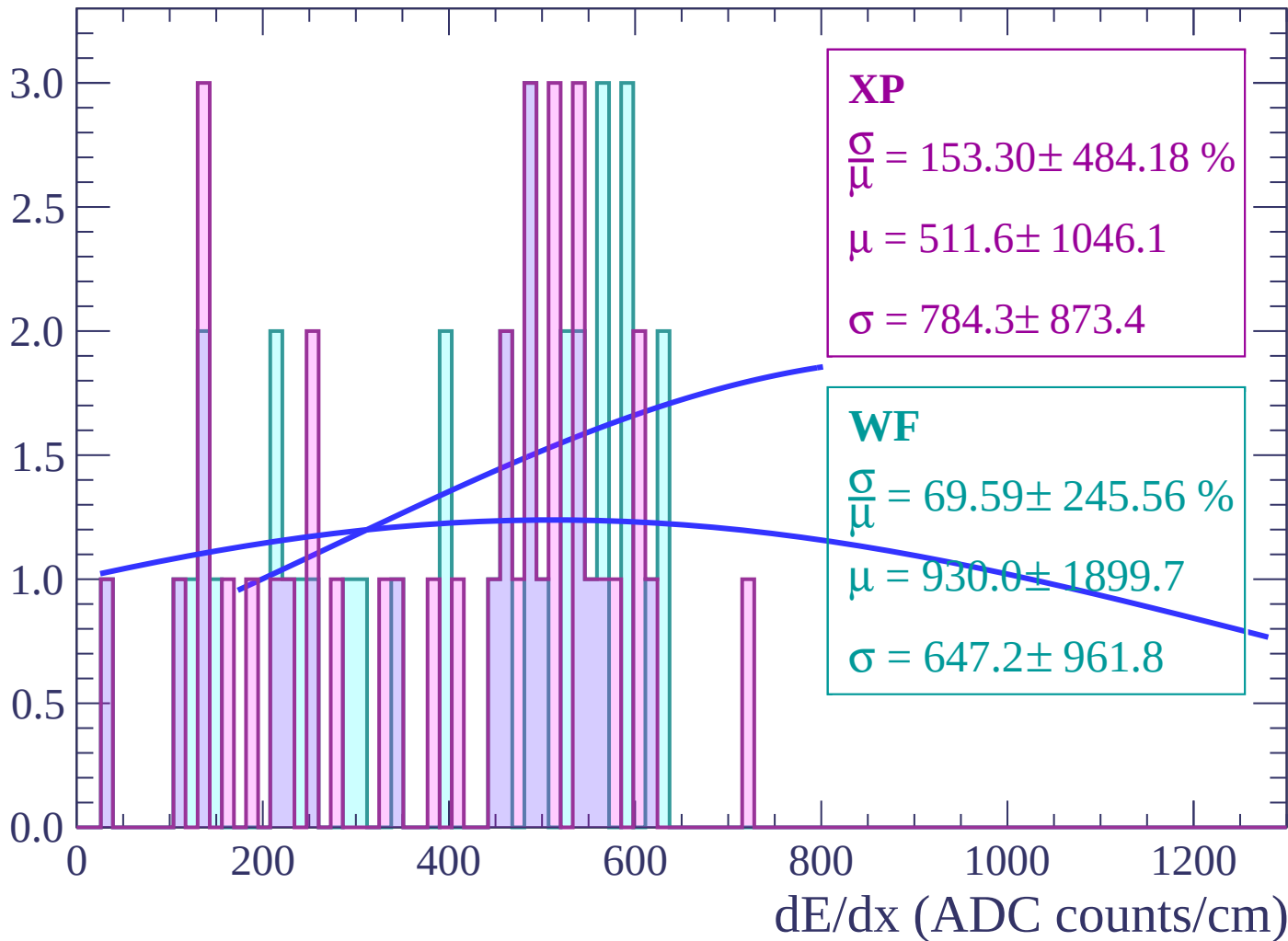
Energy loss | $68 < \theta < 70$

Count



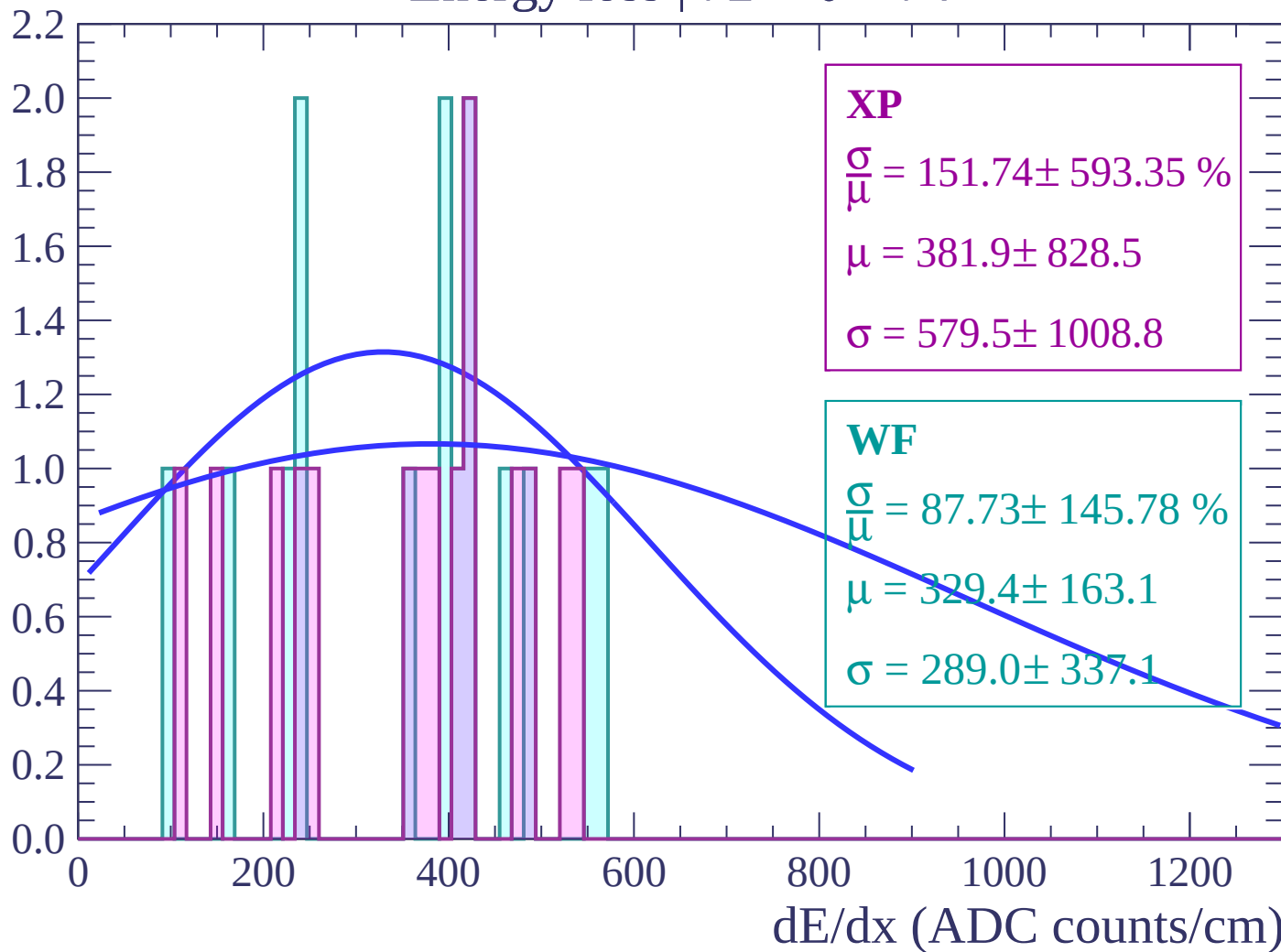
Energy loss | $70 < \theta < 72$

Count

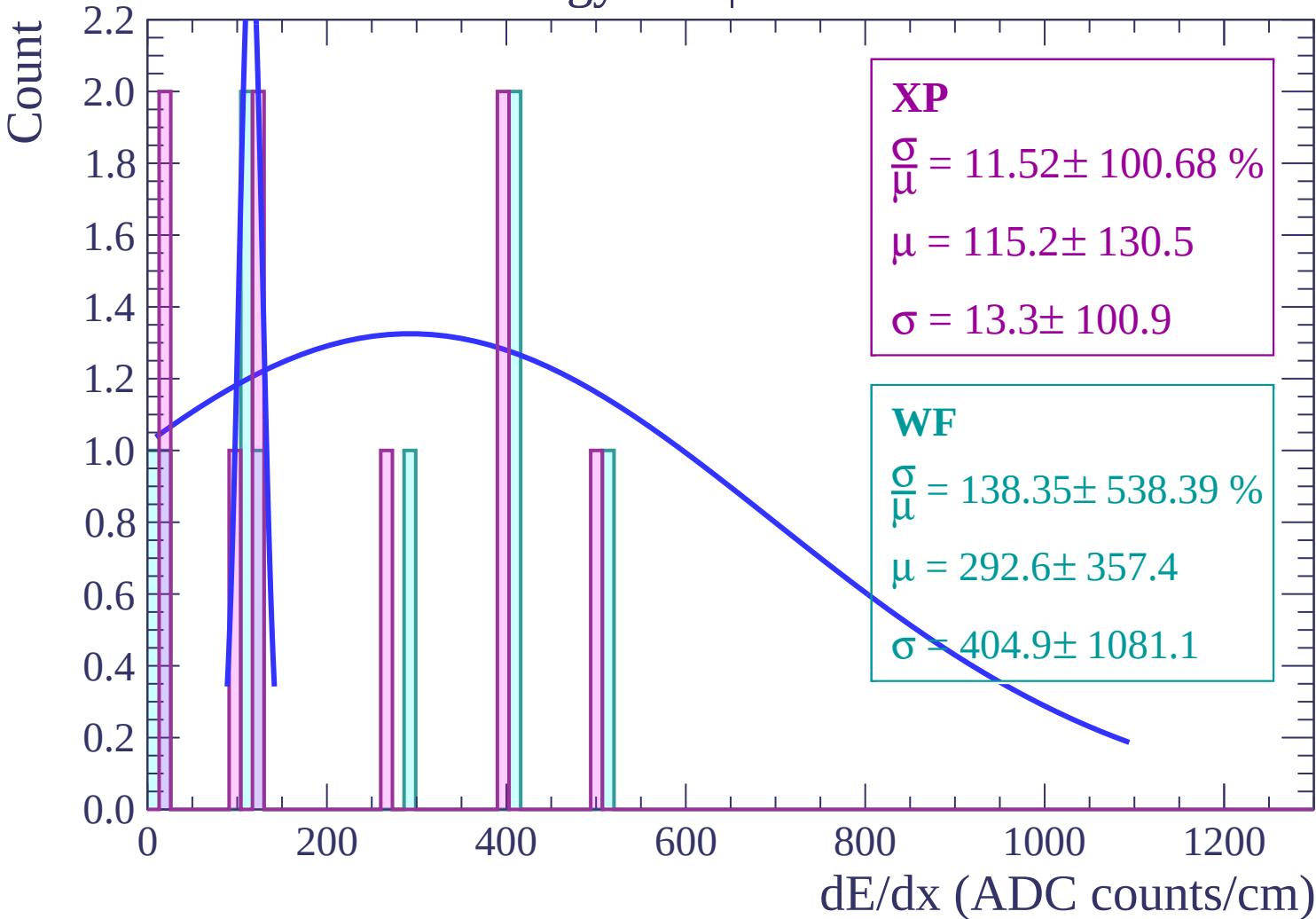


Energy loss | $72 < \theta < 74$

Count

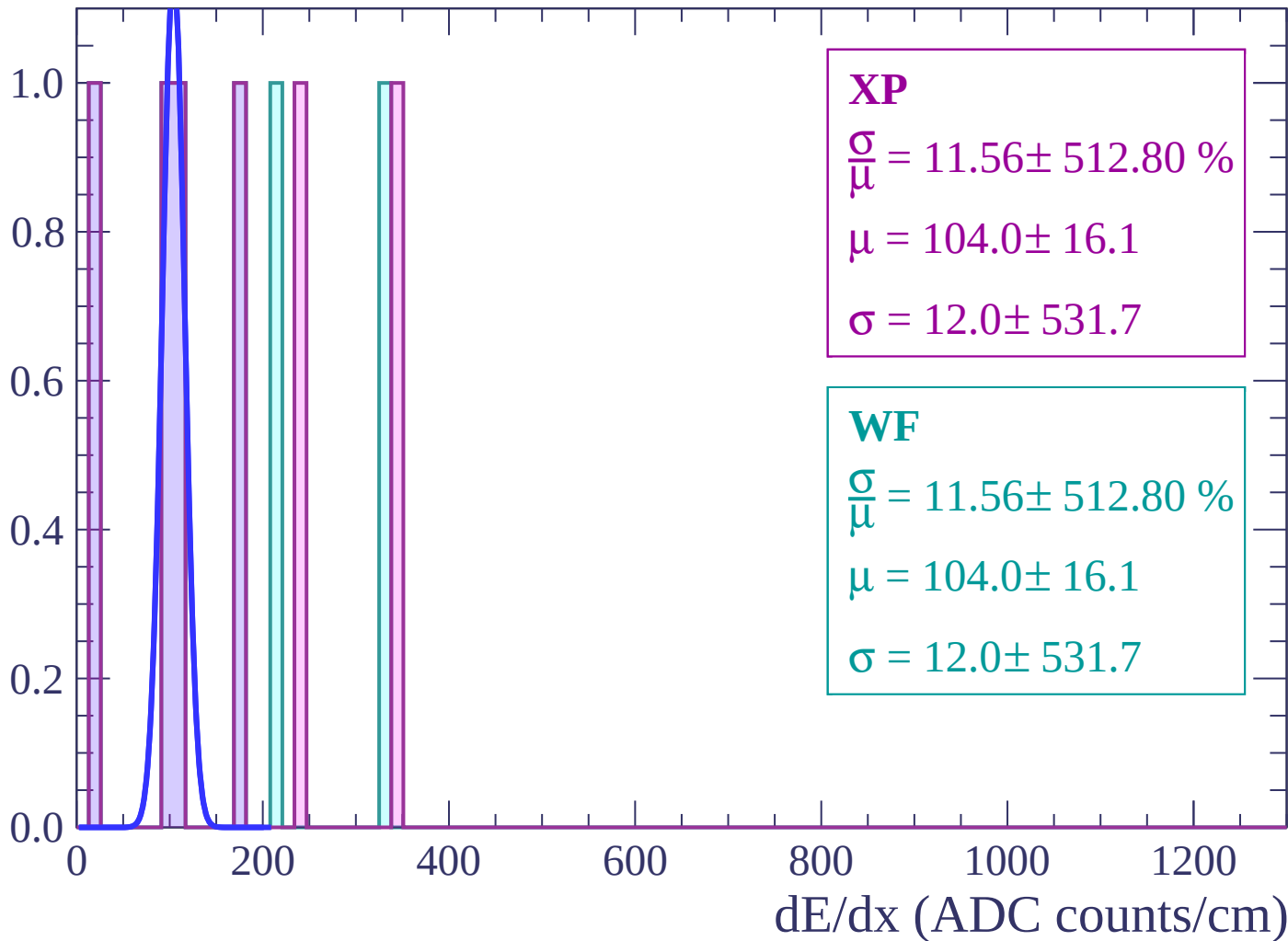


Energy loss | $74 < \theta < 76$



Energy loss | $76 < \theta < 78$

Count



Energy loss | $78 < \theta < 80$

Count

1.0

0.8

0.6

0.4

0.2

0.0

0

200

400

600

800

1000

1200

dE/dx (ADC counts/cm)

XP

$\frac{\sigma}{\bar{\mu}} = \text{-nan} \pm \text{-nan} \%$

$\mu = 0.0 \pm 0.0$

$\sigma = 0.0 \pm 0.0$

WF

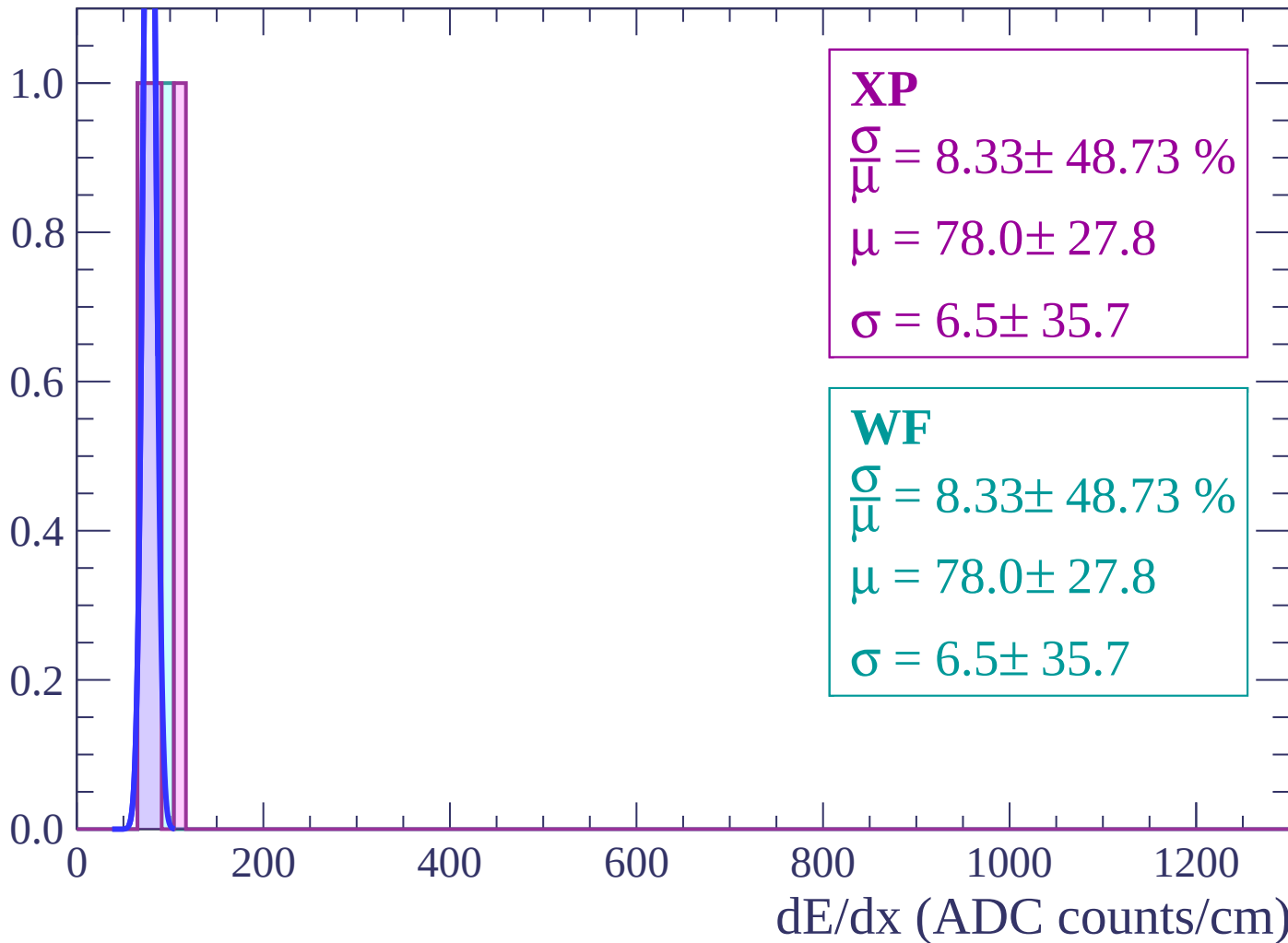
$\frac{\sigma}{\bar{\mu}} = \text{-nan} \pm \text{-nan} \%$

$\mu = 0.0 \pm 0.0$

$\sigma = 0.0 \pm 0.0$

Energy loss | $80 < \theta < 82$

Count



Energy loss | $82 < \theta < 84$

Count

1.0
0.9
0.8
0.7
0.6
0.5
0.4
0.3
0.2
0.1
0.0

0

200

400

600

800

1000

1200

dE/dx (ADC counts/cm)

XP

$\frac{\sigma}{\mu} = \text{-nan} \pm \text{-nan} \%$

$\mu = 0.0 \pm 0.0$

$\sigma = 0.0 \pm 0.0$

WF

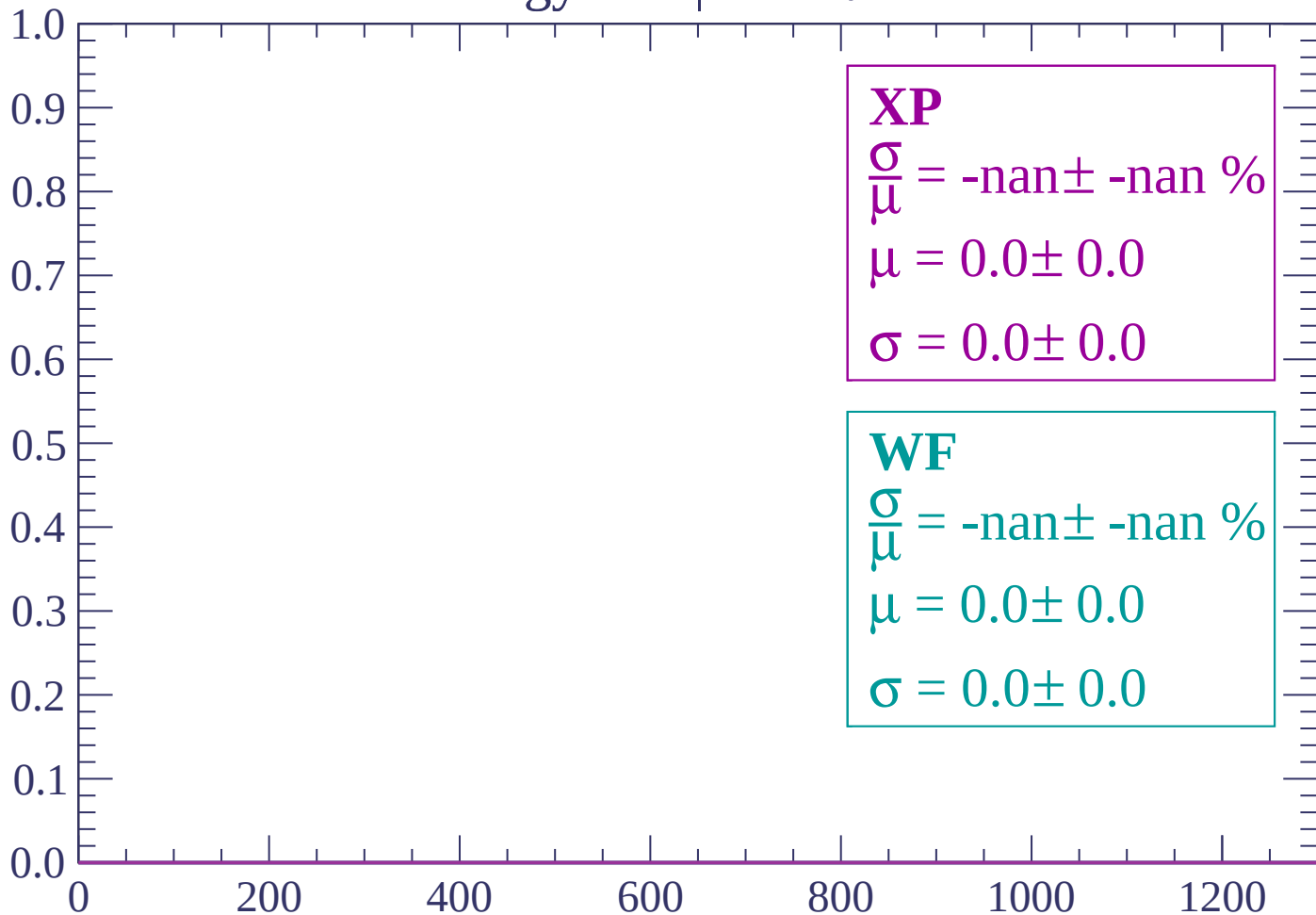
$\frac{\sigma}{\mu} = \text{-nan} \pm \text{-nan} \%$

$\mu = 0.0 \pm 0.0$

$\sigma = 0.0 \pm 0.0$

Energy loss | $84 < \theta < 86$

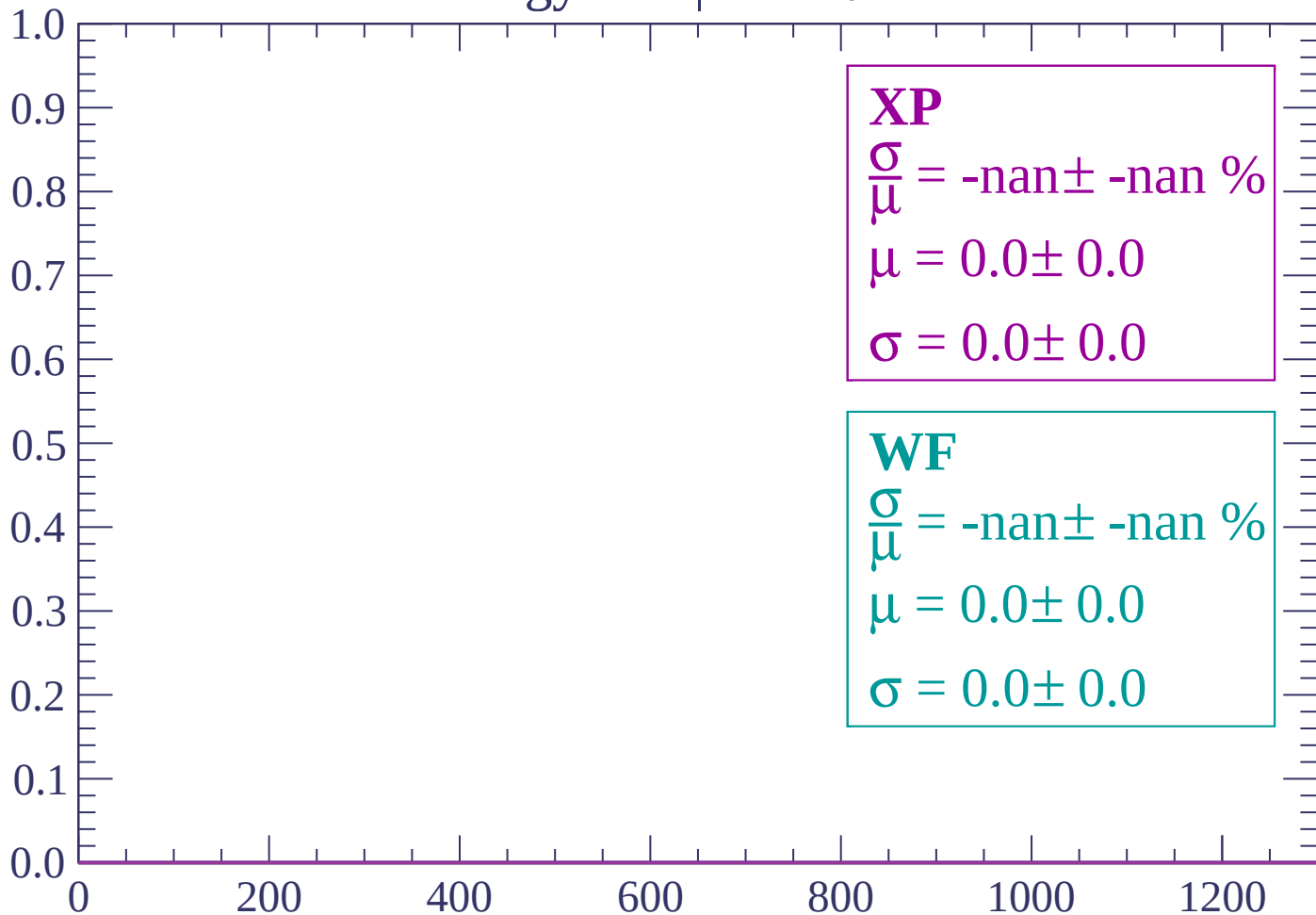
Count



dE/dx (ADC counts/cm)

Energy loss | $86 < \theta < 88$

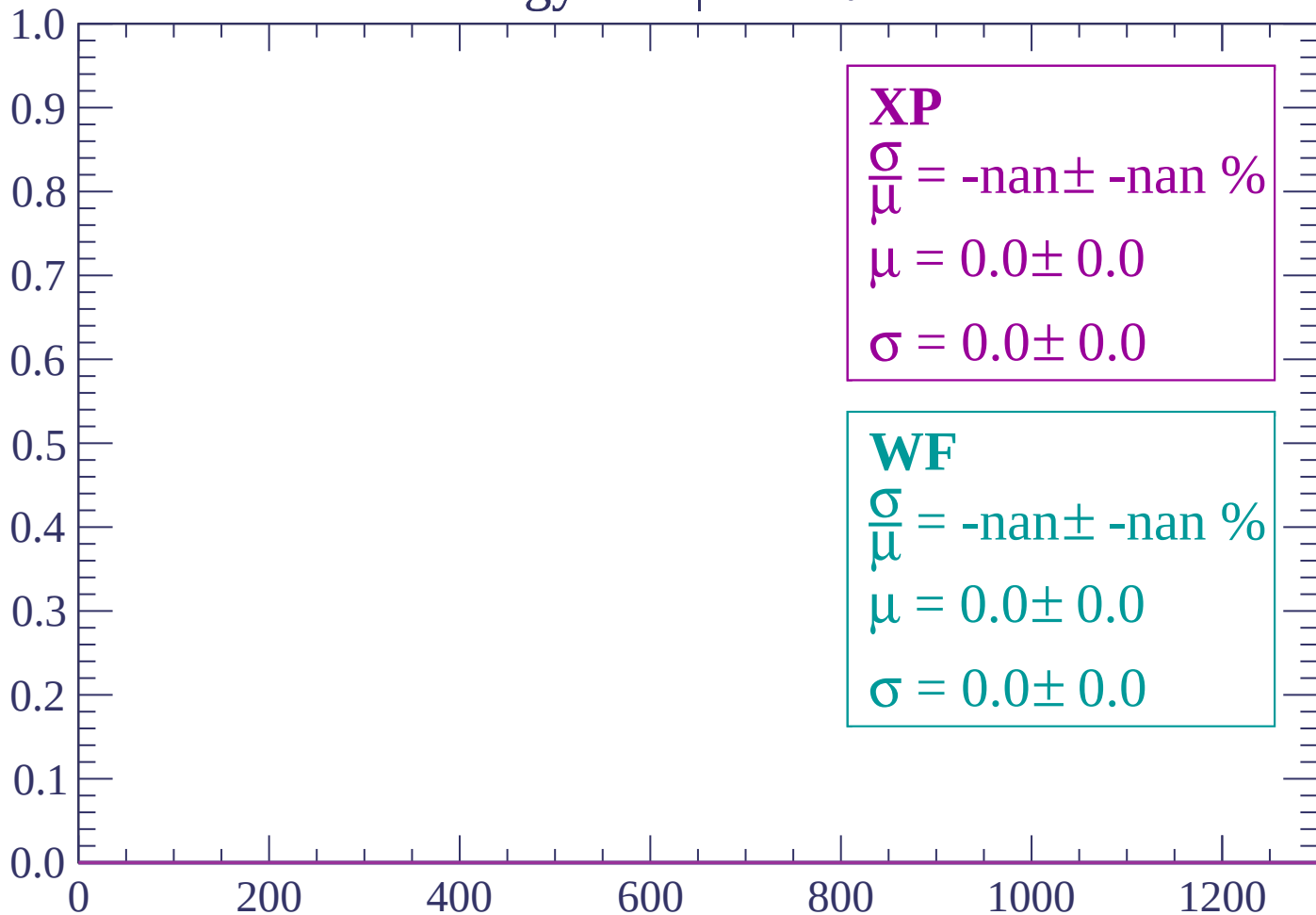
Count



dE/dx (ADC counts/cm)

Energy loss | $88 < \theta < 90$

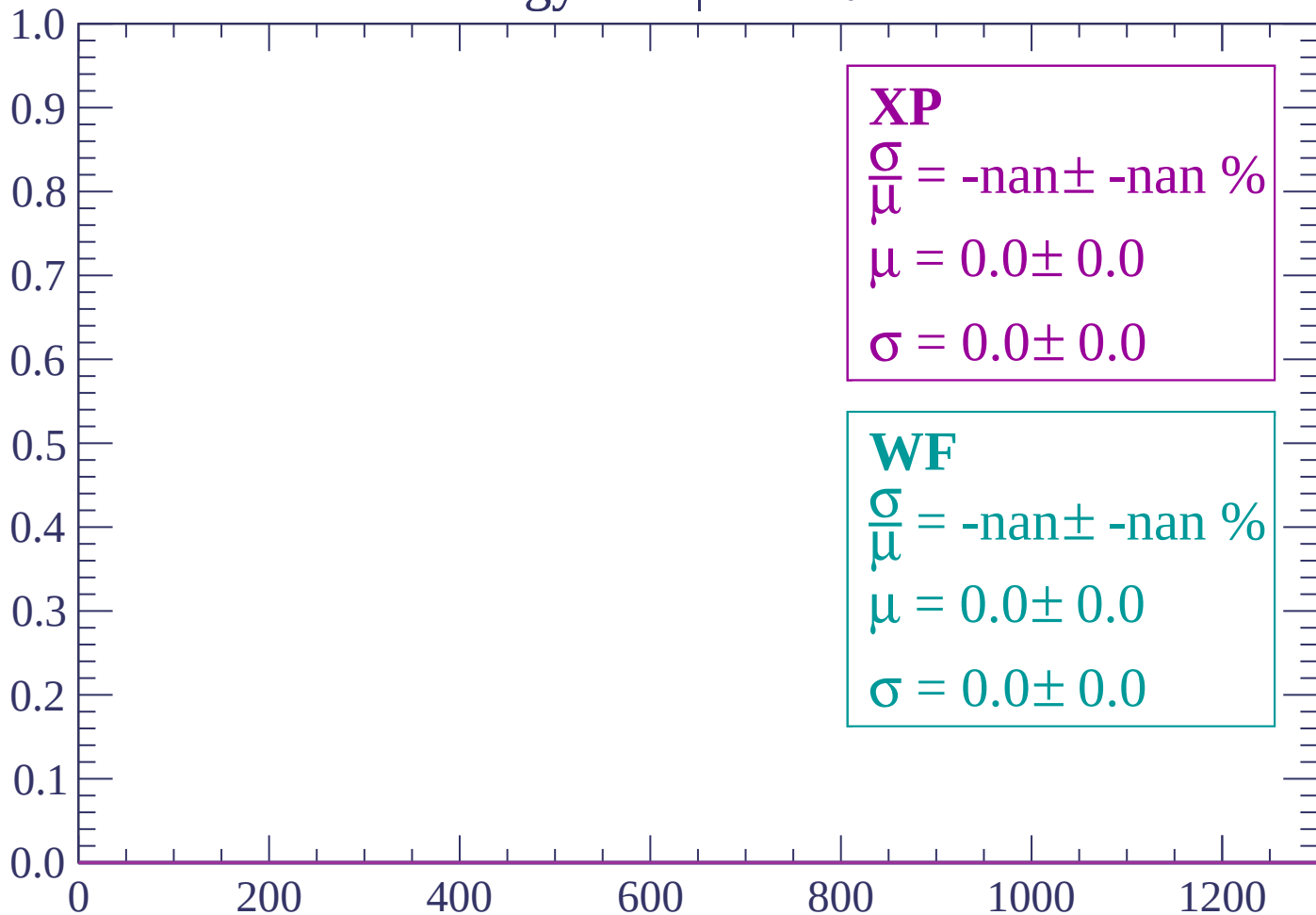
Count



dE/dx (ADC counts/cm)

Energy loss | $90 < \theta < 92$

Count



dE/dx (ADC counts/cm)

